

DERIVATIVES PRICING

A COMPUTATIONAL & THEORETICAL FRAMEWORK

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Part I

Mathematical Foundations

1 Measure Theory

1.1 Measurable Spaces

Measure theory is a branch of mathematics that studies measures and integrals on general spaces. A measure can be thought of as a generalisation of a length, area, or volume function to an arbitrary one-or multi-dimensional space.

In what follows, we let E be an arbitrary set, e.g., the interval $(0, 1]$ or the square $(0, 1] \times (0, 1]$. We wish to assign a "measure" (think of "length" or "area") to certain subsets of E . We can apply the usual set operations to subsets of E , as illustrated in Figure 1.

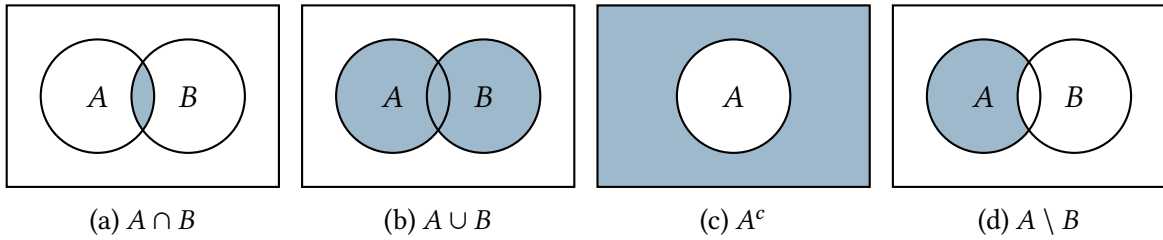


Figure 1: Venn diagrams of set operations.

Definition 1.1: Algebra

A collection $\mathcal{E}_0$ of subsets of E is called an *algebra on E* (or algebra of subsets of E) if

- (i) $E \in \mathcal{E}_0$,
- (ii) $F \in \mathcal{E}_0 \implies F^c := E \setminus F \in \mathcal{E}_0$,
- (iii) $F, G \in \mathcal{E}_0 \implies F \cup G \in \mathcal{E}_0$.

Tip 1.1: Algebra Properties

Note that $\emptyset = E^c \in \mathcal{E}_0$ and

$$F, G \in \mathcal{E}_0 \implies F \cap G = (F^c \cup G^c)^c \in \mathcal{E}_0.$$

Thus, an algebra on E is a family of subsets of E stable under finitely many set operations.

The collection of all subsets to which we wish to assign a measure is usually a σ -algebra:

Definition 1.2: σ -Algebra

A σ -algebra $\mathcal{E}$ on E is a collection of subsets of E that contains E itself, and that is closed under *complements* and *countable unions*; that is:

- (1) $E \in \mathcal{E}$.
- (2) If $A \in \mathcal{E}$, then also $A^c \in \mathcal{E}$.
- (3) If $A_1, A_2, \dots \in \mathcal{E}$, then also $\bigcup_n A_n \in \mathcal{E}$.

Thus, a σ -algebra on E is a family of subsets of E 'stable under any countable collection of set operations'.

Definition 1.3: Measurable Space

A pair $(E, \mathcal{E})$, where E is a set and $\mathcal{E}$ is a σ -algebra on E , is called a *measurable space*. An element of $\mathcal{E}$ is called a $\mathcal{E}$ -measurable subset of E .

Definition 1.4: p-System

A collection $\mathcal{C}$ of subsets of E is called a *p-system* (or π -system) if it is closed under finite intersections.

Definition 1.5: d-System

A collection $\mathcal{D}$ of subsets of E is called a *d-system* (or λ -system) if it satisfies the following three conditions:

- (1) $E \in \mathcal{D}$.
- (2) If $A, B \in \mathcal{D}$, with $A \supseteq B$, then $A \setminus B \in \mathcal{D}$.
- (3) If $(A_n) \in \mathcal{D}$, with $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots \subseteq \bigcup_n A_n := A$, then $A \in \mathcal{D}$.

Theorem 1.1: σ -Algebra as a d- and p-System

A collection of subsets of E is a σ -algebra if and only if it is both a *p-system* and a *d-system* on E .

Theorem 1.2: Monotone Class Theorem

If a d -system contains a p -system, then it also contains the σ -algebra generated by that p -system.

Definition 1.6: Product Space

The *product space* of two measurable spaces $(E, \mathcal{E})$ and $(F, \mathcal{F})$ is the measurable space $(E \times F, \mathcal{E} \otimes \mathcal{F})$, where $E \times F := \{(x, y) : x \in E, y \in F\}$ is the Cartesian product of E and F , and $\mathcal{E} \otimes \mathcal{F}$ is the *product σ -algebra* on $E \times F$ that is generated by the rectangle sets

$$A \times B, \quad A \in \mathcal{E}, B \in \mathcal{F}.$$

1.2 Measurable Functions

Now we have established the basics of measurable spaces, we are going to define a class of "well-behaved" functions between such spaces.

Definition 1.7: Measurable Function

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be two measurable spaces, and let f be a function from E to F . The function f is called $\mathcal{E}/\mathcal{F}$ -*measurable* if for all $B \in \mathcal{F}$ it holds that

$$\{x \in E : f(x) \in B\} \in \mathcal{E}. \quad (1.1)$$

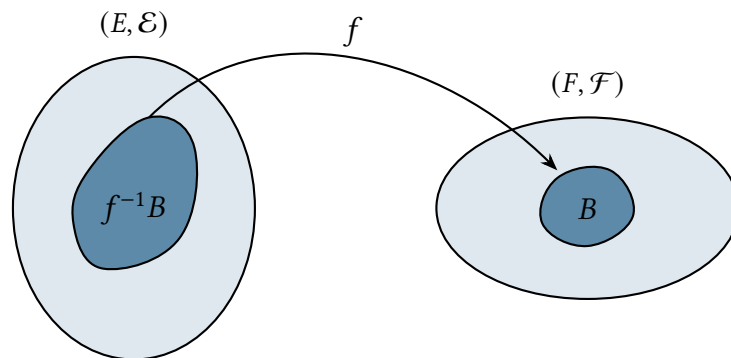


Figure 2: All elements in $f^{-1}B$ map into B .

Theorem 1.3: Measurable Function on a Generated σ -Algebra

A function $f : E \rightarrow F$ is $\mathcal{E}/\mathcal{F}$ -measurable if and only if Equation (1.1) holds for all $B \in \mathcal{F}_0$, with $\mathcal{F}_0$ generating $\mathcal{F}$.

Theorem 1.4: Composition of Measurable Functions

Let $(E, \mathcal{E})$, $(F, \mathcal{F})$, and $(G, \mathcal{G})$ be measurable spaces. If $f : E \rightarrow F$ and $g : F \rightarrow G$ are respectively $\mathcal{E}/\mathcal{F}$ - and $\mathcal{F}/\mathcal{G}$ -measurable, then $g \circ f : E \rightarrow G$ is $\mathcal{E}/\mathcal{G}$ -measurable.

Proof. Take any set $B \in \mathcal{G}$ and let $C := \{g \in B\} = g^{-1}(B)$. By the measurability of g , $C \in \mathcal{F}$. Moreover, since f is $\mathcal{E}/\mathcal{F}$ -measurable, the set $\{f \in C\}$ belongs to $\mathcal{E}$. But this set is the same as $\{g \circ f \in B\}$, as illustrated in Figure 3.

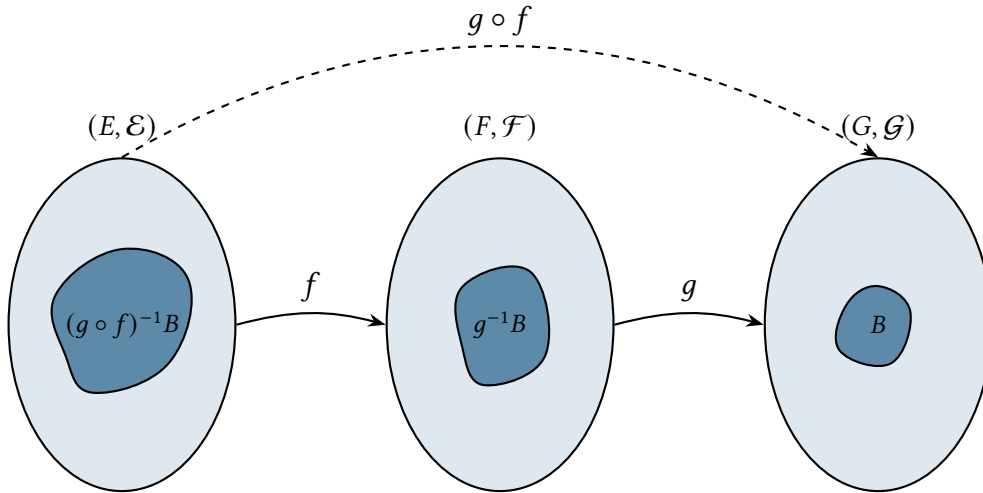


Figure 3: Composition of measurability

Thus, the latter set is a member of $\mathcal{E}$ for any $B \in \mathcal{G}$. In other words, $g \circ f$ is $\mathcal{E}/\mathcal{G}$ -measurable. $\square$

Let $(E, \mathcal{E})$ be a measurable space. We will often be interested, especially when defining integrals, in real-valued functions that are $\mathcal{E}/\mathcal{B}$ -measurable, where $\mathcal{B}$ is the Borel σ -algebra in $\mathbb{R}$. In fact, it will be convenient to consider the *extended real line* $\overline{\mathbb{R}} := [-\infty, +\infty]$, adding two “infinity” points to the set $\mathbb{R}$, and extending the arithmetic in a natural way, but leaving operations such as $\infty - \infty$ and ∞/∞ undefined. The corresponding Borel σ -algebra, denoted $\overline{\mathcal{B}}$, is generated by the p-system consisting of intervals $[-\infty, r]$, $r \in \mathbb{R}$.

Definition 1.8: Numerical Function

Let $(E, \mathcal{E})$ be a measurable space.

- A function $f : E \rightarrow \mathbb{R}$ is called a *real-valued function* on E .
- A function $f : E \rightarrow \overline{\mathbb{R}} := [-\infty, +\infty]$ is called a *numerical function* on E .
- An $\mathcal{E}/\overline{\mathcal{B}}$ -measurable numerical function f is said to be $\mathcal{E}$ -*measurable*; we write $f \in \mathcal{E}$. Also, $\mathcal{E}_+$ is the class of *positive $\mathcal{E}$ -measurable functions*.

We can further expand the collection of measurable functions that can be obtained from simple functions by taking limits of sequences of measurable functions. Let us first recall some definitions regarding sequences of real numbers.

Definition 1.9: Infimum and Supremum

Let (a_n) be a sequence of real numbers.

- (1) The *infimum*, $\inf a_n$, is the greatest element in $\overline{\mathbb{R}}$ that is less than or equal to all a_n . It can be $-\infty$.
- (2) The *supremum*, $\sup a_n$, is the smallest element in $\overline{\mathbb{R}}$ that is greater than or equal to all a_n . It can be ∞ .

If for some $N \in \mathbb{N}$ it holds that $\inf a_n = a_N$, then we can write $\inf a_n$ also as $\min a_n$. In that case, the set of indexes for which the minimum is attained is denoted by $\arg \min a_n$. In the same way, when the supremum is attained by some $N \in \arg \max a_n$, we can write $\max a_n$ instead of $\sup a_n$. Similarly, the infimum and supremum can be defined for any set $A \subseteq \overline{\mathbb{R}}$. In particular, if $A := \{a_n, n \in \mathbb{N}\}$, then $\inf A = \inf a_n$.

Definition 1.10: Liminf and Limsup

Let (a_n) be a sequence of real numbers.

- (1) The *limit inferior*, $\liminf a_n$, is the eventual lower bound for the sequence (a_n) ; that is, $\liminf a_n := \lim_{m \rightarrow \infty} \inf_{n \geq m} a_n = \sup_m \inf_{n \geq m} a_n$.
- (2) The *limit superior*, $\limsup a_n$, is the eventual upper bound for the sequence (a_n) ; that is, $\limsup a_n := \lim_{m \rightarrow \infty} \sup_{n \geq m} a_n = \inf_m \sup_{n \geq m} a_n$.

For a sequence (f_n) of numerical functions on a set E , we can define functions

$$\inf f_n, \quad \sup f_n, \quad \liminf f_n, \quad \limsup f_n$$

pointwise, considering for each x the sequence (a_n) defined by $a_n := f_n(x)$.

Obviously, $\liminf f_n \leq \limsup f_n$ (pointwise). If the two are equal, we write $\lim f_n$ for the limit, and write $f_n \rightarrow f$. If (f_n) is a sequence of functions that increases pointwise, then the limit $f := \lim f_n$ always exists. In this case we write $f_n \uparrow f$. A similar notation $f_n \downarrow f$ holds for a decreasing sequence of functions.

Proposition 1.1: Limits of Measurable Functions

Let (f_n) be a sequence of $\mathcal{E}$ -measurable numerical functions. Then, the numerical functions $\sup f_n, \inf f_n, \limsup f_n$, and $\liminf f_n$ are also $\mathcal{E}$ -measurable. In particular, limits of measurable functions (when $\limsup f_n = \liminf f_n$) are measurable as well.

Proof. (1) Let $f := \sup f_n$. Since sets of the form $[-\infty, r]$ form a p-system that generates the Borel σ -algebra on $\overline{\mathbb{R}}$, it suffices to show, by ??, that $f^{-1}[-\infty, r] \in \mathcal{E}$ for all r . The latter follows from the observation that

$$f^{-1}[-\infty, r] = \cap_n f_n^{-1}[-\infty, r],$$

where $f_n^{-1}[-\infty, r] \in \mathcal{E}$ by the measurability of f_n , and the fact that $\mathcal{E}$ is closed under countable intersections.

- (2) To show $\inf f_n \in \mathcal{E}$, we simply observe that $\inf f_n = -\sup(-f_n)$.
- (3) Since $\limsup f_n = \inf_m \sup_{n \geq m} f_n$, we can use Steps 1 and 2 above to conclude that $\limsup f_n \in \mathcal{E}$.
- (4) Similarly, since $\liminf f_n = \sup_m \inf_{n \geq m} f_n$, we have $\liminf f_n \in \mathcal{E}$.

□

Theorem 1.5: Approximation from Below

A positive numerical function on E is $\mathcal{E}$ -measurable if and only if it is the (pointwise) limit of an increasing sequence of simple functions.

Theorem 1.6: Positive and Negative Part of a Function

A function f is $\mathcal{E}$ -measurable if and only if both f^+ and f^- are.

Proof. To prove sufficiency, suppose f^+ and f^- are measurable. Consider the decomposition illustrated in Figure 4.

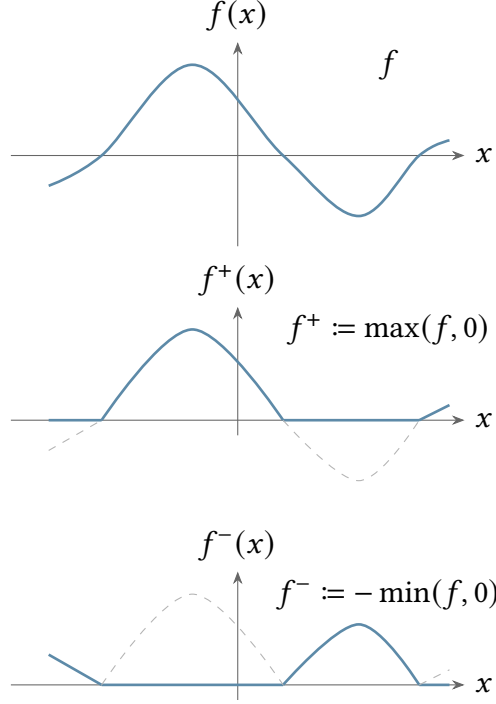


Figure 4: Decomposition of a function into positive and negative parts. Both f^+ and f^- are non-negative.

We observe the following: For $r \geq 0$, we have $\{f \leq r\} = \{f^+ \leq r\}$. For $r < 0$, we have $\{f \leq r\} = \{f^- \geq -r\}$. In both cases the inverse image of f is in $\mathcal{E}$. Hence, f is $\mathcal{E}$ -measurable.

Necessity is proved in a similar way; e.g., $\{f^+ \leq r\} = \{f \leq r\} \in \mathcal{E}$ for $r \geq 0$ and is equal to the empty set (also in $\mathcal{E}$) for $r < 0$. $\square$

The important point of the preceding results is that every measurable function can be viewed as the pointwise limit of simple functions, which themselves are constructed as linear combinations of indicator functions. As a consequence, arithmetic operations on $\mathcal{E}$ -measurable functions f and g , such as $f + g$, $f - g$, fg , f/g , $f \wedge g$, and $f \vee g$ are $\mathcal{E}$ -measurable as well, as long as these functions are well-defined (for example, $\infty - \infty$ and ∞/∞ are undefined).

As an example, if $f, g \in \mathcal{E}$, then f^+, f^-, g^+ and g^- are in $\mathcal{E}_+$. Moreover, each of these functions is the limit of a sequence of simple functions f_n^+, f_n^-, g_n^+ , and g_n^- , respectively. Thus, $f - g$ is the limit of the simple function $f_n^+ - f_n^- - g_n^+ + g_n^-$, which lies in $\mathcal{E}$. Hence, the limit is also in $\mathcal{E}$.

We conclude this section with the notion of a monotone class of functions.

Definition 1.11: Monotone Class of Functions

Let $(E, \mathcal{E})$ be a measurable space. A collection $\mathcal{M}$ of numerical functions on E is called a *monotone class* if it satisfies:

- (1) $\mathbb{1}_E \in \mathcal{M}$.
- (2) If $f, g \in \mathcal{M}$ are bounded and $a, b \in \mathbb{R}$, then $af + bg \in \mathcal{M}$.
- (3) If (f_n) is a sequence of positive functions in $\mathcal{M}$ that increases to some f , then $f \in \mathcal{M}$.

Theorem 1.7: Monotone Class Theorem for Functions

Let $\mathcal{M}$ be a monotone class of functions on E and let $\mathcal{C}$ be a p -system that generates $\mathcal{E}$.

If $\mathbb{1}_A \in \mathcal{M}$ for every $A \in \mathcal{C}$, then $\mathcal{M}$ includes

- all positive $\mathcal{E}$ -measurable functions,
- all bounded $\mathcal{E}$ -measurable functions.

1.3 Measures

Let $(E, \mathcal{E})$ be a measurable space. We wish to assign a measure to all sets in $\mathcal{E}$ that has the same properties as an area or length measure. Particularly, any measure should have the property that the measure of the union of a disjoint (i.e., non-overlapping) collection of sets is equal to the sum of the measures for the individual sets, as illustrated in [Figure 5](#).

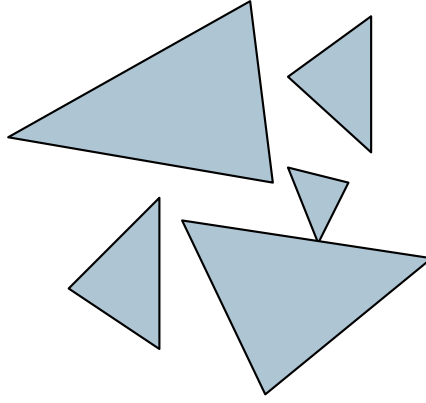


Figure 5: Any measure has the same properties as the area measure. For example, the total area of the non-overlapping triangles is the sum of the areas of the individual triangles.

Definition 1.12: Measure

Let $(E, \mathcal{E})$ be a measurable space. A *measure* μ on $\mathcal{E}$ is a set function

$$\mu : \mathcal{E} \rightarrow [0, \infty]$$

with $\mu(\emptyset) = 0$, such that for every sequence (A_n) of disjoint sets in $\mathcal{E}$,

$$\mu(\cup_n A_n) = \sum_n \mu(A_n). \quad (1.2)$$

Theorem 1.8: Carathéodory's Extension Theorem

Let E be a set with an algebra $\mathcal{E}_0$ thereon. Every pre-measure $\mu_0 : \mathcal{E}_0 \rightarrow [0, \infty]$ can be extended to a measure μ on $(E, \sigma(\mathcal{E}_0))$.

Lemma 1.1: Elementary Inequalities

Let $(E, \mathcal{E}, \mu)$ be a measure space. Then

- (a) $\mu(A \cup B) \leq \mu(A) + \mu(B) \quad (A, B \in \mathcal{E}),$
- (b) $\mu\left(\bigcup_{i \leq n} F_i\right) \leq \sum_{i \leq n} \mu(F_i) \quad (F_1, F_2, \dots, F_n \in \mathcal{E}).$

Furthermore, if $\mu(E) < \infty$, then

- (c) $\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B) \quad (A, B \in \mathcal{E}),$
- (d) (inclusion-exclusion formula): for $F_1, F_2, \dots, F_n \in \mathcal{E},$

$$\begin{aligned} \mu\left(\bigcup_{i \leq n} F_i\right) &= \sum_{i \leq n} \mu(F_i) - \sum_{i < j \leq n} \mu(F_i \cap F_j) \\ &\quad + \sum_{i < j < k \leq n} \mu(F_i \cap F_j \cap F_k) - \dots + (-1)^{n-1} \mu(F_1 \cap F_2 \cap \dots \cap F_n), \end{aligned}$$

successive partial sums alternating between over- and under-estimates.

Proposition 1.2: Properties of a Measure

Let μ be a measure on a measurable space $(E, \mathcal{E})$, and let A, B and $A_1, A_2, \dots$ be measurable sets. Then, the following hold:

- (1) (Monotonicity): $A \subseteq B \implies \mu(A) \leq \mu(B).$
- (2) (Continuity from below): If $A_1, A_2, \dots$ is increasing (i.e., $A_1 \subseteq A_2 \subseteq \dots$), then

$$\lim_n \mu(A_n) = \mu(\cup_n A_n).$$

- (3) (Countable subadditivity): $\mu(\cup_n A_n) \leq \sum_n \mu(A_n).$

Theorem 1.9: Uniqueness of Finite Measures

Let $\mathcal{C}$ be a p -system that generates $\mathcal{E}$. If μ and ν are two measures on $(E, \mathcal{E})$ with $\mu(E) = \nu(E) < \infty$, then

$$\mu(A) = \nu(A) \text{ for all } A \in \mathcal{C} \implies \mu(A) = \nu(A) \text{ for all } A \in \mathcal{E}.$$

Definition 1.13: Product Measure

Let $(E, \mathcal{E}, \mu)$ and $(F, \mathcal{F}, \nu)$ be two measure spaces. The *product measure* of μ and ν on the product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$ is defined as the measure π with

$$\pi(A \times B) := \mu(A)\nu(B), \quad A \in \mathcal{E}, B \in \mathcal{F}. \quad (1.3)$$

This measure π is often written as $\mu \otimes \nu$.

1.4 Integrals

We combine measurable functions and measures to define integrals in a unified and elegant way. We combine measurable functions and measures to define integrals in a unified and elegant way. The procedure for building up proofs for measurable functions from indicator functions, simple functions, and limits thereof is often referred to as “the standard machine”.

Proposition 1.3: Insensitivity of the Integral

Let $(E, \mathcal{E}, \mu)$ be a measure space and let f and g be positive $\mathcal{E}$ -measurable functions that are equal μ -almost everywhere. Then, $\mu f = \mu g$.

Theorem 1.10: Properties of an Integral

Below a, b are in $\mathbb{R}_+$ and f, g, f_n are in $\mathcal{E}_+$.

- (1) (Positivity and Monotonicity): $\mu f \geq 0$, and $f = 0 \implies \mu f = 0$. Also, $f \leq g \implies \mu f \leq \mu g$.
- (2) (Linearity): $\mu(af + bg) = a\mu f + b\mu g$.
- (3) (Monotone convergence): If $f_n \uparrow f$, then $\mu f_n \uparrow \mu f$.

Theorem 1.11: Measures from Linear Functionals

Let $L : \mathcal{E}_+ \rightarrow \overline{\mathbb{R}}_+$. Then, there exists a unique measure μ on $(E, \mathcal{E})$ such that $L(f) = \mu f$ if and only if

- (1) $f = 0 \implies L(f) = 0$,
- (2) $f, g \in \mathcal{E}_+$ and $a, b \in \mathbb{R}_+ \implies L(af + bg) = aL(f) + bL(g)$,
- (3) $(f_n) \in \mathcal{E}_+$ and $f_n \uparrow f \implies L(f_n) \uparrow L(f)$.

Theorem 1.12: Integrals with Densities

Let $\nu := p\mu$. Then, for every $f \in \mathcal{E}_+$, we have $\nu f = \mu(pf)$.

Theorem 1.13: Radon–Nikodym

Let ν and μ be two measures on $(E, \mathcal{E})$ such that μ is σ -finite and $\nu \ll \mu$. Then, there exists a $p \in \mathcal{E}_+$ such that $\nu = p\mu$. Moreover, p is unique up to equivalence; that is, if there is another $\tilde{p} \in \mathcal{E}_+$ such that $\nu = \tilde{p}\mu$, then $p = \tilde{p}$, μ -almost everywhere.

Theorem 1.14: Integration with Respect to Image Measures

Let $\nu = \mu \circ h^{-1}$ be the image measure of μ under h . Then, for every positive $\mathcal{F}$ -measurable function f ,

$$\nu f = (\mu \circ h^{-1})f = \int_F f d\nu = \int_E f \circ h d\mu = \mu(f \circ h).$$

Definition 1.14: Transition Kernel

Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. A *transition kernel* from $(E, \mathcal{E})$ into $(F, \mathcal{F})$ is a mapping $K : E \times \mathcal{F} \rightarrow \overline{\mathbb{R}}_+$ such that:

- (1) $K(\cdot, B)$ is $\mathcal{E}$ -measurable for every $B \in \mathcal{F}$.
- (2) $K(x, \cdot)$ is a measure on $(F, \mathcal{F})$ for every $x \in E$.

If $(F, \mathcal{F}) = (E, \mathcal{E})$ and $K(x, \cdot)$ is a probability measure for every $x \in E$, K is said to be a *probability transition kernel* on $(E, \mathcal{E})$.

Theorem 1.15: Measures and Functions from Kernels

Let K be a transition kernel from $(E, \mathcal{E})$ into $(F, \mathcal{F})$.

- (1) For all $f \in \mathcal{F}_+$ the function Kf defined by $(Kf)(x) := \int_F K(x, dy)f(y)$, $x \in E$, is in $\mathcal{E}_+$.
- (2) For every measure μ on $(E, \mathcal{E})$ the set function μK defined by $(\mu K)(B) := \int_E \mu(dx)K(x, B)$, $B \in \mathcal{F}$, is a measure on $(F, \mathcal{F})$.
- (3) For every measure μ on $(E, \mathcal{E})$ and $f \in \mathcal{F}_+$,

$$(\mu K)f = \mu(Kf) = \int_E \mu(dx) \int_F K(x, dy)f(y).$$

Theorem 1.16: Fubini

Let μ and ν be Σ -finite measures on $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively.

- (1) There exists a unique Σ -finite measure π on the product space $(E \times F, \mathcal{E} \otimes \mathcal{F})$ such that for every positive f in $\mathcal{E} \otimes \mathcal{F}$,

$$\pi f = \int_E \mu(dx) \int_F \nu(dy)f(x, y) = \int_F \nu(dy) \int_E \mu(dx)f(x, y). \quad (1.4)$$

- (2) If $f \in \mathcal{E} \otimes \mathcal{F}$ and is π -integrable, then $f(x, \cdot)$ is ν -integrable for μ -almost every x , and $f(\cdot, y)$ is μ -integrable for ν -almost every y and [Equation \(1.4\)](#) holds again.

2 Probability

2.1 Modelling Random Experiments

Probability theory is about modelling and analysing *random experiments*: experiments whose outcome cannot be determined in advance, but are nevertheless still subject to analysis. Mathematically, we can model a random experiment by defining a specific measure space $(\Omega, \mathcal{H}, \mathbb{P})$, which we call a *probability space* (**kolmogorovFoundationsTheoryProbability2013**), where the three components are: a *sample space*, a collection of *events*, and a *probability measure*.

The *sample space* Ω of a random experiment is the set of all possible outcomes of the random experiment. Sample spaces can be countable, such as $\mathbb{N}$, or uncountable, such as $\mathbb{R}^n$ or $\{0, 1\}^\infty$. The sets in the σ -algebra $\mathcal{H}$ are called *events*. These are the subsets to which we wish to assign a probability. We can view $\mathcal{H}$ as the family of events. An event A *occurs* if the outcome of the experiment is one of the elements in A . The properties of $\mathcal{H}$ are natural in the context of random experiments and events:

- If A and B are events, the set $A \cup B$ is also an event, namely the event that A or B or both occur. For a sequence $A_1, A_2, \dots$ of events, their union is the event that *at least one* of the events occurs.
- If A and B are events, the set $A \cap B$ is also an event, namely the event that A and B both occur. For a sequence $A_1, A_2, \dots$ of events, their intersection is the event that *all* of the events occur.
- If A is an event, its complement A^c is also an event, namely the event that A *does not occur*.
- The set Ω itself is an event, namely the *certain event* (it always occurs). Similarly, $\emptyset$ is an event, namely the *impossible event* (it never occurs).

When the sample space Ω is $\mathbb{R}$, the natural σ -algebra $\mathcal{H}$ is the *Borel σ -algebra* $\mathcal{B}$. Recall that this is the smallest σ -algebra on $\mathbb{R}$ that contains all the intervals of the form $(-\infty, x]$ for $x \in \mathbb{R}$. This σ -algebra of sets is big enough to contain all important sets and small enough to allow us to assign a probability to all events (**williamsProbabilityMartingales2018**).

The third ingredient in the model for a random experiment is the specification of the probability of the events. In fact, this is the crucial part of the model. Mathematically, we are looking for a *measure* $\mathbb{P}$ that assigns to each event A a number $\mathbb{P}(A)$ in $[0, 1]$ describing how likely or probable that event is.

Definition 2.1: Probability Measure

A *probability measure* is a measure on $(\Omega, \mathcal{H})$ with $\mathbb{P}(\Omega) = 1$. Thus, $\mathbb{P}$ is a mapping from $\mathcal{H}$ to $[0, 1]$ with the following properties:

- (1) $\mathbb{P}(\emptyset) = 0$ and $\mathbb{P}(\Omega) = 1$.
- (2) For any sequence $A_1, A_2, \dots$ of disjoint events,

$$\mathbb{P}(\cup_n A_n) = \sum_n \mathbb{P}(A_n). \quad (2.1)$$

Theorem 2.1: Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Monotonicity): $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$.
- (2) (Continuity from below): If $A_1, A_2, \dots$ is an increasing sequence of events, i.e., $A_1 \subseteq A_2 \subseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cup_n A_n). \quad (2.2)$$

- (3) (Countable subadditivity): $\mathbb{P}(\cup_n A_n) \leq \sum_n \mathbb{P}(A_n)$.

Theorem 2.2: Additional Properties of a Probability Measure

Let $\mathbb{P}$ be a probability measure on a measurable space $(\Omega, \mathcal{H})$, and let A, B and $A_1, A_2, \dots$ be events. Then, the following hold:

- (1) (Complement): $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- (2) (Union): $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (3) (Continuity from above): If $A_1, A_2, \dots$ is a decreasing sequence of events, i.e., $A_1 \supseteq A_2 \supseteq \dots$, then

$$\lim \mathbb{P}(A_n) = \mathbb{P}(\cap_n A_n). \quad (2.3)$$

2.2 Random Variables

Usually, the most convenient way to describe quantities of interest connected with random experiments is by using random variables.

Definition 2.2: Random Variable

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $(E, \mathcal{E})$ a measurable space. A *random variable* X taking values in E is an $\mathcal{H}/\mathcal{E}$ -measurable function; that is, a mapping $X : \Omega \rightarrow E$ that satisfies

$$X^{-1}A = \{X \in A\} = \{\omega \in \Omega : X(\omega) \in A\} \in \mathcal{H} \quad \text{for all } A \in \mathcal{E}. \quad (2.4)$$

Definition 2.3: Stochastic Process

A collection of random variables $X := \{X_t, t \in \mathbb{T}\}$, where $\mathbb{T}$ is any index set, is called a *stochastic process*. When $\mathbb{T}$ is finite, X is called a *random vector*.

Theorem 2.3: Stochastic Process

$X := \{X_t, t \in \mathbb{T}\}$ is a stochastic process taking values in $\times_{t \in \mathbb{T}} E_t$ with σ -algebra $\otimes_{t \in \mathbb{T}} \mathcal{E}_t$ if and only if each X_t is a random variable taking values in E_t with σ -algebra $\mathcal{E}_t$.

2.3 Probability Distributions

Definition 2.4: Probability Distribution

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and let X be a random variable taking values in $(E, \mathcal{E})$. The (*probability*) *distribution* of X is the image measure μ of $\mathbb{P}$ under X . That is,

$$\mu(B) := \mathbb{P}(X \in B), \quad B \in \mathcal{E}. \quad (2.5)$$

Definition 2.5: Cumulative Distribution Function

The (*cumulative*) *distribution function* of a numerical random variable X is the function $F : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F(x) := \mathbb{P}(X \leq x), \quad x \in \mathbb{R}. \quad (2.6)$$

Proposition 2.1: Properties of a Cumulative Distribution Function

- (1) (*Bounded*): $0 \leq F(x) \leq 1$.
 (2) (*Increasing*): $x \leq y \implies F(x) \leq F(y)$.
 (3) (*Right-continuous*): $\lim_{h \downarrow 0} F(x+h) = F(x)$.

Definition 2.6: Discrete Distribution

We say that X has a *discrete distribution* μ on $(E, \mathcal{E})$ if μ is a discrete measure; that is, it is of the form

$$\mu = \sum_{x \in D} f(x) \delta_x$$

for some countable set D and positive masses $\{f(x), x \in D\}$.

Definition 2.7: Distribution with a Density

We say that X has an *absolutely continuous* distribution μ on $(E, \mathcal{E})$ with respect to a measure λ on $(E, \mathcal{E})$ if there exists a positive $\mathcal{E}$ -measurable function f such that $\mu = f\lambda$; that is,

$$\mu(B) = \mathbb{P}(X \in B) = \int_B \lambda(dx) f(x), \quad B \in \mathcal{E}.$$

Definition 2.8: Independent Random Variables

Let X and Y be random variables taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$, respectively. They are said to be *independent* if for any $A \in \mathcal{E}$ and $B \in \mathcal{F}$ it holds that

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B). \quad (2.7)$$

Definition 2.9: Independency

We say that a collection of random variables $\{X_t, t \in \mathbb{T}\}$, with each X_t taking values in $(E_t, \mathcal{E}_t)$, is an *independency* if for any finite choice of indexes $t_1, \dots, t_n \in \mathbb{T}$ and sets $A_{t_1} \in \mathcal{E}_{t_1}, \dots, A_{t_n} \in \mathcal{E}_{t_n}$ it holds that

$$\mathbb{P}(X_{t_1} \in A_{t_1}, \dots, X_{t_n} \in A_{t_n}) = \mathbb{P}(X_{t_1} \in A_{t_1}) \cdots \mathbb{P}(X_{t_n} \in A_{t_n}).$$

2.4 Expectation

The expectation of a (numerical) random variable is a weighted average of all the values that the random variable can take.

Definition 2.10: Expectation

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and X a numerical random variable. The *expectation* of X is defined as the integral

$$\mathbb{E}X := \mathbb{P}X = \int X d\mathbb{P}.$$

Theorem 2.4: Properties of an Expectation

Let $a, b \in \mathbb{R}$ and let X, Y , and (X_n) be numerical random variables for which the expectations below are well-defined. Then, the following hold:

- (1) (Monotonicity): If $X \leq Y$, then $\mathbb{E}X \leq \mathbb{E}Y$.
- (2) (Linearity): $\mathbb{E}(aX + bY) = a\mathbb{E}X + b\mathbb{E}Y$.
- (3) (Monotone convergence): If $X_n \uparrow X$, then $\mathbb{E}X_n \uparrow \mathbb{E}X$.

Lemma 2.1: Fatou

For any sequence (X_n) of positive numerical random variables,

$$\mathbb{E} \liminf X_n \leq \liminf \mathbb{E}X_n.$$

Theorem 2.5: Dominated and Bounded Convergence

Let (X_n) be a sequence of numerical random variables for which $\lim X_n = X$ exists.

Then, the following hold:

- (1) (Dominated convergence): If, for each n , $|X_n| \leq Y$ for some random variable Y with $\mathbb{E}Y < \infty$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.
- (2) (Bounded convergence): If $|X_n| \leq c$ for all n for some $c \in \mathbb{R}$, then $\mathbb{E}X_n \rightarrow \mathbb{E}X < \infty$.

Theorem 2.6: Expectation and Image Measure

Let X be a random variable taking values in $(E, \mathcal{E})$ with distribution μ and let h be an $\mathcal{E}$ -measurable function. Then, provided the integral exists:

$$\mathbb{E}h(X) = \int h(X) d\mathbb{P} = \int_{\mathbb{R}} \mu(dx) h(x) = \mu h.$$

Theorem 2.7: Transformation Rule

Let X be a random vector with density f_X with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$. Let $Z := g(X)$, where $g : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is invertible with inverse g^{-1} and Jacobian $|\frac{\partial z}{\partial x}|$. Then, at $z = g(x)$ the random vector Z has the following density with respect to the Lebesgue measure on $(\mathbb{R}^n, \mathcal{B}^n)$:

$$f_Z(z) := \frac{f_X(x)}{|\frac{\partial z}{\partial x}|} = f_X(g^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|, \quad z \in \mathbb{R}^n. \quad (2.8)$$

Proposition 2.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector μ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(\Sigma A) + \mu^\top A \mu$.

2.5 L^p Spaces

Let X be a numerical random variable on

Definition 2.11: L^p Space

For each $p \in [1, \infty]$ the space L^p is comprised of all numerical random variables X for which $\|X\|_p < \infty$.

Lemma 2.2: Jensen's Inequality

Let $h : \mathcal{X} \rightarrow \mathbb{R}$ be a convex function and let X be a random variable taking values in $\mathcal{X}$, with expectation vector $\mathbb{E}X$. Then,

$$\mathbb{E}h(X) \geq h(\mathbb{E}X).$$

Theorem 2.8: Properties of the L^p Norm

Let X and Y be numerical random variables on $(\Omega, \mathcal{H}, \mathbb{P})$. Then, the following hold:

- (1) (Positivity): $\|X\|_p \geq 0$, and $\|X\|_p = 0 \iff X = 0$ almost surely.
- (2) (Multiplication by a constant): $\|cX\|_p = |c|\|X\|_p$.
- (3) (Minkowski's (triangle) inequality):

$$|\|X\|_p - \|Y\|_p| \leq \|X + Y\|_p \leq \|X\|_p + \|Y\|_p.$$

- (4) (Hölder's inequality): For $p, q, r \in [1, \infty]$ with $\frac{1}{p} + \frac{1}{q} = \frac{1}{r}$,

$$\|XY\|_r \leq \|X\|_p \|Y\|_q. \quad (2.9)$$

- (5) (Monotonicity): If $1 \leq p < q \leq \infty$, then $\|X\|_p \leq \|X\|_q$.

2.6 Integral Transforms

Expectations are used in many probabilistic analyses.

Definition 2.12: Moment Generating Function

The *moment generating function* (MGF) of a numerical random variable X with distribution μ is the function $M : \mathbb{R} \rightarrow [0, \infty]$, given by

$$M(s) := \mathbb{E}e^{sX} = \int_{-\infty}^{\infty} \mu(dx) e^{sx}, \quad s \in \mathbb{R}.$$

Theorem 2.9: Uniqueness of the MGF

A measure μ is determined by its MGF M if the latter is finite in a neighborhood of 0.

Theorem 2.10: Taylor's Theorem for MGFs

Suppose $M(s) := \mathbb{E}e^{sX}$ is finite in the neighborhood $(-s_0, s_0)$ of 0. Then, M has the Taylor expansion

$$M(s) = \sum_{k=0}^{\infty} \frac{s^k}{k!} \mathbb{E}X^k, \quad |s| < s_0.$$

In particular, all moments $\mathbb{E}X^k, k \in \mathbb{N}$ are finite.

Theorem 2.11: Product of MGFs

If X and Y are independent, then $M_{X+Y}(s) = M_X(s)M_Y(s)$.

Definition 2.13: Characteristic Function

The *characteristic function* of a numerical random variable X with distribution μ is the function $\psi : \mathbb{R} \rightarrow \mathbb{C}$, defined by

$$\psi(r) := \mathbb{E}e^{irX} = \mathbb{E} \cos(rX) + i\mathbb{E} \sin(rX) = \int_{-\infty}^{\infty} \mu(dx) e^{irx}, \quad r \in \mathbb{R}.$$

Theorem 2.12: Inversion and Uniqueness of the Characteristic Function

If the probability measure μ has characteristic function ψ , and if $\mu\{a\} = \mu\{b\} = 0$, then

$$\mu(a, b] = \lim_{t \rightarrow \infty} \frac{1}{2\pi} \int_{\mathbb{R}} dr \, \mathbb{1}_{[-t, t]}(r) \frac{e^{-ira} - e^{-irb}}{ir} \psi(r) := \lim_{t \rightarrow \infty} I_t. \quad (2.10)$$

In particular, the characteristic function determines μ uniquely. If, in addition, $\int_{\mathbb{R}} dr |\psi(r)| < \infty$, then μ has a density f with respect to the Lebesgue measure, given by

$$f(x) := \frac{1}{2\pi} \int_{-\infty}^{\infty} dr e^{-irx} \psi(r). \quad (2.11)$$

Proposition 2.3: Properties of the Characteristic Function

- (1) (Symmetry): The random variables X and $-X$ are identically distributed if and only if the characteristic function of X is real-valued.
- (2) (Convolution): If X and Y are independent, then $\psi_{X+Y} = \psi_X \psi_Y$.
- (3) (Affine transformation): If a and b are real numbers, then

$$\psi_{aX+b}(r) = e^{irb} \psi_X(ar), \quad r \in \mathbb{R}.$$

- (4) (Taylor's theorem): If $\mathbb{E}|X^n| < \infty$, then

$$\psi(r) = \sum_{k=0}^n \frac{\mathbb{E}X^k}{k!} (ir)^k + O(r^n), \quad r \in \mathbb{R}.$$

2.7 Information and Independence

In this section, we have another look at the role of σ -algebras and independence in random experiments.

Definition 2.14: σ -Algebra Generated by a Random Variable

Let X be a random variable taking values in $(E, \mathcal{E})$. The sets $\{X \in A\}, A \in \mathcal{E}$ form a σ -algebra, called the σ -algebra generated by X , and we write σX .

Theorem 2.13: σ -Algebra Generated by a Stochastic Process

For the stochastic process $X := \{X_t, t \in \mathbb{T}\}$, the σ -algebra generated by X is the smallest σ -algebra on Ω that is generated by the union of the σ -algebras $\sigma X_t, t \in \mathbb{T}$. We write $\bigvee_{t \in \mathbb{T}} \sigma X_t$ or $\sigma\{X_t, t \in \mathbb{T}\}$.

Theorem 2.14: σX -Numerical Random Variables

Let X be a random variable taking values in a measurable space $(E, \mathcal{E})$. A mapping $V : \Omega \rightarrow \overline{\mathbb{R}}$ belongs to σX if and only if $V = h \circ X$ for some $h \in \mathcal{E}$.

Definition 2.15: Filtration

Let $\mathbb{T}$ be a subset of $\mathbb{R}$. A *filtration* is an increasing collection $\{\mathcal{F}_t : t \in \mathbb{T}\}$ of sub- σ -algebras of $\mathcal{H}$; that is, $\mathcal{F}_s \subseteq \mathcal{F}_t$ whenever $s < t$.

Theorem 2.15: Stochastic Process with Arbitrary Index Set

For each t in an arbitrary set $\mathbb{T}$, let X_t be a random variable taking values in a measurable set $(E_t, \mathcal{E}_t)$. Then, the mapping $V : \Omega \rightarrow \overline{\mathbb{R}}$ belongs to $\sigma\{X_t, t \in \mathbb{T}\}$ if and only if there exists a sequence (t_n) in $\mathbb{T}$ and a function h in $\bigotimes_n \mathcal{E}_{t_n}$ such that

$$V = h(X_{t_1}, X_{t_2}, \dots).$$

Definition 2.16: Independence for σ -Algebras

Let $\{\mathcal{F}_i, i = 1, \dots, n\}$ be a collection of sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be (*mutually*) *independent* or form an *independency* if

$$\mathbb{E}V_1 \cdots V_n = \mathbb{E}V_1 \cdots \mathbb{E}V_n \quad (2.12)$$

for all positive random variables $V_i \in \mathcal{F}_i, i = 1, \dots, n$.

Theorem 2.16: Independence of σ -Algebras

Sub- σ -algebras $\mathcal{F}_1, \dots, \mathcal{F}_n$ of $\mathcal{H}$ generated by p -systems $\mathcal{C}_1, \dots, \mathcal{C}_n$ are independent if and only if for all $H_i \in \mathcal{C}_i \cup \{\Omega\}, i = 1, \dots, n$, it holds that:

$$\mathbb{P}(H_1 \cap \cdots \cap H_n) = \mathbb{P}(H_1) \cdots \mathbb{P}(H_n). \quad (2.13)$$

Theorem 2.17: Independence of Random Variables

Random variables X and Y taking values in $(E, \mathcal{E})$ and $(F, \mathcal{F})$ are independent if and only if

$$\mathbb{E}g(X)h(Y) = \mathbb{E}g(X)\mathbb{E}h(Y), \quad g \in \mathcal{E}_+, h \in \mathcal{F}_+. \quad (2.14)$$

2.8 Important Stochastic Processes

We conclude by highlighting some key classes of stochastic processes, including Gaussian processes, Poisson random measures, and Lévy processes.

Definition 2.17: Gaussian Process

A real-valued stochastic process $\{X_t, t \in \mathbb{T}\}$ is said to be *Gaussian* if all its finite-dimensional distributions are Gaussian (normal); that is, if the vector $[X_{t_1}, \dots, X_{t_n}]^\top$ is multivariate normal for any choice of n and $t_1, \dots, t_n \in \mathbb{T}$.

Definition 2.18: Poisson Random Measure

A random measure N on $(E, \mathcal{E})$ is said to be a *Poisson random measure* with mean measure μ if the following properties hold:

- (1) $N(A) \sim \text{Poi}(\mu(A))$ for any set $A \in \mathcal{E}$.
- (2) For any selection of disjoint sets $A_1, \dots, A_n \in \mathcal{E}$, the random variables $N(A_1), \dots, N(A_n)$ are independent.

Proposition 2.4: Poisson Random Measure on a Product Space

Let $X := \{X_i, i \in I\}$ be the atoms of a Poisson random measure N on $(E, \mathcal{E})$ with mean measure μ , and let $\{Y_i, i \in \mathbb{N}\}$ be i.i.d. random variables on $(F, \mathcal{F})$ with distribution π , independent of X . Then, $\{(X_i, Y_i), i \in I\}$ form the atoms of a Poisson random measure M on $(E \times F, \mathcal{E} \otimes \mathcal{F})$ with mean measure $\mu \otimes \pi$.

Theorem 2.18: Conditioning on the Total Number of Points

Let N be a Poisson random measure on $(E, \mathcal{E})$ with mean measure μ satisfying $\mu(E) := c < \infty$. Then, conditional on $N(E) = k$, the k atoms of N are i.i.d. with probability distribution μ/c .

Definition 2.19: Compound Poisson Process

The process $(X_t, t \geq 0)$ defined by

$$X_t := \int_{[0,t] \times \mathbb{R}^d} N(ds, d\mathbf{x}) \mathbf{x}, \quad t \geq 0,$$

is the *compound Poisson process* corresponding to the measure ν .

Definition 2.20: Lévy Process

The process X is said to be a *Lévy process* if

- (1) the paths of X are almost surely right-continuous and left-limited, with $X_0 = 0$, and
- (2) for every $t_1 < t_2 < t_3 < t_4$, the increments $X_{t_2} - X_{t_1}$ and $X_{t_4} - X_{t_3}$ are independent, and for every $t, u \in \mathbb{R}_+$, the increment $X_{t+u} - X_t$ has the same distribution as X_u .

Theorem 2.19: Lévy – It Decomposition

The process $X := (X_t, t \geq 0)$ is a Lévy process if and only if there exists a d -dimensional Brownian motion process $B := (B_t, t \geq 0)$ starting at 0 and a Poisson random measure N on $\mathbb{R}_+ \times \mathbb{R}^d$ that is independent of B and has mean measure $\text{Leb} \otimes \nu$, with ν satisfying

$$\int_{\mathbb{R}^d} \nu(d\mathbf{x}) (|\mathbf{x}|^2 \wedge 1) < \infty, \quad (2.15)$$

such that $X = B + U + Z$, where the summands are independent, $U := (U_t, t \geq 0)$ is the compound Poisson process

$$U_t := \int_0^t \int_{\|\mathbf{x}\| > 1} N(ds, d\mathbf{x}) \mathbf{x}, \quad t \geq 0,$$

and $Z := (Z_t, t \geq 0)$ is the limit of compensated compound Poisson processes:

$$Z := \lim_{\delta \downarrow 0} Z^\delta \quad \text{with} \quad Z_t^\delta := \int_0^t \int_{\delta \leq \|\mathbf{x}\| \leq 1} [\mathbf{x} N(ds, d\mathbf{x}) - ds \mathbf{x} \nu(d\mathbf{x})], \quad t \geq 0.$$

3 Convergence

Proposition 3.1: Cauchy Criteria for Convergence

The following statements satisfy (1) $\iff$ (2) and (3) $\implies$ (2):

- (1) (x_n) converges to some real $x := \lim x_n$.
- (2) (x_n) is Cauchy convergent, that is, $\lim_{m,n \rightarrow \infty} |x_m - x_n| = 0$.
- (3) $\sum_n \mathbb{1}_{\{|x_{n+1} - x_n| > \varepsilon_n\}} < \infty$ for any (ε_n) such that $c := \sum_n \varepsilon_n < \infty$.

Definition 3.1: Almost Sure Convergence

The sequence of random variables (X_n) converges *almost surely* (a.s.) to a random variable X if

$$\mathbb{P}(\lim X_n = X) = 1.$$

We denote the almost sure convergence by $X_n \xrightarrow{\text{a.s.}} X$.

Theorem 3.1: Almost Sure Convergence

The sequence of random variables (X_n) converges almost surely to a random variable X if and only if for every $\varepsilon > 0$

$$\lim_{m \rightarrow \infty} \mathbb{P} \left(\sup_{n \geq m} |X_n - X| > \varepsilon \right) = 0. \quad (3.1)$$

Proposition 3.2: Cauchy Criteria for Almost Sure Convergence

The following statements are equivalent:

- (1) $X_n \xrightarrow{\text{a.s.}} X$.
- (2) $\lim_{n \rightarrow \infty} \mathbb{P}(\sup_{k \geq 0} |X_{n+k} - X_n| > \varepsilon) = 0$ for any choice of $\varepsilon > 0$.
- (3) $\lim_{k \rightarrow \infty} \mathbb{P}(\sup_{m, n \geq k} |X_m - X_n| > \varepsilon) = 0$ for any choice of $\varepsilon > 0$.
- (4) $\mathbb{P}(\lim_{m, n \rightarrow \infty} |X_m - X_n| = 0) = 1$; that is, (X_n) is a.s. a Cauchy sequence.

Proposition 3.3: Markov's and Chebyshev's Inequality

For any random variable Y and $\varepsilon > 0$, Markov's inequality holds:

$$\mathbb{P}(|Y| > \varepsilon) \leq \frac{\mathbb{E}|Y|}{\varepsilon}. \quad (3.2)$$

As a consequence, we have Chebyshev's inequality, where $\mu := \mathbb{E}X$:

$$\mathbb{P}(|X - \mu| > \varepsilon) \leq \frac{\text{Var } X}{\varepsilon^2}. \quad (3.3)$$

Theorem 3.2: Kolmogorov's Inequality

Let $S_n := \sum_{i=1}^n X_i$, where $X_1, X_2, \dots$ are independent with 0 mean. Then, for every $\varepsilon > 0$:

$$\mathbb{P}\left(\sup_{k \leq n} |S_k| > \varepsilon\right) \leq \frac{\text{Var } S_n}{\varepsilon^2}. \quad (3.4)$$

Lemma 3.1: Borel–Cantelli

Let (H_n) be a sequence of events. Then, the following hold:

- (1) $\sum_n \mathbb{P}(H_n) < \infty$ implies that $\mathbb{P}(\sum_n \mathbb{1}_{H_n} < \infty) = 1$.
- (2) $\sum_n \mathbb{P}(H_n) = \infty$ implies that $\mathbb{P}(\sum_n \mathbb{1}_{H_n} < \infty) = 0$, provided that the events (H_n) are pairwise independent.

Definition 3.2: Convergence in Probability

The sequence of random variables (X_n) converges in probability to a random variable X if, for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \varepsilon) = 0.$$

We denote the convergence in probability by $X_n \xrightarrow{\mathbb{P}} X$.

Proposition 3.4: Convergence of a Subsequence

If (X_n) converges to X in probability, there is a subsequence (X_{n_k}) that converges to X almost surely.

Proposition 3.5: Cauchy Criterion for Convergence in Probability

The sequence of random variables (X_n) converges in probability if and only if for every $\varepsilon > 0$,

$$\lim_{m,n \rightarrow \infty} \mathbb{P}(|X_m - X_n| > \varepsilon) = 0. \quad (3.5)$$

Definition 3.3: Convergence in Distribution

The sequence of random variables (X_n) is said to *converge in distribution* to a random variable X with cdf F provided that:

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n \leq x) = F(x) \quad \text{for all } x \text{ such that } \lim_{a \rightarrow x} F(a) = F(x). \quad (3.6)$$

We denote the convergence in distribution by $X_n \xrightarrow{d} X$.

Theorem 3.3: Characteristic Function and Convergence in Distribution

Suppose that $\psi_{X_1}(\mathbf{r}), \psi_{X_2}(\mathbf{r}), \dots$ are the characteristic functions of the sequence of d -dimensional random vectors $X_1, X_2, \dots$ and $\psi_X(\mathbf{r})$ is the characteristic function of X .

Then, the following three statements are equivalent:

- (1) $\lim_n \psi_{X_n}(\mathbf{r}) = \psi_X(\mathbf{r})$ for all $\mathbf{r} \in \mathbb{R}^d$.
- (2) $X_n \xrightarrow{d} X$.
- (3) $\lim_n \mathbb{E}h(X_n) = \mathbb{E}h(X)$ for all bounded continuous functions $h : \mathbb{R}^d \rightarrow \mathbb{R}$.

Definition 3.4: Convergence in L^p Norm

The sequence of random variables (X_n) converges in L^p norm (for some $p \in [1, \infty]$) to a random variable X , if $\|X_n\|_p < \infty$ for all n , $\|X\|_p < \infty$, and

$$\lim_{n \rightarrow \infty} \|X_n - X\|_p = 0.$$

We denote the convergence in L^p norm by $X_n \xrightarrow{L^p} X$.

Proposition 3.6: Cauchy Criterion for Convergence in L^p Norm

The sequence of random variables (X_n) converges in L^p norm for $p \in [1, \infty]$ if and only if $\|X_n\|_p < \infty$ for all n and

$$\lim_{m,n \rightarrow \infty} \|X_m - X_n\|_p = 0.$$

Proposition 3.7: Integrable Random Variable

A real-valued random variable X is integrable if and only if

$$\lim_{b \rightarrow \infty} \mathbb{E}|X| \mathbf{1}_{\{|X| > b\}} = 0. \quad (3.7)$$

Definition 3.5: Uniform Integrability

A collection $\mathcal{K}$ of random variables is said to be *uniformly integrable* (UI) if

$$\lim_{b \rightarrow \infty} \sup_{X \in \mathcal{K}} \mathbb{E}|X| \mathbf{1}_{\{|X| > b\}} = 0.$$

Proposition 3.8: Conditions for Uniform Integrability

Let $\mathcal{K}$ be a collection of random variables.

- (1) If $\mathcal{K}$ is a finite collection of integrable random variables, then it is UI.
- (2) If $|X| \leq Y$ for all $X \in \mathcal{K}$ and some integrable Y , then $\mathcal{K}$ is UI.
- (3) $\mathcal{K}$ is UI if and only if there is an increasing convex function f such that

$$\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \infty \quad \text{and} \quad \sup_{X \in \mathcal{K}} \mathbb{E}f(|X|) < \infty.$$

- (4) $\mathcal{K}$ is UI if $\sup_{X \in \mathcal{K}} \mathbb{E}|X|^{1+\varepsilon} < \infty$ for some $\varepsilon > 0$.

- (5) $\mathcal{K}$ is UI if and only if

- (1) $\sup_{X \in \mathcal{K}} \mathbb{E}|X| < \infty$, and
- (2) for every $\varepsilon > 0$ there is a $\delta > 0$ such that for every event H :

$$\mathbb{P}(H) \leq \delta \implies \sup_{X \in \mathcal{K}} \mathbb{E}|X| \mathbf{1}_H \leq \varepsilon. \quad (3.8)$$

Theorem 3.4: L^1 Convergence and Uniform Integrability

A sequence (X_n) of real-valued integrable random variables converges in L^1 if and only if it converges in probability and is uniformly integrable.

Theorem 3.5: Weak Law of Large Numbers

If $X_1, X_2, \dots$ are i.i.d. with finite expectation μ , then for all $\varepsilon > 0$

$$\lim_{n \rightarrow \infty} \mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) = 0.$$

In other words, $\bar{X}_n \xrightarrow{\mathbb{P}} \mu$.

Theorem 3.6: Strong Law of Large Numbers

If $X_1, X_2, \dots$ are i.i.d. with expectation μ and $\mathbb{E}X^2 < \infty$, then for all $\varepsilon > 0$

$$\lim_{m \rightarrow \infty} \mathbb{P} \left(\sup_{n \geq m} |\bar{X}_n - \mu| > \varepsilon \right) = 0.$$

In other words, $\bar{X}_n \xrightarrow{\text{a.s.}} \mu$.

Theorem 3.7: Central Limit Theorem

If $X_1, X_2, \dots$ are i.i.d. with finite expectation μ and finite variance σ^2 , then for all $x \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \leq x \right) = \Phi(x),$$

where Φ is the cdf of the standard normal distribution. In other words, $\sqrt{n}(\bar{X}_n - \mu)/\sigma \xrightarrow{d} \mathcal{N}(0, 1)$.

4 Conditioning

Definition 4.1: Conditional Expectation

The conditional expectation of X given $\mathcal{F}$, denoted $\mathbb{E}_{\mathcal{F}}X$, is defined in two steps:

- (1) For $X \geq 0$, $\mathbb{E}_{\mathcal{F}}X$ is defined as any random variable $\hat{X}$ with
 - (1) $\hat{X} \in \mathcal{F}_+$.
 - (2) $\mathbb{E}VX = \mathbb{E}V\hat{X}$ for every $V \in \mathcal{F}_+$ (*projection property*).
- (2) For arbitrary X , if $\mathbb{E}X$ exists (possibly $\pm\infty$), $\mathbb{E}_{\mathcal{F}}X := \mathbb{E}_{\mathcal{F}}X^+ - \mathbb{E}_{\mathcal{F}}X^-$. Otherwise, the conditional expectation is left undefined.

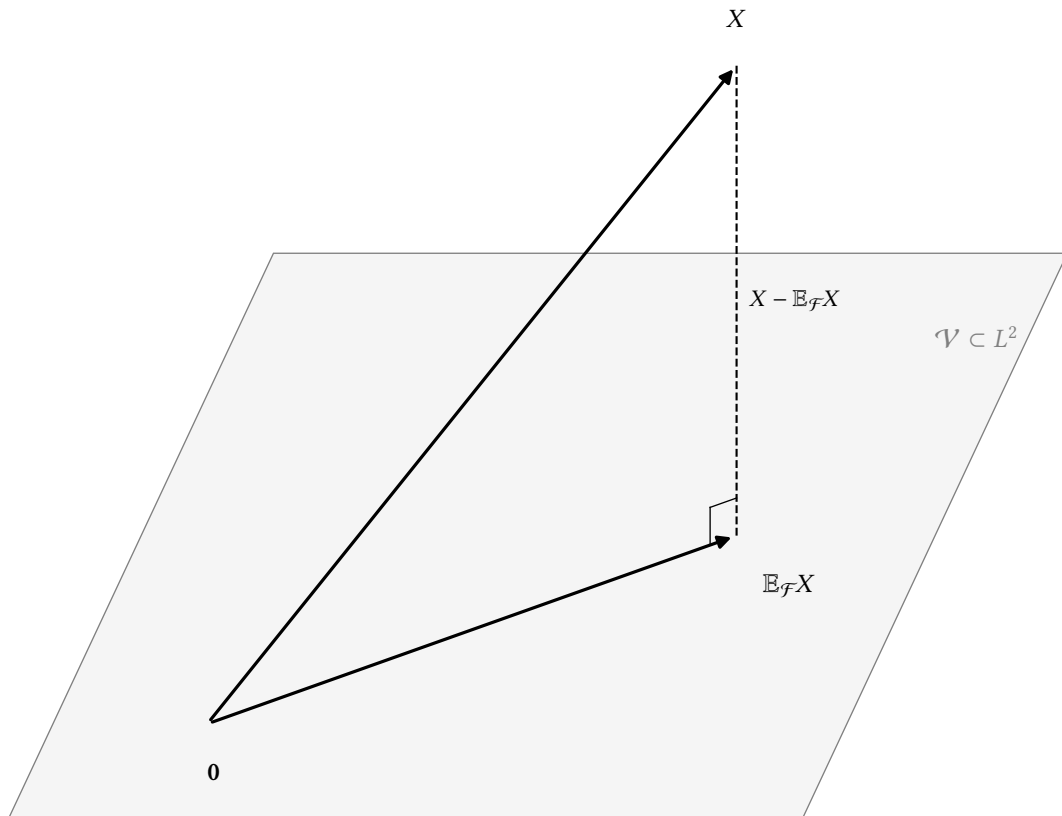


Figure 6: Geometric interpretation of Conditional Expectation $\mathbb{E}_{\mathcal{F}}X$ as an orthogonal projection.

Theorem 4.1: Existence and Uniqueness of the Conditional Expectation

Let $X \geq 0$ be a random variable and let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. Then, $\mathbb{E}_{\mathcal{F}}X$ exists and is unique up to equivalence.

Theorem 4.2: Additional Properties of the Conditional Expectation

Let $\mathcal{F}$ and $\mathcal{G}$ be sub- σ -algebras of $\mathcal{H}$. Let W and X be random variables such that $\mathbb{E}X$ and $\mathbb{E}WX$ exist. Then, the following hold:

- (1) (Taking out what is known): If $W \in \mathcal{F}$, then $\mathbb{E}_{\mathcal{F}}WX = W\mathbb{E}_{\mathcal{F}}X$.
- (2) (Repeated conditioning): If $\mathcal{F} \subseteq \mathcal{G}$, then

$$\mathbb{E}_{\mathcal{F}}\mathbb{E}_{\mathcal{G}}X = \mathbb{E}_{\mathcal{G}}\mathbb{E}_{\mathcal{F}}X = \mathbb{E}_{\mathcal{F}}X. \quad (4.1)$$

Theorem 4.3: Conditional Expectation in L^2

For $X \in L^2$, the conditional expectation $\mathbb{E}_{\mathcal{F}}X$ is the orthogonal projection of X onto the subspace $\mathcal{V}$ of $\mathcal{F}$ -measurable random variables in L^2 .

Definition 4.2: Conditional Probability for Events

Let A and B be two events with $\mathbb{P}(B) > 0$. The *conditional probability* that A occurs given that B occurs is denoted by

$$\mathbb{P}(A \mid B) := \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (4.2)$$

Theorem 4.4: Properties Involving Conditional Probabilities

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space and $A, A_1, A_2, \dots, A_n, B_1, B_2, \dots$ events such that $\mathbb{P}(A_1 \cap \dots \cap A_n) > 0$ and $\{B_i\}$ form a partition of Ω , with $\mathbb{P}(B_i) > 0$ for all i . Then, the following hold:

(1) Law of total probability: $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$.

(2) Bayes' rule: $\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(B_j) \mathbb{P}(A \mid B_j)}{\sum_i \mathbb{P}(B_i) \mathbb{P}(A \mid B_i)}$.

(3) Chain or product rule:

$$\mathbb{P}(\cap_{i=1}^n A_i) = \mathbb{P}(A_1) \mathbb{P}(A_2 \mid A_1) \cdots \mathbb{P}(A_n \mid A_1 \cap \dots \cap A_{n-1}).$$

Definition 4.3: Conditional Distribution

Let Y be a random variable on the probability space $(\Omega, \mathcal{H}, \mathbb{P})$ taking values in the measurable space $(E, \mathcal{E})$. Let $\mathcal{F}$ be a sub- σ -algebra of $\mathcal{H}$. The *conditional distribution of Y given $\mathcal{F}$* is any probability transition kernel L from $(\Omega, \mathcal{F})$ to $(E, \mathcal{E})$ such that

$$\mathbb{P}_{\mathcal{F}}(Y \in B) = L(\cdot, B), \quad B \in \mathcal{E}. \quad (4.3)$$

Theorem 4.5: Existence of the Conditional Distribution

For every standard measurable space $(E, \mathcal{E})$ there exists a version of the conditional distribution of Y given $\mathcal{F}$.

Theorem 4.6: Regular Version of a Conditional Probability

Let $(\Omega, \mathcal{H})$ be a standard measurable space and $\mathcal{F}$ a sub- σ -algebra of $\mathcal{H}$. Then, there exists a regular version of the conditional probability $\mathbb{P}_{\mathcal{F}}$.

$$\pi(dx, dy) = \mu(dx) K(x, dy). \quad (4.4)$$

Theorem 4.7: Conditional Distribution

If π is of the form [eq. \(4.4\)](#), then the kernel L defined by

$$L(\omega, B) := K(X(\omega), B), \quad \omega \in \Omega, B \in \mathcal{E} \quad (4.5)$$

is a version of the conditional distribution of Y given X .

Theorem 4.8: Disintegration

Let π be a probability measure on the product space $(D \times E, \mathcal{D} \otimes \mathcal{E})$, where $(E, \mathcal{E})$ is a standard measurable space. Then, there exists a probability measure μ on $(D, \mathcal{D})$ and a transition kernel K from $(D, \mathcal{D})$ to $(E, \mathcal{E})$ such that

$$\pi(dx, dy) = \mu(dx)K(x, dy), \quad x \in D, y \in E. \quad (4.6)$$

Definition 4.4: Conditional Independence

Let $\mathcal{F}, \mathcal{F}_1, \dots, \mathcal{F}_n$ be sub- σ -algebras of $\mathcal{H}$. The $\{\mathcal{F}_i\}$ are said to be *conditionally independent* given $\mathcal{F}$ if

$$\mathbb{E}_{\mathcal{F}} V_1 \dots V_n = \mathbb{E}_{\mathcal{F}} V_1 \dots \mathbb{E}_{\mathcal{F}} V_n \quad (4.7)$$

for all positive random variables $V_1, \dots, V_n$ in $\mathcal{F}_1, \dots, \mathcal{F}_n$, respectively.

Proposition 4.1: Conditional Independence

$\mathcal{F}_1$ and $\mathcal{F}_2$ are conditionally independent given $\mathcal{F}$ if and only if

$$\mathbb{E}_{\mathcal{F} \vee \mathcal{F}_1} V_2 = \mathbb{E}_{\mathcal{F}} V_2 \quad \text{for all positive } V_2 \in \mathcal{F}_2. \quad (4.8)$$

$$\mathbb{P}(B_0 \times \dots \times B_n \times E_{n+1} \times \dots) = \pi_n(B_0 \times \dots \times B_n), \quad B_i \in \mathcal{E}_i, i = 0, \dots, n. \quad (4.9)$$

Theorem 4.9: Ionescu–Tulcea’s Existence Theorem

There exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ such that [eq. \(4.9\)](#) holds.

Theorem 4.10: Kolmogorov's Extension Theorem

Let $(E, \mathcal{E})$ be a standard measurable space and K an index set. For each finite subset J of K , let π_J be a probability measure on the product space $(E^J, \mathcal{E}^J)$. If the (π_J) are consistent in the sense that

$$\pi_I = \pi_J \circ \text{proj}_{JI}^{-1}, \quad I \subseteq J, \quad |J| < \infty,$$

then there exists a unique probability measure $\mathbb{P}$ on $(\Omega, \mathcal{H})$ and a stochastic process X taking values in $(E, \mathcal{E})^K$ such that, with $X_J := (X_j)_{j \in J}$:

$$\mathbb{P}(X_J \in A) = \pi_J(A), \quad A \in \mathcal{E}^J, \quad |J| < \infty.$$

Definition 4.5: Hölder Continuity

A function $f : \mathbb{R}_+ \rightarrow \mathbb{R}$ is said to be *Hölder continuous* of order $\alpha > 0$ on $B \subseteq \mathbb{R}_+$ if there is a constant k such that

$$|f(t) - f(s)| \leq k|t - s|^\alpha, \quad s, t \in B.$$

Theorem 4.11: Hölder Continuous Modification

Let $X := (X_t, t \in [0, 1])$ be a stochastic process with state space $\mathbb{R}$. If there exist constants c, p, q in $(0, \infty)$ such that

$$\mathbb{E}|X_t - X_s|^p \leq c|t - s|^{1+q}, \quad s, t \in [0, 1], \quad (4.10)$$

then for every $\alpha \in (0, q/p)$ there is a modification $\tilde{X}$ of X that is almost surely Hölder continuous of order α on $[0, 1]$.

Definition 4.6: Adaptedness

A stochastic process $X := (X_t, t \in \mathbb{T})$ taking values in some measurable space $(E, \mathcal{E})$ is said to be *adapted* to a filtration $\mathcal{F} := (\mathcal{F}_t, t \in \mathbb{T})$ if for all $t \in \mathbb{T}$, X_t is $\mathcal{F}_t/\mathcal{E}$ -measurable.

Definition 4.7: Markov Property

X is said to be *Markovian* relative to $\mathcal{F}$ if for every t , the past $\mathcal{F}_t$ and the future $\mathcal{G}_t$ are conditionally independent, given the present state X_t .

Definition 4.8: Markovian Transition Function

A *Markovian transition function* on $(E, \mathcal{E})$ is a family $(P_{t,u}, t, u \in \mathbb{T}, t < u)$ of probability transition kernels on $(E, \mathcal{E})$ that satisfy the *Chapman–Kolmogorov equations*:

$$P_{s,t}P_{t,u} = P_{s,u}, \quad s < t \leq u. \quad (4.11)$$

Theorem 4.12: Generator of a Markov Jump Process

The infinitesimal generator of a Markov jump process with Lévy kernel L is given by

$$(Gf)(x) = \int_{\mathbb{R}^d} L(x, dy)[f(y) - f(x)], \quad f \in \mathcal{B}^d(\text{bounded}). \quad (4.12)$$

Proposition 4.2: Poisson Integrals

Let N be a Poisson random measure on $\mathbb{R}_+ \times E$ with mean measure μ satisfying $\mu(\{0\} \times E) = 0$. Let $R := (R(t, z), t \geq 0, z \in E)$ be a positive stochastic process in $\mathcal{F}^P \otimes \mathcal{E}$. Then,

$$\mathbb{E}NR = \mathbb{E}\mu R. \quad (4.13)$$

5 Martingales

Definition 5.1: Stopping Time

A random time $T : \Omega \rightarrow \bar{\mathbb{T}}$ is called a *stopping time* of $\mathcal{F}$ if

$$\{T \leq t\} \in \mathcal{F}_t \quad \text{for each } t \in \mathbb{T}.$$

Definition 5.2: Past until T

Let $\mathcal{F}$ be a filtration on $\bar{\mathbb{T}}$, and let T be a stopping time of it. The σ -algebra of *past information until T* is defined as

$$\mathcal{F}_T := \{H \in \mathcal{H} : H \cap \{T \leq t\} \in \mathcal{F}_t \text{ for every } t \in \bar{\mathbb{T}}\}.$$

Theorem 5.1: $\mathcal{F}_T$ Random Variables

A random variable V belongs to $\mathcal{F}_T$ if and only if

$$X_t := V \mathbb{1}_{\{T \leq t\}} \in \mathcal{F}_t \text{ for every } t \in \bar{\mathbb{T}}.$$

Definition 5.3: (Sub/Super)Martingale

Let $X := (X_t, t \in \mathbb{T})$ be a real-valued stochastic process that is adapted to a filtration $\mathcal{F}$, and for which each X_t is integrable. The process X is called an $\mathcal{F}$ -(sub/super)martingale if for all $t > s$:

$$\begin{aligned} \mathbb{E}_s[X_t - X_s] &\geq 0 \quad (\text{submartingale}), \\ \mathbb{E}_s[X_t - X_s] &= 0 \quad (\text{martingale}), \\ \mathbb{E}_s[X_t - X_s] &\leq 0 \quad (\text{supermartingale}). \end{aligned} \tag{5.1}$$

Theorem 5.2: Properties of (Sub)Martingales

- (1) When $\mathbb{T} = \mathbb{N}$, the martingale equality $\mathbb{E}_s[X_t - X_s] = 0$ holds if and only if $\mathbb{E}_k[X_{k+1} - X_k] = 0, k \in \mathbb{N}$.
- (2) If X and Y are $\mathcal{F}$ -submartingales, then so is $aX + bY$ for $a, b \geq 0$.
- (3) If X and Y are $\mathcal{F}$ -submartingales, then so is $X \vee Y$.
- (4) Let f be a convex function on $\mathbb{R}$ and X an $\mathcal{F}$ -martingale. If $f(X_t)$ is integrable for all t , then $f(X) := (f(X_t))$ is an $\mathcal{F}$ -submartingale.

Definition 5.4: Predictable Process

Let $\mathcal{F} := (\mathcal{F}_t, t \geq 0)$ be a filtration. A real-valued stochastic process F is said to be $\mathcal{F}$ -predictable (or simply *predictable*) if it is measurable with respect to the σ -algebra

$$\mathcal{F}^P := \sigma(H \times (a, b] : a, b \in \mathbb{R}_+, H \in \mathcal{F}_a) \vee \sigma(H \times \{0\} : H \in \mathcal{F}_0). \quad (5.2)$$

Definition 5.5: Discrete Predictable Process

Given a filtration $\mathcal{F} := (\mathcal{F}_n, n \in \mathbb{N})$, a process $F := (F_n, n \in \mathbb{N})$ is said to be *predictable* if $F_0 \in \mathcal{F}_0$ and $F_n \in \mathcal{F}_{n-1}, n = 1, 2, \dots$

Theorem 5.3: Integral Transform of a Martingale

Let F be a bounded predictable process and X a martingale adapted to a filtration $\mathcal{F}$. Then, the integral transform $Z_n := \int_0^n F_k dX_k, n \in \mathbb{N}$ is an $\mathcal{F}$ -martingale as well.

Definition 5.6: Stopped Stochastic Process

Let $X := (X_t, t \in \mathbb{T})$ be a stochastic process and T a random time with values in $\overline{\mathbb{T}}$. By the process X *stopped at time T* we mean the process $(X_t^\dagger, t \in \mathbb{T})$ defined by

$$X_t^\dagger(\omega) := X_{T(\omega) \wedge t}(\omega) = \begin{cases} X_t(\omega) & \text{if } t \leq T(\omega), \\ X_{T(\omega)}(\omega) & \text{if } t > T(\omega). \end{cases} \quad (5.3)$$

Theorem 5.4: Stopped Martingale

Let $M := (M_n, n \in \mathbb{N})$ be a martingale and T a stopping time with values in $\overline{\mathbb{N}}$. Then, the process M stopped at time T is again a martingale.

Proposition 5.1: Stochastic Integral for Predictable Integrands

Let N be defined as above. Then, for any positive $\mathcal{F}$ -predictable process (F_t) ,

$$\mathbb{E} \int_{\mathbb{R}_+} F_t dN_t = \mathbb{E} \int_{\mathbb{R}_+} F_t dv_t. \quad (5.4)$$

Theorem 5.5: Integral Transform of a Continuous Martingale

Let $\overline{N}$ be an $\mathcal{F}$ -martingale, as defined above. For any bounded $\mathcal{F}$ -predictable process F , the integral transform

$$Z_t := \int_0^t F_s d\overline{N}_s, \quad t \geq 0, \quad (5.5)$$

is a martingale.

Theorem 5.6: Doob's Stopping Theorem

A stochastic process $M := (M_n, n \in \mathbb{N})$ is a martingale if and only if for every pair of bounded stopping times S and T with $S \leq T$ the random variables M_S and M_T are integrable, and

$$\mathbb{E}_S(M_T - M_S) = 0. \quad (5.6)$$

Lemma 5.1: Maximum Inequality for Positive Submartingales

Let $(Y_k, k \in \mathbb{N})$ be a positive submartingale and define $Y_n^* := \max_{k \leq n} Y_k$. Then, for any $b > 0$,

$$b\mathbb{P}(Y_n^* \geq b) \leq \mathbb{E}[Y_n \mathbf{1}_{\{Y^* \geq b\}}] \leq \mathbb{E}Y_n. \quad (5.7)$$

Theorem 5.7: Doob–Kolmogorov Inequality

Let $M := (M_k)$ be a martingale in L^p for some $p \in [1, \infty)$. Then, for $b > 0$,

$$\mathbb{P}(\max_{k \leq n} |M_k| > b) \leq \frac{\mathbb{E}|M_n|^p}{b^p}. \quad (5.8)$$

Theorem 5.8: Expected Upcrossings of a Submartingale

Suppose that X is a submartingale. Then, the expected number of upcrossings $U_n(a, b)$ of (a, b) satisfies:

$$(b - a)\mathbb{E}U_n(a, b) \leq \mathbb{E}[(X_n - a)^+ - (X_0 - a)^+]. \quad (5.9)$$

Theorem 5.9: (Sub)Martingale Convergence

Let $X := (X_n, n \in \mathbb{N})$ be a submartingale. Suppose that $\sup_n \mathbb{E}X_n^+ < \infty$. Then, X converges almost surely to an integrable random variable X_∞ .

Theorem 5.10: (Sub)Martingale Convergence and Uniform Integrability

Let $X := (X_n, n \in \mathbb{N})$ be a submartingale. Then, X converges almost surely and in L^1 if and only if it is uniformly integrable. Moreover, if X converges, setting $X_\infty := \lim X_n$ extends X to a submartingale $\bar{X} := (X_n, n \in \bar{\mathbb{N}})$.

Theorem 5.11: Uniformly Integrable Martingale

A process $M := (M_n, n \in \mathbb{N})$ is a UI martingale if and only if there exists an integrable random variable Z such that

$$M_n = \mathbb{E}_n Z, \quad n \in \mathbb{N}. \quad (5.10)$$

Moreover, then M converges almost surely and in L^1 to an integrable random variable

$$M_\infty := \mathbb{E}_\infty Z, \quad (5.11)$$

and $(M_n, n \in \bar{\mathbb{N}})$ is again a UI martingale. If $Z \in \mathcal{F}_\infty$, then $M_\infty = Z$.

Theorem 5.12: Convergence for Martingales in Reversed Time

For $\mathbb{T} = \{\dots, -2, -1, 0\}$, let $M := (M_n, n \in \mathbb{T})$ be a martingale with respect to the filtration $(\mathcal{F}_n, n \in \mathbb{T})$. Then, M is UI and, moreover, as $n \downarrow -\infty$ it converges almost surely and in L^1 to the integrable random variable $M_{-\infty} := \mathbb{E}_{-\infty} M_0$, where $\mathcal{F}_{-\infty} := \cap_{n \in \mathbb{T}} \mathcal{F}_n$.

Theorem 5.13: Stopping for Uniformly Integrable Martingales

If $M := (M_n, n \in \overline{\mathbb{N}})$ is a uniformly integrable martingale, then for every pair of stopping times S and T with $S \leq T$ it holds that

$$\mathbb{E}_S M_T = M_S$$

In particular,

$$\mathbb{E} M_T = \mathbb{E} M_0.$$

Theorem 5.14: Kolmogorov's 0–1 Law

If $H \in \mathcal{T}$, then $\mathbb{P}(H)$ is either 0 or 1. Consequently, a numerical random variable Z that is $\mathcal{T}$ -measurable must be almost surely constant.

Definition 5.7: Separable σ -Algebra

A σ -algebra on Ω is said to be *separable* if it is generated by a sequence (H_n) of subsets of Ω .

Theorem 5.15: Radon–Nikodym

Let $(\Omega, \mathcal{H}, \mathbb{P})$ be a probability space, where $\mathcal{H}$ is separable, and let Q be a finite measure on $(\Omega, \mathcal{H})$ with $Q \ll \mathbb{P}$. Then, there exists a positive random variable $Z \in \mathcal{H}$, written $Z = dQ/d\mathbb{P}$, such that $Q = Z\mathbb{P}$; that is,

$$Q(H) = \int_H Z d\mathbb{P} = \mathbb{E} 1_H Z, \quad H \in \mathcal{H}. \quad (5.12)$$

Definition 5.8: Right-continuous Filtration

A filtration $\mathcal{F} := (\mathcal{F}_t)$ is said to be *right-continuous* if for all t ,

$$\bigcap_{\varepsilon > 0} \mathcal{F}_{t+\varepsilon} = \mathcal{F}_t.$$

Definition 5.9: Local Martingale

A process $(Z_t, t \geq 0)$ is called a *local martingale* if there exists a sequence of stopping times $(T_n, n \in \mathbb{N})$, called a *localizing sequence*, such that $T_n \rightarrow \infty$ almost surely and each stopped process $(Z_{t \wedge T_n}, t \geq 0)$ is a martingale.

Theorem 5.16: Properties of Local Martingales

Let $Z := (Z_t, t \geq 0)$ be a local martingale with localizing sequence $(T_n, n \in \mathbb{N})$. Then, the following hold:

- (1) If $Z \geq 0$ and $\mathbb{E}Z_0 < \infty$, then Z is a supermartingale.
- (2) If for each $t \geq 0$ the sequence $(Z_{t \wedge T_n}, n \in \mathbb{N})$ is uniformly integrable, then Z is a martingale.

Definition 5.10: Doob Martingale

Let ζ be a stopping time. A process $M := (M_t, t \in \mathbb{T})$ is called a *Doob martingale* on $[0, \zeta]$ if for all stopping times $0 \leq S \leq T \leq \zeta$ it holds that

$$\mathbb{E}_S M_T = M_S, \tag{5.13}$$

where M_S and M_T are integrable.

Theorem 5.17: Characterization of a Doob Martingale

Let ζ be a stopping time. A process $M := (M_t, t \in \mathbb{T})$ is a Doob martingale on $[0, \zeta]$ if and only if for every stopping time $T \leq \zeta$,

$$\mathbb{E}M_T = \mathbb{E}M_0. \tag{5.14}$$

Theorem 5.18: Doob's Stopping Theorem for Continuous Martingales

A martingale $M := (M_t, t \in \mathbb{R}_+)$ is a Doob martingale on $[0, b]$ for every $b \in \mathbb{R}_+$ and satisfies

$$\mathbb{E}_S M_T = M_S \quad (5.15)$$

for every pair (S, T) of bounded stopping times with $S \leq T$.

Proposition 5.2: Doob's Maximal Inequality

Let $X := (X_t, t \geq 0)$ be a submartingale that is positive and continuous. Then, for $p \geq 1$ and $b > 0$,

$$\mathbb{P}(\max_{0 \leq s \leq t} X_s \geq b) \leq \frac{\mathbb{E} X_t^p}{b^p}. \quad (5.16)$$

Proposition 5.3: Doob's Norm Inequality

Let $M := (M_t, t \geq 0)$ be a continuous martingale in L^p for some $p > 1$ and let $Z_t := \max_{s \leq t} |M_s|$. Then,

$$\mathbb{E} Z_t^p \leq q^p \mathbb{E} |M_t|^p, \text{ with } q := p/(p-1). \quad (5.17)$$

Lemma 5.2: Continuous Martingales with Finite Variation

Let $X := (X_t, t \geq 0)$ be a martingale with continuous paths and total variation $V_t < \infty$ on each interval $[0, t]$. Then, X is almost surely constant.

Proposition 5.4: Doob Martingale on $\overline{\mathbb{R}}_+$

A process M is a Doob martingale on $\overline{\mathbb{R}}_+$ if and only if it is a martingale on $\mathbb{R}_+$. If so, then

$$\mathbb{E}_S M_T = M_S \quad (5.18)$$

for arbitrary stopping times with $S \leq T$.

Theorem 5.19: Extension of UI Martingales

A martingale M on $\mathbb{R}_+$ can be extended to a Doob martingale $\bar{M}$ on $\bar{\mathbb{R}}_+$ if and only if it is uniformly integrable; that is, if and only if

$$M_t = \mathbb{E}_t Z, \quad t \in \mathbb{R}_+ \quad (5.19)$$

for some integrable random variable Z . Moreover, then, it converges almost surely and in L^1 to an integrable random variable M_∞ and can be extended to a UI martingale on $\bar{\mathbb{R}}_+$.

Proposition 5.5: Sufficient Condition for Being Doob

Let M be a martingale on $\mathbb{R}_+$ and ζ a stopping time. If, almost surely,

$$\sup_{t \in [0, \zeta] \cap \mathbb{R}_+} |M_t| \leq Z, \quad (5.20)$$

where $\mathbb{E}Z < \infty$ (i.e., M is dominated by an integrable random variable Z), then $M_\zeta = \lim M_{\zeta \wedge t}$ exists and is integrable. Moreover, M is a Doob martingale on $[0, \zeta]$.

6 Wiener and Brownian Motion Processes

We defined the Wiener process as a Gaussian process as follows:

Definition 6.1: Wiener Process

The *Wiener process* $(W_t, t \geq 0)$ is a zero-mean Gaussian process with continuous sample paths and covariance function

$$\gamma_{s,t} = \min\{s, t\} =: s \wedge t, \quad s, t \geq 0.$$

A Wiener process is a Lévy process with *continuous* sample paths, as shown in the following result, which can also serve as an equivalent definition of the Wiener process.

Theorem 6.1: Wiener Process as a Lévy Process

$W = (W_t)_{t \geq 0}$ is a Wiener process if and only if the following three properties hold:

- (1) (*Independent increments*): For any $t_1 < t_2 \leq t_3 < t_4$, the increments $W_{t_4} - W_{t_3}$ and $W_{t_2} - W_{t_1}$ are independent.
- (2) (*Gaussian stationarity*): For all $t, u \geq 0$, $W_{t+u} - W_t \sim \mathcal{N}(0, u)$.
- (3) (*Continuity of paths*): W has continuous paths, with $W_0 = 0$.

The following algorithm, [algorithm 1](#) uses the same ideas as in the previous proof to simulate the Wiener process at specific time points $t_1, \dots, t_n$

Algorithm 1 Simulating a Wiener Process

- 1: Let $0 = t_0 < t_1 < \dots < t_n$ be the times at which the process needs to be simulated.
 - 2: Simulate $Z_1, \dots, Z_n \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$.
 - 3: Return $W_0 := 0$ and $W_{t_k} := \sum_{i=1}^k Z_i \sqrt{t_i - t_{i-1}}, \quad k = 1, \dots, n$.
-

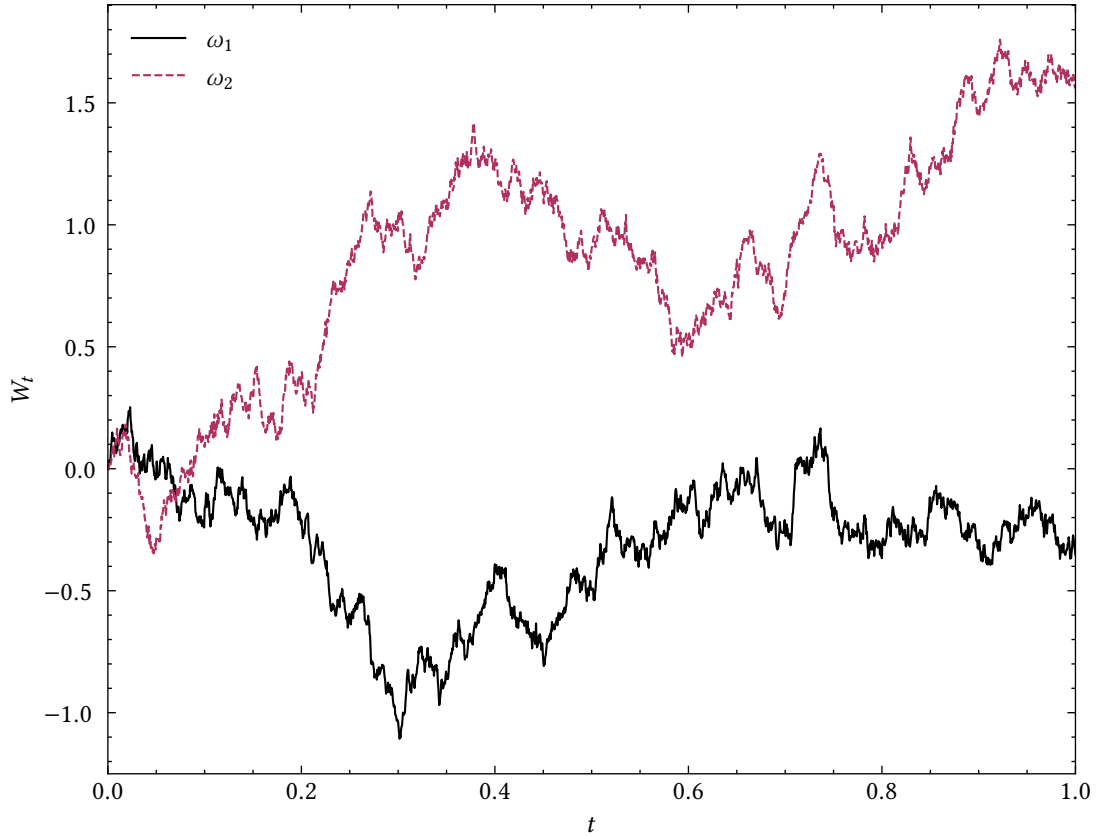


Figure 7: Two realisations of the Wiener process on the interval $[0, 1]$.

Definition 6.2: Brownian Motion

A stochastic process $(B_t, t \geq 0)$ satisfying

$$B_t = B_0 + \mu t + \sigma W_t, \quad t \geq 0,$$

where (W_t) is a Wiener process independent of B_0 , is called a *Brownian motion with drift μ and diffusion coefficient σ* .

A standard Brownian motion is one where $\mu = 0$ and $\sigma = 1$. The only difference between a Wiener process and a Brownian motion is its (random) starting position.

Algorithm 2 Simulating a Brownian Motion

- 1: Simulate the starting position B_0 .
- 2: Simulate outcomes $W_{t_1}, \dots, W_{t_n}$ of a Wiener process at times $t_1, \dots, t_n$.
- 3: Return $B_{t_i} := B_0 + \mu t_i + \sigma W_{t_i}$, $i = 1, \dots, n$ as the outcomes of the Brownian motion at times $t_1, \dots, t_n$.

Multidimensional Brownian motions are likewise obtained from an affine transformation $B_t = B_0 + \mu t + \sigma W_t$ of a multidimensional Wiener process, where σ is a matrix.

Definition 6.3: Multidimensional Wiener Process

Let $(W_{t,i}, t \geq 0)$, $i = 1, \dots, d$ be independent Wiener processes and let $W_t := [W_{t,1}, \dots, W_{t,d}]^\top$. The process $(W_t, t \geq 0)$ is called a d -dimensional Wiener process.

If (W_t) is a d -dimensional Wiener process, then W_t has a $\mathcal{N}(\mathbf{0}, t\mathbf{I}_d)$ multivariate normal distribution, where $\mathbf{I}_d$ denotes the d -dimensional identity matrix.

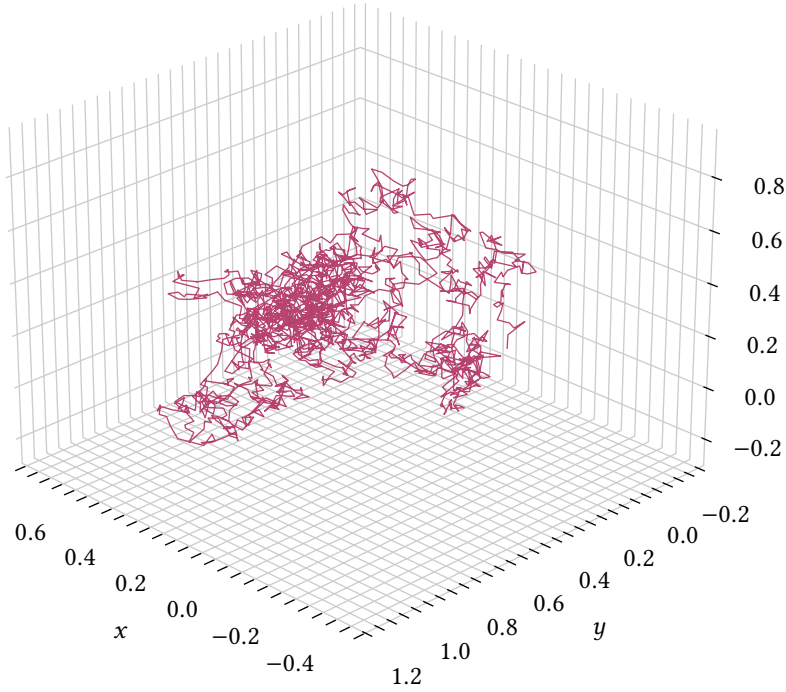


Figure 8: Three-dimensional Wiener process.

Theorem 6.2: Time-reversal and Time-shifting

If $(W_s, s \in [0, t])$ is a Wiener process on $[0, t]$, then so is the time-reversed process $(\widehat{W}_s, s \in [0, t])$ defined by $\widehat{W}_s := W_{t-s} - W_t$. Similarly, if $(W_s, s \geq 0)$ is a Wiener process, then for any $t \geq 0$ the process $(\widetilde{W}_s, s \geq 0)$ defined by $\widetilde{W}_s := W_{t+s} - W_t$ is also a Wiener process.

Theorem 6.3: Scaling

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := W_{at}/\sqrt{a}$, $t \geq 0$ for any $a > 0$.

Lemma 6.1: Probability Bound on the Supremum

For a Wiener process (W_t) , it holds for every $\varepsilon > 0$ that

$$\mathbb{P} \left(\sup_{0 \leq t \leq 1} |W_t| > \varepsilon \right) \leq 2e^{-\varepsilon^2/2}.$$

Theorem 6.4: Reciprocal Time

If (W_t) is a Wiener process, then so is (X_t) , with $X_t := tW_{1/t}$, $t > 0$, $X_0 := 0$.

Theorem 6.5: Existence of the Wiener Process

There exists a probability space $(\Omega, \mathcal{H}, \mathbb{P})$ and a zero-mean Gaussian process $W := (W_t, t \in [0, 1])$ with covariance function $\mathbb{E}W_s W_t = s \wedge t$ and uniformly continuous sample paths.

Theorem 6.6: Right-continuous Filtration

For the Wiener process, the augmentation $(\overline{\mathcal{F}}_t)$ of the natural filtration is right-continuous; that is,

$$\overline{\mathcal{F}}_t = \overline{\mathcal{F}}_t^+ := \bigcap_{\varepsilon > 0} \overline{\mathcal{F}}_{t+\varepsilon}.$$

Theorem 6.7: Wiener Process is Markov w.r.t $\mathcal{F}^+$

Let W be a Wiener process. For every $t \geq 0$, the process $(W_{t+u} - W_t, u \geq 0)$ is independent of $\mathcal{F}_t^+$. Consequently, W is Markovian with respect to $\mathcal{F}^+$.

Theorem 6.8: Strong Markov Property

For any finite stopping time T with respect to $\mathcal{F}^+$, the process $(W_{T+u} - W_T, u \geq 0)$ is a Wiener process independent of $\mathcal{F}_T^+$.

Theorem 6.9: Reflection Principle

Let T be a stopping time. Then, $(\tilde{W}_t, t \geq 0)$, defined by

$$\tilde{W}_t := W_t \mathbb{1}_{\{t \leq T\}} + (2W_T - W_t) \mathbb{1}_{\{t > T\}}, \quad t \geq 0, \quad (6.1)$$

is a Wiener process.

Theorem 6.10: Transition Kernel of the Wiener Process

The Wiener process W admits the transition function

$$P_{t,u}(x, A) = \mathbb{P}(W_u \in A \mid W_t = x) = \int_A dy p_{u-t}(y \mid x),$$

where the transition density $p_t(y \mid x)$ is given by

$$p_t(y \mid x) := \frac{1}{\sqrt{2\pi t}} e^{-\frac{(y-x)^2}{2t}}, \quad t \geq 0, \quad x, y \in \mathbb{R}. \quad (6.2)$$

Theorem 6.11: Exponential Martingale

A continuous process $X := (X_t, t \geq 0)$ is a Wiener process if and only if for each $r \in \mathbb{R}$ the process $S := (S_t)$ defined by

$$S_t := e^{rX_t - \frac{1}{2}r^2t}, \quad t \geq 0, \quad (6.3)$$

is a martingale.

Theorem 6.12: Martingales from Functions of a Brownian Motion

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable and let $\mathbf{B}$ be d -dimensional standard Brownian motion. If for all $t > 0$ and $\mathbf{x} \in \mathbb{R}^d$,

$$\mathbb{E}^{\mathbf{x}}|f(\mathbf{B}_t)| < \infty \quad \text{and} \quad \mathbb{E}^{\mathbf{x}} \int_0^t |\Delta f(\mathbf{B}_s)| ds < \infty,$$

then the process $(X_t, t \geq 0)$ defined by

$$X_t := f(\mathbf{B}_t) - \frac{1}{2} \int_0^t \Delta f(\mathbf{B}_s) ds \tag{6.4}$$

is a martingale.

7 Itô Calculus

7.1 Itô Integral

We will work with a Wiener process W on a complete probability space $(\Omega, \mathcal{H}, \mathbb{P})$ with a right-continuous filtration $\mathcal{F} := (\mathcal{F}_s, s \geq 0)$ and such that W is adapted to $\mathcal{F}$. We also use the following definition.

Definition 7.1: Simple Processes

A process $F := (F_s, s \in [0, t])$ is called *simple* if it is of the form

$$F_s := \xi_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} \xi_k \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t], \quad (7.1)$$

where $\{s_0, \dots, s_n : 0 = s_0 < \dots < s_n = t\} =: \Pi_n, n \in \mathbb{N}$, is a segmentation of $[0, t]$ and $\xi_k \in \mathcal{F}_{s_k}$ for all k .

Note that the simple process in (7.1) is right-continuous at $s = 0$ and is left-continuous at $s \in (0, t]$. Figure 9 shows a realisation of a simple process on $[0, 1]$ for the segmentation

$$\{s_0, \dots, s_{10}\} = \{0, 0.07, 0.13, 0.19, 0.41, 0.49, 0.52, 0.67, 0.71, 0.76, 1\}.$$

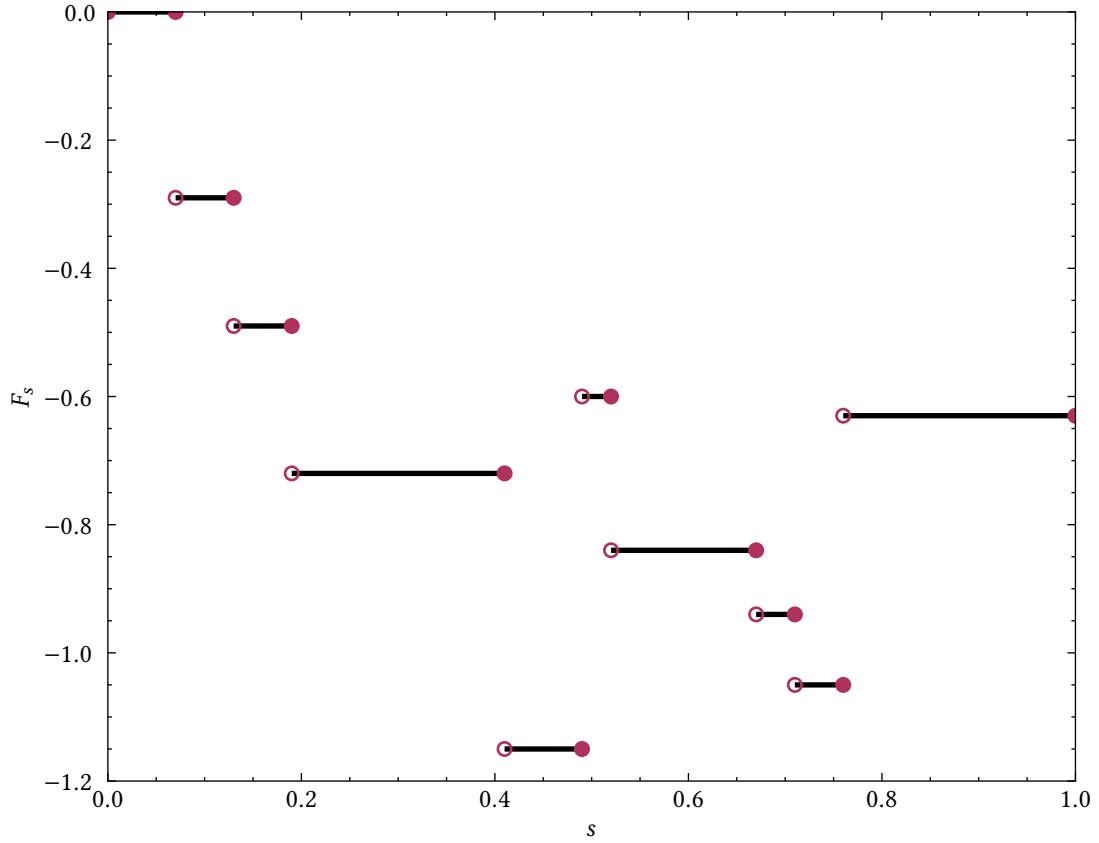


Figure 9: A realisation of a simple process.

Definition 7.2: Itô Integral for Simple Processes

Suppose that F is a simple process of the form eq. (7.1), with $\int_0^t \mathbb{E} F_s^2 ds < \infty$. Then, the *Itô integral* of F with respect to W over $[0, r]$ is defined as

$$\int_0^r F_s dW_s := \sum_{k=0}^{n-1} \xi_k (W_{s_{k+1} \wedge r} - W_{s_k \wedge r}), \quad r \in [0, t]. \quad (7.2)$$

Theorem 7.1: Properties of the Itô Integral

Assume that the Itô integral is well-defined. Then, for $\alpha, \beta \in \mathbb{R}$ and $r \in [0, t]$:

- (1) (Linearity): $\int_0^r [\alpha F_s + \beta G_s] dW_s = \alpha \int_0^r F_s dW_s + \beta \int_0^r G_s dW_s$.
- (2) (Zero mean): $\mathbb{E} \int_0^r F_s dW_s = 0$.
- (3) (Isometry): $\mathbb{E} \left[\int_0^r F_s dW_s \times \int_0^r G_s dW_s \right] = \int_0^r \mathbb{E} [F_s G_s] ds$.
- (4) (Martingale): $(\int_0^r F_s dW_s, r \in [0, t])$ is an L^2 -martingale.

Definition 7.3: Canonical Approximation

For any $\mathcal{F}$ -adapted process $F := (F_s, s \in [0, t])$, we define its *canonical approximation* as

$$F_s^{(n)} := F_0 \mathbb{1}_{\{0\}}(s) + \sum_{k=0}^{n-1} F_{s_k} \mathbb{1}_{(s_k, s_{k+1}]}(s), \quad s \in [0, t]. \quad (7.3)$$

Theorem 7.2: Properties of Left-continuous Adapted Processes

Suppose that $F := (F_s, s \in [0, t])$ is a process on $(\Omega, \mathcal{H}, \mathbb{P})$ that is adapted to the filtration $\mathcal{F}$ and is left-continuous on $(0, t]$. Then, the following hold:

- (1) (Predictable): The process F is $\mathcal{F}$ -predictable (i.e., $\mathcal{F}^P$ -measurable).
- (2) (Progressively Measurable): For each t , the mapping $F : [0, t] \times \Omega \rightarrow \mathbb{R}$ is jointly measurable with respect to $\mathcal{B}_{[0,t]} \otimes \mathcal{F}_t$.
- (3) (Indistinguishable): If $\tilde{F}$ is a modification of F and almost surely left-continuous, then $\mathbb{P}(\cap_{s \leq t} \{F_s = \tilde{F}_s\}) = 1$.

Definition 7.4: The Class of Processes $\mathcal{P}_t$

Let $\mathcal{P}_t$ be the class of stochastic processes $F := (F_s, s \in [0, t])$ that are left-continuous on $(0, t]$, $\mathcal{F}$ -adapted, and satisfy the norm condition:

$$\|F\|_{\mathcal{P}_t}^2 := \mathbb{E} \int_0^t F_s^2 ds < \infty.$$

Theorem 7.3: Approximating the Class $\mathcal{P}_t$ via Simple Processes

Suppose that $F \in \mathcal{P}_t$. Then, there exists a sequence $(F^{(n)})$ of simple processes of the form eq. (7.1) such that $\text{mesh}(\Pi_n) \rightarrow 0$ implies that $F_s^{(n)} \xrightarrow{\text{a.s.}} F_s$ for all $s \in [0, t]$ and

$$\lim_{n \rightarrow \infty} \|F^{(n)} - F\|_{\mathcal{P}_t} = 0. \quad (7.4)$$

Proposition 7.1: Deterministic Integrands and Gaussian Processes

Let $f : \mathbb{R}_+ \times \mathbb{R}_+ \rightarrow \mathbb{R}$ be left-continuous in the first argument and such that $\|f(\cdot, t)\|_{\mathcal{P}_t}^2 = \int_0^t f^2(s, t) ds < \infty$. Then, the stochastic integral process

$$Z_t := \int_0^t f(s, t) dW_s, \quad t \geq 0, \quad (7.5)$$

is a zero-mean Gaussian process with covariance function

$$\text{Cov}(Z_{t_1}, Z_{t_2}) = \int_0^{t_1 \wedge t_2} f(s, t_1) f(s, t_2) ds, \quad t_1, t_2 \geq 0. \quad (7.6)$$

$$\lim_{m \rightarrow \infty} \sup_{s \in [0, t]} \mathbb{E} F_s^2 \mathbf{1}_{\{F_s^2 > m\}} = 0. \quad (7.7)$$

Lemma 7.1: Quadratic Variation Integral

Suppose that $F \in \mathcal{P}_t$ satisfies the condition [eq. \(7.7\)](#). Then, as $\text{mesh}(\Pi_n) \rightarrow 0$:

$$\sum_{k=0}^{n-1} F_{s_k} (W_{s_{k+1}} - W_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s ds. \quad (7.8)$$

Theorem 7.4: Enlarging the Class $\mathcal{P}_t$ via Localization

Let $F := (F_s, s \in [0, t])$ be an $\mathcal{F}$ -adapted process, left-continuous on $(0, t]$, and such that (7.34) holds. Then, there exists a localizing sequence (T_n) such that the stopped process $(\int_0^{r \wedge T_n} F_s dW_s, r \in [0, t])$ is a square-integrable martingale. In addition, the Itô integral of F with respect to W on $[0, t]$ is defined as the almost sure limit:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^{t \wedge T_n} F_s dW_s. \quad (7.9)$$

Proposition 7.2: Continuous Modification and Stopping Times

If $F \in \mathcal{P}_t$ and $T \leq t$ is a stopping time with respect to $\mathcal{F}$, then almost surely:

$$\int_0^T F_s dW_s = \int_0^t \mathbf{1}_{[0, T]}(s) F_s dW_s. \quad (7.10)$$

Theorem 7.5: Continuous Modification of the Itô Integral

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Denote the limit $\int_0^r F_s^{(n)} dW_s =: Z_r^{(n)} \xrightarrow{L^2} Z_r$ as the Itô integral on $[0, r]$ for $r \leq t$. Then, there exists a modification $\tilde{Z}$ of $(Z_r, r \in [0, t])$ such that the mapping $r \mapsto \tilde{Z}_r$ is almost surely continuous. Moreover, this modification is unique up to indistinguishability.

Theorem 7.6: Existence of the Itô Integral with Integrand $F \in \mathcal{P}_t$

Suppose that $F \in \mathcal{P}_t$ and $(F^{(n)})$ is a sequence of simple processes satisfying (7.17). Then, the following limit exists in L^2 norm:

$$\int_0^t F_s dW_s := \lim_{n \rightarrow \infty} \int_0^t F_s^{(n)} dW_s, \quad r \in [0, t], \quad (7.11)$$

and it defines the Itô integral of F with respect to W on $[0, t]$.

7.2 Itô Calculus**Theorem 7.7: Itô Formula for the Wiener Process**

Let W be a Wiener process and $f : \mathbb{R} \rightarrow \mathbb{R}$ be twice continuously differentiable with derivatives f' and f'' . Then, almost surely

$$f(W_t) - f(W_0) = \int_0^t f'(W_s) dW_s + \frac{1}{2} \int_0^t f''(W_s) ds.$$

Definition 7.5: Integral With Respect To An Itô Process

Let X be an $\text{Itô}(\mu, \sigma)$ process and F a left-continuous $\mathcal{F}$ -adapted process such that almost surely $\int_0^t (|F_s \mu_s| + (F_s \sigma_s)^2) ds < \infty$ for all t . Then, the integral of F with respect to X is defined as the $\text{Itô}(F\mu, F\sigma)$ process with:

$$\int_0^t F_s dX_s := \int_0^t F_s \mu_s ds + \int_0^t F_s \sigma_s dW_s. \quad (7.12)$$

Definition 7.6: Quadratic Variation On $[0, t]$

Let $\Pi_n := \{s_k, k = 0, \dots, n\}$ be a segmentation of $[0, t]$ and let $(X^{(n)})$ be a sequence of simple processes with almost sure limit $X_s := \lim_n X_s^{(n)}$ for all $s \in [0, t]$ as $\text{mesh}(\Pi_n) \rightarrow 0$. The *quadratic variation* of $X := (X_s, s \in [0, t])$ over $[0, t]$, denoted as $\langle X \rangle_t$, is defined as the limit in probability of

$$\langle X^{(n)} \rangle_t := \sum_{k=0}^{n-1} (X_{s_{k+1}}^{(n)} - X_{s_k}^{(n)})^2. \quad (7.13)$$

Theorem 7.8: Quadratic Variation Of An Itô Process

If X is an Itô(μ, σ) process, then

$$\langle X^{(n)} \rangle_t \xrightarrow{L^2} \langle X \rangle_t = \int_0^t \sigma_s^2 ds. \quad (7.14)$$

Furthermore, if $F \in \mathcal{P}_t$ satisfies the condition (7.19) and $\text{mesh}(\Pi_n) \rightarrow 0$, then:

$$\sum_{k=0}^{n-1} F_{s_k} (X_{s_{k+1}} - X_{s_k})^2 \xrightarrow{L^2} \int_0^t F_s \sigma_s^2 ds. \quad (7.15)$$

Theorem 7.9: Itô's Formula For Itô Processes

Let X be an Itô process and let f be twice continuously differentiable. Then,

$$f(X_t) = f(X_0) + \int_0^t f'(X_s) dX_s + \frac{1}{2} \int_0^t f''(X_s) d\langle X \rangle_s. \quad (7.16)$$

Definition 7.7: Covariation Of Two Itô Processes

Let X and Y be two Itô processes adapted to the same filtration $\mathcal{F}$. Then, the *covariation* between X and Y is defined as the quantity:

$$\langle X, Y \rangle_t := \frac{\langle X + Y \rangle_t - \langle X \rangle_t - \langle Y \rangle_t}{2}. \quad (7.17)$$

Theorem 7.10: Itô's Formula In $\mathbb{R}^d$

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be twice continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) dX_{t,i} + \frac{1}{2} \sum_{i=1}^d \sum_{j=1}^d \partial_{ij} f(X_t) d\langle X_{\cdot,i}, X_{\cdot,j} \rangle_t. \quad (7.18)$$

Theorem 7.11: Itô Processes And Martingales

An Itô process in $\mathbb{R}^d$ is a local martingale if and only if its drift is 0.

7.3 Stochastic Differential Equations

Stochastic differential equations are based on the same principle as ordinary differential equations, relating an unknown function to its derivatives, but with the additional feature that part of the unknown function is driven by randomness.

Definition 7.8: Stochastic Differential Equation

A *stochastic differential equation* (SDE) for a stochastic process $(X_t, t \geq 0)$ is an expression of the form

$$dX_t = a(X_t, t)dt + b(X_t, t)dW_t, \quad (7.19)$$

where $(W_t, t \geq 0)$ is a Wiener process and a and b are deterministic functions, called the *drift* and *diffusion* functions. The resulting process $(X_t, t \geq 0)$ is called an *(Itô) diffusion*.

Consider the autonomous SDE

$$dX_t = a(X_t)dt + b(X_t)dW_t. \quad (7.20)$$

As already mentioned, the meaning of this SDE is that X is a stochastic process that satisfies the integral equation

$$X_t = X_0 + \int_0^t a(X_s)ds + \int_0^t b(X_s)dW_s. \quad (7.21)$$

However, it is not clear that such a process indeed exists. We show in this section that under mild conditions on a and b we can obtain a unique solution X of the SDE via a method of successive approximations, similar to *Picard iteration* for ordinary differential equations.

The condition that is placed on a and b is that they be *Lipschitz continuous*; that is, for all $x \in \mathbb{R}$ there exists a constant c such that

$$|a(y) - a(x)| + |b(y) - b(x)| \leq c|y - x|. \quad (7.22)$$

The successive approximation idea is very simple. Suppose that $X_0 = x$. Define $X_t^{(0)} := x$ for all t , and let

$$X_t^{(n+1)} := x + \int_0^t a(X_s^{(n)}) ds + \int_0^t b(X_s^{(n)}) dW_s \quad (7.23)$$

for $n \in \mathbb{N}$. The following is the main existence result for SDEs:

Theorem 7.12: Existence of SDE Solutions

For any $t \in [0, v]$, the sequence $(X^{(n)})$ in (7.23) converges almost surely in the uniform norm on $[0, t]$ to a process X that satisfies the integral equation (??).

Lemma 7.2: Integral Bound

Let $X^{(n)}, n = 1, 2, \dots$ be defined in (7.23), with $X^{(0)} = x$. Then, for every $v > 0$ there exists a constant α such that

$$\mathbb{E} \sup_{s \leq t} \left(X_s^{(n+1)} - X_s^{(n)} \right)^2 \leq \alpha \|X^{(n)} - X^{(n-1)}\|_{\mathcal{P}_t}^2, \quad t \leq v.$$

Definition 7.9: Itô Stratonovich Integral

Let X and Y be Itô processes such that $\int_0^t Y_s dX_s$ and $\int_0^t X_s dY_s$ are well-defined stochastic integrals. Then, the *Itô–Stratonovich integral* of Y with respect to X is defined as

$$\int_0^t Y_s \circ dX_s := \int_0^t Y_s dX_s + \frac{1}{2} \langle X, Y \rangle_t. \quad (7.24)$$

Proposition 7.3: Itô Stratonovich Formula

Let (X_t) be a d -dimensional Itô process, and let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be three times continuously differentiable in all variables. Then,

$$df(X_t) = \sum_{i=1}^d \partial_i f(X_t) \circ dX_{t,i}. \quad (7.25)$$

$$\widetilde{\mathbb{P}}^t(A) := \mathbb{E} M_t \mathbb{1}_A, \quad A \in \mathcal{F}_t. \quad (7.26)$$

$$M_t = \exp \left(\int_0^t \mu_s dW_s - \frac{1}{2} \int_0^t \mu_s^2 ds \right) = e^{Z_t - \frac{1}{2} \langle Z \rangle_t}, \quad t \geq 0, \quad (7.27)$$

$$\lim_{\varepsilon \downarrow 0} \varepsilon \ln \mathbb{E} \exp((1 - \varepsilon)\langle Z \rangle_t / 2) = 0. \quad (7.28)$$

Theorem 7.13: Girsanov's Change of Measure

Suppose that μ satisfies the Krylov condition [eq. \(7.28\)](#) and M is the exponential martingale [eq. \(7.27\)](#). Then, under the change of measure [eq. \(7.26\)](#) induced by M , the Itô process

$$\tilde{W}_t := - \int_0^t \mu_s ds + W_t, \quad t \geq 0$$

is a Wiener process.

GBM.py

```

1 N, T = 1000, 1.0
2 mu, sigma = 0.5, 0.2
3 dt = T / N
4 t = np.linspace(0, T, N + 1)
5
6 rng = np.random.default_rng(4)
7 dW = rng.normal(0, np.sqrt(dt), N)
8 W = np.r_[0, np.cumsum(dW)]
9
10 drift = (mu - 0.5 * sigma**2) * t
11 diffusion = sigma * W
12 S_t = np.exp(drift + diffusion)
13 E_St = np.exp(mu * t)

```

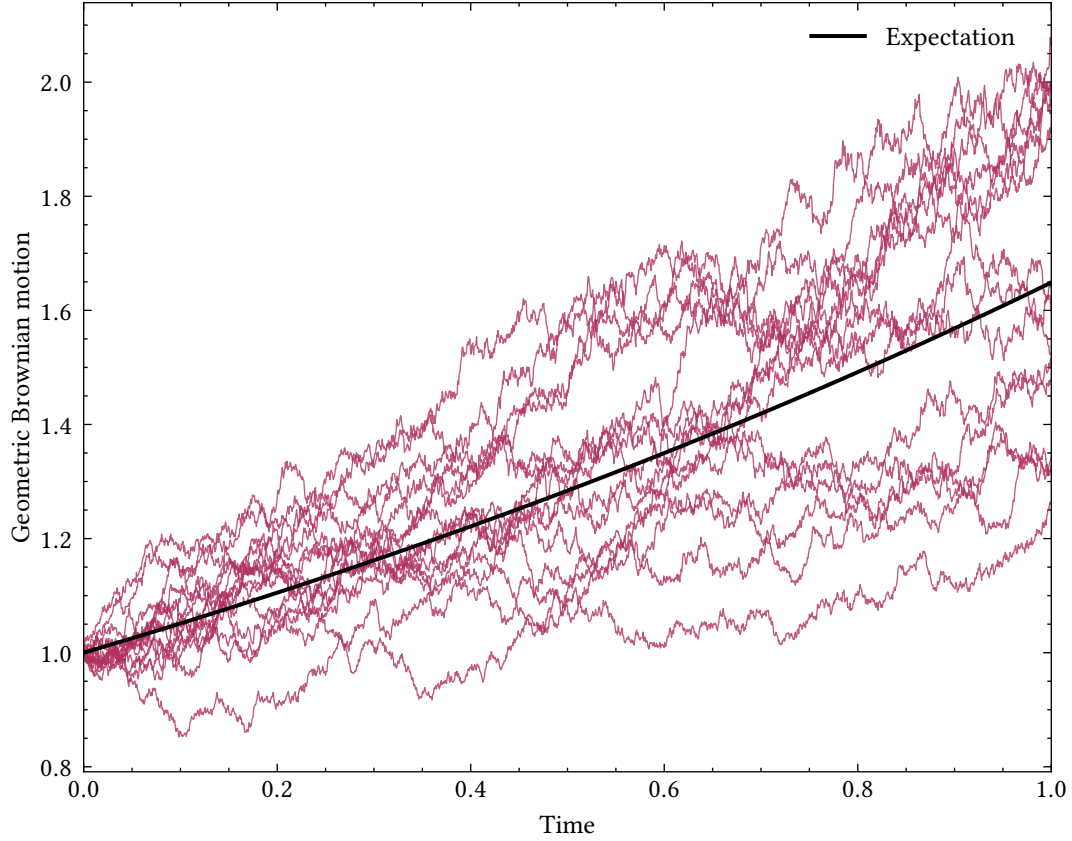


Figure 10: Geometric Brownian motion sample path and expectation for $\mu = 0.5$, $\sigma = 0.2$ and $S_0 = 1$.

Part II

Levy Processes

8 Definitions and Properties

Lévy processes serve as the fundamental building blocks of continuous-time stochastic modelling, representing the continuous analogue of random walks through their defining property of stationary and independent increments (Sato, 2013; Schilling, 2016).

Analytically, their distributional structure is isomorphic to the class of infinitely divisible distributions via the Lévy-Khintchine formula, which characterizes the characteristic exponent through a unique triplet comprising a deterministic drift, a Gaussian variance, and a Lévy measure governing jump intensity (Applebaum, 2013).

From a pathwise perspective, the Lévy-Itô decomposition explicitly resolves the trajectory into the superposition of a deterministic trend, a continuous Gaussian martingale, and a discontinuous jump component driven by a Poisson random measure (Kyprianou, 2014).

Within derivatives pricing, this framework generalizes geometric Brownian motion to endogenously capture the leptokurtosis and skewness inherent in asset returns, thereby rectifying the distributional inadequacies of diffusion-only models (Cont and Tankov, 2004).

8.1 From Random Walks to Lévy Processes

Definition 8.1: Lévy process

A cadlag¹ stochastic process $(X_t)_{t \geq 0}$ on $(\Omega, \mathcal{F}, \mathbb{P})$ with values in $\mathbb{R}^d$ such that $X_0 = 0$ is called a Lévy process if it possesses the following properties:

1. *Independent increments*: for every increasing sequence of times $t_0 \dots t_n$, the random variables $X_{t_0}, X_{t_1} - X_{t_0}, \dots, X_{t_n} - X_{t_{n-1}}$ are independent.
2. *Stationary increments*: the law of $X_{t+h} - X_t$ does not depend on t .
3. *Stochastic continuity*: $\forall \varepsilon > 0, \lim_{h \rightarrow 0} \mathbb{P}(|X_{t+h} - X_t| \geq \varepsilon) = 0$.

Definition 8.2: Infinite Divisibility

A probability distribution F on $\mathbb{R}^d$ is said to be *infinitely divisible* if for any integer $n \geq 2$, there exist n i.i.d. random variables $Y_1, \dots, Y_n$ such that $Y_1 + \dots + Y_n$ has distribution F .

Proposition 8.1: Infinite Divisibility and Lévy Processes

Let $X := (X_t)_{t \geq 0}$ be a Lévy process. Then for every t , X_t has an infinitely divisible distribution. Conversely, if F is an infinitely divisible distribution, there exists a Lévy process X such that the distribution of X_1 is given by F .

¹Some authors do not impose the cadlag (right-continuity and left limits) property in the definition of a Lévy process, but it can be shown that every Lévy process (defined without the cadlag property) has a unique modification that is cadlag; therefore, the cadlag property can be assumed without loss of generality.

Proposition 8.2: Characteristic function of a Lévy process

Let $(X_t)_{t \geq 0}$ be a Lévy process on $\mathbb{R}^d$. There exists a continuous function $\psi : \mathbb{R}^d \mapsto \mathbb{R}$ called the characteristic exponent of X , such that:

$$E \left[e^{iz \cdot X_t} \right] = e^{t\psi(z)}, \quad z \in \mathbb{R}^d. \quad (8.1)$$

9 Building Levy Processes

9.1 Model building with Lévy Processes

Let us start by some general considerations on Lévy processes, their use as models for price dynamics and different ways to specify such models.

10 Simulating Levy Processes

11 Stochastic Calculus for Jump Processes

Consider now a jump-diffusion process

$$X_t = \sigma W_t + \mu t + J_t = X^c(t) + J_t, \quad (11.1)$$

where J is a compound Poisson process and X^c is the continuous part of X :

$$J_t = \sum_{i=1}^{N_t} \Delta X_i, \quad X_t^c = \mu t + \sigma W_t. \quad (11.2)$$

Define $Y_t = f(X_t)$ where $f \in C^2(\mathbb{R})$ and denote by $T_i, i = 1 \dots N_T$ the jump times of X . On $[T_i, T_{i+1}]$, X evolves according to

$$dX_t = dX_t^c = \sigma dW_t + \mu dt, \quad (11.3)$$

so by applying the Itô formula in the Brownian case we obtain

$$\begin{aligned} Y_{T_{i+1}-} - Y_{T_i} &= \int_{T_i}^{T_{i+1}-} \frac{\sigma^2}{2} f''(X_t) dt + \int_{T_i}^{T_{i+1}-} f'(X_t) dX_t \\ &= \int_{T_i}^{T_{i+1}-} \left\{ \frac{\sigma^2}{2} f''(X_t) dt + f'(X_t) dX_t^c \right\} \end{aligned} \quad (11.4)$$

since $dX_t = dX^c(t)$ on this interval. If a jump of size ΔX_t occurs then the resulting change in Y_t is given by $f(X_{t-} + \Delta X_t) - f(X_{t-})$. The total change in Y_t can therefore be written as the sum of these two contributions:

$$\begin{aligned} f(X_t) - f(X_0) &= \int_0^t f'(X_s) dX_s^c + \int_0^t \frac{\sigma^2}{2} f''(X_s) ds \\ &\quad + \sum_{0 \leq s \leq t, \Delta X_s \neq 0} [f(X_{s-} + \Delta X_s) - f(X_{s-})]. \end{aligned} \quad (11.5)$$

Attention 11.1

replacing dX_s^c by $dX_s - \Delta X_s$ one obtains an equivalent expression:

$$\begin{aligned} f(X_t) - f(X_0) = & \int_0^t f'(X_{s-}) dX_s + \int_0^t \frac{\sigma^2}{2} f''(X_s) ds \\ & + \sum_{0 \leq s \leq t, \Delta X_s \neq 0} [f(X_{s-} + \Delta X_s) - f(X_{s-}) - \Delta X_s f'(X_{s-})]. \end{aligned} \quad (11.6)$$



When the number of jumps is finite, this form is equivalent to [eq. \(11.5\)](#). However, as we will see below, the form ?? is more general, wherein both the stochastic integral and the sum over the jumps in [eq. \(11.6\)](#) are well-defined for any semi-martingale, even if the number of jumps is infinite, while the sum in [eq. \(11.5\)](#) may not converge if jumps have infinite variation.

Here we have only used the Itô formula for diffusions, which is of course still valid if σ is replaced by a process $(\sigma_t)_{t \in [0, T]}$ belonging to the class $\mathcal{P}_T$.

Proposition 11.1: Itô Formula for Jump-Diffusion Processes

Let X be a diffusion process with jumps, defined as the sum of a drift term, a Brownian stochastic integral and a compound Poisson process:

$$X_t = X_0 + \int_0^t b_s ds + \int_0^t \sigma_s dW_s + \sum_{i=1}^{N_t} \Delta X_i,$$

where b_t and σ_t are continuous adapted processes with

$$\mathbb{E} \left[\int_0^T \sigma_t^2 dt \right] < \infty.$$

Then, for any $C^{1,2}$ function $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$, the process $Y_t = f(t, X_t)$ can be represented as

$$\begin{aligned} f(t, X_t) - f(0, X_0) &= \int_0^t \left[\frac{\partial f}{\partial s}(s, X_s) + \frac{\partial f}{\partial x}(s, X_s) b_s \right] ds \\ &\quad + \frac{1}{2} \int_0^t \sigma_s^2 \frac{\partial^2 f}{\partial x^2}(s, X_s) ds + \int_0^t \frac{\partial f}{\partial x}(s, X_s) \sigma_s dW_s \\ &\quad + \sum_{\{i \geq 1, T_i \leq t\}} [f(X_{T_i-} + \Delta X_i) - f(X_{T_i-})]. \end{aligned}$$

Expressed in differential notation

$$\begin{aligned} dY_t &= \frac{\partial f}{\partial t}(t, X_t) dt + b_t \frac{\partial f}{\partial x}(t, X_t) dt + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2}(t, X_t) dt \\ &\quad + \frac{\partial f}{\partial x}(t, X_t) \sigma_t dW_t + [f(X_{t-} + \Delta X_t) - f(X_{t-})]. \end{aligned}$$

11.1 Risk-neutral modelling with exponential Lévy Processes

12 Integro-differential equations and numerical methods

An approximate answer to the right problem is worth a good deal more than an exact answer to an approximate problem.

John Tukey

Part III

Stochastic Volatility

13 Heston Model

The Heston model is a first-generation model; it owes its popularity to the fact that, being an affine model, the Laplace transform of its moment-generating function for $\ln S$ is analytically known.

Part IV

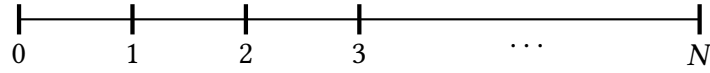
Risk-Neutral Valuation

14 Assets, Portfolios and Arbitrage

15 Discrete-time Market Model

15.1 Discrete Time Compounding

We work in a finite-horizon discrete-time model indexed by $\{0, 1, \dots, N\}$



We invest a cash flow C each year in an investment plan that carries a constant interest rate r . At the end of the N -th year, the value of the amount C invested at the beginning of year $k = 1, 2, \dots, N$ has turned into $C(1+r)^{N-k+1}$ and the total value of the plan V_N at the end of the N -th year becomes

$$\begin{aligned} V_N &= C \sum_{k=1}^N (1+r)^{N-k+1} \\ &= C \sum_{k=1}^N (1+r)^k \\ &= C(1+r) \frac{(1+r)^N - 1}{r}, \end{aligned} \quad (15.1)$$

hence the duration N of the plan satisfies

$$N+1 = \frac{1}{\log(1+r)} \log \left(1+r + \frac{rV_N}{C} \right). \quad (15.2)$$

At time $t = 0$ one borrows an amount L over a period of N years at the constant interest rate r per year.

Proposition 15.1: Constant Repayment

Assuming the loan is fully amortized at the beginning of year $N+1$, the constant annual repayment C is uniquely determined by

$$C = \frac{r(1+r)^N L}{(1+r)^N - 1} = \frac{r}{1 - (1+r)^{-N}} L. \quad (15.3)$$

Proof. Let L_k denote the outstanding principal balance at the beginning of year k , with $L_1 = L$. The evolution of the debt is governed by the linear recurrence relation

$$L_{k+1} = (1+r)L_k - C, \quad \forall k \geq 1. \quad (15.4)$$

Iteratively expanding this relation yields the value of the debt at the start of any arbitrary year k

$$\begin{aligned} L_k &= (1+r)^{k-1}L_1 - C \sum_{j=0}^{k-2} (1+r)^j \\ &= (1+r)^{k-1}L - C \frac{(1+r)^{k-1} - 1}{r}. \end{aligned} \quad (15.5)$$

The amortization condition requires that the debt is extinguished at the horizon N , such that $L_{N+1} = 0$. Substituting $k = N + 1$ into eq. (15.5) implies

$$0 = (1+r)^N L - C \frac{(1+r)^N - 1}{r}.$$

Rearranging for C yields the result in 15.1. Furthermore, substituting this expression for C back into eq. (15.5) provides the closed-form outstanding balance

$$L_k = \frac{C}{r} [1 - (1+r)^{k-1-N}] \equiv \frac{C - (1+r)^{k-1}(C - rL)}{r}. \quad (15.6)$$

□

Rearranging the result in 15.1 yields the annuity factor

$$\frac{L}{C} = \frac{1 - (1+r)^{-N}}{r}, \quad (15.7)$$

and solving for the duration N implies

$$N = \frac{1}{\log(1+r)} \log \left(\frac{C}{C - rL} \right) = -\frac{\log(1 - \frac{rL}{C})}{\log(1+r)}. \quad (15.8)$$

Tip 15.1: F



r N to be finite, we require the condition $C > rL$. This means the annual repayment must exceed the interest accrued on the initial principal; otherwise, the principal balance will never decrease.

Proposition 15.2: Repayment Decomposition

The k -th repayment C decomposes into an interest component rL_k and a principal repayment component $C - rL_k$ given by

$$rL_k = C \left(1 - \frac{1}{(1+r)^{N-k+1}} \right) = Cr \sum_{l=1}^{N-k+1} (1+r)^{-l} \quad (15.9)$$

$$C - rL_k = \frac{C}{(1+r)^{N-k+1}}, \quad k = 1, 2, \dots, N. \quad (15.10)$$

Proof. Recall from the proof of 15.1 that the outstanding balance L_k represents the present value of the remaining $N - k + 1$ payments. We express L_k as

$$L_k = \frac{C}{r} \left[1 - (1+r)^{-(N-k+1)} \right]. \quad (15.11)$$

Multiplying by r immediately yields the interest component in eq. (15.9). Consequently, the principal repayment is determined by the residual

$$\begin{aligned} C - rL_k &= C - C \left[1 - (1+r)^{-(N-k+1)} \right] \\ &= C(1+r)^{-(N-k+1)}, \end{aligned} \quad (15.12)$$

which corresponds to eq. (15.10). □

We verify that the sum of discounted payments at rate r recovers the initial loan amount

$$\sum_{l=1}^N \frac{C}{(1+r)^l} = C \frac{1 - (1+r)^{-N}}{r} = L. \quad (15.13)$$

In particular, the first interest repayment satisfies

$$rL_1 = Cr \sum_{l=1}^N \frac{1}{(1+r)^l} = C \left(1 - (1+r)^{-N} \right), \quad (15.14)$$

and the first principal repayment is

$$C - rL = \frac{C}{(1+r)^N}. \quad (15.15)$$

15.2 The Model

Definition 15.1: numéraire

A *numéraire* is a price process $(X(t))_{t=0}^T$ (a sequence of random variables), which is strictly positive for all $t \in \{0, 1, \dots, T\}$.

For the standard approach the risk-free bank account process is used as numéraire. In some applications, however, it is more convenient to use a security other than the bank account and we therefore just use S_0 without loss of generality.

Definition 15.2: Value of the Portfolio

The value of the portfolio at time t is the scalar product

$$V_\varphi(t) = \varphi(t) \cdot S(t) := \sum_{i=0}^d \varphi_i(t) S_i(t), \quad (t = 1, 2, \dots, T) \text{ and } V_\varphi(0) = \varphi(1) \cdot S(0).$$

The process $V_\varphi(t, \omega)$ is called the *wealth* or *value process* of the trading strategy φ .

The initial wealth $V_\varphi(0)$ is called the *initial investment* or *endowment* of the investor.

Now $\varphi(t) \cdot S(t-1)$ reflects the market value of the portfolio just after it has been established at time $t-1$, whereas $\varphi(t) \cdot S(t)$ is the value just after time t prices are observed, but before changes are made in the portfolio. Hence

$$\varphi(t) \cdot (S(t) - S(t-1)) = \varphi(t) \cdot \Delta S(t)$$

is the change in the market value due to changes in security prices which occur between time $t-1$ and t . This motivates the following definition.

Definition 15.3: Gains Process

The *gains process* G_φ of a trading strategy φ is given by

$$G_\varphi(t) := \sum_{\tau=1}^t \varphi(\tau) \cdot (S(\tau) - S(\tau-1)) = \sum_{\tau=1}^t \varphi(\tau) \cdot \Delta S(\tau), \quad (t = 1, 2, \dots, T).$$

Define $\tilde{S}(t) = (1, \beta(t)S_1(t), \dots, \beta(t)S_d(t))'$, the vector of discounted prices, and consider the *discounted value process*

$$\tilde{V}_\varphi(t) = \beta(t)(\varphi(t) \cdot S(t)) = \varphi(t) \cdot \tilde{S}(t), \quad (t = 1, 2, \dots, T)$$

and the *discounted gains process*

$$\tilde{G}_\varphi(t) := \sum_{\tau=1}^t \varphi(\tau) \cdot (\tilde{S}(\tau) - \tilde{S}(\tau-1)) = \sum_{\tau=1}^t \varphi(\tau) \cdot \Delta \tilde{S}(\tau), \quad (t = 1, 2, \dots, T).$$

Observe that the discounted gains process reflects the gains from trading with assets 1 to d only, which in case of the standard model (a bank account and d stocks) are the risky assets. We will only consider special classes of trading strategies.

Definition 15.4: Self-Financing Strategy

The strategy φ is *self-financing*, $\varphi \in \Phi$, if

$$\varphi(t) \cdot S(t) = \varphi(t+1) \cdot S(t) \quad (t = 1, 2, \dots, T-1). \quad (15.16)$$

Tip 15.2: W

When new prices $S(t)$ are quoted at time t , the investor adjusts his portfolio from $\varphi(t)$ to $\varphi(t+1)$, without bringing in or consuming any wealth. The following result (which is trivial in our current setting, but requires a little argument in continuous time) shows that renormalising security prices (i.e. changing the numéraire) has essentially no economic effects.

Proposition 15.3: numéraire Invariance

Let $X(t)$ be a numéraire. A trading strategy φ is self-financing with respect to $S(t)$ if and only if φ is self-financing with respect to $X(t)^{-1}S(t)$.

Proof. Since $X(t) > 0$ for all t , scalar multiplication by $X(t)^{-1}$ preserves the equality. For $t = 1, \dots, T-1$:

$$\begin{aligned} \varphi(t) \cdot S(t) &= \varphi(t+1) \cdot S(t) \\ \iff \varphi(t) \cdot (X(t)^{-1}S(t)) &= \varphi(t+1) \cdot (X(t)^{-1}S(t)). \end{aligned}$$

□

Corollary 15.1: Self-Financing Equivalence

A trading strategy φ is self-financing with respect to $S(t)$ if and only if φ is self-financing with respect to $\tilde{S}(t)$.

We now give a characterisation of self-financing strategies in terms of the discounted processes.

Proposition 15.4: Characterisation of Self-Financing Strategies

A trading strategy ϕ belongs to Φ if and only if

$$\tilde{V}_\phi(t) = V_\phi(0) + \tilde{G}_\phi(t), \quad t = 0, 1, \dots, T. \quad (15.17)$$

Proof. Assume $\phi \in \Phi$. Then, using the defining relation of the gains process, the numéraire invariance theorem, and the fact that $\tilde{S}_0^{(0)} = 1$:

$$\begin{aligned} V_\phi(0) + \tilde{G}_\phi(t) &= \phi_1 \cdot S_0 + \sum_{\tau=1}^t \phi_\tau \cdot (\tilde{S}_\tau - \tilde{S}_{\tau-1}) \\ &= \phi_1 \cdot \tilde{S}_0 + \phi_t \cdot \tilde{S}_t - \sum_{\tau=1}^{t-1} (\phi_{\tau+1} - \phi_\tau) \cdot \tilde{S}_\tau - \phi_1 \cdot \tilde{S}_0 \\ &= \phi_t \cdot \tilde{S}_t \\ &= \tilde{V}_\phi(t). \end{aligned}$$

Note that the summation term vanishes because the strategy is self-financing (i.e., $\phi_{\tau+1} \cdot \tilde{S}_\tau = \phi_\tau \cdot \tilde{S}_\tau$).

Conversely, assume that [eq. \(15.17\)](#) holds true. By the numéraire invariance theorem, it is sufficient to show the discounted version of the self-financing relation for each $t =$

$1, 2, \dots, T - 1$. Summing up to $t = 2$, [eq. \(15.17\)](#) yields:

$$\phi_2 \cdot \tilde{S}_2 = \phi_1 \cdot \tilde{S}_0 + \phi_1 \cdot (\tilde{S}_1 - \tilde{S}_0) + \phi_2 \cdot (\tilde{S}_2 - \tilde{S}_1).$$

Subtracting $\phi_2 \cdot \tilde{S}_2$ from both sides simplifies to:

$$0 = \phi_1 \cdot \tilde{S}_1 - \phi_2 \cdot \tilde{S}_1 \implies \phi_2 \cdot \tilde{S}_1 = \phi_1 \cdot \tilde{S}_1.$$

Proceeding similarly—or by induction—we can demonstrate that $\phi_t \cdot \tilde{S}_t = \phi_{t+1} \cdot \tilde{S}_t$ for $t = 2, \dots, T - 1$, as required. $\square$

We are allowed to borrow (so $\phi_t^{(0)}$ may be negative) and sell short (so $\phi_t^{(i)}$ may be negative for $i = 1, \dots, d$). Consequently, if we decide on the allocation of the risky assets, the numéraire component adjusts automatically to satisfy the budget constraint.

Proposition 15.5: Predictability of the Numéraire Component

If $(\phi_1(t), \dots, \phi_d(t))'$ is predictable and V_0 is $\mathcal{F}_0$ -measurable, there is a unique predictable process $(\phi_0(t))_{t=1}^T$ such that $\phi = (\phi_0, \phi_1, \dots, \phi_d)'$ is self-financing with initial value of the corresponding portfolio $V_\phi(0) = V_0$.

Proof. If ϕ is self-financing, then by [Proposition 15.4](#),

$$\tilde{V}_\phi(t) = V_0 + \tilde{G}_\phi(t) = V_0 + \sum_{\tau=1}^t \left(\phi_1(\tau) \Delta \tilde{S}_1(\tau) + \dots + \phi_d(\tau) \Delta \tilde{S}_d(\tau) \right).$$

On the other hand,

$$\tilde{V}_\phi(t) = \phi(t) \cdot \tilde{S}(t) = \phi_0(t) + \phi_1(t) \tilde{S}_1(t) + \dots + \phi_d(t) \tilde{S}_d(t).$$

Equate these:

$$\begin{aligned} \phi_0(t) &= V_0 + \sum_{\tau=1}^t \left(\phi_1(\tau) \Delta \tilde{S}_1(\tau) + \dots + \phi_d(\tau) \Delta \tilde{S}_d(\tau) \right) \\ &\quad - \left(\phi_1(t) \tilde{S}_1(t) + \dots + \phi_d(t) \tilde{S}_d(t) \right), \end{aligned}$$

which defines $\phi_0(t)$ uniquely. The terms in $\tilde{S}_i(t)$ are

$$\phi_i(t)\Delta\tilde{S}_i(t) - \phi_i(t)\tilde{S}_i(t) = -\phi_i(t)\tilde{S}_i(t-1),$$

which is $\mathcal{F}_{t-1}$ -measurable. So

$$\begin{aligned}\phi_0(t) = V_0 + \sum_{\tau=1}^{t-1} \left(\phi_1(\tau)\Delta\tilde{S}_1(\tau) + \cdots + \phi_d(\tau)\Delta\tilde{S}_d(\tau) \right) \\ - \left(\phi_1(t)\tilde{S}_1(t-1) + \cdots + \phi_d(t)\tilde{S}_d(t-1) \right),\end{aligned}$$

where as $\phi_1, \dots, \phi_d$ are predictable, all terms on the right-hand side are $\mathcal{F}_{t-1}$ -measurable, so ϕ_0 is predictable. $\square$

15.3 Existence of Equivalent Martingale Measures

Definition 15.5: Arbitrage Opportunity

Let $\tilde{\Phi} \subset \Phi$ be a set of self-financing strategies. A strategy $\varphi \in \tilde{\Phi}$ is called an *arbitrage opportunity* or *arbitrage strategy* with respect to $\tilde{\Phi}$ if $\mathbb{P}\{V_\varphi(0) = 0\} = 1$, and the terminal wealth of φ satisfies

$$\mathbb{P}\{V_\varphi(T) \geq 0\} = 1 \quad \text{and} \quad \mathbb{P}\{V_\varphi(T) > 0\} > 0.$$

An arbitrage opportunity is a self-financing strategy with zero initial value, which produces a non-negative final value with probability one and has a positive probability of yielding a positive final value. Note that arbitrage opportunities are always defined in relation to a specific class of trading strategies.

Definition 15.6: Arbitrage-Free Market

We say that a security market $\mathcal{M}$ is arbitrage-free if there are no arbitrage opportunities in the class Φ of trading strategies. We will allow ourselves to use ‘no-arbitrage’ in place of ‘arbitrage-free’ when convenient.

Lemma 15.1: Global Implies Local No-Arbitrage

If the market model contains no arbitrage opportunities, then for all $t \in \{0, 1, \dots, T-1\}$, for all self-financing trading strategies $\varphi \in \Phi$ and for any $A \in \mathcal{P}_t$, we have

- (i) $\mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0 \mid A) = 1 \Rightarrow \mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) = 0 \mid A) = 1,$
- (ii) $\mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \leq 0 \mid A) = 1 \Rightarrow \mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) = 0 \mid A) = 1.$

The economic meaning: no arbitrage ‘globally’ implies no arbitrage ‘locally’. Any local trading strategy can be embedded in a global strategy to which the global no-arbitrage condition applies.

Proof. We only prove (i) ((ii) is similar). Fix $t \in \{0, \dots, T-1\}$ and $\varphi \in \Phi$. Suppose $\mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0 \mid A) = 1$ for some $A \in \mathcal{P}_t$ and define a new trading strategy ψ for all times $u = 1, \dots, T$ as follows: For $u \leq t$: $\psi(u) = 0$ (‘do nothing before time t ’). For $u = t+1$: $\psi(t+1) = 0$ if $\omega \notin A$, and

$$\psi_k(t+1, \omega) = \begin{cases} \varphi_k(t+1, \omega) & \text{if } \omega \in A \text{ and } k \in \{1, \dots, d\}, \\ \varphi_0(t+1, \omega) - \tilde{V}_\varphi(t, \omega) & \text{if } \omega \in A \text{ and } k = 0. \end{cases}$$

(If $\omega \in A$ at time t , follow φ in risky assets, but adjust the numéraire to offset doing nothing when $\omega \notin A$.) For $u > t+1$: $\psi_k(u) = 0$ for $k \in \{1, \dots, d\}$ and

$$\psi_0(u, \omega) = \begin{cases} \tilde{V}_\psi(t+1, \omega) & \text{if } \omega \in A, \\ 0 & \text{if } \omega \notin A. \end{cases}$$

(Invest $\tilde{V}_\psi(t+1)$ in the numéraire if $\omega \in A$, otherwise do nothing.)

We show ψ is self-financing. By construction, ψ is predictable. For $\omega \notin A$, $\psi \equiv 0$, so consider $\omega \in A$. The relevant time is $t + 1$. Since $\psi(t) = 0$, $\psi(t) \cdot \tilde{S}(t) = 0$. Now

$$\begin{aligned}\psi(t+1) \cdot \tilde{S}(t) &= (\varphi_0(t+1) - \tilde{V}_\varphi(t))\tilde{S}_0(t) + \sum_{k=1}^d \varphi_k(t+1)\tilde{S}_k(t) \\ &= \sum_{k=0}^d \varphi_k(t+1)\tilde{S}_k(t) - \tilde{V}_\varphi(t) \\ &= \varphi(t+1) \cdot \tilde{S}(t) - \tilde{V}_\varphi(t) = \varphi(t) \cdot \tilde{S}(t) - \tilde{V}_\varphi(t) = 0,\end{aligned}$$

since φ is self-financing. For $u \leq t$, $\psi(u) \cdot \tilde{S}(u) = 0$, hence $\psi(u+1) \cdot \tilde{S}(u) = \psi(u) \cdot \tilde{S}(u)$ for all $u \leq t$. When $u > t+1$ and $\omega \in A$ we only hold the numéraire (discounted value = 1), so

$$\psi(u+1) \cdot \tilde{S}(u) = \tilde{V}_\psi(t+1) = \psi(u) \cdot \tilde{S}(u).$$

Therefore, ψ is self-financing.

We now analyze the value process of ψ . Using $\mathbb{P}(\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0 \mid A) = 1$, for all $u \geq t+1$ and $\omega \in A$,

$$\begin{aligned}\tilde{V}_\psi(u) &= \psi(u) \cdot \tilde{S}(u) = \psi(t+1) \cdot \tilde{S}(t+1) \\ &= (\varphi_0(t+1) - \tilde{V}_\varphi(t))\tilde{S}_0(t+1) + \sum_{k=1}^d \varphi_k(t+1)\tilde{S}_k(t+1) \\ &= \sum_{k=0}^d \varphi_k(t+1)\tilde{S}_k(t+1) - \tilde{V}_\varphi(t) = \tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) \geq 0.\end{aligned}$$

Since $\tilde{V}_\psi(T) = 0$ on A^c , ψ is self-financing with $\tilde{V}_\psi(0) = 0$ and $\tilde{V}_\psi(T) \geq 0$. Arbitrage-free implies $\tilde{V}_\psi(T) = 0$, hence

$$\begin{aligned}0 &= \mathbb{P}(\tilde{V}_\psi(T) > 0) = \mathbb{P}(\{\tilde{V}_\psi(T) > 0\} \cap A) \\ &= \mathbb{P}(\tilde{V}_\psi(t+1) - \tilde{V}_\psi(t) > 0 \mid A) \mathbb{P}(A).\end{aligned}$$

Therefore $\mathbb{P}(\tilde{V}_\psi(t+1) - \tilde{V}_\psi(t) = 0 \mid A) = 1$. □

Definition 15.7: Martingale Measure

A probability measure $\mathbb{P}^*$ on $(\Omega, \mathcal{F}_T)$ equivalent to $\mathbb{P}$ is called a *martingale measure* for $\tilde{S}$ if the process $\tilde{S}$ follows a $\mathbb{P}^*$ -martingale with respect to the filtration $\mathcal{F}$. We denote by $\mathcal{P}(\tilde{S})$ the class of equivalent martingale measures.

Proposition 15.6: Wealth Process Martingale

Let $\mathbb{P}^*$ be an equivalent martingale measure ($\mathbb{P}^* \in \mathcal{P}(\tilde{S})$) and $\varphi \in \Phi$ any self-financing strategy. Then the wealth process $\tilde{V}_\varphi(t)$ is a $\mathbb{P}^*$ -martingale with respect to the filtration $\mathcal{F}$.

Proof. By the self-financing property of φ (compare Proposition 15.4 and eq. (15.17)) and, we have

$$\tilde{V}_\varphi(t) = V_\varphi(0) + \tilde{G}_\varphi(t) \quad (t = 0, 1, \dots, T).$$

So

$$\tilde{V}_\varphi(t+1) - \tilde{V}_\varphi(t) = \tilde{G}_\varphi(t+1) - \tilde{G}_\varphi(t) = \varphi(t+1) \cdot (\tilde{S}(t+1) - \tilde{S}(t)).$$

So for $\varphi \in \Phi$, $\tilde{V}_t(\varphi)$ is the martingale transform of the $\mathbb{P}^*$ martingale $\tilde{S}$ by φ and hence a $\mathbb{P}^*$ martingale itself. $\square$

The next result is the key to further development.

Proposition 15.7: Equivalent Martingale Measure Implies Arbitrage-Free

If an equivalent martingale measure exists - that is, if $\mathcal{P}(\tilde{S}) \neq \emptyset$ - then the market $\mathcal{M}$ is arbitrage-free.

Proof. Assume such a $\mathbb{P}^*$ exists. For any self-financing strategy φ , we have as before

$$\tilde{V}_\varphi(t) = V_\varphi(0) + \sum_{\tau=1}^t \varphi(\tau) \cdot \Delta \tilde{S}(\tau).$$

By Proposition 15.4, $\tilde{S}(t)$ a (vector) $\mathbb{P}^*$ -martingale implies $\tilde{V}_\varphi(t)$ is a $\mathbb{P}^*$ -martingale. So the initial and final $\mathbb{P}^*$ -expectations are the same,

$$\mathbb{E}^*(\tilde{V}_\varphi(T)) = \mathbb{E}^*(\tilde{V}_\varphi(0)).$$

If the strategy is an arbitrage opportunity, its initial value - the right-hand side above - is zero. Therefore the left-hand side $\mathbb{E}^*(\tilde{V}_\varphi(T))$ is zero, but $\tilde{V}_\varphi(T) \geq 0$ (by definition). Also each $\mathbb{P}^*(\{\omega\}) > 0$ (by assumption, each $\mathbb{P}(\{\omega\}) > 0$, so by equivalence each $\mathbb{P}^*(\{\omega\}) > 0$). This and $\tilde{V}_\varphi(T) \geq 0$ force $\tilde{V}_\varphi(T) = 0$. So no arbitrage is possible. $\square$

Proposition 15.8: Arbitrage-Free Implies Existence of EMM

If the market $\mathcal{M}$ is arbitrage-free, then the class $\mathcal{P}(\tilde{S})$ of equivalent martingale measures is non-empty.

Because of the fundamental nature of this result, we will provide two proofs. The first proof is based on our previous observation that the ‘global’ no-arbitrage condition implies also no-arbitrage ‘locally’. We therefore can combine single-period results to prove the multi-period claim. The second proof uses functional-analytic techniques, i.e. a variant of the *Hahn-Banach theorem*.

Theorem 15.1: No-Arbitrage Theorem

The market $\mathcal{M}$ is arbitrage-free if and only if there exists a probability measure $\mathbb{P}^$ equivalent to $\mathbb{P}$ under which the discounted d -dimensional asset price process $\tilde{S}$ is a $\mathbb{P}^*$ -martingale.*

15.4 Pricing

We now turn to the main underlying question of this text, namely the pricing of contingent claims (i.e. financial derivatives). First, we have to model these financial instruments in our current framework. This is done in the following fashion.

Definition 15.8: Contingent Claim

A contingent claim X with maturity date T is an arbitrary non-negative $\mathcal{F}_T$ -measurable random variable (which is, by the finiteness of the probability space, bounded). We denote the class of all contingent claims by $\mathcal{X}^+$.

A typical example of a contingent claim X is an option on some underlying asset S , then (e.g. for the case of a European call option with maturity date T and strike K) we

have a functional relation $X = f(S)$ with some function f (e.g. $X = (S(T) - K)^+$). The general definition allows for more complicated relationships, which are captured by the $\mathcal{F}_T$ -measurability of X (recall that $\mathcal{F}_T$ is typically generated by the process S).

We say that the claim is *attainable* if there exists a *replicating strategy* $\varphi \in \Phi$ such that

$$V_\varphi(T) = X.$$

Therefore, the replicating strategy generates the same cash flow at time T as does X . In a highly efficient security market, we expect the law of one price to hold; that is, for a specified cash flow, there exists only one price at any given instant. Otherwise, arbitrageurs would use the opportunity to cash in a riskless profit. So the no-arbitrage condition implies that for an attainable contingent claim, its time t price must be given by the value of any replicating strategy (we say the claim is uniquely replicated in that case). This is the basic idea of the arbitrage pricing theory. Clearly, the equivalence of the no-arbitrage condition and the existence of risk-neutral probability measures implies that risk-neutral measures can be used for pricing purposes.

Proposition 15.9: Unique Replication

Suppose the market $\mathcal{M}$ is arbitrage-free. Then any attainable contingent claim X is uniquely replicated in $\mathcal{M}$.

Proof. Suppose there is an attainable contingent claim X and strategies φ and ψ such that

$$V_\varphi(T) = V_\psi(T) = X,$$

but there exists a $\tau < T$ such that

$$V_\varphi(u) = V_\psi(u) \text{ for every } u < \tau \text{ and } V_\varphi(\tau) \neq V_\psi(\tau).$$

Define $A := \{\omega \in \Omega : V_\varphi(\tau, \omega) > V_\psi(\tau, \omega)\}$, then $A \in \mathcal{F}_\tau$ and $\mathbb{P}(A) > 0$ (otherwise just rename the strategies). Define the $\mathcal{F}_\tau$ -measurable random variable $Y := V_\varphi(\tau) - V_\psi(\tau)$ and

consider the trading strategy ξ defined by

$$\xi(u) = \begin{cases} \varphi(u) - \psi(u), & u \leq \tau \\ \mathbb{1}_{A^c}(\varphi(u) - \psi(u)) + \mathbb{1}_A(Y\beta(\tau), 0, \dots, 0), & \tau < u \leq T. \end{cases}$$

Then ξ is predictable, and the self-financing condition [Equation \(17.18\)](#) is clearly true for $t \neq \tau$, and for $t = \tau$ we have using that $\varphi, \psi \in \Phi$

$$\begin{aligned} \xi(\tau) \cdot S(\tau) &= (\varphi(\tau) - \psi(\tau)) \cdot S(\tau) = V_\varphi(\tau) - V_\psi(\tau), \\ \xi(\tau + 1) \cdot S(\tau) &= \mathbb{1}_{A^c}(\varphi(\tau + 1) - \psi(\tau + 1)) \cdot S(\tau) + \mathbb{1}_A Y\beta(\tau) S_0(\tau) \\ &= \mathbb{1}_{A^c}(\varphi(\tau) - \psi(\tau)) \cdot S(\tau) + \mathbb{1}_A (V_\varphi(\tau) - V_\psi(\tau)) \beta(\tau) \beta^{-1}(\tau) \\ &= V_\varphi(\tau) - V_\psi(\tau). \end{aligned}$$

Hence ξ is a self-financing strategy with initial value equal to zero. Furthermore

$$\begin{aligned} V_\xi(T) &= \mathbb{1}_{A^c}(\varphi(T) - \psi(T)) \cdot S(T) + \mathbb{1}_A(Y\beta(\tau), 0, \dots, 0) \cdot S(T) \\ &= \mathbb{1}_A Y\beta(\tau) S_0(T) \geq 0 \end{aligned}$$

and

$$\mathbb{P}\{V_\xi(T) > 0\} = \mathbb{P}\{A\} > 0.$$

Hence, the market $\mathcal{M}$ contains an arbitrage opportunity with respect to the class Φ of self-financing strategies. But this contradicts the assumption that the market $\mathcal{M}$ is arbitrage-free. $\square$

This uniqueness property allows us now to define the important concept of an arbitrage price process.

Definition 15.9: Arbitrage Price Process

Suppose the market $\mathcal{M}$ is arbitrage-free. Let X be any attainable contingent claim with time T maturity. Then the *arbitrage price process* $\pi_X(t)$, $0 \leq t \leq T$ or simply *arbitrage price* of X in $\mathcal{M}$ is given by the value process of any replicating strategy φ for X .

The synthesis of hedging strategies that replicate the outcome of a contingent claim constitutes a fundamental objective in both theoretical and applied finance. Classical valuation frameworks postulate that an option is perfectly replicated by a dynamic self-financing portfolio of the underlying market assets (Black and Scholes, 1973; Merton, 1974). This mechanism enables the construction of a riskless hedge that neutralises stochastic exposure and informs the delta-hedging strategies employed by market participants.

The resulting pricing functional depends strictly on the assumption of non-satiation to preclude arbitrage opportunities and remains invariant to specific investor utility functions. Consequently, the value derived in an actual market must coincide with that of a hypothetical economy populated entirely by risk-neutral agents (Cox and Ross, 1976). This principle enables the valuation of derivatives as the expected discounted payoff under a specific probability measure, where discounted asset prices form martingales (Harrison and Pliska, 1981).

Proposition 15.10: Risk-Neutral Valuation Formula

The arbitrage price process of any attainable contingent claim X is given by the risk-neutral valuation formula

$$\pi_X(t) = \beta(t)^{-1} \mathbb{E}^*(X\beta(T) \mid \mathcal{F}_t) \quad \forall t = 0, 1, \dots, T, \quad (15.18)$$

where $\mathbb{E}^$ is the expectation operator with respect to an equivalent martingale measure $\mathbb{P}^*$.*

Proof. Since we assume the market is arbitrage-free there exists (at least) an equivalent martingale measure $\mathbb{P}^*$. By Proposition 15.6 the discounted value process $\tilde{V}_\varphi$ of any self-financing strategy φ is a $\mathbb{P}^*$ -martingale. So for any contingent claim X with maturity T and

any replicating trading strategy $\varphi \in \Phi$ we have for each $t = 0, 1, \dots, T$

$$\begin{aligned}
 \pi_X(t) &= V_\varphi(t) = \beta(t)^{-1} \tilde{V}_\varphi(t) \\
 &= \beta(t)^{-1} \mathbb{E}^*(\tilde{V}_\varphi(T) \mid \mathcal{F}_t) \quad (\text{as } \tilde{V}_\varphi(t) \text{ is a } \mathbb{P}^* \text{-martingale}) \\
 &= \beta(t)^{-1} \mathbb{E}^*(\beta(T)V_\varphi(T) \mid \mathcal{F}_t) \quad (\text{undoing the discounting}) \\
 &= \beta(t)^{-1} \mathbb{E}^*(\beta(T)X \mid \mathcal{F}_t) \quad (\text{as } \varphi \text{ is a replicating strategy for } X).
 \end{aligned} \tag{15.19}$$

□

15.5 Market Completeness and EMMs

The last section made clear that attainable contingent claims can be priced using an equivalent martingale measure. In this section we will discuss the question of the circumstances under which all contingent claims are attainable. This would be a very desirable property of the market $\mathcal{M}$, because we would then have solved the pricing question (at least for contingent claims) completely. Since contingent claims are merely non-negative $\mathcal{F}_T$ -measurable random variables in our setting, it should be no surprise that we can give a criterion in terms of probability measures. We start with:

Definition 15.10: Market Completeness

A market $\mathcal{M}$ is complete if every contingent claim is attainable, i.e. for every non-negative $\mathcal{F}_T$ -measurable random variable $X \in \mathcal{X}^+$ there exists a replicating self-financing strategy $\varphi \in \Phi$ such that $V_\varphi(T) = X$.

In the case of an arbitrage-free market $\mathcal{M}$ one can even insist on replicating contingent claims by an admissible strategy $\varphi \in \Phi_a$. Indeed, if φ is self-financing and $\mathbb{P}^*$ is an equivalent martingale measure under which discounted prices $\tilde{S}$ are $\mathbb{P}^*$ -martingales (such $\mathbb{P}^*$ exist since $\mathcal{M}$ is arbitrage-free and we can hence use the no-arbitrage theorem, [Theorem 15.1](#)) $\tilde{V}_\varphi(t)$ is also a $\mathbb{P}^*$ -martingale, being the martingale transform of the martingale $\tilde{S}$ by φ (see [Proposition 15.6](#)). So

$$\tilde{V}_\varphi(t) = \mathbb{E}^*(\tilde{V}_\varphi(T) \mid \mathcal{F}_t) \quad (t = 0, 1, \dots, T). \tag{15.20}$$

If φ replicates X , $V_\varphi(T) = X \geq 0$, so discounting, $\tilde{V}_\varphi(T) \geq 0$, so the above equation gives $\tilde{V}_\varphi(t) \geq 0$ for each t . Thus all the values at each time t are non-negative, not just the final value at time T , so φ is admissible.

Theorem 15.2: Completeness Theorem

An arbitrage-free market $\mathcal{M}$ is complete if and only if there exists a unique probability measure $\mathbb{P}^$ equivalent to $\mathbb{P}$ under which discounted asset prices are martingales.*

Proof. ' $\Rightarrow$ ': Assume that the arbitrage-free market $\mathcal{M}$ is complete. Then for any $\mathcal{F}_T$ -measurable random variable $X \geq 0$ (contingent claim), there exists an admissible (so self-financing) strategy φ replicating X : $X = V_\varphi(T)$. As φ is self-financing, by Proposition 15.6,

$$\beta(T)X = \tilde{V}_\varphi(T) = V_\varphi(0) + \sum_{\tau=1}^T \varphi(\tau) \cdot \Delta \tilde{S}(\tau). \quad (15.21)$$

We know by the no-arbitrage theorem (see Theorem 15.1) that an equivalent martingale measure $\mathbb{P}^*$ exists; we have to prove uniqueness. So, let $\mathbb{P}_1, \mathbb{P}_2$ be two such equivalent martingale measures. For $i = 1, 2$, $(\tilde{V}_\varphi(t))_{t=0}^T$ is a $\mathbb{P}_i$ -martingale. So,

$$\mathbb{E}_i(\tilde{V}_\varphi(T)) = \mathbb{E}_i(\tilde{V}_\varphi(0)) = V_\varphi(0), \quad (15.22)$$

since the value at time zero is non-random ($\mathcal{F}_0 = \{\emptyset, \Omega\}$) and $\beta(0) = 1$. So

$$\mathbb{E}_1(\beta(T)X) = \mathbb{E}_2(\beta(T)X). \quad (15.23)$$

Since X is arbitrary, $\mathbb{E}_1, \mathbb{E}_2$ have to agree on integrating all non-negative integrands. Taking negatives and using linearity: they have to agree on non-positive integrands also. Splitting an arbitrary integrand into its positive and negative parts: they have to agree on all integrands. Now $\mathbb{E}_i$ is expectation (i.e. integration) with respect to the measure $\mathbb{P}_i$, and measures that agree on integrating all integrands must coincide. So $\mathbb{P}_1 = \mathbb{P}_2$, giving uniqueness as required.

' $\Leftarrow$ ': Assume that the arbitrage-free market $\mathcal{M}$ is incomplete: then there exists a non-attainable $\mathcal{F}_T$ -measurable random variable $X \geq 0$ (a contingent claim). We may confine

attention to self-financing strategies φ (which will in the replicating case be automatically admissible). By [Proposition 15.5](#), we may confine attention to the risk assets $S_1, \dots, S_d$, as these suffice to tell us how to handle the numéraire S_0 .

Consider the following set of random variables

$$\tilde{V} := \left\{ Y \in \mathcal{X} : Y = Y_0 + \sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t), Y_0 \in \mathbb{R}, \varphi \text{ predictable} \right\}.$$

(Recall that Y_0 is $\mathcal{F}_0$ -measurable and set $\varphi = ((\varphi_1(t), \dots, \varphi_d(t)))_{t=1}^T$ with predictable components.) Then by the above reasoning, the discounted value $\beta(T)X$ does not belong to $\tilde{V}$, so $\tilde{V}$ is a *proper* subset of the set $\mathcal{X}$ of all random variables on Ω (which may be identified with $\mathbb{R}^{|\Omega|}$). Let $\mathbb{P}^*$ be a probability measure equivalent to $\mathbb{P}$ under which discounted prices are martingales (such $\mathbb{P}^*$ exist by the no-arbitrage theorem, [Theorem 15.1](#)). Define the scalar product

$$(Z, Y) \rightarrow \mathbb{E}^*(ZY)$$

on random variables on Ω . Since $\tilde{V}$ is a proper subset, there exists a non-zero random variable Z orthogonal to $\tilde{V}$ (since Ω is finite, $\mathbb{R}^{|\Omega|}$ is Euclidean: this is just Euclidean geometry). That is,

$$\mathbb{E}^*(ZY) = 0, \quad \forall Y \in \tilde{V}.$$

Choosing the special $Y = 1 \in \tilde{V}$ given by $\varphi_i(t) = 0, t = 1, 2, \dots, T; i = 1, \dots, d$ and $Y_0 = 1$ we find

$$\mathbb{E}^*(Z) = 0.$$

Write $\|X\|_\infty := \sup\{|X(\omega)| : \omega \in \Omega\}$, and define $\mathbb{P}^{**}$ by

$$\mathbb{P}^{**}(\{\omega\}) = \left(1 + \frac{Z(\omega)}{2\|Z\|_\infty}\right) \mathbb{P}^*(\{\omega\}).$$

By construction, $\mathbb{P}^{**}$ is equivalent to $\mathbb{P}^*$ (same null sets - actually, as $\mathbb{P}^* \sim \mathbb{P}$ and $\mathbb{P}$ has no non-empty null sets, neither do $\mathbb{P}^*, \mathbb{P}^{**}$). As Z is non-zero, $\mathbb{P}^{**}$ and $\mathbb{P}^*$ are *different*. Now

$$\begin{aligned} \mathbb{E}^{**} \left(\sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t) \right) &= \sum_{\omega \in \Omega} \mathbb{P}^{**}(\omega) \left(\sum_{t=1}^T \varphi(t, \omega) \cdot \Delta \tilde{S}(t, \omega) \right) \\ &= \sum_{\omega \in \Omega} \left(1 + \frac{Z(\omega)}{2\|Z\|_\infty} \right) \mathbb{P}^*(\omega) \left(\sum_{t=1}^T \varphi(t, \omega) \cdot \Delta \tilde{S}(t, \omega) \right). \end{aligned}$$

The ‘1’ term on the right gives

$$\mathbb{E}^* \left(\sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t) \right),$$

which is zero since this is a martingale transform of the $\mathbb{P}^*$ -martingale $\tilde{S}(t)$ (recall martingale transforms are by definition null at zero). The ‘ Z ’ term gives a multiple of the inner product

$$\left(Z, \sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t) \right),$$

which is zero as Z is orthogonal to $\tilde{V}$ and $\sum_{t=1}^T \varphi(t) \cdot \Delta \tilde{S}(t) \in \tilde{V}$. By the martingale transform lemma (Lemma 3.4.1), $\tilde{S}(t)$ is a $\mathbb{P}^{**}$ -martingale since φ is an arbitrary predictable process. Thus $\mathbb{P}^{**}$ is a second equivalent martingale measure, different from $\mathbb{P}^*$. So incompleteness implies non-uniqueness of equivalent martingale measures, as required. $\square$

15.6 Fundamental Theorem of Asset Pricing

We call a measure $\mathbb{P}^*$ under which discounted prices $\tilde{S}(t)$ are $\mathbb{P}^*$ -martingales a *martingale measure*. Such a $\mathbb{P}^*$ equivalent to the actual probability measure P is called an *equivalent martingale measure*.

Theorem 15.3: Fundamental Theorem of Asset Pricing

In an arbitrage-free complete market, there exists a unique martingale measure $\mathbb{P}^$.*

15.7 The Cox-Ross-Rubinstein Model

The development of the risk-neutral pricing formula is particularly clear in this setting since we require only elementary mathematical methods. The link to the fundamental economic principles of the arbitrage pricing method can be obtained equally straightforwardly. Moreover, binomial models, by their very construction, give rise to simple and efficient numerical procedures.

15.7.1 Model Structure

We take $d = 1$, that is, our model consists of two basic securities. Recall that the essence of the relative pricing theory is to take the price processes of these basic securities as given and price secondary securities in such a way that no arbitrage is possible.

Our time horizon is T and the set of dates in our financial market model is $t = 0, 1, \dots, T$. Assume that the first of our given basic securities is a (riskless) bond or bank account B , with price process

$$B(t) = (1 + r)^t, \quad t = 0, 1, \dots, T,$$

implying that the bond yields a riskless rate of return r in each time interval $[t, t + 1]$.

Furthermore, we have a risky asset (stock) S with price process

$$S(t + 1) = \begin{cases} uS(t) & \text{with probability } p, \\ dS(t) & \text{with probability } 1 - p, \end{cases} \quad t = 0, 1, \dots, T - 1$$

with $0 < d < u$, $S_0 \in \mathbb{R}_0^+$ (see Figure 18).

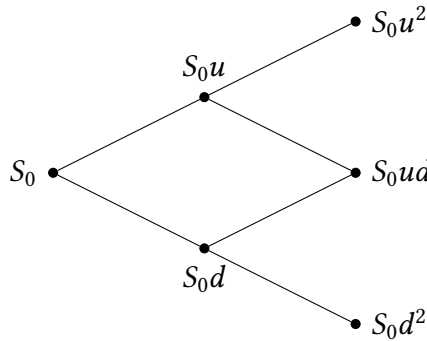


Figure 11: A two-step Binomial Tree

Alternatively we write this as

$$Z(t+1) := \frac{S(t+1)}{S(t)}, \quad t = 0, 1, \dots, T-1.$$

We set up a probabilistic model by considering the $Z(t), t = 1, \dots, T$ as random variables defined on probability spaces $(\tilde{\Omega}_t, \tilde{\mathcal{F}}_t, \tilde{\mathbb{P}}_t)$ with

$$\tilde{\Omega}_t = \tilde{\Omega} = \{d, u\},$$

$$\tilde{\mathcal{F}}_t = \tilde{\mathcal{F}} = \mathcal{P}(\tilde{\Omega}) = \{\emptyset, \{d\}, \{u\}, \tilde{\Omega}\},$$

$$\tilde{\mathbb{P}}_t = \tilde{\mathbb{P}} \text{ with } \tilde{\mathbb{P}}(\{u\}) = p, \tilde{\mathbb{P}}(\{d\}) = 1-p, \quad p \in (0, 1).$$

On these probability spaces we define

$$Z(t, u) = u \quad \text{and} \quad Z(t, d) = d, \quad t = 1, 2, \dots, T.$$

Our aim, of course, is to define a probability space on which we can model the basic securities (B, S) . Since we can write the stock price as

$$S(t) = S(0) \prod_{\tau=1}^t Z(\tau), \quad t = 1, 2, \dots, T,$$

the above definitions suggest using as the underlying probabilistic model of the financial market the *product space* $(\Omega, \mathcal{F}, \mathbb{P})$ (see e.g. [218] ch. 8), i.e.

$$\Omega = \tilde{\Omega}_1 \times \dots \times \tilde{\Omega}_T = \tilde{\Omega}^T = \{d, u\}^T,$$

with each $\omega \in \Omega$ representing the successive values of $Z(t), t = 1, 2, \dots, T$. Hence each $\omega \in \Omega$ is a T -tuple $\omega = (\tilde{\omega}_1, \dots, \tilde{\omega}_T)$ and $\tilde{\omega}_t \in \tilde{\Omega} = \{d, u\}$. For the σ -algebra we use $\mathcal{F} = \mathcal{P}(\Omega)$ and the probability measure is given by

$$\mathbb{P}(\{\omega\}) = \tilde{\mathbb{P}}_1(\{\omega_1\}) \times \dots \times \tilde{\mathbb{P}}_T(\{\omega_T\}) = \tilde{\mathbb{P}}(\{\omega_1\}) \times \dots \times \tilde{\mathbb{P}}(\{\omega_T\}).$$

The role of a product space is to model independent replication of a random experiment. The $Z(t)$ above are two-valued random variables, so can be thought of as tosses of a biased coin; we need to build a probability space on which we can model a succession of such independent tosses.

Now we redefine (with a slight abuse of notation) the $Z(t), t = 1, \dots, T$ as random variables on $(\Omega, \mathcal{F}, \mathbb{P})$ (as the t th projection)

$$Z(t, \omega) = Z(t, \omega_t).$$

Observe that by this definition (and the above construction) $Z_1, \dots, Z_T$ are independent and identically distributed with

$$\mathbb{P}(Z(t) = u) = p = 1 - \mathbb{P}(Z(t) = d).$$

To model the flow of information in the market we use the obvious filtration

$$\begin{aligned} \mathcal{F}_0 &= \{\emptyset, \Omega\} && \text{(trivial } \sigma\text{-field),} \\ \mathcal{F}_t &= \sigma(Z(1), \dots, Z(t)) = \sigma(S(1), \dots, S(t)), \\ \mathcal{F}_T &= \mathcal{F} = \mathcal{P}(\Omega) && \text{(class of all subsets of } \Omega\text{).} \end{aligned}$$

This construction emphasises again that a multi-period model can be viewed as a sequence of single-period models. Indeed, in the Cox-RossRubinstein case, we use identical and independent single-period models. As we will see in the sequel, this will make the construction of equivalent martingale measures relatively easy. Unfortunately, we can hardly defend the assumption of independent and identically distributed price movements at each time period in practical applications.

Attention 15.1: W

use this example to show how to construct the underlying probability space. Having done this in full once, we will from now on feel free to take for granted the existence of an appropriate probability space on which all relevant random variables are defined.

We now turn to the pricing of derivative assets in the Cox-Ross-Rubinstein market model. To do so we first have to discuss whether the Cox-Ross-Rubinstein model is arbitrage-free and complete.

Our first task is to find an equivalent martingale measure of the same class as $\mathbb{P}$, i.e. a probability measure $\mathbb{Q}$ defined as a product measure via a measure $\tilde{\mathbb{Q}}$ on $(\tilde{\Omega}, \tilde{\mathcal{F}})$ such that $\tilde{\mathbb{Q}}(\{u\}) = q$ and $\tilde{\mathbb{Q}}(\{d\}) = 1 - q$. We call the set of all measures of that type $\mathcal{P}$. We have:

Proposition 15.11: Martingale Measure Existence

(i) A martingale measure $\mathbb{Q} \in \mathcal{P}$ for the discounted stock price $\tilde{S}$ exists if and only if

$$d < 1 + r < u. \quad (15.24)$$

(ii) If Equation (15.24) holds true, then there is a unique such measure in $\mathcal{P}$ characterised by

$$q = \frac{1 + r - d}{u - d}. \quad (15.25)$$

Proof. Since $S(t) = \tilde{S}(t)B(t) = \tilde{S}(t)(1 + r)^t$, we have $Z(t + 1) = S(t + 1)/S(t) = (\tilde{S}(t + 1)/\tilde{S}(t))(1 + r)$. So, the discounted price $(\tilde{S}(t))$ is a $\mathbb{Q}$ -martingale if and only if for $t = 0, 1, \dots, T - 1$

$$\mathbb{E}^{\mathbb{Q}}[\tilde{S}(t + 1) | \mathcal{F}_t] = \tilde{S}(t) \iff \mathbb{E}^{\mathbb{Q}}[(\tilde{S}(t + 1)/\tilde{S}(t)) | \mathcal{F}_t] = 1 \iff \mathbb{E}^{\mathbb{Q}}[Z(t + 1) | \mathcal{F}_t] = 1 + r.$$

But $Z(1), \dots, Z(T)$ are mutually independent and hence $Z(t + 1)$ is independent of $\mathcal{F}_t = \sigma(Z(1), \dots, Z(t))$. So

$$1 + r = \mathbb{E}^{\mathbb{Q}}(Z(t + 1) | \mathcal{F}_t) = \mathbb{E}^{\mathbb{Q}}(Z(t + 1)) = uq + d(1 - q)$$

is a weighted average of u and d ; this can be $1 + r$ if and only if $1 + r \in [d, u]$. As Q is to be *equivalent* to $\mathbb{P}$ and $\mathbb{P}$ has no non-empty null sets, $r = d - 1, u - 1$ are excluded and Equation (15.24) is proved.

To prove uniqueness and to find the value of q we simply observe that under Equation (15.24)

$$u \cdot q + d \cdot (1 - q) = 1 + r$$

has a unique solution. Solving it for q leads to the above formula. $\square$

Corollary 15.2: Arbitrage-Free Model

The Cox-Ross-Rubinstein model is arbitrage-free.

Proof. By Proposition 15.11 there exists an equivalent martingale measure and this is by the no-arbitrage theorem, Theorem 15.1 enough to guarantee that the Cox-Ross-Rubinstein model is free of arbitrage. $\square$

To answer the question of market completeness we have to examine the structure of equivalent martingale measures. We already know that there is a unique martingale measure in the class $\mathcal{P}$, but can we find other equivalent martingale measures outside $\mathcal{P}$? Examining the structure of the underlying measure space (product space) again we can show that every probability measure is a product measure, and hence all probability measures are of class $\mathcal{P}$. Using this result we conclude the following:

Proposition 15.12: CRR Model Completeness

The Cox-Ross-Rubinstein model is complete.

Proof. Since by Proposition 15.11 the equivalent martingale measure is unique, the claim follows from the completeness theorem (Theorem 15.2). $\square$

We translate the above measure-theoretic argument to economic language. Since every probability measure is a product measure uniqueness of the equivalent martingale measure on the product space is equivalent to uniqueness of the individual martingale measures on the component space. (Observe that this is true for general product space situations, it

merely says that for independent random variables, the individual distributions determine the joint distribution.) We thus have the following result.

Corollary 15.3: Multi-Period Completeness

The multi-period model is complete if and only if every underlying single-period model is complete.

We can now use the risk-neutral valuation formula to price every contingent claim in the Cox-Ross-Rubinstein model.

Proposition 15.13: Arbitrage Price Process

The arbitrage price process of a contingent claim X in the Cox-Ross-Rubinstein model is given by

$$\pi_X(t) = B(t)\mathbb{E}^*(X/B(T) \mid \mathcal{F}_t) \quad \forall t = 0, 1, \dots, T,$$

where $\mathbb{E}^$ is the expectation operator with respect to the unique equivalent martingale measure $\mathbb{P}^*$ characterised by $p^* = (1 + r - d)/(u - d)$.*

Corollary 15.4: European Call Valuation

Consider a European call option with expiry T and strike price K written on (one share of) the stock S . The arbitrage price process $C(t), t = 0, 1, \dots, T$ of the option is given by (set $\tau = T - t$)

$$C(t) = (1 + r)^{-\tau} \sum_{j=0}^{\tau} \binom{\tau}{j} p^{*j} (1 - p^*)^{\tau-j} (S(t)u^j d^{\tau-j} - K)^+. \quad (15.26)$$

Since the Cox-Ross-Rubinstein model is complete we can find unique hedging strategies for replicating contingent claims. Recall that this means we can find a portfolio $\varphi(t) = (\varphi_0(t), \varphi_1(t))$, φ predictable, such that the value process $V_\varphi(t) = \varphi_0(t)B(t) + \varphi_1(t)S(t)$ satisfies

$$\Pi_X(t) = V_\varphi(t), \quad \text{for all } t = 0, 1, \dots, T.$$

Using the bond as numéraire we get the discounted equation

$$\tilde{\Pi}_X(t) = \tilde{V}_\varphi(t) = \varphi_0(t) + \varphi_1(t)\tilde{S}(t), \quad \text{for all } t = 0, 1, \dots, T.$$

By the pricing formula, [Proposition 15.13](#), we know the arbitrage price process and using the restriction of predictability of φ , this leads to a unique replicating portfolio process φ . We can compute this portfolio process at any point in time as follows. The equation $\tilde{\Pi}_X(t) = \varphi_0(t) + \varphi_1(t)\tilde{S}(t)$ has to be true for each $\omega = (\omega_1, \dots, \omega_t, \dots, \omega_T) \in \Omega$ and each $t = 1, \dots, T$ (observe that $\pi_X(0) = V_\varphi(0) = \varphi_0(1) + \varphi_1(1)S_1(0)$). Given such a t we only can use information up to (and including) time $t-1$ to ensure that φ is predictable. Therefore we know the coordinates $\omega_1, \dots, \omega_{t-1}$, but ω_t can only have one of the values d or u . This leads to the following system of equations, which can be solved for $\varphi_0(t)$ and $\varphi_1(t)$ uniquely. With a slight abuse of notation we write $\tilde{\omega} = (\omega_1, \dots, \omega_{t-1})$, $\tilde{\omega}u = (\omega_1, \dots, \omega_{t-1}, u)$ and $\tilde{\omega}d = (\omega_1, \dots, \omega_{t-1}, d)$ (recall that since $S(t)$ and $\Pi_X(t)$ are $\mathcal{F}_t$ -measurable they are constant on the corresponding partition $\mathcal{P}_t$ of Ω , which in our current setting is just given by the first t coordinates of ω , and the same argument applies for φ). Then

$$\tilde{\Pi}_X(t, \tilde{\omega}u) = \varphi_0(t, \tilde{\omega}) + \varphi_1(t, \tilde{\omega})\tilde{S}(t, \tilde{\omega}u),$$

$$\tilde{\Pi}_X(t, \tilde{\omega}d) = \varphi_0(t, \tilde{\omega}) + \varphi_1(t, \tilde{\omega})\tilde{S}(t, \tilde{\omega}d).$$

The solution is given by

$$\begin{aligned} \varphi_0(t, \tilde{\omega}) &= \frac{\tilde{S}(t, \tilde{\omega}u)\tilde{\Pi}_X(t, \tilde{\omega}d) - \tilde{S}(t, \tilde{\omega}d)\tilde{\Pi}_X(t, \tilde{\omega}u)}{\tilde{S}(t, \tilde{\omega}u) - \tilde{S}(t, \tilde{\omega}d)} \\ &= \frac{u\tilde{\Pi}_X(t, \tilde{\omega}d) - d\tilde{\Pi}_X(t, \tilde{\omega}u)}{u - d}, \\ \varphi_1(t, \tilde{\omega}) &= \frac{\tilde{\Pi}_X(t, \tilde{\omega}u) - \tilde{\Pi}_X(t, \tilde{\omega}d)}{\tilde{S}(t, \tilde{\omega}u) - \tilde{S}(t, \tilde{\omega}d)} \\ &= \frac{\tilde{\Pi}_X(t, \tilde{\omega}u) - \tilde{\Pi}_X(t, \tilde{\omega}d)}{\tilde{S}(t-1, \tilde{\omega})(u - d)}. \end{aligned}$$

Observe that we only need to know $\tilde{\omega}$ to compute $\varphi(t)$, hence φ is predictable. We make this rather abstract construction more transparent by constructing the hedge portfolio for

the European call. We use the following notation. Write

$$C(t, x) := (1 + r)^{-(T-t)} \sum_{j=0}^{T-t} \binom{T-t}{j} p^{*j} (1 - p^*)^{T-t-j} (xu^j d^{T-t-j} - K)^+.$$

Then $C(t, x)$ is value of the call at time t given that $S(t) = x$.

Proposition 15.14: Perfect Hedging Strategy

The perfect hedging strategy $\varphi = (\varphi_0, \varphi_1)$ replicating the European call option with time of expiry T and strike price K is given by

$$\begin{aligned} \varphi_1(t) &= \frac{C(t, S(t-1)u) - C(t, S(t-1)d)}{S(t-1)(u-d)}, \\ \varphi_0(t) &= \frac{uC(t, S(t-1)d) - dC(t, S(t-1)u)}{(1+r)^t(u-d)}. \end{aligned}$$

Proof. $C(t, x)$ must be the value of the portfolio at time t if the strategy $\varphi = (\varphi(t))$ replicates the claim:

$$\varphi_0(t)(1+r)^t + \varphi_1(t)S(t) = C(t, S(t)).$$

Now $S(t) = S(t-1)Z(t) = S(t-1)u$ or $S(t-1)d$, so:

$$\varphi_0(t)(1+r)^t + \varphi_1(t)S(t-1)u = C(t, S(t-1)u),$$

$$\varphi_0(t)(1+r)^t + \varphi_1(t)S(t-1)d = C(t, S(t-1)d).$$

Subtract:

$$\varphi_1(t)S(t-1)(u-d) = C(t, S(t-1)u) - C(t, S(t-1)d).$$

So $\varphi_1(t)$ in fact depends only on $S(t-1)$, thus yielding the predictability of φ , and

$$\varphi_1(t) = \frac{C(t, S(t-1)u) - C(t, S(t-1)d)}{S(t-1)(u-d)}.$$

Using any of the equations in the above system and solving for $\varphi_0(t)$ completes the proof. $\square$

Notice that the numerator in the equation for $\varphi_1(t)$ is the difference of two values of $C(t, x)$, with the larger value of x in the first term (recall $u > d$). When the payoff function $C(t, x)$ is an increasing function of x , as for the European call option considered here, this is non-negative. In this case, the Proposition gives $\varphi_1(t) \geq 0$: the replicating strategy does not involve short-selling. We record this as:

Corollary 15.5: Monotone Payoff Hedging

When the payoff function is a non-decreasing function of the asset price $S(t)$, the perfect-hedging strategy replicating the claim does not involve short-selling of the risky asset.

If we do not use the pricing formula from [Proposition 15.13](#) (i.e. the information on the price process), but only the final values of the option (or more generally of a contingent claim) we are still able to compute the arbitrage price and to construct the hedging portfolio by backward induction. We outline this procedure for the European call starting with the last period $(T - 1, T]$. We have to choose a replicating portfolio $\varphi(T) = (\varphi_0(T), \varphi_1(T))$ based on the information available at time $T - 1$ (and so $\mathcal{F}_{T-1}$ -measurable). So for each $\omega \in \Omega$ the following equation has to hold:

$$C(T, \omega) = \varphi_0(T, \omega)B(T, \omega) + \varphi_1(T, \omega)S(T, \omega).$$

Given the information $\mathcal{F}_{T-1}$, we know all but the last coordinate of ω , and this gives rise to two equations (with the same notation as above, $\tilde{\omega}$ denotes the first $T - 1$ coordinates)

$$C(T, \tilde{\omega}u) = \varphi_0(T, \tilde{\omega})(1 + r)^T + \varphi_1(T, \tilde{\omega})S(T, \tilde{\omega}u),$$

$$C(T, \tilde{\omega}d) = \varphi_0(T, \tilde{\omega})(1 + r)^T + \varphi_1(T, \tilde{\omega})S(T, \tilde{\omega}d).$$

Since we know the payoff structure of the call at time T , e.g. $C(T, \tilde{\omega}u) = (uS(T - 1, \tilde{\omega}) - K)^+$ and $C(T, \tilde{\omega}d) = (dS(T - 1, \tilde{\omega}) - K)^+$, we can solve the above system and obtain

$$\begin{aligned}\varphi_1(T, \tilde{\omega}) &= \frac{(uS(T - 1, \tilde{\omega}) - K)^+ - (dS(T - 1, \tilde{\omega}) - K)^+}{S(T - 1, \tilde{\omega})(u - d)}, \\ \varphi_0(T, \tilde{\omega}) &= \frac{u(dS(T - 1, \tilde{\omega}) - K)^+ - d(uS(T - 1, \tilde{\omega}) - K)^+}{(1 + r)(u - d)}.\end{aligned}$$

Using this portfolio one can compute the arbitrage price of the European call at time $T - 1$ in state of the world $\tilde{\omega}$ as

$$C(T - 1, \tilde{\omega}) = \varphi_0(T, \tilde{\omega})(1 + r)^{T-1} + \varphi_1(T, \tilde{\omega})S(T - 1, \tilde{\omega}).$$

Now that the arbitrage prices at time $T - 1$ are known, one can repeat the procedure to successively compute the prices at $T - 2, \dots, 1, 0$.

The advantage of our risk-neutral pricing procedure over this approach is that we have a single formula for the price of the contingent claim at all times t , and don't have to resort to backward induction to compute a price at a special time t .

Using the pricing formula for the European call we see that the price process depends on the current stock price $S(t)$, the strike price K , the expiry date T , the interest rate r and (hidden in u, d) the volatility of the stock price. We thus have the determining factors of the European call price explicit in [Equation \(15.26\)](#) and can examine their effects explicitly. We use the notation $C(F)$ to mean that we consider the price C of the call as a function of the factor F , keeping all others fixed.

15.7.1.1 Changes in Option Price Relative to the Underlying Stock Price. If the current stock price S is changed by an amount ΔS we see that

$$\begin{aligned} \Delta C &= C(S + \Delta S) - C(S) \\ &= (1 + r)^{-T} \sum_{j=0}^T \left(\binom{T}{j} p^{*j} (1 - p^*)^{T-j} \right. \\ &\quad \left. \{ ((S + \Delta S)u^j d^{T-j} - K)^+ - (Su^j d^{T-j} - K)^+ \} \right) \\ &\geq 0, \end{aligned}$$

since the function $(x - K)^+$ is increasing in x . Hence $C(S + \Delta S) \geq C(S)$ and the value of the option increases as the underlying stock price increases.

The quotient $\Delta C/\Delta S$ measures the change in the option price for an infinitesimal change in the stock price, keeping everything else fixed. If $\Delta S \rightarrow 0$ this quotient converges (under conditions of regularity) to $\partial C/\partial S$ and is therefore often referred to as the option's delta.

Looking at [Proposition 15.14](#) we see the relation of the delta to the hedging portfolio for the option. The number of stocks to be held in the portfolio during period $(t-1, t]$ is given by the difference quotient

$$\varphi_1(t) = \frac{C(t, S(t-1)u) - C(t, S(t-1)d)}{S(t-1)u - S(t-1)d}$$

and this quotient is called the hedge ratio. This is an expression of the above type and we see that the number of stocks in the hedge portfolio is given approximately by the corresponding derivative, the option's delta.

15.7.1.2 Changes in Option Price Relative to the Strike Price K . Since the strike price K is only present in the expression $(x - K)^+$ and this function is non-increasing in K , we immediately see that the options value is a non-increasing function of the strike price.

[Equation \(15.26\)](#) can be used to discuss the other determining factors of the option price in a similar (although quite tedious) manner.

15.8 Binomial Approximations

Suppose we observe financial assets during a continuous time period $[0, T]$. To construct a stochastic model of the price processes of these assets (to e.g., value contingent claims) one basically has two choices

- (i) one could model the processes as continuous-time stochastic processes
- (ii) one could construct a sequence of discrete-time models in which the continuous-time price processes are approximated by discrete-time stochastic processes in a suitable sense.

We describe the second approach now by examining the asymptotic properties of a sequence of Cox-Ross-Rubinstein models.

15.8.1 Model Structure

We assume that all random variables subsequently introduced are defined on a suitable probability space $(\Omega, \mathcal{F}, \mathbb{P})$. We want to model two assets, a riskless bond B and a risky stock S , which we now observe in a continuous-time interval $[0, T]$. To transfer the continuous-time framework into a binomial structure we make the following adjustments. Looking at the n th Cox-Ross-Rubinstein model in our sequence, there is a prespecified number k_n of trading dates. We set $\Delta_n = T/k_n$ and divide $[0, T]$ in k_n subintervals of length Δ_n , namely $I_j = [j\Delta_n, (j+1)\Delta_n]$, $j = 0, \dots, k_n - 1$. We suppose that trading occurs only at the equidistant time points $t_{n,j} = j\Delta_n$, $j = 0, \dots, k_n - 1$. We fix r_n as the riskless interest rate over each interval I_j , and hence the bond process (in the n th model) is given by

$$B(t_{n,j}) = (1 + r_n)^j, \quad j = 0, \dots, k_n.$$

In the continuous-time model we compound continuously with spot rate $r \geq 0$ and hence the bond price process $B(t)$ is given by $B(t) = e^{rt}$. In order to approximate this process in the discrete-time framework, we choose r_n such that

$$1 + r_n = e^{r\Delta_n}. \quad (15.27)$$

With this choice we have for any $j = 0, \dots, k_n$ that $(1 + r_n)^j = \exp(rj\Delta_n) = \exp(rt_{n,j})$. Thus we have approximated the bond process exactly at the time points of the discrete model.

Next we model the one-period returns $S(t_{n,j+1})/S(t_{n,j})$ of the stock by a family of random variables $Z_{n,i}$; $i = 1, \dots, k_n$ taking values $\{d_n, u_n\}$ with

$$\mathbb{P}(Z_{n,i} = u_n) = p_n = 1 - \mathbb{P}(Z_{n,i} = d_n)$$

for some $p_n \in (0, 1)$ (which we specify later). With these $Z_{n,j}$ we model the stock price process S_n in the n th Cox-Ross-Rubinstein model as

$$S_n(t_{n,j}) = S_n(0) \prod_{i=1}^j Z_{n,i}, \quad j = 0, 1, \dots, k_n.$$

With the specification of the one-period returns we get a complete description of the discrete dynamics of the stock price process in each Cox-Ross-Rubinstein model. We call such a finite sequence $Z_n = (Z_{n,i})_{i=1}^{k_n}$ a lattice or tree. The parameters u_n, d_n, p_n, k_n differ from lattice to lattice, but remain constant throughout a specific lattice. In the triangular array $(Z_{n,i}), i = 1, \dots, k_n; n = 1, 2, \dots$ we assume that the random variables are row-wise independent (but we allow dependence between rows). The approximation of a continuous-time setting by a sequence of lattices is called the lattice approach.

It is important to stress that for each n we get a different discrete stock price process $S_n(t)$ and that in general these processes do not coincide on common time points (and are also different from the price process $S(t)$).

Turning back to a specific Cox-Ross-Rubinstein model, we now have a discrete-time bond and stock price process. We want arbitrage-free financial market models and therefore have to choose the parameters u_n, d_n, p_n accordingly. An arbitrage-free financial market model is guaranteed by the existence of an equivalent martingale measure, and by [Proposition 15.11\(i\)](#) the (necessary and) sufficient condition for that is

$$d_n < 1 + r_n < u_n.$$

The risk-neutrality approach implies that the expected (under an equivalent martingale measure) one-period return must equal the one-period return of the riskless bond and hence we get (see [Proposition 15.11\(ii\)](#))

$$p_n = \frac{(1 + r_n) - d_n}{u_n - d_n}. \quad (15.28)$$

So the only parameters to choose freely in the model are u_n and d_n . In the next sections we consider some special choices.

We now choose the parameters in the above lattice approach in a special way. Assuming the risk-free rate of interest r is given, we have by [Equation \(15.27\)](#)

$$1 + r_n = e^{r\Delta_n},$$

and the remaining degrees of freedom are resolved by choosing u_n and d_n . We use the following choice:

$$u_n = e^{\sigma\sqrt{\Delta_n}}, \quad d_n = u_n^{-1} = e^{-\sigma\sqrt{\Delta_n}}.$$

By condition [Equation \(15.28\)](#) the risk-neutral probabilities for the corresponding single-period models are given by

$$p_n^* = \frac{1 + r_n - d_n}{u_n - d_n} = \frac{e^{r\Delta_n} - e^{-\sigma\sqrt{\Delta_n}}}{e^{\sigma\sqrt{\Delta_n}} - e^{-\sigma\sqrt{\Delta_n}}}.$$

We can now price contingent claims in each Cox-Ross-Rubinstein model using the expectation operator with respect to the (unique) equivalent martingale measure characterised by the probabilities p_n^* . In particular, we can compute the price $C(t)$ at time t of a European call on the stock S with strike K and expiry T by [Equation \(15.26\)](#) of ?? . Let us reformulate this formula slightly. We define

$$a_n = \min\{j \in \mathbb{N}_0 \mid S(0)u_n^j d_n^{k_n-j} > K\}. \quad (15.29)$$

Then we can rewrite the pricing formula [Equation \(15.26\)](#) for $t = 0$ in the setting of the n th Cox-Ross-Rubinstein model as

$$\begin{aligned} C(0) &= (1 + r_n)^{-k_n} \sum_{j=a_n}^{k_n} \binom{k_n}{j} p_n^{*j} (1 - p_n^*)^{k_n-j} (S(0)u_n^j d_n^{k_n-j} - K) \\ &= S(0) \left[\sum_{j=a_n}^{k_n} \binom{k_n}{j} \left(\frac{p_n^* u_n}{1 + r_n} \right)^j \left(\frac{(1 - p_n^*) d_n}{1 + r_n} \right)^{k_n-j} \right] \\ &\quad - K(1 + r_n)^{-k_n} \left[\sum_{j=a_n}^{k_n} \binom{k_n}{j} p_n^{*j} (1 - p_n^*)^{k_n-j} \right]. \end{aligned}$$

Denoting the binomial cumulative distribution function with parameters (n, p) as $B^{n,p}(\cdot)$ we see that the second bracketed expression is

$$\bar{B}^{k_n, p_n^*}(a_n) = 1 - B^{k_n, p_n^*}(a_n).$$

Also the first bracketed expression is $\bar{B}^{k_n, \hat{p}_n}(a_n)$ with

$$\hat{p}_n = \frac{p_n^* u_n}{1 + r_n}.$$

That $\hat{p}_n$ is indeed a probability can be shown straightforwardly. Using this notation we have in the n th Cox-Ross-Rubinstein model for the price of a European call at time $t = 0$ the following formula:

$$C_n(0) = S_n(0) \bar{B}^{k_n, \hat{p}_n}(a_n) - K(1 + r_n)^{-k_n} \bar{B}^{k_n, p_n^*}(a_n). \quad (15.30)$$

(We stress again that the underlying is $S_n(t)$, dependent on n , but $S_n(0) = S(0)$ for all n .)

We now look at the limit of this expression.

Proposition 15.15: Binomial Model Limit Relation

We have the following limit relation:

$$\lim_{n \rightarrow \infty} C_n(0) = C_{BS}(0)$$

with $C_{BS}(0)$ given by the Black-Scholes formula (we use $S = S(0)$ to ease the notation)

$$C_{BS}(0) = SN(d_1(S, T)) - Ke^{-rT}N(d_2(S, T)). \quad (15.31)$$

The functions $d_1(s, t)$ and $d_2(s, t)$ are given by

$$d_1(s, t) = \frac{\log(s/K) + (r + \frac{\sigma^2}{2})t}{\sigma\sqrt{t}},$$

$$d_2(s, t) = d_1(s, t) - \sigma\sqrt{t} = \frac{\log(s/K) + (r - \frac{\sigma^2}{2})t}{\sigma\sqrt{t}}$$

and $N(\cdot)$ is the standard normal cumulative distribution function.

Proof. Since $S_n(0) = S$ (say) all we have to do to prove the proposition is to show

- (i) $\lim_{n \rightarrow \infty} \bar{B}^{k_n, \hat{p}_n}(a_n) = N(d_1(S, T)),$
- (ii) $\lim_{n \rightarrow \infty} \bar{B}^{k_n, p_n^*}(a_n) = N(d_2(S, T)).$

These statements involve the convergence of distribution functions.

To show (i) we interpret

$$\bar{B}^{k_n, \hat{p}_n}(a_n) = \mathbb{P}(a_n \leq Y_n \leq k_n)$$

with (Y_n) a sequence of random variables distributed according to the binomial law with parameters $(k_n, \hat{p}_n)$. We normalise Y_n to

$$\tilde{Y}_n = \frac{Y_n - \mathbb{E}(Y_n)}{\sqrt{\text{Var}(Y_n)}} = \frac{Y_n - k_n \hat{p}_n}{\sqrt{k_n \hat{p}_n (1 - \hat{p}_n)}} = \frac{\sum_{j=1}^{k_n} (B_{j,n} - \hat{p}_n)}{\sqrt{k_n \hat{p}_n (1 - \hat{p}_n)}},$$

where $B_{j,n}$, $j = 1, \dots, k_n$; $n = 1, 2, \dots$ are row-wise independent Bernoulli random variables with parameter $\hat{p}_n$. Now using the central limit theorem we know that for $\alpha_n \rightarrow \alpha$, $\beta_n \rightarrow \beta$ we have

$$\lim_{n \rightarrow \infty} \mathbb{P}(\alpha_n \leq \tilde{Y}_n \leq \beta_n) = N(\beta) - N(\alpha).$$

By definition we have

$$\mathbb{P}(a_n \leq Y_n \leq k_n) = \mathbb{P}(\alpha_n \leq \tilde{Y}_n \leq \beta_n)$$

with

$$\alpha_n = \frac{a_n - k_n \hat{p}_n}{\sqrt{k_n \hat{p}_n (1 - \hat{p}_n)}} \quad \text{and} \quad \beta_n = \frac{k_n(1 - \hat{p}_n)}{\sqrt{k_n \hat{p}_n (1 - \hat{p}_n)}}.$$

Using the following limiting relations:

$$\lim_{n \rightarrow \infty} \hat{p}_n = \frac{1}{2}, \quad \lim_{n \rightarrow \infty} k_n(1 - 2\hat{p}_n)\sqrt{\Delta_n} = -T \left(\frac{r}{\sigma} + \frac{\sigma}{2} \right),$$

and the defining relation for a_n , [Equation \(15.29\)](#), we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \alpha_n &= \lim_{n \rightarrow \infty} \frac{\frac{\log(K/S) + k_n \sigma \sqrt{\Delta_n}}{2\sigma \sqrt{\Delta_n}} - k_n \hat{p}_n}{\sqrt{k_n \hat{p}_n (1 - \hat{p}_n)}} \\ &= \lim_{n \rightarrow \infty} \frac{\log(K/S) + \sigma k_n \sqrt{\Delta_n} (1 - 2\hat{p}_n)}{2\sigma \sqrt{k_n \Delta_n \hat{p}_n (1 - \hat{p}_n)}} \\ &= \frac{\log(K/S) - (r + \frac{\sigma^2}{2})T}{\sigma \sqrt{T}} = -d_1(S, T). \end{aligned}$$

Furthermore we have

$$\lim_{n \rightarrow \infty} \beta_n = \lim_{n \rightarrow \infty} \sqrt{k_n \hat{p}_n^{-1} (1 - \hat{p}_n)} = +\infty.$$

So $N(\beta_n) \rightarrow 1$, $N(\alpha_n) \rightarrow N(-d_1) = 1 - N(d_1)$, completing the proof of (i).

To prove (ii) we can argue in very much the same way and arrive at parameters α_n^* and β_n^* with $\hat{p}_n$ replaced by p_n^* . Using the following limiting relations:

$$\lim_{n \rightarrow \infty} p_n^* = \frac{1}{2}, \quad \lim_{n \rightarrow \infty} k_n (1 - 2p_n^*) \sqrt{\Delta_n} = T \left(\frac{\sigma}{2} - \frac{r}{\sigma} \right),$$

we get

$$\begin{aligned} \lim_{n \rightarrow \infty} \alpha_n^* &= \lim_{n \rightarrow \infty} \frac{\log(K/S) + \sigma n \sqrt{\Delta_n} (1 - 2p_n^*)}{2\sigma \sqrt{n \Delta_n} p_n^* (1 - p_n^*)} \\ &= \frac{\log(K/S) - (r - \frac{\sigma^2}{2})T}{\sigma \sqrt{T}} = -d_2(s, T). \end{aligned}$$

For the upper limit we get

$$\lim_{n \rightarrow \infty} \beta_n^* = \lim_{n \rightarrow \infty} \sqrt{k_n (p_n^*)^{-1} (1 - p_n^*)} = +\infty,$$

whence (ii) follows similarly. □

By the above proposition we have derived the classical Black-Scholes European call option valuation formula as an asymptotic limit of option prices in a sequence of Cox-Ross-Rubinstein type models with a special choice of parameters. We will therefore call these models discrete Black-Scholes models. Let us mention here that in the continuous-time Black-Scholes model the dynamics of the (stochastic) stock price process $S(t)$ are modelled by a geometric Brownian motion (or exponential Wiener process). The sample paths of this stochastic price process are almost all continuous and the probability law of $S(t)$ at any time t is lognormal. In particular the time T distribution of $\log\{S(T)/S(0)\}$ is $N(T\mu, T\sigma^2)$. Looking back at the construction of our sequence of Cox-Ross-Rubinstein models we see that

$$\log \frac{S_n(T)}{S(0)} = \sum_{i=1}^{k_n} \log(Z_{n,i}). \quad (15.32)$$

with $\log(Z_{n,i})$ Bernoulli random variables with

$$\mathbb{P}(\log(Z_{n,i}) = \sigma\sqrt{\Delta_n}) = p_n = 1 - \mathbb{P}(\log(Z_{n,i}) = -\sigma\sqrt{\Delta_n}). \quad (15.33)$$

By the (triangular array version) of the central limit theorem we know that the $\log \frac{S_n(T)}{S(0)}$ properly normalised converges in distribution to a random variable with standard normal distribution. Doing similar calculations as in the above proposition we can compute the normalising constants and get

$$\lim_{n \rightarrow \infty} \log \frac{S_n(T)}{S(0)} \sim \mathcal{N}(T(r - \sigma^2/2), T\sigma^2), \quad (15.34)$$

i.e. $S_n(T)/S(0)$ is in the limit lognormally distributed.

Using the terminology of weak convergence we can therefore say that the probability measures $\mathbb{P}^n$ induced by the distributions of $S_n(T)/S(0)$ converge to the probability measure $\mathbb{Q}$ induced by $\mathcal{N}(T(r - \sigma^2/2), T\sigma^2)$. Now the probability measures $\mathbb{P}^n$ are the risk-neutral equivalent probability measures in the n th Cox-Ross-Rubinstein model, and we will show in Chapter 6 that in continuous-time market models, similarly to discrete-time financial market modelling, contingent claims can be priced using risk-neutral equivalent martingale measures. In particular we will show that in the above Black-Scholes market the (in this case unique) equivalent martingale measure is given by $\mathbb{Q}$.

Proposition 15.16: Binomial Approximation Pricing Limit

Let X be a contingent claim of the form $X = h(S(T))$ with h a bounded, uniformly continuous real function. Denote by Π_X^n resp. Π_X the time $t = 0$ price of X in the n th discrete-time resp. the continuous-time Black-Scholes market model. Then

$$\lim_{n \rightarrow \infty} \Pi_X^n = \Pi_X.$$

Proof. Writing the pricing formula for the contingent claim using the expectation operator with respect to the risk-neutral probability measures, we have

$$\Pi_X^n = \mathbb{E}_{\mathbb{P}^n}(h(S_n(T))) = \int h \, d\mathbb{P}^n, \quad \text{resp.} \quad \Pi_X = \mathbb{E}_{\mathbb{Q}}(h(S(T))) = \int h \, d\mathbb{Q}$$

(since the σ -field at $t = 0$ is assumed to be trivial, we can use expectation instead of conditional expectation), and the portmanteau theorem gives the result. $\square$

16 Continuous-time Market Model

We start with a general model of a frictionless security market where investors are allowed to trade continuously up to some fixed finite planning horizon T . Uncertainty in the financial market is modelled by a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and a filtration $\mathcal{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ satisfying the usual conditions of right-continuity and completeness. We assume that the σ -field $\mathcal{F}_0$ is trivial, i.e. for every $A \in \mathcal{F}_0$ either $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$, and that $\mathcal{F}_T = \mathcal{F}$.

There are $d + 1$ primary traded assets, whose price processes are given by stochastic processes $S_0, \dots, S_d$. We assume that $S = (S_0, \dots, S_d)$ follows an adapted, right-continuous with left-limits (RCLL) and strictly positive semimartingale on $(\Omega, \mathcal{F}, \mathbb{P}, \mathcal{F})$. We assume that the processes $S_0, \dots, S_d$ represent the prices of some traded assets (stocks, bonds, or options). We will occasionally use $\mathcal{M}(\mathbb{P})$ (or just $\mathcal{M}$ if reference to $\mathbb{P}$ is not needed) to denote a financial market model of the above type.

We have not emphasised so far that there was an implicit numéraire behind the prices $S_0, \dots, S_d$; it is the numéraire relevant for domestic transactions at time t . The formal definition of a numéraire is very much as in the discrete setting.

Definition 16.1: numéraire

A numéraire is a price process $X(t)$ almost surely strictly positive for each $t \in [0, T]$.

We assume now that $S_0(t)$ is a non-dividend paying asset, which is (almost surely) strictly positive and we use S_0 as numéraire. *Historically* the money market account $B(t)$, given by $B(t) = e^{rt}$ with a positive deterministic process $r(t)$ and $r(0) = 0$, was used as numéraire, and the reader may think of $S_0(t)$ as being $B(t)$.

Our principal task will be the pricing and hedging of contingent claims, which we model as $\mathcal{F}_T$ -measurable random variables. This implies that the contingent claims specify a stochastic cash-flow at time T and that they may depend on the whole path of the underlying in $[0, T]$, because $\mathcal{F}_T$ contains all that information. We will often have to impose further integrability restrictions on the contingent claims under consideration. The fundamental concept in (arbitrage) pricing and hedging contingent claims is the interplay

of self-financing replicating portfolios and risk-neutral probabilities. Although the current setting is on a much higher level of sophistication, the key ideas remain the same.

We call an $\mathbb{R}^{d+1}$ -valued predictable, locally bounded process

$$\varphi(t) = (\varphi_0(t), \dots, \varphi_d(t)), \quad t \in [0, T],$$

a trading strategy (or dynamic portfolio process). (The conditions ensure that the stochastic integral $\int_0^t \varphi(u) dS(u)$ exists.) Here $\varphi_i(t)$ denotes the number of shares of asset i held in the portfolio at time t , to be determined on the basis of information available before time t ; i.e. the investor selects his time t portfolio after observing the prices $S(t_-)$. The components $\varphi_i(t)$ may assume negative as well as positive values, reflecting the fact that we allow short sales and assume that the assets are perfectly divisible.

Definition 16.2: Trading Strategy, Value And Gains Processes

- (i) The value of the portfolio φ at time t is given by the scalar product

$$V_\varphi(t) = \varphi(t) \cdot S(t) = \sum_{i=0}^d \varphi_i(t) S_i(t), \quad t \in [0, T]. \quad (16.1)$$

The process $V_\varphi(t)$ is called the value process, or wealth process, of the trading strategy φ .

- (ii) The gains process $G_\varphi(t)$ is defined by

$$G_\varphi(t) = \int_0^t \varphi(u) dS(u) = \sum_{i=0}^d \int_0^t \varphi_i(u) dS_i(u). \quad (16.2)$$

- (iii) A trading strategy φ is called self-financing if the wealth process $V_\varphi(t)$ satisfies

$$V_\varphi(t) = V_\varphi(0) + G_\varphi(t) \quad \text{for all } t \in [0, T]. \quad (16.3)$$

Tip 16.1: Interpretation of Self-Financing Condition

- (i) The financial implications of the above equations are that all changes in the wealth of the portfolio are due to capital gains, as opposed to withdrawals of cash or injections of new funds.
- (ii) The definition of a trading strategy includes regularity assumptions in order to ensure the existence of stochastic integrals. In the current rather general setup these conditions are flexible enough to allow a discussion of the no-arbitrage condition; however, they prove too restrictive for the question of market completeness (see below).



Our objective is to be very flexible with regard to the chosen numéraire. The next results underline this.

Proposition 16.1: Numéraire Invariance Theorem

Self-financing portfolios remain self-financing after a numéraire change.

Proof. From a financial viewpoint this property is immediately clear. For a mathematical deduction we use the product rule to get

$$d\left(\frac{S_k(t)}{X(t)}\right) = S_k(t) d\left(\frac{1}{X(t)}\right) + \frac{1}{X(t)} dS_k(t) + d\langle S_k, 1/X \rangle(t), \quad (16.4)$$

where $\langle S_k, 1/X \rangle(t)$ denotes the quadratic covariation between the semimartingales S_k and $1/X$. In the same manner

$$d\left(\frac{V_\varphi(t)}{X(t)}\right) = V_\varphi(t) d\left(\frac{1}{X(t)}\right) + \frac{1}{X(t)} dV_\varphi(t) + d\langle V_\varphi, 1/X \rangle(t). \quad (16.5)$$

The self-financing condition implies that

$$\begin{aligned} d\left(\frac{V_\varphi(t)}{X(t)}\right) &= \sum_{k=1}^d \varphi_k(t) \left\{ S_k(t) d\left(\frac{1}{X(t)}\right) + \frac{1}{X(t)} dS_k(t) + d\langle S_k, 1/X \rangle(t) \right\} \\ &= \sum_{k=1}^d \varphi_k(t) d\left(\frac{S_k(t)}{X(t)}\right). \end{aligned} \quad (16.6)$$

Thus the portfolio expressed in the new numéraire remains self-financing. $\square$

Turning back to the special numéraire $S_0(t)$ we consider the discounted price process

$$\tilde{S}(t) := \frac{S(t)}{S_0(t)} = (1, \tilde{S}_1(t), \dots, \tilde{S}_d(t)), \quad (16.7)$$

with $\tilde{S}_i(t) := S_i(t)/S_0(t)$, $i = 1, 2, \dots, d$. Furthermore, the discounted wealth process $\tilde{V}_\varphi(t)$ is given by

$$\tilde{V}_\varphi(t) := \frac{V_\varphi(t)}{S_0(t)} = \varphi_0(t) + \sum_{i=1}^d \varphi_i(t) \tilde{S}_i(t). \quad (16.8)$$

and the discounted gains process $\tilde{G}_\varphi(t)$ is

$$\tilde{G}_\varphi(t) := \sum_{i=1}^d \int_0^t \varphi_i(u) d\tilde{S}_i(u). \quad (16.9)$$

Observe that $\tilde{G}_\varphi(t)$ does not depend on the numéraire component φ_0 .

It is convenient to reformulate the self-financing condition in terms of the discounted processes:

Proposition 16.2: Self-Financing In Discounted Terms

Let φ be a trading strategy. Then φ is self-financing if and only if

$$\tilde{V}_\varphi(t) = \tilde{V}_\varphi(0) + \tilde{G}_\varphi(t). \quad (16.10)$$

Of course, $V_\varphi(t) \geq 0$ if and only if $\tilde{V}_\varphi(t) \geq 0$.

Tip 16.2: Construction of Self-Financing Strategies

The above result shows that a self-financing strategy is completely determined by its initial value and the components $\varphi_1, \dots, \varphi_d$. In other words, any set of predictable processes $\varphi_1, \dots, \varphi_d$ such that the stochastic integrals $\int \varphi_i d\tilde{S}_i$ exist can be uniquely extended to a self-financing strategy φ with specified initial value $\tilde{V}_\varphi(0) = v$ by setting





$$\varphi_0(t) = v + \sum_{i=1}^d \int_0^t \varphi_i(u) d\tilde{S}_i(u) - \sum_{i=1}^d \varphi_i(t) \tilde{S}_i(t), \quad t \in [0, T].$$

16.1 Equivalent Martingale Measures

We develop a relative pricing theory for contingent claims. Again the underlying concept is the link between the no-arbitrage condition and certain probability measures.

Definition 16.3: Arbitrage Opportunity

A self-financing trading strategy φ is called an arbitrage opportunity if the wealth process V_φ satisfies the following set of conditions:

$$V_\varphi(0) = 0, \quad \mathbb{P}(V_\varphi(T) \geq 0) = 1, \quad \text{and} \quad \mathbb{P}(V_\varphi(T) > 0) > 0.$$

Arbitrage opportunities represent the limitless creation of wealth through risk-free profit and thus should not be present in a well-functioning market.

The main tool in investigating arbitrage opportunities is the concept of equivalent martingale measures:

Definition 16.4: Equivalent Martingale Measure

We say that a probability measure Q defined on $(\Omega, \mathcal{F})$ is a *(strong) equivalent martingale measure* if:

- (i) Q is equivalent to $\mathbb{P}$,
- (ii) the discounted price process $\tilde{S}$ is a Q -local martingale (martingale).

We denote the set of martingale measures by $\mathcal{P}$.

Lemma 16.1: Growth Rates

Assume $S_0(t) = B(t) = e^{r(t)}$, then $Q \sim \mathbb{P}$ is a martingale measure if and only if every asset price process S_i has price dynamics under Q of the form

$$dS_i(t) = r(t)S_i(t)dt + dM_i(t),$$

where M_i is a Q -local martingale.

In order to proceed we have to impose further restrictions on the set of trading strategies.

Definition 16.5: Tame Strategy

A self-financing trading strategy φ is called *tame* (relative to the numéraire S_0) if

$$\tilde{V}_\varphi(t) \geq 0 \quad \text{for all } t \in [0, T].$$

We use the notation Φ for the set of tame trading strategies.

We next analyse the value process under equivalent martingale measures for such strategies.

Proposition 16.3: Value Process Martingale

For $\varphi \in \Phi$, $\tilde{V}_\varphi(t)$ is a non-negative local martingale, and also a supermartingale, under each $Q \in \mathcal{P}$.

Proof. Since $\tilde{S}(t)$ is a local martingale and $\varphi \in \Phi$ is predictable and locally bounded, it follows that the stochastic integral

$$\tilde{G}_\varphi(t) = \int_0^t \varphi(u) d\tilde{S}(u)$$

is a local martingale. Now $\tilde{V}_\varphi(t) = \tilde{V}_\varphi(0) + \tilde{G}_\varphi(t)$, so $\tilde{V}_\varphi(t)$ is a local martingale, and non-negative since φ is tame. But non-negative local martingales are supermartingales. $\square$

This observation is the key to our first central result:

Theorem 16.1: Market Model Arbitrage

Assume $\mathcal{P} \neq \emptyset$. Then the market model contains no arbitrage opportunities in Φ .

Proof. For any $\varphi \in \Phi$ and under any $Q \in \mathcal{P}$ $\tilde{V}_\varphi(t)$ is a supermartingale by [Proposition 16.3](#).

That is,

$$\mathbb{E}_Q \left(\tilde{V}_\varphi(t) | \mathcal{F}_u \right) \leq \tilde{V}_\varphi(u), \text{ for all } u \leq t \leq T.$$

For $\varphi \in \Phi$ to be an arbitrage opportunity we must have $\tilde{V}_\varphi(0) = V_\varphi(0) = 0$. So

$$\mathbb{E}_Q \left(\tilde{V}_\varphi(t) \right) \leq 0, \text{ for all } 0 \leq t \leq T.$$

Now φ is tame, so $\tilde{V}_\varphi(t) \geq 0, 0 \leq t \leq T$, implying $\mathbb{E}_Q \left(\tilde{V}_\varphi(t) \right) = 0, 0 \leq t \leq T$, and in particular $\mathbb{E}_Q \left(\tilde{V}_\varphi(T) \right) = 0$. But an arbitrage opportunity φ also has to satisfy $\mathbb{P}(V_\varphi(T) \geq 0) = 1$, and since $Q \sim \mathbb{P}$, this means $Q(V_\varphi(T) \geq 0) = 1$. Both together yield

$$Q(V_\varphi(T) > 0) = \mathbb{P}(V_\varphi(T) > 0) = 0,$$

and hence the result follows. □

The following fundamental theorem of asset pricing is proved in [55].

Theorem 16.2: Fundamental Theorem of Asset Pricing

There exists an equivalent martingale measure for the financial market model $\mathcal{M}(\mathbb{P})$ if and only if the condition NFLVR holds true.

16.2 Risk-neutral Pricing

We now assume that there exists a strong equivalent martingale measure $\mathbb{P}^*$ (which implies that there are no arbitrage opportunities with respect to Φ in the financial market model $\mathcal{M}$.) Until further notice we use $\mathbb{P}^*$ as our reference measure, and when using the terms martingale and local martingale we always assume that the underlying probability measure is $\mathbb{P}^*$. In particular, we restrict our attention to contingent claims X such that $X/S_0(T) \in L^1(\mathcal{F}, \mathbb{P}^*)$.

We now define a further subclass of trading strategies:

Definition 16.6: Admissible Strategy

A self-financing trading strategy φ is called $(\mathbb{P}^*)$ -admissible if the relative gains process

$$\tilde{G}_\varphi(t) = \int_0^t \varphi(u) d\tilde{S}(u)$$

is a $(\mathbb{P}^*)$ -martingale. The class of all $(\mathbb{P}^*)$ -admissible trading strategies is denoted $\Phi(\mathbb{P}^*)$.

By definition $\tilde{S}$ is a martingale, and $\tilde{G}$ is the stochastic integral with respect to $\tilde{S}$. By [Tip 16.2](#) we see that any sufficiently integrable processes $\varphi_1, \dots, \varphi_d$ give rise to $\mathbb{P}^*$ -admissible trading strategies.

We do not assume that admissible trading strategies are tame. However, since we only used the supermartingale property of $\tilde{G}_\varphi$ in the proof of [Theorem 16.1](#), we can repeat the above argument to obtain

Theorem 16.3: No Arbitrage for Admissible Strategies

The financial market model $\mathcal{M}$ contains no arbitrage opportunities in $\Phi(\mathbb{P}^)$.*

Under the assumption that no arbitrage opportunities exist, the question of pricing and hedging a contingent claim reduces to the existence of replicating self-financing trading strategies. Formally:

Definition 16.7: Attainable Claims and Market Completeness

- (i) A contingent claim X is called *attainable* if there exists an admissible trading strategy $\varphi \in \Phi(\mathbb{P}^*)$ such that

$$V_\varphi(T) = X.$$

Such a φ is called a *replicating strategy* for X .

- (ii) The financial market model $\mathcal{M}$ is *complete* if every contingent claim is attainable.

Again we emphasise that this depends on the class of trading strategies. On the other hand, it does not depend on the numéraire: it is an easy exercise to show that if a contingent claim is attainable in a given numéraire it is also attainable in any other numéraire and the replicating strategies are the same.

If a contingent claim X is attainable, X can be replicated by a portfolio $\varphi \in \Phi(\mathbb{P}^*)$. This means that holding the portfolio and holding the contingent claim are equivalent from a financial point of view. In the absence of arbitrage the (arbitrage) price process $\Pi_X(t)$ of the contingent claim must therefore satisfy

$$\Pi_X(t) = V_\varphi(t).$$

Of course the questions arise of what will happen if X can be replicated by more than one portfolio, and what the relation of the price process to the equivalent martingale measure(s) is. The following central theorem is the key to answering these questions:

Theorem 16.4: Risk-Neutral Valuation Formula

The arbitrage price process of any attainable claim is given by the risk-neutral valuation formula

$$\Pi_X(t) = S_0(t) \mathbb{E}_{\mathbb{P}^*} \left[\frac{X}{S_0(T)} \middle| \mathcal{F}_t \right]. \quad (16.11)$$

Proof. Since X is attainable, there exists a replicating strategy $\varphi \in \Phi(\mathbb{P}^*)$ such that $V_\varphi(T) = X$ and $\Pi_X(t) = V_\varphi(t)$. Since $\varphi \in \Phi(\mathbb{P}^*)$ the discounted value process $\tilde{V}_\varphi(t)$ is a martingale, and hence

$$\begin{aligned} \Pi_X(t) &= V_\varphi(t) = S_0(t) \tilde{V}_\varphi(t) \\ &= S_0(t) \mathbb{E}_{\mathbb{P}^*} \left[\tilde{V}_\varphi(T) \middle| \mathcal{F}_t \right] = S_0(t) \mathbb{E}_{\mathbb{P}^*} \left[\frac{V_\varphi(T)}{S_0(T)} \middle| \mathcal{F}_t \right] \\ &= S_0(t) \mathbb{E}_{\mathbb{P}^*} \left[\frac{X}{S_0(T)} \middle| \mathcal{F}_t \right]. \end{aligned}$$

□

The uniqueness question is immediate from the above theorem:

Corollary 16.1: Uniqueness of Replicating Portfolio Value

For any two replicating portfolios $\varphi, \psi \in \Phi(\mathbb{P}^*)$ we have

$$V_\varphi(t) = V_\psi(t).$$

Attention 16.1: Choice of Martingale Measure

The above argument addresses only the case of different replicating portfolios for X in $\Phi(\mathbb{P}^*)$. Of course the choice of $\mathbb{P}^*$ was itself completely arbitrary, and we could have several - indeed, many - equivalent martingale measures $\mathbf{Q}$. This makes the definition of admissible strategies with respect to different martingale measures somewhat unsatisfactory. However, if we combine our previous two definitions and call a self-financing trading strategy φ admissible, if φ is tame and $\mathbf{Q}$ -admissible for some $\mathbf{Q} \in \mathcal{P}$, and X attainable, if there exists an admissible φ replicating it, we show that for admissible $\varphi_1 \in \Phi(\mathbf{Q}_1)$ and $\varphi_2 \in \Phi(\mathbf{Q}_2)$ replicating X we have

$$\mathbb{E}_{\mathbf{Q}_1} \left[\frac{X}{S_0(T)} \middle| \mathcal{F}_t \right] = \mathbb{E}_{\mathbf{Q}_2} \left[\frac{X}{S_0(T)} \middle| \mathcal{F}_t \right],$$

implying that the arbitrage price process is unique. More generally, one can even show that

$$\Pi_X(0) = \sup_{\mathbf{Q} \in \mathcal{P}} S_0(0) \mathbb{E}_{\mathbf{Q}} \left[\frac{X}{S_0(T)} \middle| \mathcal{F}_t \right] = \inf_{\varphi \in \theta(X)} V_\varphi(0),$$

where $\theta(X)$ is the class of all admissible (in the wider sense) trading strategies replicating X .

Tip 16.3: Existence of Equivalent Martingale Measures

The question of existence of (strong) equivalent martingale measures for a given stochastic process S in our market model is closely related to properties of the stochastic (Doléans) exponential $\mathcal{E}(U)$. Loosely speaking, a probability measure $\mathbf{Q}$ on $(\Omega, \mathcal{F}_T)$ is equivalent to the underlying probability measure $\mathbb{P}$ if the Radon-Nikodým derivative $d\mathbf{Q}/d\mathbb{P}$ is of the form $\mathcal{E}(U)$ for some local martingale U .

For pure pricing purposes it is sufficient to find a strong equivalent martingale measure, but for risk-management purposes it is more important to find the replicating portfolio. We have the following useful result, which shows that hedging is equivalent to the existence of a stochastic integral representation of the discounted claim.

Lemma 16.2: Integral Representation Implies Attainability

Assume that the discounted contingent claim $X/S_0(T)$ is $\mathbb{P}^*$ -integrable. If the $(\mathbb{P}^*)$ -martingale M defined by

$$M(t) = \mathbb{E}_{\mathbb{P}^*} \left[\frac{X}{S_0(T)} \middle| \mathcal{F}_t \right]$$

admits an integral representation of the form

$$M(t) = x + \sum_{i=1}^d \int_0^t \varphi_i(u) d\tilde{S}_i(u),$$

with $\varphi_1, \dots, \varphi_d$ predictable and locally bounded, then X is attainable.

Proof. We are looking for a trading strategy $\varphi \in \Phi(\mathbb{P}^*)$ such that $V_\varphi(T) = X$. We use the components $\varphi_1, \dots, \varphi_d$ from the definition of the integral representation of M and then define φ_0 as in [Tip 16.2](#), with $v = x$. Then $\varphi = (\varphi_0, \varphi_1, \dots, \varphi_d)$ is a self-financing trading strategy with

$$\tilde{V}_\varphi(t) = v + \tilde{G}_\varphi(t) = x + \sum_{i=1}^d \int_0^t \varphi_i(u) d\tilde{S}_i(u) = M(t).$$

By definition M is a non-negative martingale, so $\varphi \in \Phi(\mathbb{P}^*)$ follows, and also by definition

$$V_\varphi(T) = S_0(T) \tilde{V}_\varphi(T) = S_0(T) M(T) = S_0(T) \frac{X}{S_0(T)} = X.$$

□

We thus see that (given some integrability conditions) attainability, and hence completeness, is closely linked to martingale representation theorems. Statement and proofs of the martingale representation property require rather technical definitions involving the fine structure of the filtration $\mathcal{F}$. Moreover, in a general set-up the conditions on the process φ (the integrand) no longer guarantee that φ may be used as a suitable trading strategy.

We close by stating the following partial analogue of the completeness theorem in the discrete setting:

Theorem 16.5: Restricted Completeness

If the strong martingale measure $\mathbb{P}^$ is the unique martingale measure for the financial market model $\mathcal{M}$, then $\mathcal{M}$ is complete, in the restricted sense that every contingent claim X satisfying*

$$\frac{X}{S_0(T)} \in L^1(\mathcal{F}, \mathbb{P}^*)$$

is attainable.

16.3 Change of Numéraire

We now develop the technique of change of numéraire, which will be extremely useful in the valuation of contingent claims, and also for deduction of theoretical results.

We put ourselves in the following framework. Assume that we used $S_0(t)$ as numéraire, and that there exists an equivalent martingale measure $\mathbb{P}^*$.

Theorem 16.6: Numéraire Change Measure

Let $X(t)$ be a non-dividend-paying numéraire such that $X(t)/S_0(t)$ is a $\mathbb{P}^$ -martingale. Then there exists a probability measure $\mathbb{Q}_X$, defined by its Radon-Nikodým derivative $\eta(T)$ with respect to $\mathbb{P}^*$,*

$$\eta(t) = \frac{d\mathbb{Q}_X}{d\mathbb{P}^*} \Big|_{\mathcal{F}_t} = \frac{X(t)}{X(0)S_0(t)},$$

such that:

- (i) *the basic security prices discounted with respect to X are $\mathbb{Q}_X$ -local martingales;*
- (ii) *if a contingent claim Y is attainable under $(S_0(t), \mathbb{P}^*)$, then it is attainable under $(X(t), \mathbb{Q}_X)$, and the replicating portfolio is the same. So the arbitrage price processes, given by the risk-neutral valuation formula, coincide.*

Proof. (i) Denote by $\bar{Z} = Z(t)/X(t)$ the relative price of a security Z (it may be one of the components S_i) with respect to the numéraire $X(t)$. Since $\eta(t) = X(t)/X(0)S_0(t)$

is a $\mathbb{P}^*$ -martingale we can use the Bayes formula and the $\mathbb{P}^*$ -martingale property of $Z(t)/S_0(t)$ to get for $0 \leq u \leq t$

$$\begin{aligned}\mathbb{E}_{Q_X} [\bar{Z}(t) | \mathcal{F}_u] &= \frac{1}{\eta(u)} \mathbb{E}_{\mathbb{P}^*} [\bar{Z}(t) \eta(t) | \mathcal{F}_u] \\ &= \frac{X(0)S_0(u)}{X(u)} \mathbb{E}_{\mathbb{P}^*} \left[\frac{Z(t)}{X(t)} \frac{X(t)}{X(0)S_0(t)} \middle| \mathcal{F}_u \right] \\ &= \frac{S_0(u)}{X(u)} \frac{Z(u)}{S_0(u)} = \frac{Z(u)}{X(u)} = \bar{Z}(u).\end{aligned}$$

So the basic securities are martingales, and hence so is any portfolio.

- (ii) If Y is attainable under $\mathbb{P}^*$, then there exists a $\mathbb{P}^*$ -admissible portfolio φ such that $V_\varphi(T) = Y$, and by the risk-neutral valuation formula

$$V_\varphi(t) = S_0(t) \mathbb{E}_{\mathbb{P}^*} \left[\frac{Y}{S_0(T)} \middle| \mathcal{F}_t \right].$$

Using again the Bayes formula we obtain

$$\begin{aligned}\mathbb{E}_{Q_X} \left[\frac{Y}{X(T)} \middle| \mathcal{F}_t \right] &= \frac{1}{\eta(t)} \mathbb{E}_{\mathbb{P}^*} \left[\frac{Y}{X(T)} \eta(T) \middle| \mathcal{F}_t \right] \\ &= \frac{X(0)S_0(t)}{X(t)} \mathbb{E}_{\mathbb{P}^*} \left[\frac{Y}{X(T)} \frac{X(T)}{X(0)S_0(T)} \middle| \mathcal{F}_t \right] \\ &= \frac{V_\varphi(t)}{X(t)}.\end{aligned}$$

Since self-financing portfolios remain self-financing under a change of numéraire (by the numéraire invariance theorem), and since by part (i) $V_\varphi(t)$ is a Q_X -martingale, it follows that φ is Q_X -admissible and the result follows. □

Corollary 16.2: Change of Numéraire Expectation Formula

Let $X(t), Y(t)$ be numéraires satisfying the assumptions of the above proposition and Z a contingent claim. Then we have the change of numéraire formula

$$X(t)\mathbb{E}_{Q_X} \left[\frac{Z}{X(T)} \middle| \mathcal{F}_t \right] = Y(t)\mathbb{E}_{Q_Y} \left[\frac{Z}{Y(T)} \middle| \mathcal{F}_t \right]. \quad (16.12)$$

Proof. We can use the second part of the above proposition, or use the Radon-Nikodým derivative

$$\frac{dQ_X}{dQ_Y} \bigg|_{\mathcal{F}_T} = \frac{dQ_X}{d\mathbb{P}^*} \bigg/ \frac{dQ_Y}{d\mathbb{P}^*} = \frac{X(T)}{X(0)S_0(T)} \bigg/ \frac{Y(T)}{Y(0)S_0(T)} = \frac{X(T)/Y(T)}{X(0)/Y(0)}$$

and repeat the argument. $\square$

The main application of changes of numéraire is its usefulness in pricing contingent claims. In short, we claim there is a best numéraire for any given problem, and it pays to find it and work with it.

Proposition 16.4: Option Pricing under Numéraire Change

Denote by T and K respectively the maturity and exercise price of a European call option on an asset Z . The time $t = 0$ price $C(0)$ of the call can be written as

$$\frac{C(0)}{p(0, T)} = \mathbb{E}_{Q^T} [(Z(T) - K)^+],$$

, or

$$C(0) = Z(0)Q_Z(A) - Kp(0, T)Q^T(A),$$

where $A = \{\omega : Z(T, \omega) > K\}$.

Proof. The first expression is immediate from [Theorem 16.6](#). To derive the second write

$$\frac{C(0)}{p(0, T)} = \mathbb{E}_{Q^T} [(Z(T) - K)^+] = \mathbb{E}_{Q^T} [Z(T)\mathbb{1}_A] - KQ^T(A).$$

Using Z as a numéraire in the middle expectation, we have by the change of numéraire, Equation (16.12)

$$p(0, T) \mathbb{E}_{Q^T} \left[\frac{Z(T)}{p(T, T)} \mathbb{1}_A \right] = Z(0) \mathbb{E}_Z [\mathbb{1}_A],$$

which gives the second expression. $\square$

Part V

Martingale Methods

17 The Black-Scholes Model

We start by describing processes that model the prices of primary securities, a savings account (equivalently, a risk-free bond), and a common stock.

17.1 Risk-free Bond

The first primary security represents in our model an accumulation factor corresponding to a *savings account* (a *money market account*). We assume that the *short-term interest rate* r is constant (but not necessarily nonnegative) over the trading interval $[0, T^*]$. The risk-free security is assumed to continuously compound in value at the rate r ; that is, $dB_t = rB_t dt$. We adopt the usual convention that $B_0 = 1$, so that its price equals $B_t = e^{rt}$ for every $t \in [0, T^*]$ (Merton, 1973). When dealing with the Black-Scholes model, we may equally well replace the savings account by the *risk-free bond*. A unit zero-coupon bond maturing at time T is a security that pays its holder 1 unit of cash at a predetermined date T in the future, known as the bond's *maturity date*. Let $B(t, T)$ stand for the price at time $t \in [0, T]$ of a bond maturing at time T (Brennan and Schwartz, 1979). It is easily seen that to replicate the payoff 1 at time T it suffices to invest B_t/B_T units of cash at time t in the savings account B . This shows that, in the absence of arbitrage opportunities, the price of the bond satisfies

$$B(t, T) = e^{-r(T-t)}, \quad \forall t \in [0, T]. \quad (17.1)$$

Note that for any fixed T , the bond price solves the ordinary differential equation

$$dB(t, T) = rB(t, T)dt, \quad B(0, T) = e^{-rT}. \quad (17.2)$$

We consider here a *risk-free bond*, meaning that its issuer will not default on its obligation to pay to the bondholder the face value at maturity date (Bielecki, Jeanblanc, and Rutkowski, 2004).

17.2 Common Stock Price

Let us fix the horizon date $T^* > 0$. The continuous-time evolution of the stock price process $(S_t)_{t \in [0, T^*]}$ is described by the following linear *stochastic differential equation* (SDE)

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (17.3)$$

where $\mu \in \mathbb{R}$ is a constant *appreciation rate* of the stock price, $\sigma > 0$ is a constant *volatility*, and S_0 is the initial value of the stock price (Black and Scholes, 1973). As we shall see in what follows, the restrictive condition that μ and σ are constant can be weakened substantially (Heston, 1993; Bensoussan, Crouhy, and Galai, 1994).

The driving noise W_t , $t \in [0, T^*]$, is a one-dimensional standard Brownian motion defined on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

We refer to Equation (17.3) as an SDE; however, expression Equation (17.3) is merely a shorthand notation for the following *stochastic integral equation* (see Subsection 7.3)

$$S_t = S_0 + \int_0^t \mu S_u du + \int_0^t \sigma S_u dW_u, \quad \forall t \in [0, T^*]. \quad (17.4)$$

In other words, by a stochastic differential equation of the form Equation (17.3) we mean in fact the stochastic integral equation Equation (17.4). The definition of the second integral in Equation (17.4), interpreted here as the *Itô stochastic integral*, can be found in Subsection 7.3. Let us only mention here that if X is a continuous, $\mathcal{F}$ -adapted process that satisfies suitable

integrability conditions, then

$$\int_0^t X_u dW_u = \lim_{n \rightarrow +\infty} \sum_{j=0}^{n-1} X_{t_j^n} (W_{t_{j+1}^n} - W_{t_j^n}), \quad (17.5)$$

where the limit is in fact taken over an arbitrary sequence of finite partitions $\{t_0^n = 0 < t_1^n < \dots < t_n^n = t\}$ of the interval $[0, t]$ such that $\lim_{n \rightarrow \infty} \max_{k=0, \dots, n-1} (t_{k+1}^n - t_k^n) = 0$.

Proposition 17.1: Solution of the SDE

The process S given by the formula

$$S_t = S_0 \exp \left(\sigma W_t + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right), \quad \forall t \in [0, T^*], \quad (17.6)$$

is the unique solution of the stochastic differential equation [Equation \(17.3\)](#), or equivalently, the stochastic integral equation [Equation \(17.4\)](#).

Proof. We will first check that the process given by [Equation \(17.6\)](#) satisfies [Equation \(17.3\)](#). As customary, we use the differential notation, rather than the integral notation. Let us introduce the auxiliary process X by setting

$$dX_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t$$

with $X_0 = 0$. Then the process S given by [Equation \(17.6\)](#) satisfies $S_t = g(X_t)$ for $t \in [0, T^*]$, where the function $g : \mathbb{R} \rightarrow \mathbb{R}$ is defined as $g(x) = S_0 e^x$. Obviously, $g(x) = g'(x) = g''(x)$ and thus Itô's formula yields

$$\begin{aligned} dS_t &= dg(X_t) \\ &= g'(X_t) \left(\mu - \frac{1}{2} \sigma^2 \right) dt + g'(X_t) \sigma dW_t + \frac{1}{2} g''(X_t) \sigma^2 dt \\ &= g(X_t) (\mu dt + \sigma dW_t) \\ &= S_t (\mu dt + \sigma dW_t), \end{aligned}$$

as expected. Uniqueness of solutions to Equation (17.4) is an immediate consequence of the general result, due to Itô, stating that an SDE with Lipschitz continuous coefficients has a unique solution.

Alternatively, one can prove the uniqueness by applying the Itô formula to the ratio of two solutions to Equation (17.4). Let us assume that Y is another solution to Equation (17.3) with the initial value $Y_0 = S_0$. We wish to show that the Itô differential of the ratio Y/S is equal to zero. To this end, we will first compute the Itô differential of the process $1/S$. Using Itô's formula again, we obtain (note that now we take $g(s) = 1/s$)

$$\begin{aligned} d\left(\frac{1}{S_t}\right) &= -\frac{1}{S_t^2}dS_t + \frac{1}{S_t^3}d\langle S \rangle_t = -\frac{1}{S_t^2}(\mu S_t dt + \sigma S_t dW_t) + \frac{1}{S_t^3}\sigma^2 S_t^2 dt \\ &= -\frac{1}{S_t}((\mu - \sigma^2)dt + \sigma dW_t), \end{aligned}$$

where we used the fact that the quadratic variation of S equals

$$\langle S \rangle_t = \int_0^t \sigma^2 S_u^2 d\langle W \rangle_u = \int_0^t \sigma^2 S_u^2 du. \quad (17.7)$$

By applying Itô's integration by parts formula to processes Y and $1/S$, we get

$$\begin{aligned} d\left(\frac{Y_t}{S_t}\right) &= \frac{1}{S_t}dY_t + Y_t d\left(\frac{1}{S_t}\right) + d\langle Y, 1/S \rangle_t \\ &= \frac{Y_t}{S_t}(\mu dt + \sigma dW_t) - \frac{Y_t}{S_t}((\mu - \sigma^2)dt + \sigma dW_t) - \frac{Y_t}{S_t}\sigma^2 dt \\ &= 0, \end{aligned}$$

since the cross-variation of Y and $1/S$ equals (recall that $\langle W \rangle_t = t$)

$$\langle Y, 1/S \rangle_t = - \int_0^t \sigma^2 \frac{Y_u}{S_u} d\langle W \rangle_u = - \int_0^t \sigma^2 \frac{Y_u}{S_u} du. \quad (17.8)$$

Since the Itô differential of the ratio Y/S is zero, this process is constant, and thus it equals its initial value, which is 1. We conclude that $Y_t = S_t$ for every $t \in [0, T^*]$, and thus the uniqueness of a solution to Equation (17.4) is established. $\square$

We make the standing assumption that the underlying filtration $\mathcal{F} = (\mathcal{F}_t)_{t \in [0, T^*]}$ is the natural filtration $\mathcal{F}^W$ of the driving Brownian motion. More precisely, $\mathcal{F}$ is the standard augmentation of the natural filtration of W , but it will still be denoted by $\mathcal{F}^W$. For any fixed $t \in [0, T^*]$, we have $S_t = f(W_t)$ for some function $f : \mathbb{R} \rightarrow \mathbb{R}_+$ that determines the one-to-one correspondence between the values S_t of the stock price and the values W_t of the Brownian motion. Specifically, the strictly increasing function f is given by the formula

$$f(x) = S_0 \exp \left(\sigma x + \left(\mu - \frac{1}{2} \sigma^2 \right) t \right), \quad \forall x \in \mathbb{R}. \quad (17.9)$$

Consequently, for any natural n , any dates $0 \leq t_1 < t_2 < \dots < t_n \leq T^*$, and any real numbers $x_1, x_2, \dots, x_n$, we have

$$\omega \in \Omega \mid S_{t_1} < x_1, \dots, S_{t_n} < x_n = \omega \in \Omega \mid W_{t_1} < y_1, \dots, W_{t_n} < y_n, \quad (17.10)$$

where we set $y_i = f^{-1}(x_i)$. It is thus clear that we have, for every $t \in [0, T^*]$,

$$\mathcal{F}_t^W = \sigma W_u \mid u \leq t = \sigma S_u \mid u \leq t = \mathcal{F}_t^S. \quad (17.11)$$

We conclude that the filtration generated by the stock price coincides with the natural filtration of the driving noise W , and thus $\mathcal{F}^S = \mathcal{F}^W = \mathcal{F}$. This means that the information structure of the model is based on observations of the stock price process only.

It is readily verified that the Brownian motion W is a martingale with respect to the underlying filtration $\mathcal{F}$. Indeed, for any t , the random variable W_t is $\mathbb{P}$ -integrable, and for any $u \leq t$:

$$\mathbb{E}_{\mathbb{P}}(W_t \mid \mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(W_t - W_u \mid \mathcal{F}_u) + \mathbb{E}_{\mathbb{P}}(W_u \mid \mathcal{F}_u) = W_u. \quad (17.12)$$

Define an auxiliary process M by setting:

$$M_t = \exp \left(\sigma W_t - \frac{1}{2} \sigma^2 t \right), \quad \forall t \in [0, T^*]. \quad (17.13)$$

Lemma 17.1: Martingale Property

The process M is a martingale under $\mathbb{P}$ with respect to the filtration $\mathcal{F}$.

Proof. In order to show that M is a martingale under $\mathbb{P}$ with respect to $\mathcal{F}$, it suffices to note that

$$\mathbb{E}_{\mathbb{P}}(e^{\sigma(W_t - W_u)} \mid \mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(e^{\sigma(W_t - W_u)}) = \exp\left(\frac{1}{2}\sigma^2(t - u)\right), \quad (17.14)$$

where the first equality is a consequence of the independence of the increment $W_t - W_u$ of the σ -field $\mathcal{F}_u$ and the second follows from the fact that $W_t - W_u$ has the Gaussian distribution $\mathcal{N}(0, t - u)$. $\square$

Since $S_t = e^{\mu t} M_t$, we have, for every $u \leq t \leq T^*$,

$$\mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) = \mathbb{E}_{\mathbb{P}}(e^{\mu t} M_t \mid \mathcal{F}_u) = e^{\mu t} M_u = S_u e^{\mu(t-u)}. \quad (17.15)$$

Corollary 17.1: Martingale Property of Stock Price

The stock price S follows a martingale under the actual probability $\mathbb{P}$ with respect to the filtration $\mathcal{F}$ if and only if its appreciation rate $\mu = 0$. If $\mu > 0$ (respectively $\mu < 0$) then $\mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) > S_u$ (respectively $\mathbb{E}_{\mathbb{P}}(S_t \mid \mathcal{F}_u) < S_u$) for every $t > u$ and thus the stock price S is a submartingale (respectively a supermartingale) under the actual probability $\mathbb{P}$.

It is apparent from Equation (17.6) that the logarithmic stock returns are Gaussian, meaning that the random variable $\ln(S_t/S_u)$ has under $\mathbb{P}$ a Gaussian probability distribution for any choice of dates $u < t \leq T^*$. Moreover, S is a *time-homogeneous Markov process* under $\mathbb{P}$ with respect to the filtration $\mathcal{F}$ with the transition probability density function, for $x, y > 0$ and $t > u$,

$$p_S(u, x, t, y) = \frac{1}{\sqrt{2\pi(t-u)}\sigma y} \exp\left\{-\frac{(\ln(y/x) - \mu(t-u) + \frac{1}{2}\sigma^2(t-u))^2}{2\sigma^2(t-u)}\right\} \quad (17.16)$$

where formally $p_S(u, x, t, y) = \mathbb{P}\{S_t = y \mid S_u = x\}$. It is important to observe that even if the appreciation rate μ is replaced by an adapted stochastic process μ_t and the stock price S is

no longer Markovian under the actual probability $\mathbb{P}$, it will always be a time-homogeneous Markov process under the martingale measure $\mathbb{P}^*$ for the discounted stock price.

17.3 Self-financing Trading Strategies

By a *trading strategy* we mean a pair $\phi = (\phi^1, \phi^2)$ of $\mathcal{F}$ -progressively measurable stochastic processes on the underlying probability space $(\Omega, \mathcal{F}, \mathbb{P})$. The concept of a self-financing trading strategy in the Black–Scholes market is formally based on the notion of the Itô integral. Intuitively, this choice of stochastic integral is supported by the fact that, in the case of the Itô integral (as opposed, for instance, to the Fisk–Stratonovich integral), the underlying process is integrated in a predictable way, meaning that we take its values on the left-hand end of each (infinitesimal) time interval. Unless explicitly stated otherwise, we will assume that a stock S does not pay dividends (at least during the option's lifetime).

Definition 17.1: Self-Financing Trading Strategy

We say that a trading strategy $\phi = (\phi^1, \phi^2)$ over the time interval $[0, T]$ is *self-financing* if its wealth process $V(\phi)$, which is set to equal

$$V_t(\phi) = \phi_t^1 S_t + \phi_t^2 B_t, \quad \forall t \in [0, T], \quad (17.17)$$

satisfies the following condition

$$V_t(\phi) = V_0(\phi) + \int_0^t \phi_u^1 dS_u + \int_0^t \phi_u^2 dB_u, \quad \forall t \in [0, T], \quad (17.18)$$

where the first integral is understood in the Itô sense and the second is the path-wise Riemann (or Lebesgue) integral.

17.4 Martingale Measure for the Black-Scholes Model

We find it convenient to introduce the concept of the *admissibility* of a trading strategy directly in terms of a martingale measure. For this reason, our next goal is to study the concept of a martingale measure within the Black-Scholes set-up.

Let $S^* = S/B$ denote the discounted stock price. Using the geometric Brownian motion solution Equation (17.6) and $B_t = e^{rt}$, we obtain for every $t \in [0, T^*]$

$$S_t^* = S_0^* \exp \left(\sigma W_t + \left(\mu - r - \frac{1}{2} \sigma^2 \right) t \right). \quad (17.19)$$

Equivalently, the process S^* is the unique solution to the SDE

$$dS_t^* = (\mu - r) S_t^* dt + \sigma S_t^* dW_t, \quad S_0^* = S_0. \quad (17.20)$$

The martingale property of S^* under $\mathbb{P}$ allows for analysis identical to the stock price S .

Corollary 17.2: Martingale Property of Discounted Stock

The discounted stock price S^ is a martingale under the actual probability $\mathbb{P}$ with respect to the filtration $\mathcal{F}$ if and only if $\mu = r$. If $\mu > r$ (respectively $\mu < r$), then $\mathbb{E}_{\mathbb{P}}(S_t^* | \mathcal{F}_u) > S_u^*$ (respectively $\mathbb{E}_{\mathbb{P}}(S_t^* | \mathcal{F}_u) < S_u^*$) for every $t > u$, implying S^* is a sub-martingale (respectively a super-martingale) under $\mathbb{P}$.*

The subsequent definition directly addresses the martingale property of the discounted stock price S^* . We seek a probability measure under which the drift term in the governing equation Equation (17.20) disappears.

Definition 17.2: Martingale Measure for S^*

A probability measure $\mathbb{Q}$ on $(\Omega, \mathcal{F}_{T^*})$, equivalent to $\mathbb{P}$, is called a *martingale measure* for S^* if S^* is a local martingale under $\mathbb{Q}$.

To analyse the no-arbitrage property of the model and derive the risk-neutral valuation formula, we focus on the martingale property of the discounted wealth process.

Definition 17.3: Spot Martingale Measure

A probability measure $\mathbb{P}^*$ on $(\Omega, \mathcal{F}_{T^*})$, equivalent to $\mathbb{P}$, is called a *spot martingale measure* if the discounted wealth $V_t^*(\phi) = V_t(\phi)/B_t$ of any self-financing trading strategy ϕ is a local martingale under $\mathbb{P}^*$.

These notions coincide within the present setup, provided the stock S pays no dividends during the period $[0, T^*]$. The case of a dividend-paying stock is analysed in subsequent sections.

Lemma 17.2: Equivalence of Martingale Measures

A probability measure is a spot martingale measure if and only if it is a martingale measure for the discounted stock price S^ .*

Proof. Let ϕ be a self-financing strategy and let $V^* = V^*(\phi)$. Using Itô's integration by parts formula, we get (note that $dB_t^{-1} = -rB_t^{-1} dt$)

$$\begin{aligned} dV_t^* &= d(V_t B_t^{-1}) = V_t dB_t^{-1} + B_t^{-1} dV_t \\ &= (\phi_t^1 S_t + \phi_t^2 B_t) dB_t^{-1} + B_t^{-1} (\phi_t^1 dS_t + \phi_t^2 dB_t) \\ &= \phi_t^1 (B_t^{-1} dS_t + S_t dB_t^{-1}) = \phi_t^1 dS_t^*. \end{aligned}$$

We thus obtain the following equality

$$V_t^*(\phi) = V_0^*(\phi) + \int_0^t \phi_u^1 dS_u^*, \quad \forall t \in [0, T^*].$$

It suffices to make use of the local martingale property of the Itô integral. □

In the Black-Scholes setting, the martingale measure for the discounted stock price is unique and can be explicitly derived using the Girsanov theorem.

Lemma 17.3: Unique Martingale Measure

(i) The unique martingale measure $\mathbb{Q}$ for the discounted stock price process S^* is given by the Radon-Nikodým derivative

$$\frac{d\mathbb{Q}}{d\mathbb{P}} \stackrel{\text{a.s.}}{=} \exp \left(\frac{r - \mu}{\sigma} W_{T^*} - \frac{1}{2} \left(\frac{r - \mu}{\sigma} \right)^2 T^* \right) \quad (17.21)$$

(ii) Under the martingale measure $\mathbb{Q}$, the discounted stock price S^* satisfies

$$dS_t^* = \sigma S_t^* dW_t^*, \quad (17.22)$$

where the continuous, $\mathcal{F}$ -adapted process W^* defined by

$$W_t^* = W_t - \frac{r - \mu}{\sigma} t, \quad \forall t \in [0, T^*], \quad (17.23)$$

is a standard Brownian motion on the probability space $(\Omega, \mathcal{F}, \mathbb{Q})$.

Proof. Let us first prove part (i).

$$\left. \frac{d\mathbb{Q}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \mathcal{E}_t \left(\int_0^t \gamma_u dW_u - \frac{1}{2} \int_0^t \gamma_u^2 du \right), \quad \mathbb{P} \text{ a.s.}$$

Moreover, by Girsanov's theorem, the process W^* defined by

$$W_t^* = W_t - \int_0^t \gamma_u du, \quad \forall t \in [0, T^*],$$

is a standard Brownian motion under $\mathbb{Q}$. Substituting $dW_t = dW_t^* + \gamma_t dt$ into the dynamics of S^* in Equation (17.20), we obtain

$$dS_t^* = (\mu - r + \gamma_t \sigma) S_t^* dt + \sigma S_t^* dW_t^*, \quad S_0^* = S_0.$$

□

Definition 17.4: Admissible Trading Strategy

A trading strategy $\phi \in \Phi$ is called $\mathbb{P}^*$ -admissible if the discounted wealth process

$$V_t^*(\phi) = B_t^{-1}V_t(\phi), \quad \forall t \in [0, T],$$

follows a martingale under $\mathbb{P}^*$. We write $\Phi(\mathbb{P}^*)$ to denote the class of all $\mathbb{P}^*$ -admissible trading strategies. The triple $\mathcal{M}_{BS} = (S, B, \Phi(\mathbb{P}^*))$ is called the *arbitrage-free Black-Scholes model* of a financial market, or briefly, the *Black-Scholes model*.

For brevity, we shall say that ϕ is *admissible* rather than $\mathbb{P}^*$ -admissible if no confusion may arise. It is easy to check that by restricting our attention to the class of all admissible strategies, we have ensured the absence of arbitrage opportunities in the Black-Scholes model.

We say that a European contingent claim X that settles at time $T \leq T^*$ is *attainable* in the Black-Scholes model if it can be replicated by means of an admissible strategy. Given an attainable European contingent claim, we can uniquely define its arbitrage price in the Black-Scholes model.

Definition 17.5: Arbitrage Price Process

Let X be an attainable European contingent claim that settles at time $T \leq T^*$. Then its *arbitrage price process* in the Black-Scholes model $\mathcal{M}_{BS}$, denoted as $\pi_t(X)$, $t \in [0, T]$, is defined as the wealth process $V_t(\phi)$, $t \in [0, T]$, of any admissible trading strategy ϕ that replicates X , that is, satisfies the equality $V_T(\phi) = X$.

If no replicating admissible strategy exists², the arbitrage price of such a claim is not defined. Conforming to the definition of an arbitrage price, to value a derivative security, we typically first search for its replicating strategy. The risk-neutral valuation approach to the pricing problem is also possible, as the following simple result shows.

²This happens only if a claim is not integrable under $\mathbb{P}^*$

Corollary 17.3: Risk-Neutral Valuation Formula

Let X be an attainable European contingent claim that settles at time T . Then the arbitrage price $\pi_t(X)$ at time $t \in [0, T]$ in $\mathcal{M}_{BS}$ is given by the risk-neutral valuation formula

$$\pi_t(X) = B_t \mathbb{E}_{\mathbb{P}^*}(B_T^{-1}X \mid \mathcal{F}_t), \quad \forall t \in [0, T]. \quad (17.24)$$

17.5 Black-Scholes Option Pricing Formula

Two alternative justifications of the option valuation formula are provided. The first relies on the fact that the risk-free return can be replicated by holding a continuously adjusted position in the underlying stock and an option. In other words, if an option is not priced in accordance with the Black-Scholes formula, there is a sure profit to be made by some combination of either short or long sales of the option and the underlying asset. The second method of derivation is based on equilibrium arguments, which require, in particular, that the option earns an expected rate of return commensurate with the risk involved in holding the option as an asset. The first approach is usually referred to as the *risk-free portfolio* method, while the second is known as the *equilibrium derivation* of the Black-Scholes formula.

The replication approach presented below is based on the observation that in the Black-Scholes setting the option value can be mimicked by holding a continuously rebalanced position in the underlying stock and risk-free bonds. Recall that we assume throughout that the financial market we are dealing with is perfect (partially, this was already implicit in the definition of a self-financing trading strategy). We shall first consider a European call option written on a stock S , with expiry date T and strike price K . Hence the payoff (and the price) at time T equals $(S_T - K)^+$. Let the function $c : \mathbb{R}_+ \times (0, T] \rightarrow \mathbb{R}$ be given by the formula

$$c(s, t) = s\Phi(d_1(s, t)) - Ke^{-rt}\Phi(d_2(s, t)), \quad (17.25)$$

where

$$d_1(s, t) = \frac{\ln(s/K) + (r + \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}, \quad (17.26)$$

and $d_2(s, t) = d_1(s, t) - \sigma\sqrt{t}$, or explicitly

$$d_2(s, t) = \frac{\ln(s/K) + (r - \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}. \quad (17.27)$$

Furthermore, let Φ stand for the standard Gaussian cumulative distribution function

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-z^2/2} dz, \quad \forall x \in \mathbb{R}.$$

We adopt the following notational convention

$$d_{1,2}(s, t) = \frac{\ln(s/K) + (r \pm \frac{1}{2}\sigma^2)t}{\sigma\sqrt{t}}.$$

Let us denote by C_t the arbitrage price at time t of a European call option in the Black-Scholes model.

Theorem 17.1: Arbitrage Price of European Call Option

The arbitrage price at time $t < T$ of the European call option with expiry date T and strike price K in the Black-Scholes model is given by the formula

$$C_t = c(S_t, T - t), \quad \forall t \in [0, T], \quad (17.28)$$

where the function $c : \mathbb{R}_+ \times (0, T] \rightarrow \mathbb{R}$ is given by Equation (17.25). More explicitly, we have, for any $t < T$

$$C_t = S_t \Phi(d_1(S_t, T - t)) - KB(t, T) \Phi(d_2(S_t, T - t)),$$

where d_1, d_2 are given by Equations (17.26) and (17.27). Moreover, the unique admissible replicating strategy $\phi = (\phi^1, \phi^2)$ of the call option satisfies, for every $t \in [0, T]$,

$$\phi_t^1 = \frac{\partial c}{\partial s}(S_t, T - t) = \Phi(d_1(S_t, T - t))$$

and

$$\phi_t^2 = e^{-rt}(c(S_t, T - t) - \phi_t^1 S_t).$$

17.6 Put-Call Parity for Spot Options

We shall show that if no arbitrage opportunities exist, puts and calls with the same strike and maturity must at all times during their lives obey the put-call parity relationship. Let us mention, in this regard, that empirical studies of the put-call parity have been reported (gouldTransactionsCostsRelationship1974; klemkoskyPutCallParityMarket1979).

A point worth stressing is that equality (17.29) does not rely on specific assumptions imposed on the stock price model. Indeed, it is satisfied in any arbitrage-free, continuous-time model of a security market, provided that the savings account B is given by the formula $B_t = e^{rt}$.

Proposition 17.2: Put-Call Parity

The arbitrage prices of European call and put options with the same expiry date T and strike price K satisfy the put-call parity relationship

$$C_t - P_t = S_t - Ke^{-r(T-t)} \quad (17.29)$$

for every $t \in [0, T]$.

The put-call parity can be used to derive a closed-form expression for the arbitrage price of a European put option. Let us denote by $p : \mathbb{R}_+ \times (0, T] \rightarrow \mathbb{R}$ the function

$$p(s, t) = Ke^{-rt}\Phi(-d_2(s, t)) - s\Phi(-d_1(s, t)), \quad (17.30)$$

with $d_1(s, t)$ and $d_2(s, t)$ given by [Equations \(17.26\) and \(17.27\)](#). The following result is an immediate consequence of [Proposition 17.2](#) combined with [Theorem 17.1](#).

Corollary 17.4: Black-Scholes Price of European Put Option

The Black-Scholes price at time $t < T$ of a European put option with strike price K equals $P_t = p(S_t, T - t)$, where the function $p : \mathbb{R}_+ \times [0, T] \rightarrow \mathbb{R}$ is given by [Equation \(17.30\)](#). More explicitly, the price at time $t < T$ of a European put option equals

$$P_t = KB(t, T)\Phi(-d_2(S_t, T - t)) - S_t\Phi(-d_1(S_t, T - t)). \quad (17.31)$$

17.7 Black-Scholes PDE

Random payoffs of such a simple form are termed *path-independent* claims, as opposed to *path-dependent* contingent claims

Path-dependent payoffs correspond to the various kinds of OTC options, known under the generic name of *exotic options*.

Lemma 17.4: PDE for Brownian Functionals

Let W be a one-dimensional Brownian motion defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

For a Borel-measurable function $h : \mathbb{R} \rightarrow \mathbb{R}$, we define the function $u : \mathbb{R} \times [0, T] \rightarrow \mathbb{R}$ by setting

$$u(x, t) = \mathbb{E}_{\mathbb{P}}(e^{-r(T-t)} h(W_T) \mid W_t = x), \quad \forall (x, t) \in \mathbb{R} \times [0, T]. \quad (17.32)$$

Suppose that

$$\int_{-\infty}^{+\infty} e^{-ax^2} |h(x)| dx < \infty \quad (17.33)$$

for some $a > 0$. Then the function u is defined for $0 < T - t < 1/2a$ and $x \in \mathbb{R}$, and has derivatives of all orders. In particular, it belongs to the class $C^{2,1}(\mathbb{R} \times (0, T))$ and satisfies the following PDE

$$-\frac{\partial u}{\partial t}(x, t) = \frac{1}{2} \frac{\partial^2 u}{\partial x^2}(x, t) - ru(x, t), \quad \forall (x, t) \in \mathbb{R} \times (0, T), \quad (17.34)$$

with the terminal condition $u(x, T) = h(x)$ for $x \in \mathbb{R}$.

18 A Dividend-paying Stock

Up to now, we have studied arbitrage pricing within the Black-Scholes framework under the assumption that the stock upon which an option is written pays no dividends during the option's lifetime. We assume that the dividends (or the dividend rate) that will be paid to the shareholders during an option's lifetime can be predicted with certainty.

18.1 Case of a Constant Dividend Yield

We will first examine the case when the dividend yield, rather than the dividend payoffs, is assumed to be known.

The dividend payments should be used in full, either to purchase additional shares of stock or to invest in risk-free bonds (however, intertemporal consumption or infusion of funds is not allowed).

Definition 18.1: Self-Financing Strategy

A trading strategy $\phi = (\phi^1, \phi^2)$ is said to be *self-financing* when its wealth process $V(\phi)$, which equals $V_t(\phi) = \phi_t^1 S_t + \phi_t^2 B_t$, satisfies

$$dV_t(\phi) = \phi_t^1 dS_t + \kappa \phi_t^1 S_t dt + \phi_t^2 dB_t. \quad (18.1)$$

In view of the postulated dynamics of the stock price, formula [Equation \(18.1\)](#) can also be represented as follows

$$dV_t(\phi) = \phi_t^1 (\mu + \kappa) S_t dt + \phi_t^1 \sigma S_t dW_t + \phi_t^2 dB_t. \quad (18.2)$$

We find it convenient to introduce an auxiliary process $\tilde{S}_t = e^{\kappa t} S_t$, whose dynamics are given by the SDE

$$d\tilde{S}_t = e^{\kappa t} (dS_t + \kappa S_t dt) = \mu_\kappa \tilde{S}_t dt + \sigma \tilde{S}_t dW_t, \quad (18.3)$$

where we write $\mu_\kappa = \mu + \kappa$. In terms of the process $\tilde{S}$, we obtain

$$V_t(\phi) = \phi_t^1 e^{-\kappa t} \tilde{S}_t + \phi_t^2 B_t \quad (18.4)$$

and

$$dV_t(\phi) = \phi_t^1 e^{-\kappa t} d\tilde{S}_t + \phi_t^2 dB_t. \quad (18.5)$$

Also, it is not difficult to check that the discounted wealth $V^*(\phi)$ satisfies

$$dV_t^*(\phi) = \phi_t^1 e^{-\kappa t} d\tilde{S}_t^*, \quad (18.6)$$

where we denote $\tilde{S}_t^* = \tilde{S}_t B_t^{-1}$. More explicitly, we have that

$$dV_t^*(\phi) = \sigma \phi_t^1 e^{-\kappa t} \tilde{S}_t^* (dW_t + \sigma^{-1}(\mu_\kappa - r)dt). \quad (18.7)$$

18.2 Case of Known Dividends

We will now focus on valuing European options on a stock with declared dividends during the option's lifetime. We assume that the amount of dividends to be paid before the option's expiry and the dates of their payments are known in advance.

As usual, it is sufficient to consider the case $t = 0$. Assume that the dividends $\kappa_1, \dots, \kappa_m$ are to be paid at the dates $0 < T_1 < \dots < T_m < T$. We assume, in addition, that dividend payments are known in advance; that is, $\kappa_1, \dots, \kappa_m$ are real numbers. The present value of all future dividend payments equals

$$\tilde{I}_t = \sum_{j=1}^m \kappa_j e^{-r(T_j-t)} \mathbb{1}_{[t,T]}(T_j), \quad \forall t \in [0, T], \quad (18.8)$$

and the value of all the dividends paid after time t and compounded at the risk-free rate to the option's expiry date T is given by the expression

$$I_t = \sum_{j=1}^m \kappa_j e^{r(T-T_j)} \mathbb{1}_{[t,T]}(T_j), \quad \forall t \in [0, T]. \quad (18.9)$$

In modelling the stock price, we will separate the capital gains process from the impact of dividend payments. The capital gains process G is assumed to follow the usual stochastic

differential equation (Heath and Jarrow, 1988).

$$dG_t = \mu G_t dt + \sigma G_t dW_t, \quad G_0 > 0, \quad (18.10)$$

hence G follows a geometric Brownian motion. The process S , representing the price of a stock that pays dividends $\kappa_1, \dots, \kappa_m$ at times $T_1, \dots, T_m$ may be introduced by setting

$$S_t = G_t - \sum_{j=1}^m \kappa_j e^{r(t-T_j)} \mathbb{1}_{[T_j, T]}(t) = G_t - D_t, \quad (18.11)$$

where the process D , given by the equality

$$D_t = \sum_{j=1}^m \kappa_j e^{r(t-T_j)} \mathbb{1}_{[T_j, T]}(t), \quad (18.12)$$

accounts for the impact of ex-dividend stock price decline. Alternatively, one may put, for $t \in [0, T]$,

$$S_t = G_t + \sum_{j=1}^m \kappa_j e^{-r(T_j-t)} \mathbb{1}_{[0, T_j]}(t) = G_t + \tilde{D}_t, \quad (18.13)$$

where $\tilde{D}$ now satisfies

$$\tilde{D}_t = \sum_{j=1}^m \kappa_j e^{-r(T_j-t)} \mathbb{1}_{[0, T_j]}(t). \quad (18.14)$$

Note that in the first case we have $G_0 = S_0$ and $G_T = S_T + D_T = S_T + I_0$, while in the second case $G_0 = S_0 - \tilde{D}_0 = S_0 - \tilde{I}_0$ and $G_T = S_T$.

Not surprisingly, the option valuation formulas corresponding to various specifications of the stock price, that is, to Equations (18.11) and (18.13) are not identical.

As usual, the first step in option valuation is to introduce the notion of a self-financing trading strategy. Under the present assumptions, it is natural to assume that if $\phi = (\phi^1, \phi^2)$ is a self-financing trading strategy, then all dividends received during the option's lifetime are immediately reinvested in stocks or bonds. Hence the wealth process of any trading strategy ϕ , which equals $V_t(\phi) = \phi_t^1 S_t + \phi_t^2 B_t$, satisfies, as usual,

$$dV_t(\phi) = \phi_t^1 dG_t + \phi_t^2 dB_t. \quad (18.15)$$

Intuitively, since the decline of the stock price equals the dividend, the wealth of a portfolio is not influenced by dividend payments. It turns out that the resulting option valuation formula will not agree with the standard Black-Scholes result, however. In view of Equation (18.15), the martingale measure $\mathbb{P}^*$ for the security market model corresponds to the unique martingale measure for the discounted capital gains process $G_t^* = G_t B_t^{-1}$. Hence $\mathbb{P}^*$ can be found in exactly the same way as in the proof of Theorem 17.1. We can also price options using the standard risk-neutral valuation approach.

Proposition 18.1: Valuation with Deterministic Dividends

Consider a European call option with strike price K and expiry date T , written on a stock S that pays deterministic dividends $\kappa_1, \dots, \kappa_m$ at times $T_1, \dots, T_m$. Assume that the stock price S satisfies Equations (18.10) and (18.11). Then the arbitrage price at time t of this option equals

$$C_t^K = S_t N(h_1(S_t, T - t)) - e^{-r(T-t)} (K + I_t) N(h_2(S_t, T - t)), \quad (18.16)$$

where I_t is given by expression Equation (18.9), and

$$h_{1,2}(S_t, T - t) = \frac{\ln S_t - \ln(K + I_t) + (r \pm \frac{1}{2}\sigma^2)(T - t)}{\sigma\sqrt{T - t}}. \quad (18.17)$$

19 Robustness of the Black-Scholes Approach

19.1 Uncertain Volatility

20 Discretisation Schemes

20.1 Stochastic Taylor Expansion

Deterministic Taylor expansions are well known. To begin, we shall consider the solution X_t of a 1-dimensional ordinary differential equation

$$\frac{d}{dt}X_t = a(X_t), \quad (20.1)$$

with initial value X_{t_0} , for $t \in [t_0, T]$ where $0 \leq t_0 < T$. We can write this in the equivalent integral equation form

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds. \quad (20.2)$$

To justify the following constructions, we shall require the function a to satisfy appropriate properties, for example, to be sufficiently smooth and to have a linear growth bound. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuously differentiable function. Then, by the chain rule, we have

$$\frac{d}{dt}f(X_t) = a(X_t) \frac{\partial}{\partial x}f(X_t), \quad (20.3)$$

which, using the operator

$$L = a \frac{\partial}{\partial x}, \quad (20.4)$$

we can express as the integral relation

$$f(X_t) = f(X_{t_0}) + \int_{t_0}^t Lf(X_s) ds \quad (20.5)$$

for all $t \in [t_0, T]$. When $f(x) \equiv x$ we have $Lf = a$, $L^2f = La, \dots$ and (20.5) reduces to

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds, \quad (20.6)$$

that is to equation (??).

If we now apply the relation (20.5) to the function $f = a$ in the integral in (20.6), we obtain

$$\begin{aligned} X_t &= X_{t_0} + \int_{t_0}^t \left(a(X_{t_0}) + \int_{t_0}^s La(X_z) dz \right) ds \\ &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + \int_{t_0}^t \int_{t_0}^s La(X_z) dz ds, \end{aligned} \quad (20.7)$$

which is the simplest nontrivial Taylor expansion for X_t . We can apply (20.5) again to the function $f = La$ in the double integral to derive

$$X_t = X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + La(X_{t_0}) \int_{t_0}^t \int_{t_0}^s dz ds + R_3 \quad (20.8)$$

with remainder

$$R_3 = \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z L^2 a(X_u) du dz ds, \quad (20.9)$$

for $t \in [t_0, T]$. For a general $r + 1$ times continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ this method gives the classical Taylor formula in integral form:

$$\begin{aligned} f(X_t) &= f(X_{t_0}) + \sum_{l=1}^r \frac{(t - t_0)^l}{l!} L^l f(X_{t_0}) \\ &\quad + \int_{t_0}^t \cdots \int_{t_0}^{s_2} L^{r+1} f(X_{s_1}) ds_1 \cdots ds_{r+1} \end{aligned} \quad (20.10)$$

for $t \in [t_0, T]$ and $r = 1, 2, 3, \dots$

The Taylor formula (20.10) has proven to be a very useful tool in both theoretical and practical investigations, particularly in numerical analysis. It allows the approximation of a sufficiently smooth function in a neighbourhood of a given point to any desired order

of accuracy. This expansion depends on the values of the function and some of its higher derivatives at the expansion point, weighted by corresponding multiple time integrals. In addition, there is a remainder term which contains the next multiple time integral, but now with a time-dependent integrand.

We shall indicate it here for the solution X_t of the 1-dimensional Ito stochastic differential equation in integral form

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds + \int_{t_0}^t b(X_s) dW_s \quad (20.11)$$

for $t \in [t_0, T]$, where the second integral in (20.11) is an Ito stochastic integral and the coefficients a and b are sufficiently smooth real-valued functions satisfying a linear growth bound. Then, for any twice continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ the Ito formula eq. (20.12) gives

$$\begin{aligned} f(X_t) &= f(X_{t_0}) \\ &+ \int_{t_0}^t \left(a(X_s) \frac{\partial}{\partial x} f(X_s) + \frac{1}{2} b^2(X_s) \frac{\partial^2}{\partial x^2} f(X_s) \right) ds \\ &+ \int_{t_0}^t b(X_s) \frac{\partial}{\partial x} f(X_s) dW_s \\ &= f(X_{t_0}) + \int_{t_0}^t L^0 f(X_s) ds + \int_{t_0}^t L^1 f(X_s) dW_s, \end{aligned} \quad (20.12)$$

for $t \in [t_0, T]$. Here we have introduced the operators

$$L^0 = a \frac{\partial}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2}{\partial x^2} \quad (20.13)$$

and

$$L^1 = b \frac{\partial}{\partial x}. \quad (20.14)$$

Obviously, for $f(x) \equiv x$ we have $L^0 f = a$ and $L^1 f = b$, in which case (20.12) reduces to the original Ito equation for X_t , that is to

$$X_t = X_{t_0} + \int_{t_0}^t a(X_s) ds + \int_{t_0}^t b(X_s) dW_s. \quad (20.15)$$

In analogy with the above deterministic expansions, if we apply the Ito formula (20.12) to the functions $f = a$ and $f = b$ in (20.15) we obtain

$$\begin{aligned} X_t &= X_{t_0} \\ &+ \int_{t_0}^t \left(a(X_{t_0}) + \int_{t_0}^s L^0 a(X_z) dz + \int_{t_0}^s L^1 a(X_z) dW_z \right) ds \\ &+ \int_{t_0}^t \left(b(X_{t_0}) + \int_{t_0}^s L^0 b(X_z) dz + \int_{t_0}^s L^1 b(X_z) dW_z \right) dW_s \\ &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t dW_s + R \end{aligned} \quad (20.16)$$

with remainder

$$\begin{aligned} R &= \int_{t_0}^t \int_{t_0}^s L^0 a(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s L^1 a(X_z) dW_z ds \\ &+ \int_{t_0}^t \int_{t_0}^s L^0 b(X_z) dz dW_s + \int_{t_0}^t \int_{t_0}^s L^1 b(X_z) dW_z dW_s. \end{aligned} \quad (20.17)$$

This is the simplest nontrivial Ito-Taylor expansion. We can continue it, for instance, by applying the Ito formula (20.12) to $f = L^1 b$ in (20.16), in which case we get

$$\begin{aligned} X_t &= X_{t_0} + a(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t dW_s \\ &+ L^1 b(X_{t_0}) \int_{t_0}^t \int_{t_0}^s dW_z dW_s + \bar{R} \end{aligned} \quad (20.18)$$

with remainder

$$\begin{aligned}
\bar{R} = & \int_{t_0}^t \int_{t_0}^s L^0 a(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s L^1 a(X_z) dW_z ds \\
& + \int_{t_0}^t \int_{t_0}^s L^0 b(X_z) dz dW_s + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z L^0 L^1 b(X_u) du dW_z dW_s \\
& + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z L^1 L^1 b(X_u) dW_u dW_z dW_s.
\end{aligned} \tag{20.19}$$

We shall now consider another representation for the increments of a function of an Ito process X , which we shall call the *Stratonovich-Taylor expansion*. For this we start with the 1-dimensional Stratonovich stochastic differential equation in integral form

$$X_t = X_{t_0} + \int_{t_0}^t \underline{a}(X_s) ds + \int_{t_0}^t b(X_s) \circ dW_s \tag{20.20}$$

for $t \in [t_0, T]$, where the second integral in (20.20) is a Stratonovich stochastic integral and the coefficients $\underline{a}$ and b are sufficiently smooth real valued functions satisfying a linear growth bound. Consider that

$$\underline{a} = a - \frac{1}{2}bb'$$

the eq. (20.11) and the Stratonovich equation (20.20) are equivalent and have the same solutions. The solution of a Stratonovich SDE transforms according to the deterministic chain rule, so for any twice continuously differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ we have

$$\begin{aligned}
f(X_t) &= f(X_{t_0}) + \int_{t_0}^t \underline{a}(X_s) \frac{\partial}{\partial x} f(X_s) ds + \int_{t_0}^t b(X_s) \frac{\partial}{\partial x} f(X_s) \circ dW_s, \\
&= f(X_{t_0}) + \int_{t_0}^t \underline{L}^0 f(X_s) ds + \int_{t_0}^t \underline{L}^1 f(X_s) \circ dW_s,
\end{aligned} \tag{20.21}$$

for $t \in [t_0, T]$, with the operators

$$\underline{L}^0 = a \frac{\partial}{\partial x} \tag{20.22}$$

and

$$\underline{L}^1 = b \frac{\partial}{\partial x}. \quad (20.23)$$

Obviously, for $f(x) \equiv x$ we have $\underline{L}^0 f = \underline{a}$ and $\underline{L}^1 f = b$, in which case (20.21) reduces to the original Stratonovich equation

$$X_t = X_{t_0} + \int_{t_0}^t \underline{a}(X_s) ds + \int_{t_0}^t b(X_s) \circ dW_s. \quad (20.24)$$

Analogously with the Ito case just considered, we can apply (20.21) to the integrand functions $f = \underline{a}$ and $f = b$ in (20.22). This gives

$$\begin{aligned} X_t &= X_{t_0} \\ &+ \int_{t_0}^t \left(\underline{a}(X_{t_0}) + \int_{t_0}^s \underline{L}^0 \underline{a}(X_z) dz + \int_{t_0}^s \underline{L}^1 \underline{a}(X_z) \circ dW_z \right) ds \\ &+ \int_{t_0}^t \left(b(X_{t_0}) + \int_{t_0}^s \underline{L}^0 b(X_z) dz + \int_{t_0}^s \underline{L}^1 b(X_z) \circ dW_z \right) \circ dW_s \\ &= X_{t_0} + \underline{a}(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t \circ dW_s + R \end{aligned} \quad (20.25)$$

with remainder

$$\begin{aligned} R &= \int_{t_0}^t \int_{t_0}^s \underline{L}^0 \underline{a}(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s \underline{L}^1 \underline{a}(X_z) \circ dW_z ds \\ &+ \int_{t_0}^t \int_{t_0}^s \underline{L}^0 b(X_z) dz \circ dW_s + \int_{t_0}^t \int_{t_0}^s \underline{L}^1 b(X_z) \circ dW_z \circ dW_s. \end{aligned} \quad (20.26)$$

This is the simplest nontrivial Stratonovich-Taylor expansion of $f(X_t)$. We can continue expanding, for instance, by applying (20.21) to the integrand $f = \underline{L}^1 b$ in (20.25) to obtain

$$\begin{aligned} X_t &= X_{t_0} + \underline{a}(X_{t_0}) \int_{t_0}^t ds + b(X_{t_0}) \int_{t_0}^t \circ dW_s \\ &+ \underline{L}^1 b(X_{t_0}) \int_{t_0}^t \int_{t_0}^s \circ dW_z \circ dW_s + \bar{R} \end{aligned} \quad (20.27)$$

with remainder

$$\begin{aligned}
\bar{R} = & \int_{t_0}^t \int_{t_0}^s \underline{L}^0 \underline{a}(X_z) dz ds + \int_{t_0}^t \int_{t_0}^s \underline{L}^1 \underline{a}(X_z) \circ dW_z ds \\
& + \int_{t_0}^t \int_{t_0}^s \underline{L}^0 b(X_z) dz \circ dW_s \\
& + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z \underline{L}^0 \underline{L}^1 b(X_u) du \circ dW_z \circ dW_s \\
& + \int_{t_0}^t \int_{t_0}^s \int_{t_0}^z \underline{L}^1 \underline{L}^1 b(X_u) \circ dW_u \circ dW_z \circ dW_s.
\end{aligned} \tag{20.28}$$

Similar expansions hold for multi-dimensional Ito processes satisfying nonautonomous stochastic differential equations. We shall refer to the nonautonomous d -dimensional Ito equation

$$X_t = X_{t_0} + \int_{t_0}^t a(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) dW_s^j \tag{20.29}$$

for $t \in [t_0, T]$ and the equivalent Stratonovich equation

$$X_t = X_{t_0} + \int_{t_0}^t \underline{a}(s, X_s) ds + \sum_{j=1}^m \int_{t_0}^t b^j(s, X_s) \circ dW_s^j \tag{20.30}$$

for $t \in [t_0, T]$, where

$$\underline{a}^i = a^i - \frac{1}{2} \sum_{j=1}^m \sum_{k=1}^d b^{k,j} \frac{\partial b^{i,j}}{\partial x^k} \tag{20.31}$$

for $i = 1, 2, \dots, d$. In both of these SDEs $W = \{W_t, t \in [0, T]\}$ is a standard m -dimensional Wiener process adapted to an increasing family of sub- σ -algebras $\{\mathcal{A}_t, t \in [0, T]\}$.

20.2 Strong Convergence Criterion

In simulation, filtering, and estimation of Itô processes, the sample paths of the approximation must closely track those of the underlying process. The primary criterion for this proximity is the absolute error at the final time instant T .

$$\epsilon(\delta) = \mathbb{E}(|X_T - Y_N|). \tag{20.32}$$

This error is bounded by the root mean square error via the *Lyapunov inequality*:

$$\epsilon(\delta) = \mathbb{E}(|X_T - Y_N|) \leq \sqrt{\mathbb{E}(|X_T - Y_N|^2)}. \quad (20.33)$$

Definition 20.1: Strong Order of Convergence

An approximating process Y converges in the *strong sense with order* $\gamma \in (0, \infty]$ if there exists a finite constant K and a positive constant δ_0 such that

$$\mathbb{E}(|X_T - Y_N|) \leq K\delta^\gamma \quad (20.34)$$

for any time discretization with maximum step size $\delta \in (0, \delta_0)$.

Tip 20.1: I



the deterministic limit where diffusion vanishes ($b \equiv 0$), the strong convergence criterion reduces to the standard error criterion for ordinary differential equations (ODEs). Stochastic schemes generically exhibit lower convergence orders than their deterministic counterparts because the regularity of the Wiener process dictates that increments ΔW_n scale as $\delta^{1/2}$ in the root mean square sense rather than δ . This intrinsic degradation results in the Euler approximation achieving a strong order of $\gamma = 0.5$ for SDEs in contrast to the order 1.0 obtained for ODEs.

20.3 Weak Convergence Criterion

In many practical scenarios, such as moment estimation or option pricing, a precise path-wise approximation of the Itô process is not required. Instead, the focus is often on the expectation of a function of the process at a final time T , such as the first two moments $\mathbb{E}(X_T)$ and $\mathbb{E}((X_T)^2)$, or more generally $\mathbb{E}(g(X_T))$ for some function g . In these cases, it suffices to approximate the probability distribution of the random variable X_T rather than the sample paths. This leads to a weaker convergence criterion than the strong sense.

Definition 20.2: Weak Order of Convergence

A time discrete approximation Y converges in the *weak sense with order* $\beta \in (0, \infty]$ if for any polynomial g there exists a finite constant K and a positive constant δ_0 such that

$$|\mathbb{E}(g(X_T)) - \mathbb{E}(g(Y_N))| \leq K\delta^\beta \quad (20.35)$$

for any time discretization with maximum step size $\delta \in (0, \delta_0)$.

Tip 20.2: I

the limit of vanishing diffusion where $b \equiv 0$, the weak convergence criterion with $g(x) \equiv x$ collapses to the standard deterministic convergence criterion for ordinary differential equations. For stochastic systems, the weak order β frequently diverges from the strong order γ , exemplified by the Euler approximation achieving a higher weak order of $\beta = 1.0$ compared to its strong order of $\gamma = 0.5$. This distinction becomes critical when the objective involves estimating moments or probability distributions rather than simulating specific sample paths.

20.4 Euler-Maruyama

To apply the Monte Carlo method in option pricing, we generate a sequence $(Y_1, \dots, Y_N)$ of sample realisations of a random variable, such that the empirical mean

$$\mathbb{E}[\phi(X)] \approx \frac{1}{N} \sum_{i=1}^N \phi(Y_i) \quad (20.36)$$

converges to the expected value $\mathbb{E}[\phi(X)]$ by the Strong Law of Large Numbers.

Random samples for the solution of a stochastic differential equation defined on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ of the form

$$dX_t = b(t, X_t)dt + a(t, X_t)dW_t, \quad (20.37)$$

where $(W_t)_{t \in \mathbb{R}_+}$ is a standard Brownian motion, can be generated by time discretisation on a grid $\pi_\delta = \{0 = t_0 < t_1 < \dots < t_M = T\}$.

More precisely, the Euler-Maruyama discretisation scheme for the stochastic differential equation, [eq. \(20.37\)](#), is given by the recursive approximation Y^δ :

$$\begin{aligned} Y_{t_{k+1}}^\delta &= Y_{t_k}^\delta + \int_{t_k}^{t_{k+1}} b(t_k, Y_{t_k}^\delta) ds + \int_{t_k}^{t_{k+1}} a(t_k, Y_{t_k}^\delta) dW_s \\ &\approx Y_{t_k}^\delta + b(t_k, Y_{t_k}^\delta)(t_{k+1} - t_k) + a(t_k, Y_{t_k}^\delta)(W_{t_{k+1}} - W_{t_k}), \end{aligned} \quad (20.38)$$

where the increments satisfy $W_{t_{k+1}} - W_{t_k} \sim \mathcal{N}(0, t_{k+1} - t_k)$ for $k = 0, 1, \dots, M-1$.

Let $T > 0$ and consider the stochastic differential equation [eq. \(20.37\)](#) on the time interval $[0, T]$. Suppose that the initial data satisfies

$$\mathbb{E}(|X_0|^2) < \infty, \quad (20.39)$$

$$\mathbb{E}\left(|X_0 - Y_0^\delta|^2\right)^{1/2} \leq K_1 \delta^{1/2}, \quad (20.40)$$

and that the coefficients $a : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^{d \times m}$ and $b : [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}^d$ satisfy the global Lipschitz condition

$$|a(t, x) - a(t, y)| + |b(t, x) - b(t, y)| \leq K_2 |x - y| \quad (20.41)$$

and the linear growth condition

$$|a(t, x)| + |b(t, x)| \leq K_3(1 + |x|) \quad (20.42)$$

for all $t \in [0, T]$ and $x, y \in \mathbb{R}^d$. Furthermore, assume the coefficients satisfy the Hölder continuity condition in time:

$$|a(s, x) - a(t, x)| + |b(s, x) - b(t, x)| \leq K_4(1 + |x|)|s - t|^{1/2} \quad (20.43)$$

for all $s, t \in [0, T]$ and $x \in \mathbb{R}^d$. The constants $K_1, \dots, K_4$ must not depend on the discretisation step size δ .

Theorem 20.1: Strong Convergence of the Euler-Maruyama Scheme

Then, for the Euler-Maruyama approximation Y^δ defined on the grid π_δ , the following strong error estimate holds:

$$\mathbb{E}(|X_T - Y^\delta(T)|) \leq C\delta^{1/2}, \quad (20.44)$$

where the constant C depends only on $K_1, \dots, K_4$ and T , but does not depend on δ .

We illustrate Euler's method via the geometric Brownian motion SDE

eulerMaruyama.py

```

1  T, N, R = 1.0, 2**8, 4
2  dt = T / N
3  mu, sigma, X0 = 1.0, 0.5, 1.0
4
5  np.random.seed(4)
6  dB = np.sqrt(dt) * np.random.randn(N)
7
8  t = np.linspace(0, T, N + 1)
9  B = np.r_[0, np.cumsum(dB)]
10 Y = X0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * B)
11
12 X = np.r_[X0, X0 * np.cumprod(1 + mu * dt + sigma * dB)]
13
14 dB_c = dB.reshape(-1, R).sum(axis=1)
15 X_c = np.r_[X0, X0 * np.cumprod(1 + mu*dt*R + sigma*dB_c)]

```

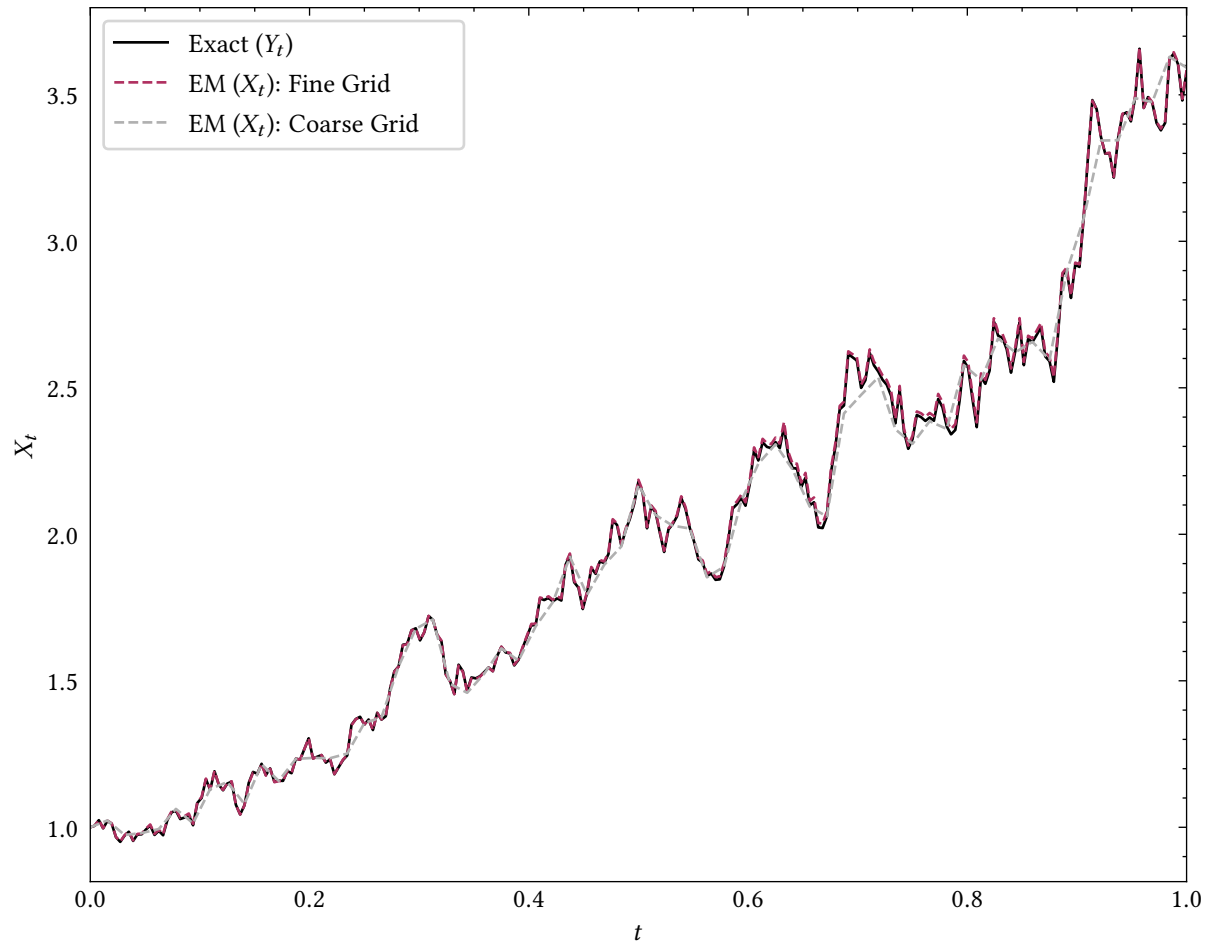



Figure 12: Comparison of Euler-Maruyama approximations against the exact analytical solution for a GBM. The simulation uses a fine grid step size of $\Delta t = 2^{-8}$ and a coarse grid step size of $\Delta t_{\text{coarse}} = 4\Delta t$. Parameters: $\mu = 1.0$, $\sigma = 0.5$, $X_0 = 1.0$.

20.5 Milstein

In the Milstein scheme we use [eq. \(20.37\)](#) to expand $a(X_s)$ as

$$\begin{aligned} a(X_s) &\simeq a(X_{t_k}) + a'(X_{t_k})(X_s - X_{t_k}) \\ &\simeq a(X_{t_k}) + a'(X_{t_k})(b(X_{t_k})(s - t_k) + a(X_{t_k})(W_s - W_{t_k})), \end{aligned}$$

$0 \leq t_q \leq s$. As a consequence, we get

$$\begin{aligned} \hat{X}_{t_{k+1}}^N &= \hat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s)ds + \int_{t_k}^{t_{k+1}} a(X_s)dW_s \\ &\simeq \hat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s)ds + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\ &\quad + a'(X_{t_k})b(X_{t_k}) \int_{t_k}^{t_{k+1}} (s - t_k)dW_s \\ &\quad + a'(X_{t_k})a(X_{t_k}) \int_{t_k}^{t_{k+1}} (W_s - W_{t_k})dW_s \\ &\simeq \hat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s)ds + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\ &\quad + a'(X_{t_k})a(X_{t_k}) \int_{t_k}^{t_{k+1}} (W_s - W_{t_k})dW_s. \end{aligned}$$

Next, using Ito's formula, we note that

$$(W_{t_{k+1}} - W_{t_k})^2 = 2 \int_{t_k}^{t_{k+1}} (W_s - W_{t_k})dW_s + \int_{t_k}^{t_{k+1}} ds,$$

hence

$$\int_{t_k}^{t_{k+1}} (W_s - W_{t_k})dW_s = \frac{1}{2}((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k))$$

and

$$\begin{aligned}
\hat{X}_{t_{k+1}}^N &\simeq \hat{X}_{t_k}^N + \int_{t_k}^{t_{k+1}} b(X_s) ds + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\
&\quad + \frac{1}{2} a'(X_{t_k}) a(X_{t_k}) ((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)) \\
&\simeq \hat{X}_{t_k}^N + b(X_{t_k})(t_{k+1} - t_k) + a(X_{t_k})(W_{t_{k+1}} - W_{t_k}) \\
&\quad + \frac{1}{2} a'(X_{t_k}) a(X_{t_k}) ((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)).
\end{aligned}$$

As a consequence, the Milstein scheme is written as

$$\begin{aligned}
\hat{X}_{t_{k+1}}^N &\simeq \hat{X}_{t_k}^N + b(\hat{X}_{t_k}^N)(t_{k+1} - t_k) + a(\hat{X}_{t_k}^N)(W_{t_{k+1}} - W_{t_k}) \\
&\quad + \frac{1}{2} a'(\hat{X}_{t_k}^N) a(\hat{X}_{t_k}^N) ((W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)),
\end{aligned}$$

i.e., in the Milstein scheme, we take into account the "small" difference

$$(W_{t_{k+1}} - W_{t_k})^2 - (t_{k+1} - t_k)$$

existing between $(\Delta W_t)^2$ and Δt . Taking $(\Delta W_t)^2$ equal to Δt brings us back to the Euler scheme.

Let $T > 0$. Suppose that the initial data satisfies

$$\mathbb{E}(|X_0|^2) < \infty, \quad (20.45)$$

$$\mathbb{E}\left(|X_0 - Y_0^\delta|^2\right)^{1/2} \leq K_1 \delta^{1/2}. \quad (20.46)$$

We assume higher-order regularity on the coefficients compared to the Euler scheme. Specifically, let the differential operator $\mathcal{L}^j$ be defined appropriately for the diffusion terms. As-

sume the coefficients satisfy the global Lipschitz conditions:

$$\begin{aligned} |b(t, x) - b(t, y)| &\leq K_2|x - y| \\ |a^j(t, x) - a^j(t, y)| &\leq K_2|x - y| \\ |\mathcal{L}^{j_1} a^{j_2}(t, x) - \mathcal{L}^{j_1} a^{j_2}(t, y)| &\leq K_2|x - y| \end{aligned} \quad (20.47)$$

and the linear growth conditions:

$$\begin{aligned} |b(t, x)| + |\mathcal{L}^j b(t, x)| &\leq K_3(1 + |x|) \\ |a^j(t, x)| + |\mathcal{L}^j a^{j_2}(t, x)| &\leq K_3(1 + |x|) \\ |\mathcal{L}^{j_1} \mathcal{L}^{j_2} a^{j_3}(t, x)| &\leq K_3(1 + |x|) \end{aligned} \quad (20.48)$$

Finally, assume the coefficients satisfy the Hölder continuity condition in time:

$$\begin{aligned} |b(s, x) - b(t, x)| &\leq K_4(1 + |x|)|s - t|^{1/2} \\ |a^j(s, x) - a^j(t, x)| &\leq K_4(1 + |x|)|s - t|^{1/2} \\ |\mathcal{L}^{j_1} a^{j_2}(s, x) - \mathcal{L}^{j_1} a^{j_2}(t, x)| &\leq K_4(1 + |x|)|s - t|^{1/2} \end{aligned} \quad (20.49)$$

for all $s, t \in [0, T]$, $x, y \in \mathbb{R}^d$, and indices $j, j_1, j_2, j_3 = 1, \dots, m$. The constants $K_1, \dots, K_4$ must not depend on δ .

Theorem 20.2: Strong Convergence of the Milstein Scheme

Under the assumptions above, for the Milstein approximation Y^δ , the following strong error estimate holds:

$$\mathbb{E} \left(|X_T - Y^\delta(T)| \right) \leq K_5 \delta, \quad (20.50)$$

where the constant K_5 does not depend on δ .

milstein.py

```
1  T, N, R = 1.0, 2**8, 4
2  dt = T / N
3  mu, sigma, X0 = 1.0, 0.5, 1.0
4
5  np.random.seed(4)
6  dB = np.sqrt(dt) * np.random.randn(N)
7
8  t = np.linspace(0, T, N + 1)
9  B = np.r_[0, np.cumsum(dB)]
10 Y = X0 * np.exp((mu - 0.5 * sigma**2) * t + sigma * B)
11
12 X = np.r_[X0, X0 * np.cumprod(1 + mu * dt + sigma * dB + 0.5 *
    sigma**2 * (dB**2 - dt))]
13
14 dB_c = dB.reshape(-1, R).sum(axis=1)
15 dt_c = dt * R
16 X_c = np.r_[X0, X0 * np.cumprod(1 + mu * dt_c + sigma * dB_c + 0.5
    * sigma**2 * (dB_c**2 - dt_c))]
```

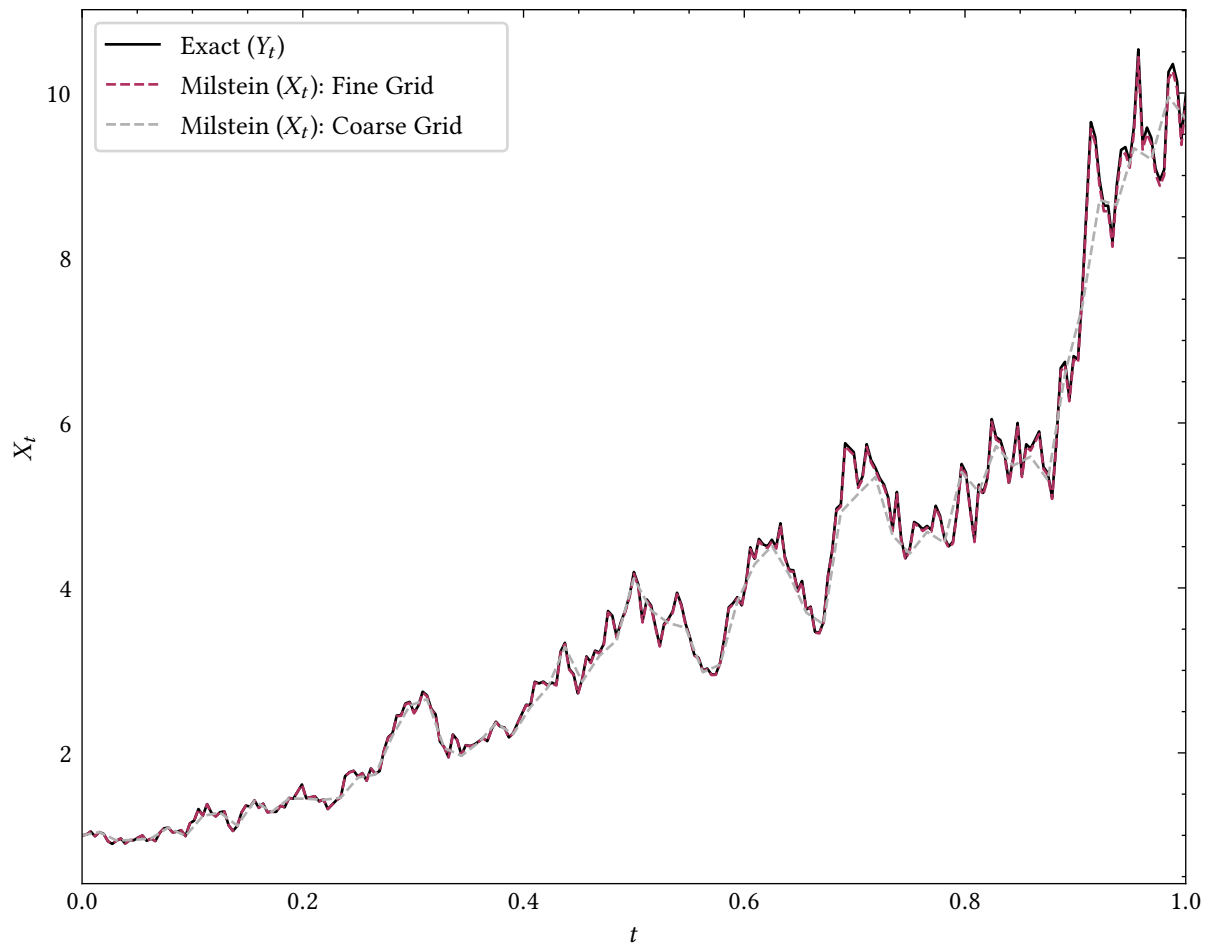


Figure 13: Comparison of Milstein approximations against the exact analytical solution for a GBM. The simulation uses a fine grid step size of $\Delta t = 2^{-8}$ and a coarse grid step size of $\Delta t_{\text{coarse}} = 4\Delta t$. Parameters: $\mu = 1.0$, $\sigma = 0.5$, $X_0 = 1.0$.

21 Connection to Partial Differential Equations

21.1 Introduction

Part VI

Monte Carlo Methods

SIMULATION techniques are invaluable in managing complex financial scenarios that involve path-dependent payoffs or distributions without analytical expressions. One prominent application is pricing derivative securities, estimating value at risk, and emulating hedging strategies. Monte Carlo simulation, in particular, proves highly effective for such tasks, offering a blend of simplicity and adaptability as its primary advantages. However, the method's drawback lies in its considerable computational demands, which can be mitigated to some extent. Our approach employs Monte Carlo simulation to generate simulated option price values, which are then compared against outcomes derived from analytical formulas.

22 Generating Random Numbers and Random Variables

At the core of nearly all Monte Carlo simulations is a sequence of apparently random numbers used to drive the simulation. In analysing Monte Carlo methods, we will treat this driving sequence as though it were genuinely random. This is a convenient fiction that allows us to apply tools from probability and statistics to analyse Monte Carlo computations. It is convenient because modern Pseudo-random number generators are sufficiently good at mimicking genuine randomness to make this analysis informative. Nevertheless, we should be aware that the apparently random numbers at the heart of a simulation are, in fact, produced by completely deterministic algorithms.

Because a random number generator may be called thousands or even millions of times in a single simulation, it must be fast.

Before discussing sequences that appear to be random but are not, we should specify what we mean by a generator of genuinely random numbers: we mean a mechanism for producing a sequence of random variables $U_1, U_2, \dots$ with the property that

- (i) each U_i is uniformly distributed between 0 and 1;
- (ii) the U_i are mutually independent.

Property (i) is a convenient but arbitrary normalisation; values uniformly distributed between 0 and $1/2$ would be just as useful, as would values from nearly any other simple distribution. Uniform random variables on the unit interval can be transformed into samples from essentially any other distribution. Property (ii) is the more important one. It implies, in particular, that all pairs of values should be uncorrelated and, more generally, that the value of U_i should not be predictable from $U_1, \dots, U_{i-1}$.

A random number generator (often called a *pseudo-random* number generator to emphasise that it only mimics randomness) produces a finite sequence of numbers $u_1, u_2, \dots, u_K$ in the unit interval. Typically, the values generated depend in part on input parameters specified by the user. *Any* such sequence constitutes a set of possible outcomes of independent uniforms $U_1, \dots, U_K$. A good random number generator satisfies the admittedly vague requirement that small (relative to K) segments of the sequence $u_1, \dots, u_K$ should be difficult to distinguish from a realisation of independent uniforms.

An effective generator, therefore, produces values that appear consistent with properties (i) and (ii) above. If the number of values K is large, the fraction of values falling in any sub-interval of the unit interval should be approximately the length of the sub-interval; this is uniformity. Independence suggests that there should be no discernible pattern among the values. To put this only slightly more precisely, statistical tests for independence should not easily reject segments of the sequence $u_1, \dots, u_K$.

Most random number generators at present are based on linear recurrence relations. One of the most important random number generators is the *multiple-recursive generator* (MRG) of *order* k , which generates a sequence of integers $X_k, X_{k+1}, \dots$ via the linear recurrence

$$X_t = (a_1 X_{t-1} + \dots + a_k X_{t-k}) \bmod m, \quad t = k, k+1, \dots \quad (22.1)$$

for some *modulus* m and *multipliers* $\{a_i, i = 1, \dots, k\}$. Here “mod” refers to the modulo operation: $n \bmod m$ is the remainder when n is divided by m . The recurrence is initialised by specifying k “seeds”, $X_0, \dots, X_{k-1}$. To yield fast algorithms, all but a few of the multipliers should be 0. When m is a large integer, one can obtain a stream of pseudo-random numbers $U_k, U_{k+1}, \dots$ between 0 and 1 from the sequence $X_k, X_{k+1}, \dots$, simply by setting $U_t = X_t/m$. It is also possible to set a small modulus, in particular $m = 2$. The output function for such *modulo 2 generators* is then typically of the form

$$U_t = \sum_{i=1}^w X_{tw+i-1} 2^{-i} \quad (22.2)$$

for some $w \leq k$, e.g., $w = 32$ or 64 . Examples of modulo 2 generators are the *feedback shift register generators*, the most popular of which are the *Mersenne twisters*.

MRGs with excellent statistical properties can be implemented efficiently by combining several simpler MRGs and carefully choosing their respective moduli and multipliers. One of the most successful is L’Ecuyer’s MRG32k3a generator, see **([lecuyerGoodParametersImplementations19](#))**

mrg32k5a.py

```

1  M1, M2, NORM = 4294949027.0, 4294934327.0, 2.3283163396834613e-10
2  A12, A14, A15 = 1154721.0, 1739991.0, 1108499.0
3  A21, A23, A25 = 1776413.0, 865203.0, 1641052.0
4
5  state = np.random.SeedSequence(42).generate_state(10).astype(np.
        float64)
6  s1, s2, out = state[:5], state[5:], np.empty(1000, dtype=np.float64)
7
8  for i in range(1000):
9      p1 = (A12 * s1[2] + A14 * s1[1] - A15 * s1[0]) % M1
10     s1[:-1], s1[-1] = s1[1:], p1
11
12     p2 = (A21 * s2[4] + A23 * s2[2] - A25 * s2[0]) % M2
13     s2[:-1], s2[-1] = s2[1:], p2
14
15     out[i] = (p1 - p2 + M1) * NORM if p1 <= p2 else (p1 - p2) * NORM

```

Simulating a random variable X from an arbitrary (that is, not necessarily uniform) distribution invariably involves the following two steps.

Algorithm 1 Simulation of Random Variable X

- 1: Simulate uniform random numbers $U_1, \dots, U_k$ on $(0, 1)$ for some $k = 1, 2, \dots$
 - 2: Return $X = g(U_1, \dots, U_k)$, where g is some real-valued function.
-

22.1 Simulating Random Variables

22.1.1 Inverse-Transform Method

Let X be a random variable with cumulative distribution function (cdf) F . Let F^{-1} denote the inverse of F and $U \sim \mathcal{U}(0, 1)$. Then,

$$\mathbb{P}[F^{-1}(U) \leq x] = \mathbb{P}[U \leq F(x)] = F(x). \quad (22.3)$$

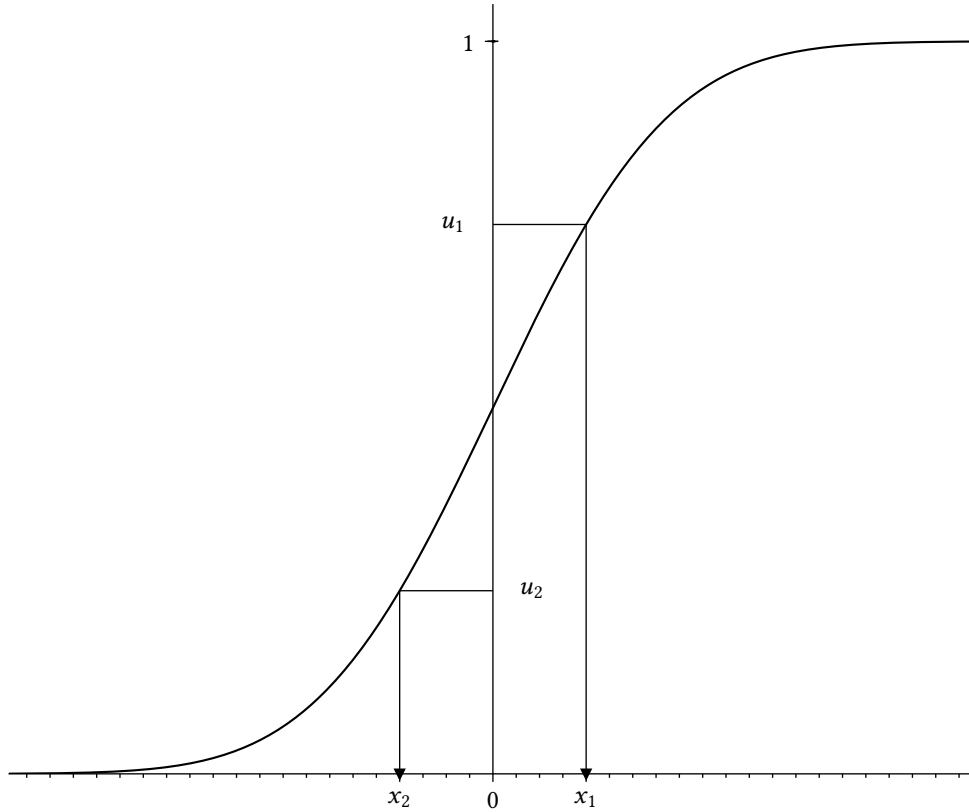


Figure 14: Inverse transform method.

This leads to the following method to simulate a random variable X with cdf F :

²Every cdf has a unique inverse function defined by $F^{-1}(u) = \inf\{x : F(x) \geq u\}$. If, for each u , the equation $F(x) = u$ has a unique solution x , this definition coincides with the usual interpretation of the inverse function.

Algorithm 2 Inverse-Transform Method**Require:** Cumulative distribution function F .**Ensure:** Random variable X distributed according to F .1: Generate U from $\mathcal{U}(0, 1)$.2: $X \leftarrow F^{-1}(U)$ 3: **return** X **Tip 22.1:** T

e inverse-transform method is applicable to both continuous and discrete distributions. After importing numpy as `np`, simulating numbers $0, \dots, k-1$ according to probabilities $p_0, \dots, p_{k-1}$ can be done via `np.min(np.where(np.cumsum(p) > np.random.rand()))`, where `p` is the vector of the probabilities.

22.1.2 Acceptance-Rejection Method

The acceptance-rejection method is used to sample from a “difficult” probability density function (pdf) $f(x)$ by generating instead from an “easy” pdf $g(x)$ satisfying $f(x) \leq Cg(x)$ for some constant $C \geq 1$ (for example, via the inverse-transform method), and then accepting or rejecting the drawn sample with a certain probability. [algorithm 3](#) gives the pseudo-code.

The idea of the algorithm is to generate a point (X, Y) under the graph of the function Cg uniformly, by first drawing $X \sim g$ and then $Y \sim \mathcal{U}(0, Cg(X))$. If this point lies under the graph of f , then we accept X as a sample from f ; otherwise, we try again. The efficiency of the acceptance-rejection method is usually expressed in terms of the probability of acceptance, which is $1/C$.

Algorithm 3 Acceptance-Rejection Method**Require:** Pdf g and constant C such that $Cg(x) \geq f(x)$ for all x .**Ensure:** Random variable X distributed according to pdf f .

```

1: found  $\leftarrow$  false
2: while not found do
3:   Generate  $X$  from  $g$ .
4:   Generate  $U$  from  $\mathcal{U}(0, 1)$  independently of  $X$ .
5:    $Y \leftarrow UCg(X)$ 
6:   if  $Y \leq f(X)$  then
7:     found  $\leftarrow$  true
8:   end if
9: end while
10: return  $X$ 

```

22.2 Simulating Random Vectors and Processes

TECHNIQUES for generating random vectors and processes are as diverse as the class of random processes themselves. We highlight a few general scenarios.

When $X_1, \dots, X_n$ are *independent* random variables with pdfs $f_i, i = 1, \dots, n$, so that their joint pdf is $f(\mathbf{x}) = f_1(x_1) \cdots f_n(x_n)$, the random vector $X = [X_1, \dots, X_n]^\top$ can be simply simulated by drawing each component $X_i \sim f_i$ individually, for example, via the inverse-transform method or acceptance-rejection.

For *dependent* components $X_1, \dots, X_n$, we can, as a consequence of the *product rule* of probability, represent the joint pdf $f(\mathbf{x})$ as

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = f_1(x_1)f_2(x_2|x_1) \cdots f_n(x_n|x_1, \dots, x_{n-1}), \quad (22.4)$$

where $f_1(x_1)$ is the marginal pdf of X_1 and $f_k(x_k|x_1, \dots, x_{k-1})$ is the conditional pdf of X_k given $X_1 = x_1, X_2 = x_2, \dots, X_{k-1} = x_{k-1}$. Provided the conditional pdfs are known, one can generate X by first generating X_1 , then, given $X_1 = x_1$, generate X_2 from $f_2(x_2|x_1)$, and so on, until generating X_n from $f_n(x_n|x_1, \dots, x_{n-1})$.

22.2.1 Normal Random Variables and Vectors

Normal random variables are the building blocks of many financial simulation models

$$\Phi(x) := \mathbb{P}(X \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy, \quad x \in \mathbb{R},$$

denotes the standard Gaussian Cumulative Distribution Function (CDF) of a standard normal random variable $X \simeq \mathcal{N}(0, 1)$, which posits the relation

$$\Phi(-x) = 1 - \Phi(x), \quad x \in \mathbb{R}. \quad (22.5)$$

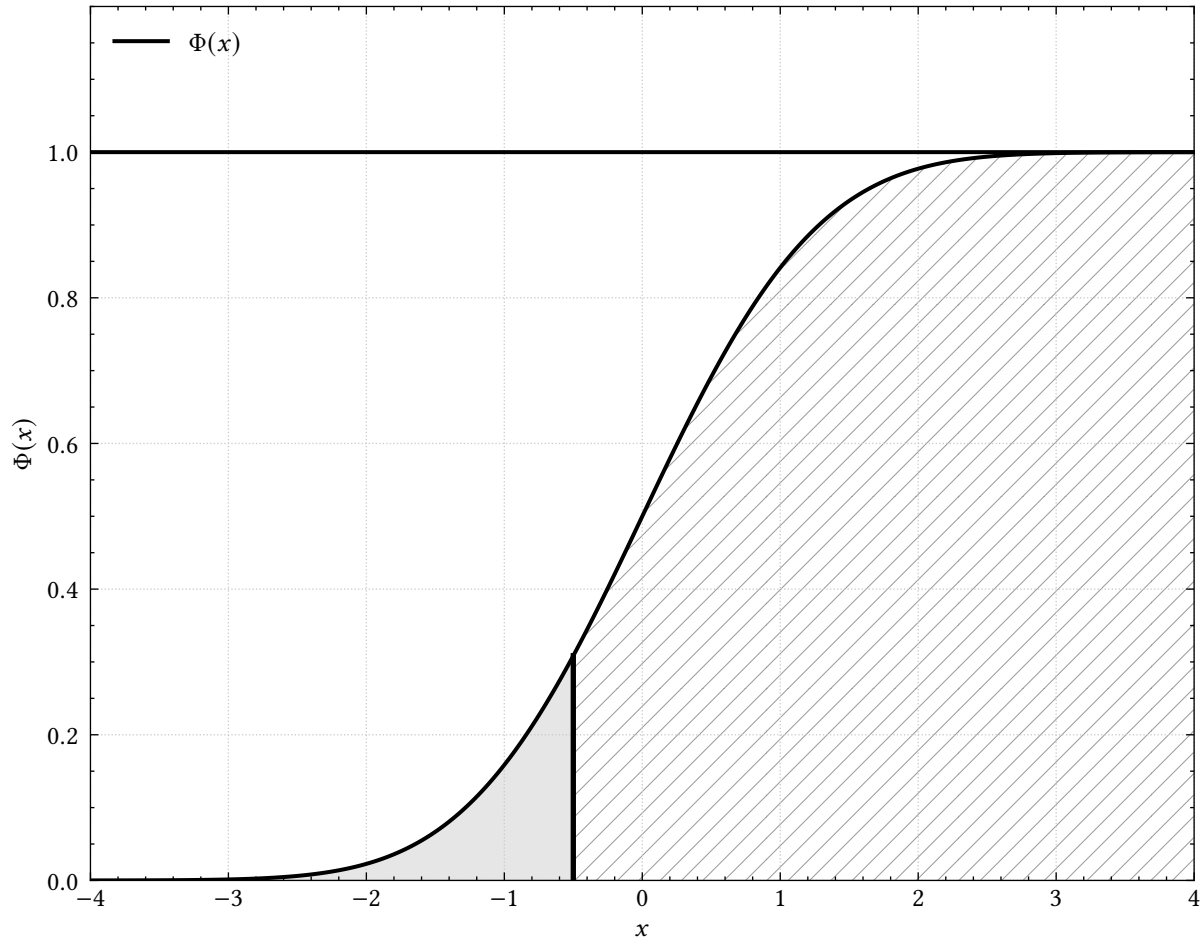


Figure 15: Visualisation of the Gaussian Cumulative Distribution Function.

Algorithm 4 Normal Random Vector Simulation**Require:** Mean vector μ , Covariance matrix Σ **Ensure:** Random vector $X \sim \mathcal{N}(\mu, \Sigma)$

- 1: Compute Cholesky decomposition $\Sigma = BB^\top$
- 2: Generate standard normal vector $Z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
- 3: $X \leftarrow \mu + BZ$
- 4: **return** X

BivariateNormal.py

```

1  rng = np.random.default_rng(4)
2  N, rhos = 1000, [0.0, 0.8]
3  Z = rng.standard_normal((2, N))
4  Ls = [np.linalg.cholesky([[1, r], [r, 1]]) for r in rhos]
5  Xs = [L @ Z for L in Ls]

```

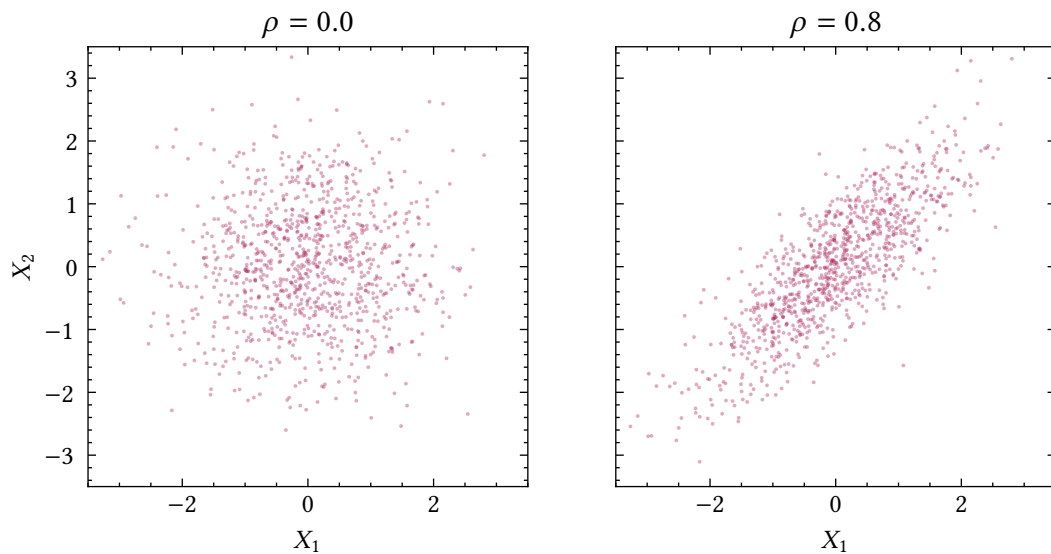


Figure 16: $N = 1000$ realisations of $X \sim \mathcal{N}(\mathbf{0}, \Sigma)$ with $\Sigma_{ii} = 1$ for $\rho = 0$ (left) and $\rho = 0.8$ (right).

23 Quasi Monte Carlo

IT is customary in this setting to focus on the problem of numerical integration over the unit hypercube.

The digital constructions are similar to feedback shift register random number generators, while lattice rules are similar to congruential generators. One useful way to think of QMC is that we are utilising a small random number generator in its entirety. The tools used to develop and analyse low-discrepancy methods are very different from those used in ordinary Monte Carlo, as they draw on number theory and abstract algebra rather than probability and statistics.

23.1 Discrepancy Measures

23.2 Low-discrepancy Sequences

23.2.1 Halton and Hammersley

23.2.2 Faure

23.2.3 Sobol

23.3 Koksma-Hlawka

24 Variance Reduction

The estimation of performance measures in Monte Carlo simulation can be made more efficient by utilising known information about the simulation model. Variance reduction techniques include antithetic variables, control variables, and importance sampling.

Suppose Y is the output of a simulation experiment. A random variable $\tilde{Y}$, obtained from the same simulation run, is called a *control variable* for Y if Y and $\tilde{Y}$ are correlated (negatively or positively) and the expectation of $\tilde{Y}$ is known. The use of control variables for variance reduction is based on the following theorem.

Theorem 24.1: Control Variable Estimation

Let $Y_1, \dots, Y_N$ be the output of N independent simulation runs and let $\tilde{Y}_1, \dots, \tilde{Y}_N$ be the corresponding control variables, with $\mathbb{E}\tilde{Y}_k = \tilde{\mu}$ known. Let $\varrho_{Y, \tilde{Y}}$ be the correlation coefficient between each Y_k and $\tilde{Y}_k$. For each $\alpha \in \mathbb{R}$ the estimator

$$\hat{\mu}^{(c)} = \frac{1}{N} \sum_{k=1}^N \left[Y_k - \alpha (\tilde{Y}_k - \tilde{\mu}) \right] \quad (24.1)$$

is an unbiased estimator for $\mu = \mathbb{E}Y$. The minimal variance of $\hat{\mu}^{(c)}$ is

$$\text{Var } \hat{\mu}^{(c)} = \frac{1}{N} (1 - \varrho_{Y, \tilde{Y}}^2) \text{Var } Y, \quad (24.2)$$

which is obtained for $\alpha = \varrho_{Y, \tilde{Y}} \sqrt{\text{Var } Y / \text{Var } \tilde{Y}}$.

24.0.1 Antithetics

When we use Monte Carlo averages of quantities $f(X_i)$, the randomness in the algorithm leads to some error cancellation. In antithetic sampling, we aim to achieve even greater cancellation. An antithetic sample is one that somehow gives the opposite value of $f(\mathbf{x})$, being low when $f(\mathbf{x})$ is high and vice versa. Ordinarily, we get an opposite f by sampling at a point $\tilde{\mathbf{x}}$ that is somehow opposite to $\mathbf{x}$.

Let $\mu = \mathbb{E}(X)$ for $X \sim p$, where p is a symmetric density on the symmetric set $\mathcal{D}$. Here, symmetry is with respect to reflection through the center point $\mathbf{c}$ of $\mathcal{D}$. If we reflect $\mathbf{x} \in \mathcal{D}$ through $\mathbf{c}$ we get the point $\tilde{\mathbf{x}}$ with $\tilde{\mathbf{x}} - \mathbf{c} = -(\mathbf{x} - \mathbf{c})$, that is $\tilde{\mathbf{x}} = 2\mathbf{c} - \mathbf{x}$. Symmetry means that $p(\tilde{\mathbf{x}}) = p(\mathbf{x})$ including the constraint that $\mathbf{x} \in \mathcal{D}$ if and only if $\tilde{\mathbf{x}} \in \mathcal{D}$. For basic examples, when p is $\mathcal{N}(0, \Sigma)$ then $\tilde{\mathbf{x}} = -\mathbf{x}$, and when p is $\mathcal{U}(0, 1)^d$ we have $\tilde{\mathbf{x}} = 1 - \mathbf{x}$ componentwise. The antithetic counterpart of a random curve could be its reflection in the horizontal axis. From the symmetry it follows that $\tilde{\tilde{\mathbf{x}}} = \mathbf{x}$.

The *antithetic sampling* estimate of μ is

$$\hat{\mu}_{\text{anti}} = \frac{1}{n} \sum_{i=1}^{n/2} (f(\mathbf{X}_i) + f(\tilde{\mathbf{X}}_i)), \quad (24.3)$$

where $X_i \stackrel{\text{iid}}{\sim} p$, and n is an even number.

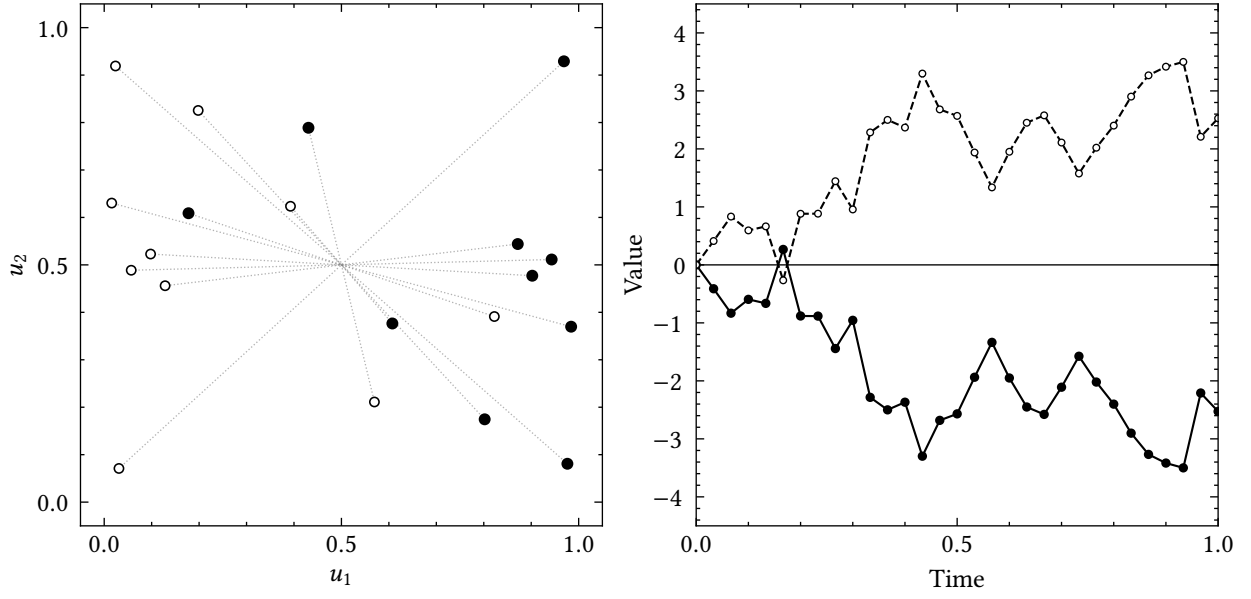


Figure 17: The left panel shows 10 points $\mathbf{u}_i \in [0, 1]^2$ as solid points, connected to their antithetic counterparts $\tilde{\mathbf{u}}_i = 1 - \mathbf{u}_i$, shown as open circles. The right panel shows one random trajectory of 30 points joined by solid lines and connected to the origin, along with its antithetic mirror image in open points.

The rationale for antithetic sampling is that each value of $\mathbf{x}$ is balanced by its opposite $\tilde{\mathbf{x}}$ satisfying $(\mathbf{x} + \tilde{\mathbf{x}})/2 = \mathbf{c}$. Whether this balance is helpful depends on f . Clearly, if f is nearly linear, we could obtain a large improvement. Suppose that $\sigma^2 = \mathbb{E}((f(\mathbf{X}) - \mu)^2) < \infty$. Then the variance in antithetic sampling is

$$\begin{aligned}
 \text{Var}(\hat{\mu}_{\text{anti}}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^{n/2} f(\mathbf{X}_i) + f(\tilde{\mathbf{X}}_i)\right) \\
 &= \frac{n/2}{n^2} \text{Var}(f(\mathbf{X}) + f(\tilde{\mathbf{X}})) \\
 &= \frac{1}{2n} \left(\text{Var}(f(\mathbf{X})) + \text{Var}(f(\tilde{\mathbf{X}})) + 2 \text{Cov}(f(\mathbf{X}), f(\tilde{\mathbf{X}})) \right) \\
 &= \frac{\sigma^2}{n} (1 + \rho)
 \end{aligned} \tag{24.4}$$

where $\rho = \text{Corr}(f(\mathbf{X}), f(\tilde{\mathbf{X}}))$.

From $-1 \leq \rho \leq 1$ we obtain $0 \leq \sigma^2(1 + \rho) \leq 2\sigma^2$. In the best case, antithetic sampling gives the exact answer from just one pair of function evaluations. In the worst case, it doubles the variance. Both cases do arise.

It is clear that a negative correlation is favourable. If f happens to be monotone in all d components of $\mathbf{x}$, then it is known that $\rho < 0$. Monotonicity of f is a safe harbour, wherein if f is monotone, then we're sure antithetic sampling will reduce the variance. We can often establish monotonicity theoretically, for example, by differentiating f . But $\rho < 0$ can hold without f being monotone in any of its inputs. Conversely, ρ can be just barely negative when f is monotone. As a result, monotonicity alone is not a reliable guide to determining whether antithetic sampling will yield a significant gain.

To get a qualitative understanding of antithetic sampling, break f into even and odd parts via

$$\begin{aligned} f(\mathbf{x}) &= \frac{f(\mathbf{x}) + f(\tilde{\mathbf{x}})}{2} + \frac{f(\mathbf{x}) - f(\tilde{\mathbf{x}})}{2} \\ &\equiv f_E(\mathbf{x}) + f_O(\mathbf{x}). \end{aligned} \quad (24.5)$$

The even part satisfies $f_E(\mathbf{x}) = f_E(\tilde{\mathbf{x}})$ and $\int_{\mathcal{D}} f_E(\mathbf{x})p(\mathbf{x})d\mathbf{x} = \mu$. The odd part satisfies $f_O(\mathbf{x}) = -f_O(\tilde{\mathbf{x}})$ and $\int_{\mathcal{D}} f_O(\mathbf{x})p(\mathbf{x})d\mathbf{x} = 0$.

The even and odd parts of f are orthogonal. This is not a surprise, because the product $f_O(\mathbf{x})f_E(\mathbf{x})$ is an odd function. But to be careful and rule out $\mathbb{E}(|f_O(\mathbf{X})f_E(\mathbf{X})|) = \infty$, we compute directly that

$$\begin{aligned} \int_{\mathcal{D}} f_E(\mathbf{x})f_O(\mathbf{x})p(\mathbf{x})d\mathbf{x} &= \int_{\mathcal{D}} \left(\frac{f(\mathbf{x}) + f(\tilde{\mathbf{x}})}{2} \right) \left(\frac{f(\mathbf{x}) - f(\tilde{\mathbf{x}})}{2} \right) p(\mathbf{x})d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathcal{D}} (f(\mathbf{x})^2 - f(\tilde{\mathbf{x}})^2)p(\mathbf{x})d\mathbf{x} = 0. \end{aligned}$$

Now it follows easily that $\sigma^2 = \sigma_E^2 + \sigma_O^2$ where $\sigma_E^2 = \int_{\mathcal{D}} (f_E(\mathbf{x}) - \mu)^2 p(\mathbf{x})d\mathbf{x}$ and $\sigma_O^2 = \int_{\mathcal{D}} f_O(\mathbf{x})^2 p(\mathbf{x})d\mathbf{x}$.

Reworking eq. (24.5) yields $\hat{\mu}_{\text{anti}} = (2/n) \sum_{i=1}^{n/2} f_E(\mathbf{X}_i)$. Therefore $\text{Var}(\hat{\mu}_{\text{anti}}) = 2\sigma_E^2/n$ and we can combine this with the variance of ordinary Monte Carlo sampling as follows

$$\begin{pmatrix} V(\hat{\mu}) \\ V(\hat{\mu}_{\text{anti}}) \end{pmatrix} = \frac{1}{n} \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} \sigma_E^2 \\ \sigma_O^2 \end{pmatrix}. \quad (24.6)$$

We see from eq. (24.6) that antithetic sampling eliminates the variance contribution of f_O but doubles the contribution from f_E . Antithetic sampling is extremely beneficial for integrands that are primarily odd functions of their inputs, having $\sigma_O^2 \gg \sigma_E^2$. The connection to correlation is via $\rho = (\sigma_E^2 - \sigma_O^2)/(\sigma_E^2 + \sigma_O^2)$.

The analysis above shows that antithetic sampling reduces variance if $\rho = \text{Corr}(f(X), f(\tilde{X})) < 0$, or equivalently, if $\sigma_O^2 > \sigma_E^2$. That analysis is appropriate when the most of the computation is in evaluating f and there is no economy in evaluating both $f(X)$ and $f(\tilde{X})$.

Variance reduction is only part of the story because the cost of antithetic sampling using n points could well be smaller than the cost of plain Monte Carlo with n points. That will happen if it is expensive to generate X , compared to the cost of computing f , but inexpensive to generate $\tilde{X}$. For example, X might be a carefully constructed and expensive sample path from a Gaussian process while $\tilde{X} = -X$.

We can explore this effect by letting c_x be the cost of generating X , and c_f be the cost of computing $f(X)$ once we have X . We also let $\tilde{c}_x$ and $\tilde{c}_f$ be the corresponding costs for the antithetic sample. For illustration, suppose that to a reasonable approximation $\tilde{c}_x = 0$ and $\tilde{c}_f = c_f$. In special circumstances $\tilde{c}_f < c_f$ because it may be possible to reuse some computation.

Under the assumptions we are exploring, the efficiency of antithetic sampling relative to plain Monte Carlo is

$$E_{\text{anti}} = \frac{2c_x + 2c_f}{c_x + 2c_f} \cdot \frac{\sigma_O^2 + \sigma_E^2}{2\sigma_E^2}. \quad (24.7)$$

Then antithetic sampling is more efficient than plain Monte Carlo if

$$\frac{\sigma_O^2}{\sigma_E^2} > \frac{c_f}{c_x + c_f}.$$

If generating x costs ten times as much as computing f then antithetic sampling pays off when $\sigma_O^2/\sigma_E^2 > 1/11$.

Because antithetic samples have dependent values within pairs, the usual variance estimate must be modified. The most straightforward approach is to analyze the data as a sample of size $m = n/2$ values of $f_E(X)$. Let $Y_i = f_E(X_i) = (f(X_i) + f(\tilde{X}_i))/2$ for

$i = 1, \dots, m = n/2$. Then take

$$\hat{\mu}_{\text{anti}} = \frac{1}{m} \sum_{i=1}^m Y_i, \quad \text{and}$$

$$s_{\text{anti}}^2 = \frac{1}{m-1} \sum_{i=1}^m (Y_i - \hat{\mu}_{\text{anti}})^2,$$

and use s_{anti}^2/m as the estimate of $\text{Var}(\hat{\mu}_{\text{anti}})$.

24.0.2 Common Random Numbers

Suppose that f and g are closely related functions and that we want to find $\mathbb{E}(f(X) - g(X))$ for $X \sim p$. Perhaps $f(x) = h(x, \theta)$ for a parameter $\theta \in \mathbb{R}^p$, and then to study the effect of θ we look at $g(x) = h(x, \tilde{\theta})$ for some $\tilde{\theta} \neq \theta$. We assume at first that neither f nor g (nor h) makes any use of random numbers other than X . Later we relax that assumption.

Because $\mathbb{E}(f(X) - g(X)) = \mathbb{E}(f(X)) - \mathbb{E}(g(X))$ we clearly have two different ways to go. We could estimate the difference by

$$\widehat{D}_{\text{com}} = \frac{1}{n} \sum_{i=1}^n f(X_i) - g(X_i), \quad (24.8)$$

for $X_i \stackrel{\text{iid}}{\sim} p$, or by differencing averages

$$\widehat{D}_{\text{ind}} = \frac{1}{n_1} \sum_{i=1}^{n_1} f(X_{i1}) - \frac{1}{n_2} \sum_{i=1}^{n_2} g(X_{i2}) \quad (24.9)$$

for $X_{ij} \stackrel{\text{iid}}{\sim} p$. Taking $n = n_1 = n_2$ makes the computing costs in (24.8) and (24.9) comparable, assuming that costs of computing f and g dominate those of generating X .

The sampling variances of these methods are

$$\begin{aligned} \text{Var}(\widehat{D}_{\text{com}}) &= \frac{1}{n} \left(\sigma_f^2 + \sigma_g^2 - 2\rho\sigma_f\sigma_g \right) \\ \text{Var}(\widehat{D}_{\text{ind}}) &= \frac{1}{n} \left(\sigma_f^2 + \sigma_g^2 \right), \end{aligned} \quad (24.10)$$

where σ_f^2 and σ_g^2 are individual function variances and $\rho = \text{Corr}(f(X), g(X))$. When $\rho > 0$ we are better off using common random numbers. There is no guarantee that $\rho > 0$. When

f and g compute similar quantities then we anticipate that $\rho > 0$, and if so, then $\widehat{D}_{\text{com}}$ is more effective than $\widehat{D}_{\text{ind}}$.

Most people would instinctively use the common variates. So at first sight, the method looks more like avoiding a variance increase than engineering a variance decrease. Later, when we relax the rule forbidding f , g , and h to use other sources of randomness, we will find that retaining some common random numbers requires considerable care in synchronization. The added complexity might well tip the balance against using common random numbers.

Much the same problem arises if we are comparing $\mathbb{E}(f(X))$ for $X \sim p$ and $\mathbb{E}(f(\widetilde{X}))$ for $\widetilde{X} \sim \tilde{p}$. Sometimes we can rewrite that problem in terms of common random variables that get transformed to a different distribution before f is applied. For instance, if the first simulation has $X_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ and the second has $\widetilde{X}_i \stackrel{\text{iid}}{\sim} \mathcal{N}(\tilde{\mu}, \tilde{\sigma}^2)$ then we can sample $Z_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ and use

$$\widehat{D}_{\text{com}} = \frac{1}{n} \sum_{i=1}^n f(\mu + \sigma Z_i) - f(\tilde{\mu} + \tilde{\sigma} Z_i).$$

More generally, when X_i is generated via a transformation $\Psi(U_i; \theta)$ of $U_i \sim \mathcal{U}(0, 1)^s$ then we can average $f(\Psi(U_i; \theta)) - f(\Psi(U_i; \tilde{\theta}))$. Acceptance-rejection sampling of X does not fit cleanly into this framework, because the number s of needed uniform random variables is not fixed and may vary with θ .

The construction above is a *coupling* of the random vectors X and $\widetilde{X}$. Any joint distribution on $(X, \widetilde{X})$ with $X \sim p$ and $\widetilde{X} \sim \tilde{p}$ is a coupling. Common random numbers provide a particularly close coupling between X and $\widetilde{X}$.

We want to estimate $\mu_j = \mathbb{E}(h(X, \theta_j))$ for $j = 1, \dots, m$ using n random inputs X_i , for $i = 1, \dots, n$. The content uniformity example had a large value of m but in the simplest case, $m = 2$ and we're interested in $\mu_1 - \mu_2$. We still assume that h really is a function of X and θ and in particular our implementation of h does not cause more random numbers to be generated.

Algorithm 1 Common Random Numbers Algorithm I

```

1: SETSEED(seed)
2:  $\hat{\mu}_j \leftarrow 0, \quad 1 \leq j \leq m$ 
3: for  $i \leftarrow 1, n$  do
4:    $X_i \sim p$ 
5:    $\hat{\mu}_j \leftarrow \hat{\mu}_j + h(X_i, \theta_j), \quad 1 \leq j \leq m$ 
6: end for
7:  $\hat{\mu}_j \leftarrow \hat{\mu}_j / n, \quad 1 \leq j \leq m$ 
8: return  $\hat{\mu}_1, \dots, \hat{\mu}_m$ 

```

We can run a nested loop over samples indexed by i and parameter values indexed by j . There are two main approaches that we can take, depending on which is the outer loop.

[algorithm 1](#) shows common random numbers with the outer loop over X_i for $i = 1, \dots, n$. When X_i is multi-dimensional we have to make sure that every component of X needed for any value of θ_j is provided. In the content uniformity problem (page 18) we generate Z_{11} for every simulated batch even though some only use $Z_1, \dots, Z_{10}$.

A vectorized implementation of [algorithm 1](#) is advantageous. It uses a function H that takes X and a list $\Theta = (\theta_1, \dots, \theta_m)$ of parameter values. This H returns a list $(h(X, \theta_1), \dots, h(X, \theta_m))$ and the simulation computes

$$(\hat{\mu}_1, \dots, \hat{\mu}_m) = \frac{1}{n} \sum_{i=1}^n H(X_i, \Theta). \quad (24.11)$$

This vectorized H makes it easier to separate the code that creates Θ from that which evaluates h .

24.0.3 Control Variates

Control variates provide another way to exploit closed-form results. With control variates, we use some other problem, quite similar to our given one, but for which an exact answer is known. The precise meaning of 'similar' depends on how we intend to use this other problem, and more than one method is provided below. As for 'exact', we will mean it

literally, but in practice it may just mean known with an error negligible compared to Monte Carlo errors.

Suppose first that we want to find $\mu = \mathbb{E}(f(\mathbf{X}))$ and that we know the value $\theta = \mathbb{E}(h(\mathbf{X}))$ where $h(\mathbf{X}) \approx f(\mathbf{X})$. Letting $\hat{\mu} = (1/n) \sum_{i=1}^n f(\mathbf{X}_i)$ and $\hat{\theta} = (1/n) \sum_{i=1}^n h(\mathbf{X}_i)$ we can estimate μ by the *difference estimator*

$$\hat{\mu}_{\text{diff}} = \frac{1}{n} \sum_{i=1}^n (f(\mathbf{X}_i) - h(\mathbf{X}_i)) + \theta = \hat{\mu} - \hat{\theta} + \theta. \quad (24.12)$$

The expected value of $\hat{\mu}_{\text{diff}}$ is μ . The variance of $\hat{\mu}_{\text{diff}}$ is

$$\text{Var}(\hat{\mu}_{\text{diff}}) = \frac{1}{n} \text{Var}(f(\mathbf{X}) - h(\mathbf{X})). \quad (24.13)$$

So if h is similar to f in the sense that the difference $f(\mathbf{X}) - h(\mathbf{X})$ has smaller variance than $f(\mathbf{X})$ has, we will get reduced variance by using $\hat{\mu}_{\text{diff}}$. In this setting $h(\mathbf{X})$, the random variable whose mean is known, is the *control variate*. The difference estimator is not the only way to use a control variate. The ratio and product estimators

$$\hat{\mu}_{\text{ratio}} = \hat{\mu}/\hat{\theta}, \quad \text{and} \quad (24.14)$$

$$\hat{\mu}_{\text{prod}} = \hat{\mu}\hat{\theta}/\theta \quad (24.15)$$

respectively, are also used. These estimators are undefined when $\theta = 0$, but otherwise they generally converge to μ as $n \rightarrow \infty$. The ratio and product estimators are usually biased because $\mathbb{E}(\hat{\mu}/\hat{\theta}) \neq \mathbb{E}(\hat{\mu})/\mathbb{E}(\hat{\theta})$ in general. It is possible to generalize the control variate method in very complicated ways. Maybe we could use $\hat{\mu} \cos(\hat{\theta} - \theta)$ or some more imaginative quantity. But we don't. By far the most common way of using a control variate is through the regression estimator, considered next. For a value $\beta \in \mathbb{R}$, the *regression estimator* of μ is

$$\hat{\mu}_{\beta} = \frac{1}{n} \sum_{i=1}^n (f(\mathbf{X}_i) - \beta h(\mathbf{X}_i)) + \beta \theta. \quad (24.16)$$

Taking $\beta = 0$ yields the simple MC estimator $\hat{\mu}$ and $\beta = 1$ gives the difference estimator. The regression estimator is unbiased: $\mathbb{E}(\hat{\mu}_{\beta}) = \mu$ for all β because $\mathbb{E}(h(\mathbf{X}) - \theta) = 0$.

The variance of the regression estimator is

$$\text{Var}(\hat{\mu}_\beta) = \frac{1}{n} (\text{Var}(f(\mathbf{X})) - 2\beta \text{Cov}(f(\mathbf{X}), h(\mathbf{X})) + \beta^2 \text{Var}(h(\mathbf{X}))) . \quad (24.17)$$

By differentiating, we find that the best value of β is

$$\beta_{\text{opt}} = \frac{\text{Cov}(f(\mathbf{X}), h(\mathbf{X}))}{\text{Var}(h(\mathbf{X}))} = \frac{\mathbb{E}((f(\mathbf{X}) - \mu)(h(\mathbf{X}) - \theta))}{\mathbb{E}((h(\mathbf{X}) - \theta)^2)}, \quad (24.18)$$

and after some algebra, the resulting minimal variance is

$$\text{Var}(\hat{\mu}_{\beta_{\text{opt}}}) = \frac{\sigma^2}{n} (1 - \rho^2), \quad (24.19)$$

where $\rho = \text{Corr}(f(\mathbf{X}), h(\mathbf{X}))$. In the regression estimator, any control variate that correlates with f is helpful, even one that correlates negatively.

Algorithm 2 Control Variates by Regression

Require: Given $f(\mathbf{X}_i), h_j(\mathbf{X}_i), \theta_j$ for $i = 1, \dots, n, j = 1, \dots, J$

- 1: $\bar{f} \leftarrow \frac{1}{n} \sum f(\mathbf{X}_i); \quad \bar{h}_j \leftarrow \frac{1}{n} \sum h_j(\mathbf{X}_i)$
 - 2: $Y_i \leftarrow f(\mathbf{X}_i); \quad Z_{ij} \leftarrow h_j(\mathbf{X}_i) - \bar{h}_j \quad \text{for all } i, j$
 - 3: $\hat{\mu} \leftarrow$ estimated intercept from MLR
 - 4: $\hat{\beta} \leftarrow$ estimated slope vector from MLR
 - 5: $\hat{\mu} \leftarrow \hat{\mu} + \hat{\beta}^\top (\boldsymbol{\theta} - \bar{\mathbf{h}})$
 - 6: **return** $\hat{\mu}$ and intercept standard error from MLR
-

This algorithm demonstrates how to utilise linear regression software for control variate computation. It is *essential* to centre the control variates. It may be necessary to drop one or more control variates if they are linearly dependent in the sample.

To minimize [eq. \(24.17\)](#) with respect to β is a least squares problem and the solution vector β satisfies $\text{Var}(h(\mathbf{X}))\beta = \text{Cov}(h(\mathbf{X}), f(\mathbf{X}))$. If the J by J matrix $\text{Var}(h(\mathbf{X}))$ is singular, then one of the h_j is a linear combination of the other $J - 1$ control variates. In the former

case, the best value of β is

$$\beta_{\text{opt}} = \text{Var}(h(X))^{-1} \text{Cov}(h(X), f(X)) - \text{Var}(h(X))^{-1} \mathbb{E}(h(X) - \theta) \mathbb{E}(f(X)). \quad (24.20)$$

In applications, we ordinarily do not know β_{opt} . The usual way to estimate it is by replacing expectations by sample averages:

$$\hat{\beta} = \left(\frac{1}{n} \sum_{i=1}^n (h(X_i) - \bar{H})(h(X_i) - \bar{H})^\top \right)^{-1} \frac{1}{n} \sum_{i=1}^n (h(X_i) - \bar{H})f(X_i). \quad (24.21)$$

Equation [eq. \(24.21\)](#) is the least squares estimate of β_{opt} . The usual estimate of μ with control variates is $\hat{\mu}_{\hat{\beta}}$. The estimated variance is

$$\widehat{\text{Var}}(\hat{\mu}_{\hat{\beta}}) = \frac{1}{n^2} \sum_{i=1}^n (f(X_i) - \hat{\mu}_{\hat{\beta}} - \hat{\beta}^\top (h(X_i) - \bar{h}))^2, \quad (24.22)$$

and its standard error $\sqrt{\widehat{\text{Var}}(\hat{\mu}_{\hat{\beta}})}$, can be computed using standard multiple linear regression software. See [algorithm 2](#). The key insight is to treat μ as the intercept in a multiple linear regression relating $f(X)$ to predictors $h_j(X) - \theta_j$. The regression formula is $f(X) \approx \mu + \beta^\top (h(X) - \theta)$. It is crucial to subtract θ_j from the random variable in order to make $\mathbb{E}(f(X)) = \mu$ match the regression intercept.

The error of the regression estimator using $\hat{\beta} = \hat{\beta}$ is

$$\hat{\mu}_{\hat{\beta}} - \mu = (\hat{\beta} - \beta_{\text{opt}})^\top (\theta - \bar{H}) + \hat{\mu}_{\beta_{\text{opt}}} - \mu. \quad (24.23)$$

The first term in [eq. \(24.23\)](#) is the product of two components of mean zero, while the second term is the error in the unknown optimal regression estimator. The second term has a mean of zero, but the first does not in general, because the expected value of a product is not necessarily the same as the product of the expected values. As a result, the control variate estimator is usually biased.

Although estimating β from the sample data brings a bias, that bias is ordinarily negligible. Each of the factors $\hat{\beta} - \beta_{\text{opt}}$ and $\bar{h} - \theta$ is of order $O(n^{-1/2})$. For applications with $J \ll \sqrt{n}$, it is common to neglect the bias from using estimated control variate coefficients.

When an unbiased estimator is required, then we can get one by using an estimate of β_{opt} that is independent of the X_i used in $\hat{\mu}_{\hat{\beta}}$. For example $\hat{\beta}$ can be computed from [eq. \(24.21\)](#) using only a pilot sample $\tilde{X}_1, \dots, \tilde{X}_m \sim p$ independent of the X_i . Then $\hat{\mu}_{\hat{\beta}}$ can be computed by [eq. \(24.16\)](#) using $X_1, \dots, X_n$ and taking $\beta = \hat{\beta}$. Now $\mathbb{E}(\hat{\mu}_{\hat{\beta}}) = \mu$ and

$$\text{Var}(\hat{\mu}_{\hat{\beta}}) = \mathbb{E}(\text{Var}(\hat{\mu}_{\hat{\beta}} \mid \tilde{X}_1, \dots, \tilde{X}_m)) = \frac{1}{n} \mathbb{E}(\sigma_{\hat{\beta}}^2). \quad (24.24)$$

If $f(X_i)$ and $h(X_i)$ have finite fourth moments then $\hat{\beta} = \beta_{\text{opt}} + O(1/\sqrt{m})$. The variance with a control variate is $\sigma^2(1 - \rho^2)/n$, which is never worse than σ^2/n and usually better. Whether the control variate is helpful ultimately depends on how much it costs to use it. Suppose that the total cost of generating X_i and then computing $f(X_i)$ is, on average, c_f . Let c_h be the average extra cost incurred by the control variate. That includes the cost to evaluate $h(X_i)$ but not the cost of sampling X_i . We will suppose that the cost to compute $\hat{\beta}$ is small. If not, then c_h should be increased to reflect it. Control variates improve efficiency when $(1 - \rho^2)(c_f + c_h) < c_f$, that is when $|\rho| > \sqrt{c_h/(c_f + c_h)}$. For illustration, if $c_h = c_f$ then we need $|\rho| > \sqrt{1/2} \approx 0.71$ in order to benefit from the control variate.

The estimated variance of $\hat{\mu}_{\hat{\beta}}$ is

$$\widehat{\text{Var}}(\hat{\mu}_{\hat{\beta}}) = \frac{1}{n^2} \sum_{i=1}^n (f(X_i) - \hat{\mu}_{\hat{\beta}} - \hat{\beta}^\top (h(X_i) - \bar{h}))^2, \quad (24.25)$$

and a 99% confidence interval is $\hat{\mu}_{\hat{\beta}} \pm 2.58 \sqrt{\widehat{\text{Var}}(\hat{\mu}_{\hat{\beta}})}$.

24.0.4 Importance Sampling

Suppose that our problem is to find $\mu = \mathbb{E}(f(X)) = \int_{\mathcal{D}} f(x)p(x) \, dx$ where p is a probability density function on $\mathcal{D} \subseteq \mathbb{R}^d$ and f is the integrand. We take $p(x) = 0$ for all $x \notin \mathcal{D}$. If q is

a positive probability density function on $\mathbb{R}^d$, then

$$\mu = \int_{\mathcal{D}} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} = \int_{\mathcal{D}} \frac{f(\mathbf{x})p(\mathbf{x})}{q(\mathbf{x})} q(\mathbf{x}) \, d\mathbf{x} = \mathbb{E}_q \left(\frac{f(\mathbf{X})p(\mathbf{X})}{q(\mathbf{X})} \right), \quad (24.26)$$

where $\mathbb{E}_q(\cdot)$ denotes expectation for $\mathbf{X} \sim q$. We also write $\text{Var}_q(\cdot)$, $\text{Cov}_q(\cdot, \cdot)$, and $\text{Corr}_q(\cdot, \cdot)$ for variance, covariance and correlation when $\mathbf{X} \sim q$. Our original goal then is to find $\mathbb{E}_p(f(\mathbf{X}))$. By making a multiplicative adjustment to f we compensate for sampling from q instead of p . The adjustment factor $p(\mathbf{x})/q(\mathbf{x})$ is called the *likelihood ratio*. The distribution q is the *importance distribution* and p is the *nominal distribution*.

The importance distribution q does not have to be positive everywhere. It is enough to have $q(\mathbf{x}) > 0$ whenever $f(\mathbf{x})p(\mathbf{x}) \neq 0$. That is, for $Q = \{\mathbf{x} \mid q(\mathbf{x}) > 0\}$ we have $\mathbf{x} \in Q$ whenever $f(\mathbf{x})p(\mathbf{x}) \neq 0$. So if $\mathbf{x} \in \mathcal{D} \cap Q^c$ we know that $f(\mathbf{x}) = 0$, while if $\mathbf{x} \in Q \cap \mathcal{D}^c$ we have $p(\mathbf{x}) = 0$. Now

$$\begin{aligned} \mathbb{E}_q \left(\frac{f(\mathbf{X})p(\mathbf{X})}{q(\mathbf{X})} \right) &= \int_Q \frac{f(\mathbf{x})p(\mathbf{x})}{q(\mathbf{x})} q(\mathbf{x}) \, d\mathbf{x} = \int_Q f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} \\ &= \int_{\mathcal{D}} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} + \int_{Q \cap \mathcal{D}^c} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} - \int_{\mathcal{D} \cap Q^c} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} \\ &= \int_{\mathcal{D}} f(\mathbf{x})p(\mathbf{x}) \, d\mathbf{x} = \mu. \end{aligned} \quad (24.27)$$

It is natural to wonder what happens for $\mathbf{x}$ with $q(\mathbf{x}) = 0$ in the denominator. The answer is that there are no such points $\mathbf{x} \in Q$ and we will never see one when sampling $\mathbf{X} \sim q$. Later we will see examples where $q(\mathbf{x})$ close to 0 causes extreme difficulties, but $q(\mathbf{x}) = 0$ is not a problem if $f(\mathbf{x})p(\mathbf{x}) = 0$ too.

When we want q to work for many different functions f_j then we need $q(\mathbf{x}) > 0$ at every $\mathbf{x}$ where any $f_j(\mathbf{x})p(\mathbf{x}) \neq 0$. Then, a density q with $q(\mathbf{x}) > 0$ whenever $p(\mathbf{x}) > 0$ will suffice, allowing us to add new functions f_j to our list after the sample has been drawn.

The *importance sampling estimate* of $\mu = \mathbb{E}_p(f(\mathbf{X}))$ is

$$\hat{\mu}_q = \frac{1}{n} \sum_{i=1}^n \frac{f(\mathbf{X}_i)p(\mathbf{X}_i)}{q(\mathbf{X}_i)}, \quad \mathbf{X}_i \sim q. \quad (24.28)$$

To use (24.28) we must be able to compute $f p/q$. Assuming that we can compute f , this estimate requires that we can compute $p(x)/q(x)$ at any x we might sample. When p or q has an unknown normalization constant, then we will resort to a ratio estimate. For now, we assume that p/q is computable, and study the variance of $\hat{\mu}_q$. Exponential tilting is one way to choose q with computable p/q even when p is unnormalized.

Theorem 24.2: Variance of Importance Sampling

Let $\hat{\mu}_q$ be given by (24.28) where $\mu = \int_{\mathcal{D}} f(x)p(x) dx$ and $q(x) > 0$ whenever $f(x)p(x) \neq 0$. Then $\mathbb{E}_q(\hat{\mu}_q) = \mu$, and $\text{Var}_q(\hat{\mu}_q) = \sigma_q^2/n$ where

$$\sigma_q^2 = \int_Q \frac{(f(x)p(x))^2}{q(x)} dx - \mu^2 = \int_Q \frac{(f(x)p(x) - \mu q(x))^2}{q(x)} dx, \quad (24.29)$$

where $Q = \{x \mid q(x) > 0\}$. The domain Q can be replaced by $\mathcal{D}$ in the above integrals.

Proof. That $\mathbb{E}_q(\hat{\mu}_q) = \mu$ follows directly from (24.27). Next,

$$\text{Var}_q(\hat{\mu}_q) = \frac{1}{n} \left[\int_Q \left(\frac{f(x)p(x)}{q(x)} \right)^2 q(x) dx - \mu^2 \right].$$

The first expression in equation (24.29) follows by the definition of σ_q^2 and the second by simple rearrangement. Because $q > 0$ whenever $f p \neq 0$, we can replace Q by $\mathcal{D}$ in the integrals. $\square$

To form a confidence interval for μ , we need to estimate σ_q^2 . From the second expression in (24.29) we find that

$$\sigma_q^2 = \mathbb{E}_q((f(X)p(X) - \mu q(X))^2 / q(X)^2).$$

Because $\mathbf{x}_i$ are sampled from q , the natural variance estimate is

$$\hat{\sigma}_q^2 = \frac{1}{n} \sum_{i=1}^n \left(\frac{f(\mathbf{x}_i)p(\mathbf{x}_i)}{q(\mathbf{x}_i)} - \hat{\mu}_q \right)^2 = \frac{1}{n} \sum_{i=1}^n (w_i f(\mathbf{x}_i) - \hat{\mu}_q)^2, \quad (24.30)$$

where $w_i = p(\mathbf{x}_i)/q(\mathbf{x}_i)$. Then an approximate 99% confidence interval for μ is $\hat{\mu}_q \pm 2.58\hat{\sigma}_q/\sqrt{n}$.

Theorem 24.2 guides us in selecting a good importance sampling rule. The first expression in (24.29) is simpler to study. A better q is one that gives a smaller value of $\int_{\mathcal{D}} (fp)^2/q \, d\mathbf{x}$. When we want upper bounds on σ_q^2 we can bound $\int_{\mathcal{D}} (fp)^2/q \, d\mathbf{x}$.

The second integral expression in (24.29) illustrates how importance sampling can succeed or fail. The numerator in the integrand at the right of (24.29) is small when $f(\mathbf{x})p(\mathbf{x}) - \mu q(\mathbf{x})$ is close to zero, that is, when $q(\mathbf{x})$ is nearly proportional to $f(\mathbf{x})p(\mathbf{x})$. From the denominator, we see that regions with small values of $q(\mathbf{x})$ greatly magnify whatever lack of proportionality appears in the numerator.

Suppose that $f(\mathbf{x}) \geq 0$ for all $\mathbf{x}$, and to rule out a trivial case, assume that $\mu > 0$ too. Then $q_{\text{opt}}(\mathbf{x}) = f(\mathbf{x})p(\mathbf{x})/\mu$ is a probability density, and it has $\sigma_{q_{\text{opt}}}^2 = 0$. It is optimal, but not really usable because $\hat{\mu}_{q_{\text{opt}}}$ becomes an average of $f(\mathbf{x}_i)p(\mathbf{x}_i)/q(\mathbf{x}_i) = \mu$ meaning that we could compute μ directly from f , p , and q without any sampling. Likewise, for $f(\mathbf{x}) \leq 0$ with $\mu < 0$, a zero variance is obtained for the density $q = -fp/\mu$.

Although zero-variance importance sampling densities are not usable, they provide valuable insight into the design of an effective importance sampling scheme. Suppose that $f(\mathbf{x}) \geq 0$. It may be beneficial for q to have spikes in the same places as f or p , but it is even better to have them where fp does. Moreover, the best q is proportional to fp , not $\sqrt{fp}$ or f^2p or some other combination.

Choosing a good importance sampling distribution requires some educated guessing and possibly numerical search. In many applications, there is domain knowledge about where the spikes are. In a financial setting, we may know which stock fluctuations will cause an option to go to its maximum value. For a queuing system, it may be easy to determine which combination of arrivals will cause the system to become overloaded.

In general, the density q^* that minimizes σ_q^2 is proportional to $|f(\mathbf{x})|p(\mathbf{x})$ (`kahnMethodsReducingSamp` outside of trivial cases where $\int |f(\mathbf{x})|p(\mathbf{x})d\mathbf{x} = 0$). This optimal density does not have $\sigma_{q^*}^2 = 0$ on problems where fp can be positive for some $\mathbf{x}$ and negative for other $\mathbf{x}$. When

fp takes both positive and negative values, a zero variance method is still possible, but it requires sampling at two points.

To prove that $q^*(\mathbf{x}) = |f(\mathbf{x})|p(\mathbf{x})/\mathbb{E}_p(|f(\mathbf{X})|)$ is optimal, let q be any density that is positive when $fp \neq 0$. Then

$$\begin{aligned}\mu^2 + \sigma_{q^*}^2 &= \int \frac{f(\mathbf{x})^2 p(\mathbf{x})^2}{q^*(\mathbf{x})} d\mathbf{x} = \int \frac{f(\mathbf{x})^2 p(\mathbf{x})^2}{|f(\mathbf{x})|p(\mathbf{x})/\mathbb{E}_p(|f(\mathbf{X})|)} d\mathbf{x} \\ &= \mathbb{E}_p(|f(\mathbf{X})|)^2 = \mathbb{E}_q(|f(\mathbf{X})|p(\mathbf{X})/q(\mathbf{X}))^2 \\ &\leq \mathbb{E}_q(f(\mathbf{X})^2 p(\mathbf{X})^2 / q(\mathbf{X})^2) = \mu^2 + \sigma_q^2,\end{aligned}$$

and so $\sigma_{q^*}^2 \leq \sigma_q^2$. This proof is a straightforward consequence of the Cauchy-Schwarz inequality. However, it requires that we know the optimal density before applying the Cauchy-Schwarz inequality. For a principled way to find candidates for q^* in problems like importance sampling, we can turn to the calculus of variations.

We may use the likelihood ratio $w(\mathbf{x}) = p(\mathbf{x})/q(\mathbf{x})$ as a means to understand which importance sampling densities are good or bad choices. The first term in σ_q^2 is $\int f(\mathbf{x})^2 p(\mathbf{x})^2 / q(\mathbf{x}) d\mathbf{x}$. We may write this term as $\mathbb{E}_p(f(\mathbf{X})^2 w(\mathbf{X})) = \mathbb{E}_q(f(\mathbf{X})^2 w(\mathbf{X})^2)$. The appearance of q in the denominator of w means that light-tailed importance densities q are dangerous. If we are clever or lucky, then f might be small just where it needs to be to offset the small denominator. But we often need to use the same sample with multiple integrands f , and so, as a rule, q should have tails at least as heavy as p does.

When p is a Gaussian distribution, then a common tactic is to take q to be a Student's t distribution. Such a q has heavier tails than p . It even has heavier tails than fp for integrands of the form $f(\mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\theta})$, because then fp is proportional to another Gaussian density.

The reverse practice, of using a Gaussian importance distribution q for a student's t nominal distribution, can easily lead to $\sigma_q^2 = \infty$. Even when q is nearly proportional to fp we may still have $\sigma_q^2 = \infty$. The irony is that the infinite variance could be largely attributable to the unimportant region of $\mathcal{D}$ where $|f|p$ is small (but q is extremely small).

Bounded weights are especially valuable. If $w(\mathbf{x}) \leq c$ the $\sigma_q^2 \leq c\sigma_p^2$.

The likelihood ratio also reveals a dimension effect for importance sampling. Suppose that $\mathbf{x} \in \mathbb{R}^d$ and to keep things simple, that the components of $\mathbf{X}$ are independent under both p and q . Then $w(\mathbf{X}) = \prod_{j=1}^d p_j(X_j)/q_j(X_j)$. Though f plays a role, the value of σ_q^2 is also driven largely by $\mathbb{E}_q(w(\mathbf{X})^2) = \prod_{j=1}^d \mathbb{E}_q(w_j(X_j)^2)$ where $w_j = p_j/q_j$. Each $\mathbb{E}_q(w_j(X_j)) = 1$ and $\mathbb{E}_q(w(\mathbf{X})^2) = \prod_{j=1}^d (1 + \text{Var}_q(w_j(X_j)))$. This quantity will grow exponentially with d unless we arrange for the variances of $w_j(X_j)$ to eventually decrease sharply as j increases.

Although we can get a lot of insight from the likelihood ratio $w(\mathbf{x})$, the efficiency of importance sampling compared to alternatives still depends on f . We will see another example of that phenomenon in §9.3 where we discuss some measures of effective sample size for importance sampling.

Sometimes we can only compute an unnormalized version of p , $p_u(\mathbf{x}) = cp(\mathbf{x})$ where $c > 0$ is unknown. The same may be true of q . Suppose that we can compute $q_u(\mathbf{x}) = bq(\mathbf{x})$ where $b > 0$ might be unknown. If we are fortunate or clever enough to have $b = c$, then $p(\mathbf{x})/q(\mathbf{x}) = p_u(\mathbf{x})/q_u(\mathbf{x})$ and we can still use (24.28). Otherwise we may compute the ratio $w_u(\mathbf{x}) = p_u(\mathbf{x})/q_u(\mathbf{x}) = (c/b)p(\mathbf{x})/q(\mathbf{x})$ and consider the *self-normalized importance sampling estimate*

$$\tilde{\mu}_q = \frac{\sum_{i=1}^n f(\mathbf{X}_i) w_u(\mathbf{X}_i)}{\sum_{i=1}^n w_u(\mathbf{X}_i)} \quad (24.31)$$

where $\mathbf{X}_i \sim q$ are independent. The factor c/b cancels from numerator and denominator in (24.31), leading to the same estimate as if we had used the desired ratio $w(\mathbf{x}) = p(\mathbf{x})/q(\mathbf{x})$ instead of the computable alternative $w_u(\mathbf{x})$. As a result we can write

$$\tilde{\mu}_q = \frac{\frac{1}{n} \sum_{i=1}^n f(\mathbf{X}_i) w(\mathbf{X}_i)}{\frac{1}{n} \sum_{i=1}^n w(\mathbf{X}_i)}. \quad (24.32)$$

Theorem 24.3: Consistency of Self-Normalized Importance Sampling

Let p be a probability density function on $\mathbb{R}^d$ and let $f(\mathbf{x})$ be a function such that $\mu = \int f(\mathbf{x})p(\mathbf{x})d\mathbf{x}$ exists. Suppose that $q(\mathbf{x})$ is a probability density function on $\mathbb{R}^d$ with $q(\mathbf{x}) > 0$ whenever $p(\mathbf{x}) > 0$. Let $X_1, \dots, X_n \sim q$ be independent and let $\tilde{\mu}_q$ be the self-normalized importance sampling estimate (24.31). Then

$$\mathbb{P}\left(\lim_{n \rightarrow \infty} \tilde{\mu}_q = \mu\right) = 1.$$

Proof. We may use expression (24.32) for $\tilde{\mu}_q$. The numerator in (24.32) is the average $\hat{\mu}_q$ of $f(X_i)p(X_i)/q(X_i)$ under sampling from q . These are i.i.d. with mean μ by (24.27). The strong law of large numbers gives $\mathbb{P}(\lim_{n \rightarrow \infty} \hat{\mu}_q = \mu) = 1$. The denominator of (24.32) converges to 1, using the argument for the numerator on the function f with $f(\mathbf{x}) \equiv 1$. $\square$

Importance sampling uses unequally weighted observations. The weights are $w_i = p(X_i)/q(X_i) \geq 0$ for $i = 1, \dots, n$. In extreme settings, one of the w_i may be vastly larger than all the others and then we have effectively only got one observation. We would like to have a diagnostic to tell when the weights are problematic. It is even possible that $w_1 = w_2 = \dots = w_n = 0$. In that case, importance sampling has clearly failed and we don't need a diagnostic to tell us so. Hence, we may assume that $\sum_{i=1}^n w_i > 0$.

Consider a hypothetical linear combination

$$S_w = \frac{\sum_{i=1}^n w_i Z_i}{\sum_{i=1}^n w_i} \quad (24.33)$$

where Z_i are independent random variables with a common mean and common variance $\sigma^2 > 0$, and $\mathbf{w} \in [0, \infty)^n$ is the vector of weights. The unweighted average of n_e independent random variables Z_i has variance σ^2/n_e . Setting $\text{Var}(S_w) = \sigma^2/n_e$ and solving for n_e yields the *effective sample size*

$$n_e = \frac{(\sum_{i=1}^n w_i)^2}{\sum_{i=1}^n w_i^2} = \frac{n\bar{w}^2}{\bar{w}^2} \quad (24.34)$$

where $\bar{w} = (1/n) \sum_{i=1}^n w_i$ and $\bar{w}^2 = (1/n) \sum_{i=1}^n w_i^2$. If the weights are too imbalanced, then the result is similar to averaging only $n_e \ll n$ observations and might therefore be

unreliable. The point at which n_e becomes alarmingly small is hard to specify, because it is application-specific.

The effective sample size [eq. \(24.34\)](#) is computed from the generated values. For simple enough distributions, we can obtain a population version of n_e , as

$$n_e^* = \frac{n\mathbb{E}_q(\mathbf{w})^2}{\mathbb{E}_q(\mathbf{w}^2)} = \frac{n}{\mathbb{E}_q(\mathbf{w}^2)} = \frac{n}{\mathbb{E}_p(\mathbf{w})}. \quad (24.35)$$

If n_e^* is too small, then we know q will produce imbalanced weights. Another effective sample size formula is

$$\frac{n}{1 + \text{cv}(\mathbf{w})^2} \quad (24.36)$$

where $\text{cv}(\mathbf{w}) = ((n-1)^{-1} \sum_{i=1}^n (w_i - \bar{w})^2)^{1/2} / \bar{w}$ is the coefficient of variation of the weights. If we replace the $n-1$ by n in $\text{cv}(\mathbf{w})$ and substitute into $n/(1 + \text{cv}(\mathbf{w})^2)$, the result is n_e of [eq. \(24.34\)](#), so the two formulas are essentially the same.

Instead of using an effective sample size, we might just estimate the variance of $\hat{\mu}_q$ (or of $\tilde{\mu}_q$) and use the variance as a diagnostic. When that variance is quite large, we would conclude that the importance sampling had not worked well. Unfortunately, the variance estimate is based on the same weights as the estimate itself. Badly skewed weights can lead to a badly estimated mean, along with a poor variance estimate that masks the problem.

The variance estimate for $\tilde{\mu}_q$ uses a weighted average of $(f(\mathbf{x}_i) - \tilde{\mu}_q)^2$ with weights equal to w_i^2 . Similarly, for $\hat{\mu}_q$ we have $\hat{\sigma}_q^2 = (1/n) \sum_{i=1}^n w_i^2 f(\mathbf{x}_i)^2 - \hat{\mu}_q^2$ which again depends on weights w_i^2 . We can compute an effective sample size for variance estimates by

$$n_{e,\sigma} = \frac{(\sum_{i=1}^n w_i^2)^2}{\sum_{i=1}^n w_i^4}. \quad (24.37)$$

If $n_{e,\sigma}$ is small, then we cannot trust the variance estimate. Estimating a variance well is typically a harder problem than estimating a mean well, and here we have $n_{e,\sigma} \leq n_{e,\mu}$.

The accuracy of the central limit theorem for a mean depends on the skewness of the random variables being summed. To gauge the reliability of the CLT for $\hat{\mu}_q$ and $\tilde{\mu}_q$ we find

that the skewness of S_w matches that of the average of

$$n_{e,\gamma} = \frac{(\sum w_i^2)^3}{(\sum w_i^3)^2}$$

equally weighted observations.

Effective sample sizes are imperfect diagnostics. When they are too small, then we have a sign that the importance sampling weights are problematic. When they are large, we still cannot conclude that importance sampling has worked. It remains possible that some important region was missed by all of $X_1, \dots, X_n$.

The diagnostics above used the weights w_i only through their relative values $w_i / \sum_j w_j$. In cases where w_i is computable, it is the observed value of a random variable with mean $\mathbb{E}_q(p(X)/q(X)) = 1$. If the sample mean of the weights is far from 1, then that is a sign that q was poorly chosen. That is, $\bar{w} = (1/n) \sum_{i=1}^n w_i$ is another diagnostic.

These weight-based diagnostics are the same for every function f . They are convenient summaries, but the effectiveness of importance sampling also depends on f . If every $f(X_i)p(X_i)/q(X_i) = 0$, it is likely that our importance sampling has failed, and no further diagnostic is needed. Otherwise, for a diagnostic that is specific to f let

$$\tilde{w}_i(f) = \frac{|f(X_i)|p(X_i)/q(X_i)}{\sum_{j=1}^n |f(X_j)|p(X_j)/q(X_j)}.$$

After some algebra, the sample coefficient of variation of these weights is $\text{cv}(\mathbf{w}, f) = \sqrt{n \sum_{i=1}^n \tilde{w}_i^2 - 1}$. An effective sample size customised to f is

$$n_e(f) = \frac{n}{1 + \text{cv}(\mathbf{w}, f)^2} = \frac{1}{\sum_{i=1}^n \tilde{w}_i(f)^2}. \quad (24.38)$$

Part VII

Grid & Lattice Methods

25 Lattice Methods

Lattice methods are computational techniques used to value options and other derivatives by constructing a lattice, essentially a grid-like representation of the underlying asset's possible price movements over discrete time intervals (**coxOptionPricingSimplified1979a**). In a lattice, each node represents a possible price level of the underlying asset at a specific point in time. Starting from the initial price, the lattice evolves by considering possible upward and downward movements based on the asset's volatility.

Pricing barrier options can be challenging due to the added complexity introduced by the barrier conditions. In a lattice, the barrier levels can be incorporated into the calculations by adjusting the probabilities of price movements, which will be analysed with the Bino-Trinomial model.

At each step of the tree, the asset price can over the time step Δt either move up by a fixed amount u with a probability p_u or decrease by a fixed amount d with a probability $p_d = 1 - p_u$. We assume there are n steps in the tree.

We define i and j as the number of price and time steps until the considered node (j, i) . As a consequence, the number of paths that lead to the node is $\binom{j}{i}$ with probability $\binom{j}{i} p_u^i p_d^{j-i}$. At each time point j , the random variable S_n (describing the stock price) takes value on $\{S_0 u^j, S_0 u^{j-1} d, \dots, S_0 d^j\}$. The discrete stock price process follows

$$S_n = S_0 \prod_{i=1}^n \xi_i$$

where ξ_i are iid random variables defined as follows:

$$\xi_i = \begin{cases} u & \text{with probability } q_u \\ d & \text{with probability } q_d \end{cases}$$

A simple approach uses backward induction to price the option with a binomial tree.

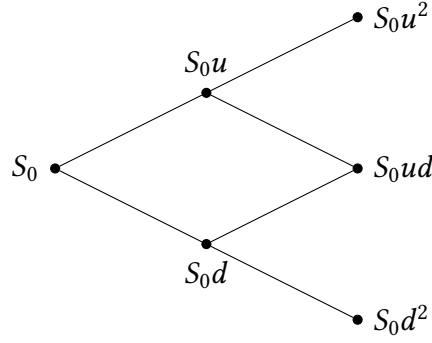


Figure 18: A two-step Binomial Tree

Theorem 25.1: Backward Induction

Consider $V^j = e^{-r_d(n-j)\Delta t} \mathbb{E}[g(S_n) | \mathcal{F}_j]$, the time- j fair value of the European call option.

We have the following recursive algorithm:

$$V^j = \begin{cases} g(S_n) & j = n \\ e^{-r_d\Delta t} \mathbb{E}[V^{j+1} | \mathcal{F}_j] & j \leq n-1 \end{cases}$$

The up and down factors u and d , along with their corresponding probabilities, must match the first two moments of the price distribution. There are only two equations and three parameters to determine (knowing one probability p_u also makes it possible to know $p_d = 1 - p_u$). Cox-Ross-Rubinstein (CRR) set the additional equation to $ud = 1$. The CRR model has parameters:

$$u = e^{\sigma\sqrt{\Delta t}} \quad (25.1)$$

$$d = e^{-\sigma\sqrt{\Delta t}} \quad (25.2)$$

$$p_u = \frac{e^{(r-q)\Delta t} - d}{u - d} \quad (25.3)$$

Under this configuration and with the previous theorem, we define the following recursive formula:

$$V_i^j = e^{-r_d\Delta t} \left[p_u V_{i+1}^{j+1} + p_d V_i^{j+1} \right] \quad (25.4)$$

At the end of the recursive algorithm, we obtain the time-zero option value.

In the standard CRR tree, the local volatility is constant for each time step and converges to the input volatility as n becomes large:

$$\sigma_i^j = \frac{1}{\sqrt{\Delta t}} \sqrt{p_u p_d} \ln u^2 \quad (25.5)$$

Theorem 25.2: Closed-Form Binomial Price for Vanilla Option

The price of a Vanilla option for an $n \in \mathbb{N}$ step tree is given by,

$$V = e^{-r_d n \Delta t} \left[\sum_{j=0}^n \binom{n}{j} p_u^j p_d^{n-j} V^j \right].$$

Attention 25.1: O

e remark to make with that CRR model is that in some cases where $\sigma < |(r-q)/\sqrt{\Delta t}|$, negative probabilities can occur. First, it is inconsistent from a probability theory perspective, but additionally, it may prevent the model from capturing relevant events.

Proposition 25.1: Cox-Ross-Rubinstein Valuation

For every $m = 1, 2, \dots, T$, the arbitrage price of a European call option at time $t = T - m$ is given by the Cox-Ross-Rubinstein valuation formula

$$C_{T-m} = S_{T-m} \sum_{j=a}^m \binom{m}{j} \bar{p}^j (1 - \bar{p})^{m-j} - \frac{K}{\hat{r}^m} \sum_{j=a}^m \binom{m}{j} p_*^j (1 - p_*)^{m-j}, \quad (25.6)$$

where $a = a_m(S_{T-m})$, $p_* = (\hat{r} - d)/(u - d)$ and $\bar{p} = p_* u / \hat{r}$. At time $t = T - m - 1$, the unique replicating strategy $\phi_{T-m-1} = (\alpha_{T-m-1}, \beta_{T-m-1})$ equals

$$\begin{aligned} \alpha_{T-m-1} &= \sum_{j=a^d}^m \binom{m}{j} \bar{p}^j (1 - \bar{p})^{m-j} + \frac{\delta \Delta_m(u S_{T-m-1}, a^u)}{S_{T-m-1}(u - d)}, \\ \beta_{T-m-1} &= -\frac{K}{\hat{r}^{m+1}} \sum_{j=a^d}^m \binom{m}{j} p_*^j (1 - p_*)^{m-j} - \frac{\delta d \Delta_m(u S_{T-m-1}, a^u)}{\hat{r}(u - d)}, \end{aligned}$$

where $a^d = a_m(d S_{T-m-1})$, $a^u = a_m(u S_{T-m-1})$ and $\delta = a^d - a^u$.

26 Finite Difference Methods

27 Contract Sensitivities

CONSIDER a model that depends on a parameter θ ranging over some interval of the real line. Suppose that for each value of θ we have a mechanism for generating a random variable $Y(\theta)$, representing the output of the model at parameter θ . Let

$$\alpha(\theta) = \mathbb{E}[Y(\theta)]$$

The derivative estimation problem consists of finding a way to estimate $\alpha'(\theta)$, the derivative of α with respect to θ .

In the application to option pricing, $Y(\theta)$ is the discounted payoff of an option, $\alpha(\theta)$ is its price, and θ could be any of the many model or market parameters that influence the price. When θ is the initial price of an underlying asset, then $\alpha'(\theta)$ is the option's delta (with respect to that asset). The second derivative $\alpha''(\theta)$ is the option's gamma. When θ is a volatility parameter, $\alpha'(\theta)$ is often called "vega." For interest rate derivatives, the sensitivities of prices to the initial term structure (as represented by, e.g., a yield curve or forward curve) are important.

27.1 Bias and Variance

An obvious approach to derivative estimation proceeds as follows. Simulate independent replications $Y_1(\theta), \dots, Y_n(\theta)$ of the model at parameter θ and n additional replications $Y_1(\theta+h), \dots, Y_n(\theta+h)$ at $\theta+h$, for some $h > 0$. Average each set of replications to get $\bar{Y}_n(\theta)$ and $\bar{Y}_n(\theta+h)$ and form the forward-difference estimator.

$$\hat{\Delta}_F = \hat{\Delta}_F(n, h) = \frac{\bar{Y}_n(\theta+h) - \bar{Y}_n(\theta)}{h} \quad (27.1)$$

This estimator has an expectation.

$$\mathbb{E}[\hat{\Delta}_F] = h^{-1}[\alpha(\theta+h) - \alpha(\theta)]. \quad (27.2)$$

We have not specified what relation, if any, holds between the outcomes $Y_i(\theta)$ and $Y_i(\theta + h)$; this will be important when we consider variance, but Equation (27.19) is purely a property of the marginal distributions of the outcomes at θ and $\theta + h$.

If α is twice differentiable at θ , then

$$\alpha(\theta + h) = \alpha(\theta) + \alpha'(\theta)h + \frac{1}{2}\alpha''(\theta)h^2 + O(h^2).$$

In this case, it follows from Equation (27.19) that the bias in the forward-difference estimator is

$$\text{Bias}(\hat{\Delta}_F) = \mathbb{E}[\hat{\Delta}_F - \alpha'(\theta)] = \frac{1}{2}\alpha''(\theta)h + O(h). \quad (27.3)$$

By simulating at $\theta - h$ and $\theta + h$, we can form a central-difference estimator

$$\hat{\Delta}_C \equiv \hat{\Delta}_C(n, h) = \frac{\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta - h)}{2h}. \quad (27.4)$$

This estimator is often more costly than the forward-difference estimator in the following sense. If we are ultimately interested in estimating $\alpha(\theta)$ and $\alpha'(\theta)$, then we would ordinarily simulate at θ to estimate $\alpha(\theta)$, so the forward-difference estimator requires simulating at just one additional point $\theta + h$. Conversely, the central-difference estimator requires simulating at two additional points. However, this additional computational effort improves the convergence rate of the bias. If α is at least twice differentiable in a neighbourhood of θ , then

$$\alpha(\theta + h) = \alpha(\theta) + \alpha'(\theta)h + \alpha''(\theta)h^2/2 + O(h^2)$$

$$\alpha(\theta - h) = \alpha(\theta) - \alpha'(\theta)h + \alpha''(\theta)h^2/2 + O(h^2)$$

so subtraction eliminates the second-order terms, leading

$$\text{Bias}(\hat{\Delta}_C) = \frac{\alpha(\theta + h) - \alpha(\theta - h)}{2h} - \alpha'(\theta) = O(h), \quad (27.5)$$

which is of smaller order than Equation (27.20). If α'' is itself differentiable at θ , we can refine Equation (27.22) to

$$\text{Bias}(\hat{\Delta}_C) = \frac{1}{6}\alpha'''(\theta)h^2 + O(h^2). \quad (27.6)$$

27.1.0.1 Variance The form of the bias of the forward- and central-difference estimators would lead us to take ever smaller values of h to improve accuracy, at least if we ignore the limits of machine precision. But the effect of h on bias must be weighed against its effect on variance.

The variance of the forward-difference estimator Equation (27.18) is

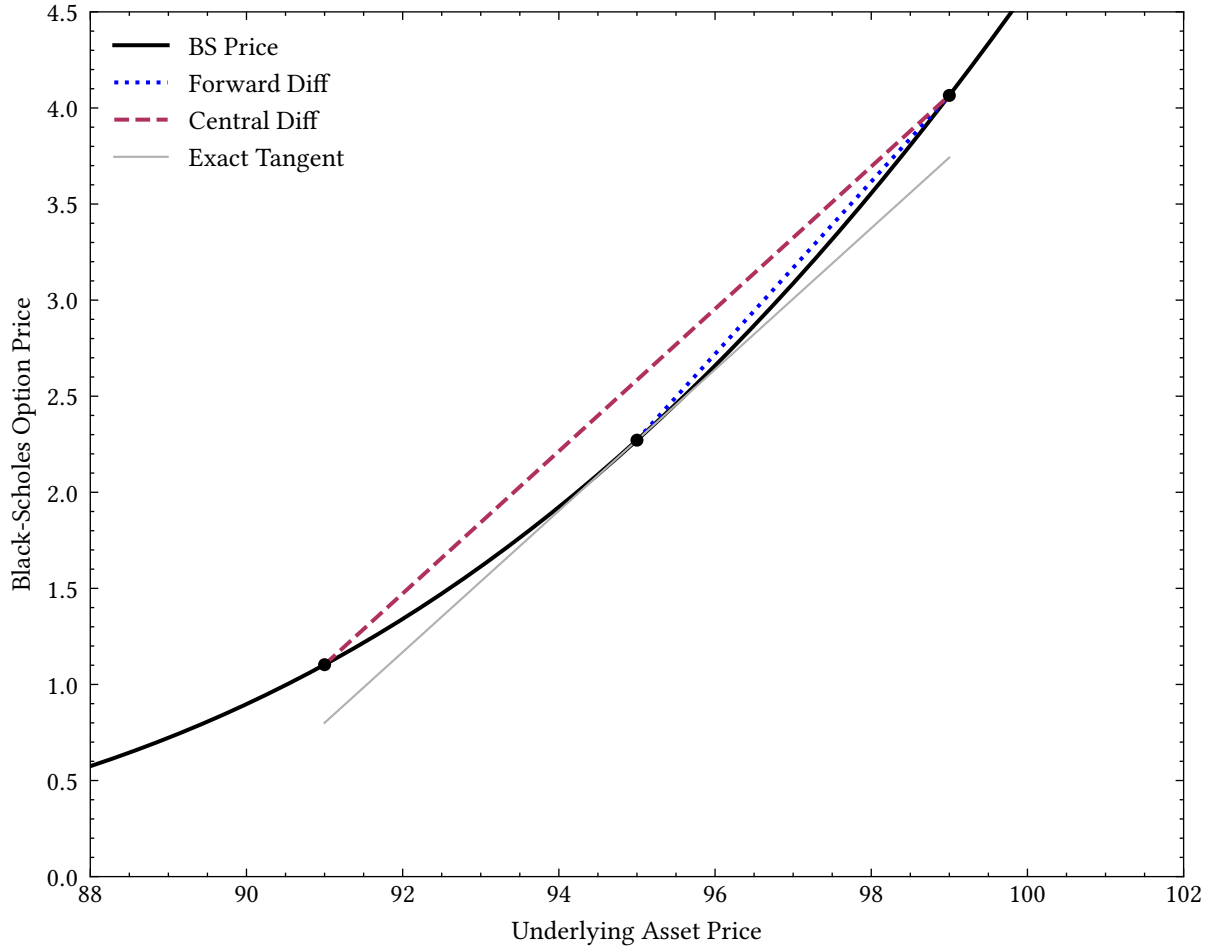


Figure 19: Comparison of central-difference approximations (dashed) and forward-difference approximation (dotted) with exact tangent value to the Black-Scholes formula.

$$\text{Var}[\hat{\Delta}_F(n, h)] = h^{-2} \text{Var}[\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)] \quad (27.7)$$

and a corresponding expression holds for the central-difference estimator Equation (27.21). In both cases, the factor h^{-2} alerts us to possibly disastrous consequences of taking h to be very small. Equation (27.24) also makes clear that the dependence between values simulated at different values of θ affects the variance of a finite-difference estimator.

Suppose, for simplicity, that the pairs $(Y(\theta), Y(\theta + h))$ and $(Y_i(\theta), Y_i(\theta + h))$, $i = 1, 2, \dots$, are i.i.d. so that

$$\text{Var}[\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)] = \frac{1}{n} \text{Var}[Y(\theta + h) - Y(\theta)].$$

How the variance in Equation (27.24) changes with h is determined by the dependence of $\text{Var}[Y(\theta + h) - Y(\theta)]$ on h .

Three cases of primary importance arise in practice:

$$\text{Var}[Y(\theta + h) - Y(\theta)] = \begin{cases} \mathcal{O}(1), & \text{Case (i)} \\ \mathcal{O}(h), & \text{Case (ii)} \\ \mathcal{O}(h^2), & \text{Case (iii).} \end{cases} \quad (27.8)$$

Case (i) applies if we simulate $Y(\theta)$ and $Y(\theta + h)$ independently; for in this case we have

$$\text{Var}[Y(\theta + h) - Y(\theta)] = \text{Var}[Y(\theta + h)] + \text{Var}[Y(\theta)] \rightarrow 2 \text{Var}[Y(\theta)],$$

under the minor assumption that $\text{Var}[Y(\theta)]$ is continuous in θ . Case (ii) is the typical consequence of simulating $Y(\theta + h)$ and $Y(\theta)$ using *common random numbers*; i.e., generating them from the same sequence $U_1, U_2, \dots$ of $\text{Unif}[0, 1]$ random variables. (In practice, this is accomplished by starting the simulations at θ and $\theta + h$ with the same seed for the random number generator). For Case (iii) to hold, we generally need not only that $Y(\theta)$ and $Y(\theta + h)$ use the same random numbers, but also that for (almost) all values of the random numbers, the output $Y(\cdot)$ is continuous in the input θ .

27.2 Optimal Mean Square Error

Because decreasing h can increase variance while decreasing bias, minimizing mean square error (MSE) requires balancing these two considerations. Increasing the number of replications n decreases variance with no effect on bias, whereas h affects both bias and variance; our objective is to find the optimal relation between the two.

Consider the forward-difference estimator with independent simulation at θ and $\theta + h$. We can reasonably assume that Case (i) of Equation (27.25) applies, and for emphasis we denote this estimator by $\hat{\Delta}_{F,i} = \hat{\Delta}_{F,i}(n, h)$. Squaring the bias in Equation (27.20) and adding it to the variance in Equation (27.24), we get

$$\text{MSE}(\hat{\Delta}_{F,i}(n, h)) = O(h^2) + O(n^{-1}h^{-2}),$$

from which we see that minimal conditions for convergence are $h \rightarrow 0$ and $nh^2 \rightarrow \infty$.

To derive a more precise conclusion, we strengthen Cases (i) and (ii) of Equation (27.25). We have four estimators to consider: forward-difference and central-difference using independent sampling or common random numbers at different values of θ . We can give a unified treatment of these cases by considering a generic estimator $\hat{\Delta} = \hat{\Delta}(n, h)$ for which

$$\mathbb{E}[\hat{\Delta} - \alpha'(\theta)] = bh^\beta + o(h^\beta), \quad \text{Var}[\hat{\Delta}] = \frac{\sigma^2}{nh^\eta} + o(h^{-\eta}), \quad (27.9)$$

for some positive β, η, σ , and some nonzero b . Forward- and central-difference estimators typically have $\beta = 1$ and $\beta = 2$; taking $\eta = 2$ sharpens Case (i) of Equation (27.25) and taking $\eta = 1$ sharpens Case (ii).

Consider a sequence of estimators $\hat{\Delta}(n, h_n)$ with

$$h_n = h_* n^{-\gamma} \quad (27.10)$$

for some positive h_* and γ . From our assumptions on bias and variance, we get

$$\text{MSE}(\hat{\Delta}) = b^2 h_n^{2\beta} + \frac{\sigma^2}{nh_n^\eta}, \quad (27.11)$$

up to terms that are of higher order in h_n . The value of γ that maximizes the rate of decrease of the MSE is $\gamma = 1/(2\beta + \eta)$, from which we draw the natural implication that with a smaller bias (larger β), we can use a larger increment h . If we substitute this value of γ into Equation (27.28) and take the square root, we find that

$$\text{RMSE}(\hat{\Delta}) = \mathcal{O}\left(n^{-\frac{\beta}{2\beta+\eta}}\right), \quad (27.12)$$

and this is a reasonable measure of the convergence rate of the estimator.

Taking this analysis one step further, we find that

$$n^{2\beta/(2\beta+\eta)} \text{MSE}(\hat{\Delta}) \rightarrow b^2 h_*^{2\beta} + \sigma^2 h_*^{-\eta}; \quad (27.13)$$

minimizing over h_* yields an optimal value of

$$h_* = \left(\frac{\eta \sigma^2}{2\beta b^2} \right)^{\frac{1}{2\beta+\eta}}. \quad (27.14)$$

The results of this analysis are summarized in Table 2 for the forward- and central-difference estimators using independent sampling or common random numbers. The variance and bias columns display leading terms only and abbreviate the more complete expressions in Equation (27.26). The table should be understood as follows: if the leading terms of the variance and bias are as indicated in the second and third columns, then the conclusions in the last three columns hold. The requirement that b not equal zero in Equation (27.26) translates to $\alpha''(\theta) \neq 0$ and $\alpha'''(\theta) \neq 0$ for the forward and central estimators, respectively. The results in the table indicate that, at least asymptotically, $\hat{\Delta}_{C,ii}$ dominates the other three estimators because it exhibits the fastest convergence.

Table 1: Convergence rates of finite-difference estimators with optimal increment h_n . The estimators use either forward (F) or central (C) differences and either independent sampling (i) or common random numbers (ii).

Estimator	Variance	Bias	Optimal h_n	Convergence	h_*
$\hat{\Delta}_{F,i}$	$\frac{\sigma_{F,i}^2}{nh^2}$	$\frac{1}{2}\alpha''(\theta)h$	$O(n^{-1/4})$	$O(n^{-1/4})$	$\left(\frac{4\sigma_{F,i}^2}{\alpha''(\theta)^2}\right)^{1/4}$
$\hat{\Delta}_{C,i}$	$\frac{\sigma_{C,i}^2}{nh^2}$	$\frac{1}{6}\alpha'''(\theta)h^2$	$O(n^{-1/6})$	$O(n^{-1/3})$	$\left(\frac{18\sigma_{C,i}^2}{\alpha'''(\theta)^2}\right)^{1/6}$
$\hat{\Delta}_{F,ii}$	$\frac{\sigma_{F,ii}^2}{nh}$	$\frac{1}{2}\alpha''(\theta)h$	$O(n^{-1/3})$	$O(n^{-1/3})$	$\left(\frac{2\sigma_{F,ii}^2}{\alpha''(\theta)^2}\right)^{1/3}$
$\hat{\Delta}_{C,ii}$	$\frac{\sigma_{C,ii}^2}{nh}$	$\frac{1}{6}\alpha'''(\theta)h^2$	$O(n^{-1/5})$	$O(n^{-2/5})$	$\left(\frac{9\sigma_{C,ii}^2}{\alpha'''(\theta)^2}\right)^{1/5}$

Glynn [153] proves a central limit theorem for each of the cases in Table 2. These results take the form

$$n^{\frac{\beta}{2\beta+\eta}} [\hat{\Delta}(n, h_n) - \alpha'(\theta)] \Rightarrow \mathcal{N}\left(bh_*^\beta, \frac{\sigma^2}{h_*^\eta}\right) \quad (27.15)$$

with h_n as in Equation (27.27) and $\gamma = 1/(2\beta + \eta)$. The limit holds for any $h_* > 0$; the optimal values of h_* in Table 2 minimize the second moment of the limiting normal random variable.

For the forward-difference estimator with independent samples, the variance parameter $\sigma_{F,i}^2$ is given by

$$\sigma_{F,i}^2 = \lim_{h \rightarrow 0} (\text{Var}[Y(\theta + h)] + \text{Var}[Y(\theta)]) = 2 \text{Var}[Y(\theta)] \quad (27.16)$$

under the minimal assumption of continuity of $\text{Var}[Y(\theta)]$. Because the central-difference estimator has a denominator of $2h$, $\sigma_{C,i}^2 = \text{Var}[Y(\theta)]/2$. The parameters $\sigma_{F,ii}^2$ and $\sigma_{C,ii}^2$ do not admit a simple description. Their values depend on the joint distribution of $(Y(\theta -$

h), $Y(\theta)$, $Y(\theta + h)$), which depends in part on the particular algorithm used for simulation — different algorithms may respond differently to changes in an input parameter with the random numbers held fixed. In contrast, $\text{Var}[Y(\theta)]$ is determined by the marginal distribution of $Y(\theta)$ so all algorithms that legitimately sample from this distribution produce the same variance. The variance of the finite-difference estimators appear in the optimal values of h_* ; though these are unlikely to be known in advance, they can be estimated from preliminary runs and potentially combined with rough estimates of the derivatives of α to approximate h_* .

Case (iii) of Equation (27.25) is not reflected in Table 2. When it applies, the mean square error takes the form

$$\text{MSE}(\hat{\Delta}) = b^2 h_n^{2\beta} + \frac{\sigma^2}{n}, \quad (27.17)$$

and there is no tradeoff between bias and variance. We should take h_n as small as possible, and so long as $nh_n^{2\beta}$ is bounded, the RMSE is $O(n^{-1/2})$. This dominates all the convergence rates in Table 2.

The distinction between Cases (ii) and (iii) is illustrated in Figure 7.2. The figure compares RMS relative errors of forward-difference estimators of delta for a standard call option paying $(S(T) - K)^+$ and a digital option paying $\mathbb{1}\{S(T) > K\}$, with model parameters $K = S(0) = 100$, $\sigma = 0.30$, $r = 0.05$, and $T = 0.25$. The forward-difference estimators use common random numbers, which in this example simply means using the same draw from the normal distribution to generate $S(T)$ from both $S(0)$ and $S(0) + h$. This example is simple enough to allow exact calculation of the RMSE. To simplify comparison, in the figure we divide each RMSE by the true value of delta to get a relative error. The figure shows the effect of varying h with the number of replications n fixed at 5000.

The standard call option fits in Case (iii) of Equation (27.25). As expected, its RMS relative error decreases with decreasing h . The digital option fits in Case (ii), so its relative error explodes as h approaches zero. From the figure, we see that the relative error for the digital option is minimised at a surprisingly large value of about 4. The figure also shows that the cost of taking h too large is much smaller than the cost of taking it too small.

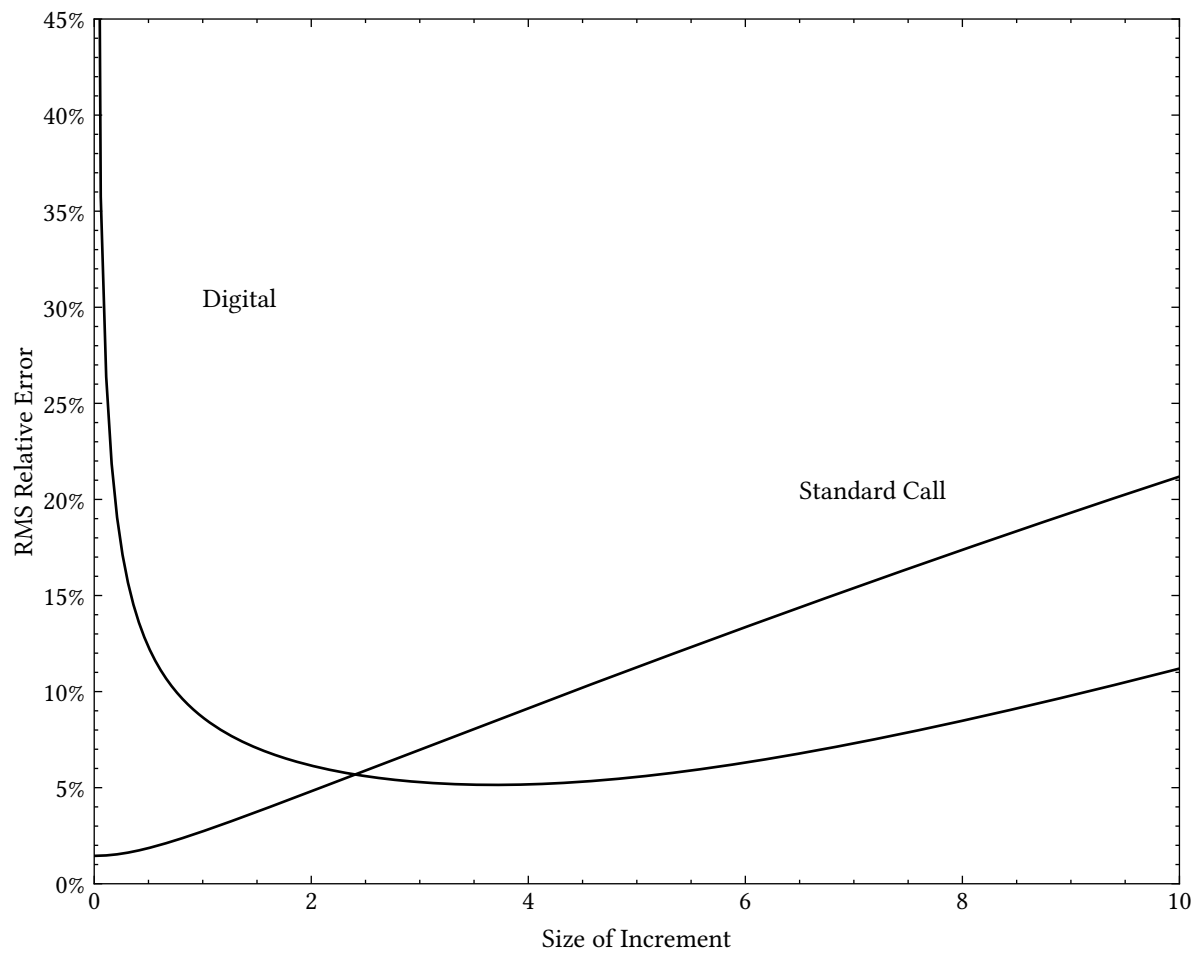


Figure 20: RMS relative errors in forward-difference delta estimates for a standard call and a digital option as a function of the increment h , with $n = 5000$

27.3 Likelihood Ratio

27.4 Pathwise Derivatives

27.5 Finite-Difference Approximations

CONSIDER a model that depends on a parameter θ ranging over some interval of the real line. Suppose that for each value of θ we have a mechanism for generating a random variable $Y(\theta)$, representing the output of the model at parameter θ . Let

$$\alpha(\theta) = \mathbb{E}[Y(\theta)]$$

The derivative estimation problem consists of finding a way to estimate $\alpha'(\theta)$, the derivative of α with respect to θ .

In the application to option pricing, $Y(\theta)$ is the discounted payoff of an option, $\alpha(\theta)$ is its price, and θ could be any of the many model or market parameters that influence the price. When θ is the initial price of an underlying asset, then $\alpha'(\theta)$ is the option's delta (with respect to that asset). The second derivative $\alpha''(\theta)$ is the option's gamma. When θ is a volatility parameter, $\alpha'(\theta)$ is often called "vega." For interest rate derivatives, the sensitivities of prices to the initial term structure (as represented by, e.g., a yield curve or forward curve) are important.

27.6 Bias and Variance

An obvious approach to derivative estimation proceeds as follows. Simulate independent replications $Y_1(\theta), \dots, Y_n(\theta)$ of the model at parameter θ and n additional replications $Y_1(\theta+h), \dots, Y_n(\theta+h)$ at $\theta+h$, for some $h > 0$. Average each set of replications to get $\bar{Y}_n(\theta)$ and $\bar{Y}_n(\theta+h)$ and form the forward-difference estimator.

$$\hat{\Delta}_F = \hat{\Delta}_F(n, h) = \frac{\bar{Y}_n(\theta+h) - \bar{Y}_n(\theta)}{h} \quad (27.18)$$

This estimator has an expectation.

$$\mathbb{E}[\hat{\Delta}_F] = h^{-1}[\alpha(\theta+h) - \alpha(\theta)]. \quad (27.19)$$

We have not specified what relation, if any, holds between the outcomes $Y_i(\theta)$ and $Y_i(\theta + h)$; this will be important when we consider variance, but Equation (27.19) is purely a property of the marginal distributions of the outcomes at θ and $\theta + h$.

If α is twice differentiable at θ , then

$$\alpha(\theta + h) = \alpha(\theta) + \alpha'(\theta)h + \frac{1}{2}\alpha''(\theta)h^2 + O(h^2).$$

In this case, it follows from Equation (27.19) that the bias in the forward-difference estimator is

$$\text{Bias}(\hat{\Delta}_F) = \mathbb{E}[\hat{\Delta}_F - \alpha'(\theta)] = \frac{1}{2}\alpha''(\theta)h + O(h). \quad (27.20)$$

By simulating at $\theta - h$ and $\theta + h$, we can form a central-difference estimator

$$\hat{\Delta}_C \equiv \hat{\Delta}_C(n, h) = \frac{\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta - h)}{2h}. \quad (27.21)$$

This estimator is often more costly than the forward-difference estimator in the following sense. If we are ultimately interested in estimating $\alpha(\theta)$ and $\alpha'(\theta)$, then we would ordinarily simulate at θ to estimate $\alpha(\theta)$, so the forward-difference estimator requires simulating at just one additional point $\theta + h$. Conversely, the central-difference estimator requires simulating at two additional points. However, this additional computational effort improves the convergence rate of the bias. If α is at least twice differentiable in a neighbourhood of θ , then

$$\alpha(\theta + h) = \alpha(\theta) + \alpha'(\theta)h + \alpha''(\theta)h^2/2 + O(h^2)$$

$$\alpha(\theta - h) = \alpha(\theta) - \alpha'(\theta)h + \alpha''(\theta)h^2/2 + O(h^2)$$

so subtraction eliminates the second-order terms, leading

$$\text{Bias}(\hat{\Delta}_C) = \frac{\alpha(\theta + h) - \alpha(\theta - h)}{2h} - \alpha'(\theta) = O(h), \quad (27.22)$$

which is of smaller order than Equation (27.20). If α'' is itself differentiable at θ , we can refine Equation (27.22) to

$$\text{Bias}(\hat{\Delta}_C) = \frac{1}{6}\alpha'''(\theta)h^2 + O(h^2). \quad (27.23)$$

27.6.0.1 Variance The form of the bias of the forward- and central-difference estimators would lead us to take ever smaller values of h to improve accuracy, at least if we ignore the limits of machine precision. But the effect of h on bias must be weighed against its effect on variance.

The variance of the forward-difference estimator Equation (27.18) is

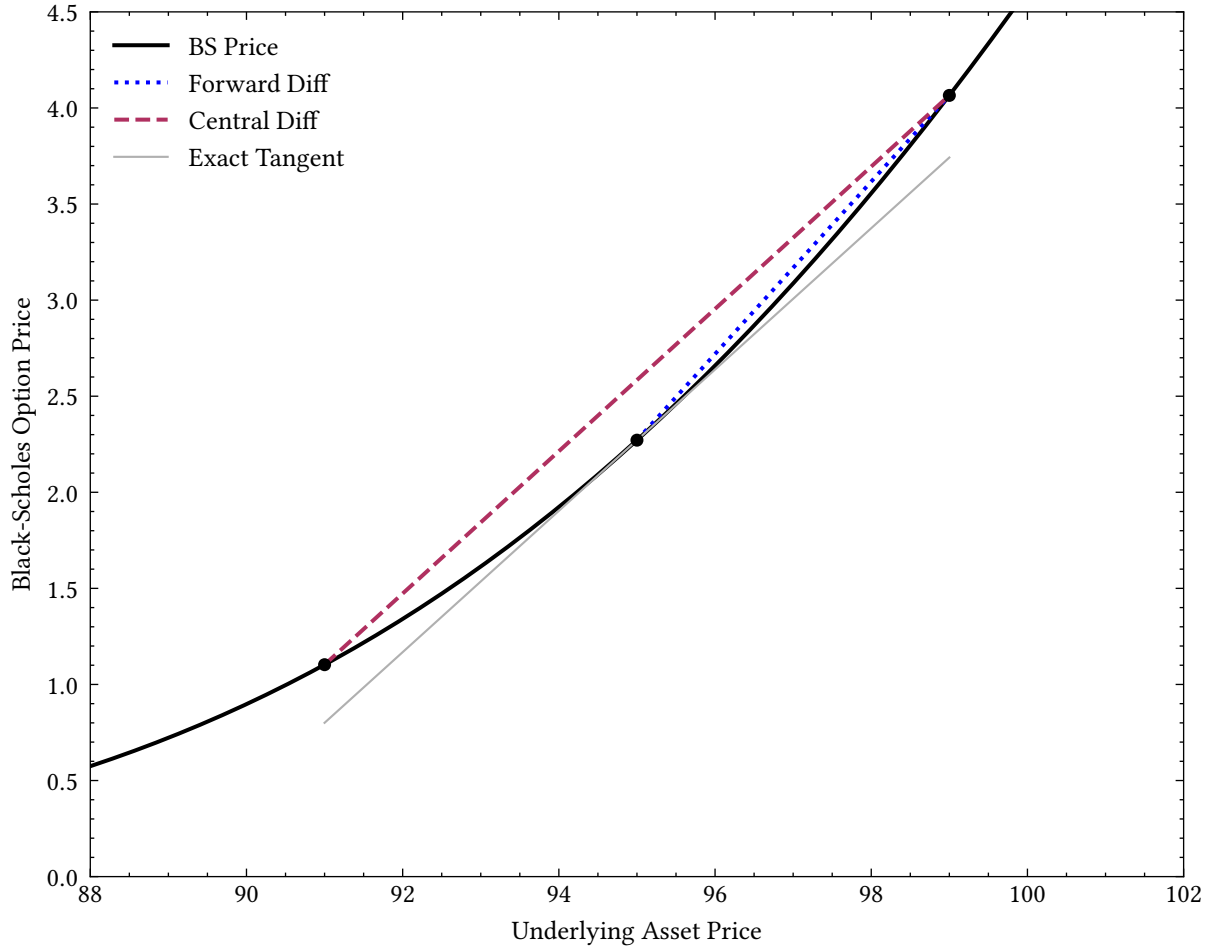


Figure 21: Comparison of central-difference approximations (dashed) and forward-difference approximation (dotted) with exact tangent value to the Black-Scholes formula.

$$\text{Var}[\hat{\Delta}_F(n, h)] = h^{-2} \text{Var}[\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)] \quad (27.24)$$

and a corresponding expression holds for the central-difference estimator Equation (27.21). In both cases, the factor h^{-2} alerts us to possibly disastrous consequences of taking h to be very small. Equation (27.24) also makes clear that the dependence between values simulated at different values of θ affects the variance of a finite-difference estimator.

Suppose, for simplicity, that the pairs $(Y(\theta), Y(\theta + h))$ and $(Y_i(\theta), Y_i(\theta + h))$, $i = 1, 2, \dots$, are i.i.d. so that

$$\text{Var}[\bar{Y}_n(\theta + h) - \bar{Y}_n(\theta)] = \frac{1}{n} \text{Var}[Y(\theta + h) - Y(\theta)].$$

How the variance in Equation (27.24) changes with h is determined by the dependence of $\text{Var}[Y(\theta + h) - Y(\theta)]$ on h .

Three cases of primary importance arise in practice:

$$\text{Var}[Y(\theta + h) - Y(\theta)] = \begin{cases} \mathcal{O}(1), & \text{Case (i)} \\ \mathcal{O}(h), & \text{Case (ii)} \\ \mathcal{O}(h^2), & \text{Case (iii).} \end{cases} \quad (27.25)$$

Case (i) applies if we simulate $Y(\theta)$ and $Y(\theta + h)$ independently; for in this case we have

$$\text{Var}[Y(\theta + h) - Y(\theta)] = \text{Var}[Y(\theta + h)] + \text{Var}[Y(\theta)] \rightarrow 2 \text{Var}[Y(\theta)],$$

under the minor assumption that $\text{Var}[Y(\theta)]$ is continuous in θ . Case (ii) is the typical consequence of simulating $Y(\theta + h)$ and $Y(\theta)$ using *common random numbers*; i.e., generating them from the same sequence $U_1, U_2, \dots$ of $\text{Unif}[0, 1]$ random variables. (In practice, this is accomplished by starting the simulations at θ and $\theta + h$ with the same seed for the random number generator). For Case (iii) to hold, we generally need not only that $Y(\theta)$ and $Y(\theta + h)$ use the same random numbers, but also that for (almost) all values of the random numbers, the output $Y(\cdot)$ is continuous in the input θ .

27.7 Optimal Mean Square Error

Because decreasing h can increase variance while decreasing bias, minimizing mean square error (MSE) requires balancing these two considerations. Increasing the number of replications n decreases variance with no effect on bias, whereas h affects both bias and variance; our objective is to find the optimal relation between the two.

Consider the forward-difference estimator with independent simulation at θ and $\theta + h$. We can reasonably assume that Case (i) of Equation (27.25) applies, and for emphasis we denote this estimator by $\hat{\Delta}_{F,i} = \hat{\Delta}_{F,i}(n, h)$. Squaring the bias in Equation (27.20) and adding it to the variance in Equation (27.24), we get

$$\text{MSE}(\hat{\Delta}_{F,i}(n, h)) = O(h^2) + O(n^{-1}h^{-2}),$$

from which we see that minimal conditions for convergence are $h \rightarrow 0$ and $nh^2 \rightarrow \infty$.

To derive a more precise conclusion, we strengthen Cases (i) and (ii) of Equation (27.25). We have four estimators to consider: forward-difference and central-difference using independent sampling or common random numbers at different values of θ . We can give a unified treatment of these cases by considering a generic estimator $\hat{\Delta} = \hat{\Delta}(n, h)$ for which

$$\mathbb{E}[\hat{\Delta} - \alpha'(\theta)] = bh^\beta + o(h^\beta), \quad \text{Var}[\hat{\Delta}] = \frac{\sigma^2}{nh^\eta} + o(h^{-\eta}), \quad (27.26)$$

for some positive β, η, σ , and some nonzero b . Forward- and central-difference estimators typically have $\beta = 1$ and $\beta = 2$; taking $\eta = 2$ sharpens Case (i) of Equation (27.25) and taking $\eta = 1$ sharpens Case (ii).

Consider a sequence of estimators $\hat{\Delta}(n, h_n)$ with

$$h_n = h_* n^{-\gamma} \quad (27.27)$$

for some positive h_* and γ . From our assumptions on bias and variance, we get

$$\text{MSE}(\hat{\Delta}) = b^2 h_n^{2\beta} + \frac{\sigma^2}{nh_n^\eta}, \quad (27.28)$$

up to terms that are of higher order in h_n . The value of γ that maximizes the rate of decrease of the MSE is $\gamma = 1/(2\beta + \eta)$, from which we draw the natural implication that with a smaller bias (larger β), we can use a larger increment h . If we substitute this value of γ into Equation (27.28) and take the square root, we find that

$$\text{RMSE}(\hat{\Delta}) = \mathcal{O}\left(n^{-\frac{\beta}{2\beta+\eta}}\right), \quad (27.29)$$

and this is a reasonable measure of the convergence rate of the estimator.

Taking this analysis one step further, we find that

$$n^{2\beta/(2\beta+\eta)} \text{MSE}(\hat{\Delta}) \rightarrow b^2 h_*^{2\beta} + \sigma^2 h_*^{-\eta}; \quad (27.30)$$

minimizing over h_* yields an optimal value of

$$h_* = \left(\frac{\eta \sigma^2}{2\beta b^2} \right)^{\frac{1}{2\beta+\eta}}. \quad (27.31)$$

The results of this analysis are summarized in Table 2 for the forward- and central-difference estimators using independent sampling or common random numbers. The variance and bias columns display leading terms only and abbreviate the more complete expressions in Equation (27.26). The table should be understood as follows: if the leading terms of the variance and bias are as indicated in the second and third columns, then the conclusions in the last three columns hold. The requirement that b not equal zero in Equation (27.26) translates to $\alpha''(\theta) \neq 0$ and $\alpha'''(\theta) \neq 0$ for the forward and central estimators, respectively. The results in the table indicate that, at least asymptotically, $\hat{\Delta}_{C,ii}$ dominates the other three estimators because it exhibits the fastest convergence.

Table 2: Convergence rates of finite-difference estimators with optimal increment h_n . The estimators use either forward (F) or central (C) differences and either independent sampling (i) or common random numbers (ii).

Estimator	Variance	Bias	Optimal h_n	Convergence	h_*
$\hat{\Delta}_{F,i}$	$\frac{\sigma_{F,i}^2}{nh^2}$	$\frac{1}{2}\alpha''(\theta)h$	$O(n^{-1/4})$	$O(n^{-1/4})$	$\left(\frac{4\sigma_{F,i}^2}{\alpha''(\theta)^2}\right)^{1/4}$
$\hat{\Delta}_{C,i}$	$\frac{\sigma_{C,i}^2}{nh^2}$	$\frac{1}{6}\alpha'''(\theta)h^2$	$O(n^{-1/6})$	$O(n^{-1/3})$	$\left(\frac{18\sigma_{C,i}^2}{\alpha'''(\theta)^2}\right)^{1/6}$
$\hat{\Delta}_{F,ii}$	$\frac{\sigma_{F,ii}^2}{nh}$	$\frac{1}{2}\alpha''(\theta)h$	$O(n^{-1/3})$	$O(n^{-1/3})$	$\left(\frac{2\sigma_{F,ii}^2}{\alpha''(\theta)^2}\right)^{1/3}$
$\hat{\Delta}_{C,ii}$	$\frac{\sigma_{C,ii}^2}{nh}$	$\frac{1}{6}\alpha'''(\theta)h^2$	$O(n^{-1/5})$	$O(n^{-2/5})$	$\left(\frac{9\sigma_{C,ii}^2}{\alpha'''(\theta)^2}\right)^{1/5}$

Glynn [153] proves a central limit theorem for each of the cases in Table 2. These results take the form

$$n^{\frac{\beta}{2\beta+\eta}} [\hat{\Delta}(n, h_n) - \alpha'(\theta)] \Rightarrow \mathcal{N}\left(bh_*^\beta, \frac{\sigma^2}{h_*^\eta}\right) \quad (27.32)$$

with h_n as in Equation (27.27) and $\gamma = 1/(2\beta + \eta)$. The limit holds for any $h_* > 0$; the optimal values of h_* in Table 2 minimize the second moment of the limiting normal random variable.

For the forward-difference estimator with independent samples, the variance parameter $\sigma_{F,i}^2$ is given by

$$\sigma_{F,i}^2 = \lim_{h \rightarrow 0} (\text{Var}[Y(\theta + h)] + \text{Var}[Y(\theta)]) = 2 \text{Var}[Y(\theta)] \quad (27.33)$$

under the minimal assumption of continuity of $\text{Var}[Y(\theta)]$. Because the central-difference estimator has a denominator of $2h$, $\sigma_{C,i}^2 = \text{Var}[Y(\theta)]/2$. The parameters $\sigma_{F,ii}^2$ and $\sigma_{C,ii}^2$ do not admit a simple description. Their values depend on the joint distribution of $(Y(\theta -$

h), $Y(\theta)$, $Y(\theta + h)$), which depends in part on the particular algorithm used for simulation — different algorithms may respond differently to changes in an input parameter with the random numbers held fixed. In contrast, $\text{Var}[Y(\theta)]$ is determined by the marginal distribution of $Y(\theta)$ so all algorithms that legitimately sample from this distribution produce the same variance. The variance of the finite-difference estimators appear in the optimal values of h_* ; though these are unlikely to be known in advance, they can be estimated from preliminary runs and potentially combined with rough estimates of the derivatives of α to approximate h_* .

Case (iii) of Equation (27.25) is not reflected in Table 2. When it applies, the mean square error takes the form

$$\text{MSE}(\hat{\Delta}) = b^2 h_n^{2\beta} + \frac{\sigma^2}{n}, \quad (27.34)$$

and there is no tradeoff between bias and variance. We should take h_n as small as possible, and so long as $nh_n^{2\beta}$ is bounded, the RMSE is $O(n^{-1/2})$. This dominates all the convergence rates in Table 2.

The distinction between Cases (ii) and (iii) is illustrated in Figure 7.2. The figure compares RMS relative errors of forward-difference estimators of delta for a standard call option paying $(S(T) - K)^+$ and a digital option paying $\mathbb{1}\{S(T) > K\}$, with model parameters $K = S(0) = 100$, $\sigma = 0.30$, $r = 0.05$, and $T = 0.25$. The forward-difference estimators use common random numbers, which in this example simply means using the same draw from the normal distribution to generate $S(T)$ from both $S(0)$ and $S(0) + h$. This example is simple enough to allow exact calculation of the RMSE. To simplify comparison, in the figure we divide each RMSE by the true value of delta to get a relative error. The figure shows the effect of varying h with the number of replications n fixed at 5000.

The standard call option fits in Case (iii) of Equation (27.25). As expected, its RMS relative error decreases with decreasing h . The digital option fits in Case (ii), so its relative error explodes as h approaches zero. From the figure, we see that the relative error for the digital option is minimised at a surprisingly large value of about 4. The figure also shows that the cost of taking h too large is much smaller than the cost of taking it too small.

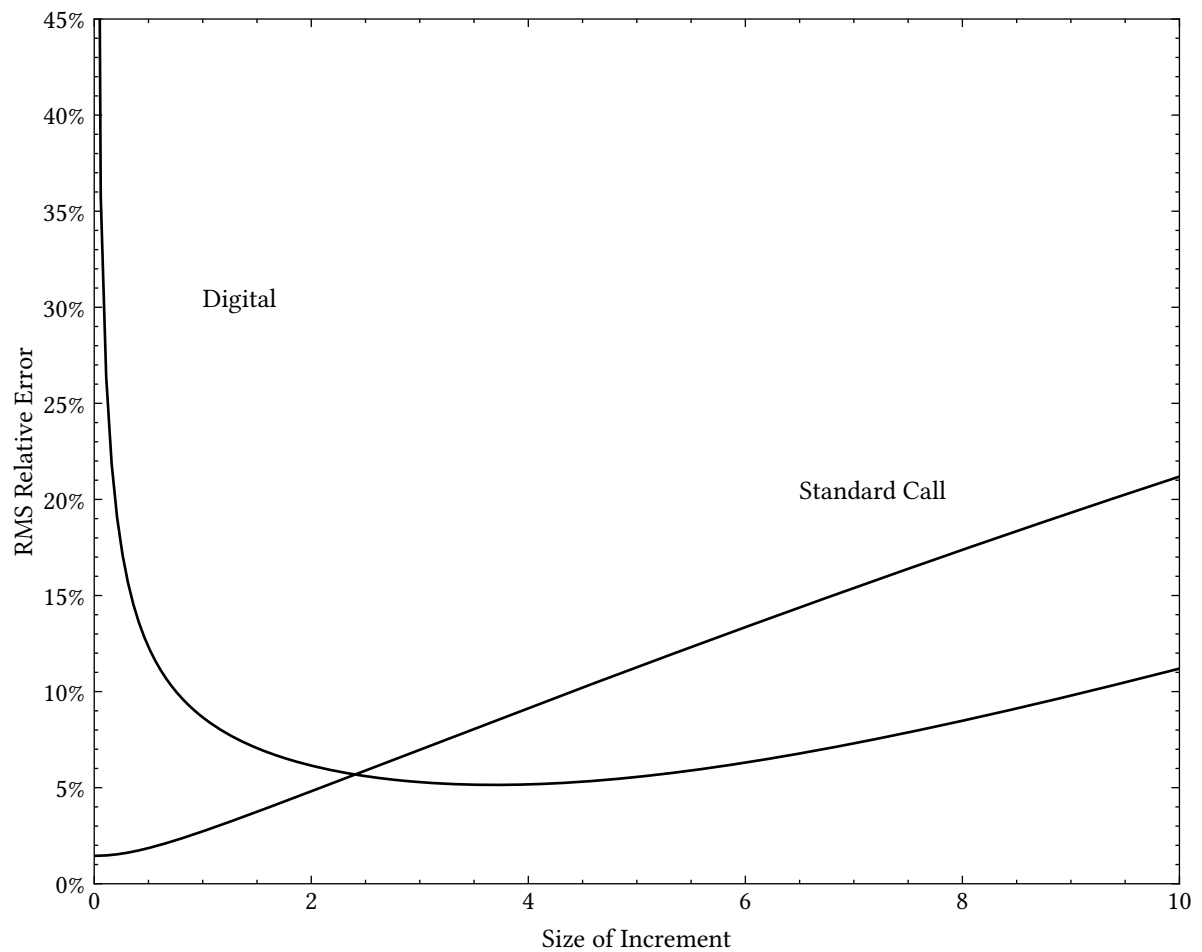


Figure 22: RMS relative errors in forward-difference delta estimates for a standard call and a digital option as a function of the increment h , with $n = 5000$

Part VIII

Financial Contracts

28 European Options

29 Exotic Options

29.1 Introduction

The European calls and puts considered thus far are sometimes called *vanilla* or even *plain vanilla* options. Their payoffs depend only on the final value of the underlying asset.

Options whose payoffs depend on the path of the underlying asset are called *path-dependent* or *exotic*.

We present three types of exotic options on a geometric Brownian motion asset and work out a detailed analysis for one option of each type. The types considered are *barrier options*, *lookback options*, and *Asian options*. We work out the standard partial differential equations governing the option price. The first two options have explicit pricing formulas, which are based on the reflection principle for Brownian motion. Such a formula for Asian options is not known. However, for the Asian option there is a change-of-numéraire argument that reduces the pricing partial differential equation to a simple form that can easily be solved numerically.

29.2 Maximum of Brownian Motion with Drift

Figure 23 represents the running maximum process

$$M_0^t := \max_{s \in [0, t]} W_s, \quad t \geq 0,$$

of Brownian motion $(W_t)_{t \in \mathbb{R}_+}$.

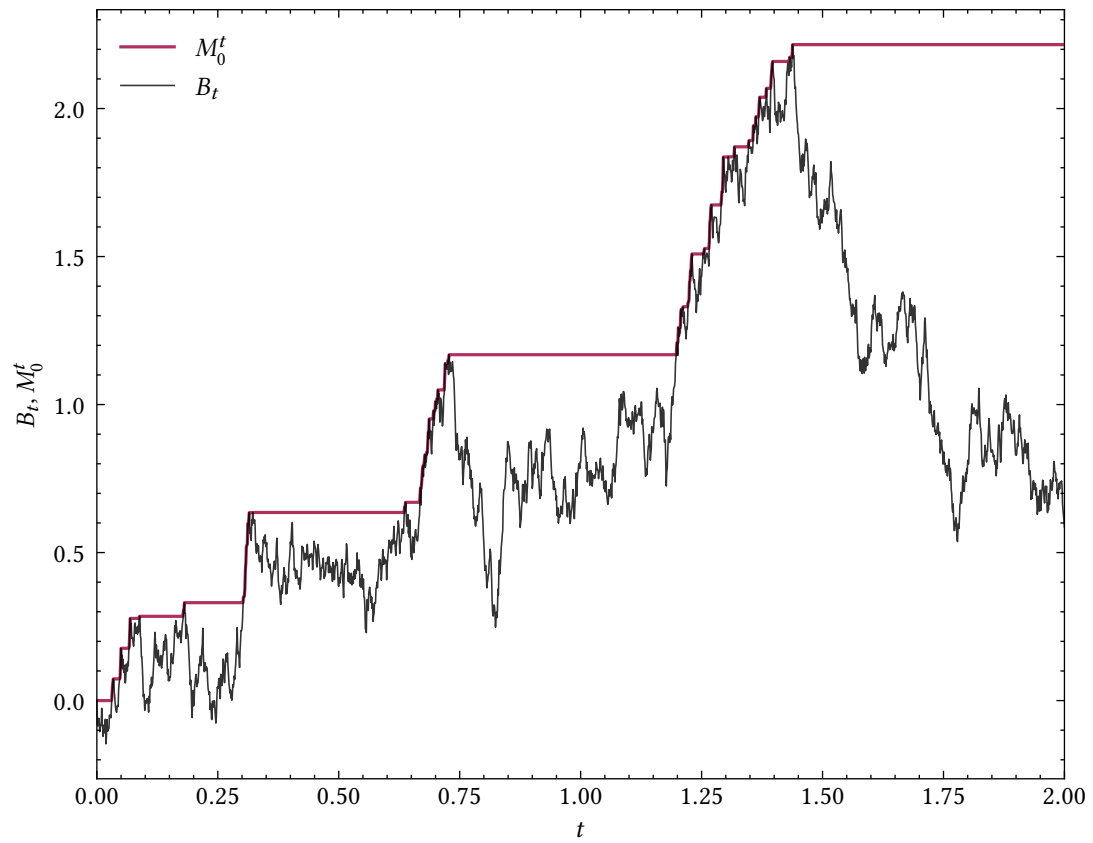


Figure 23: A Wiener process path B_t and its running maximum M_0^t .

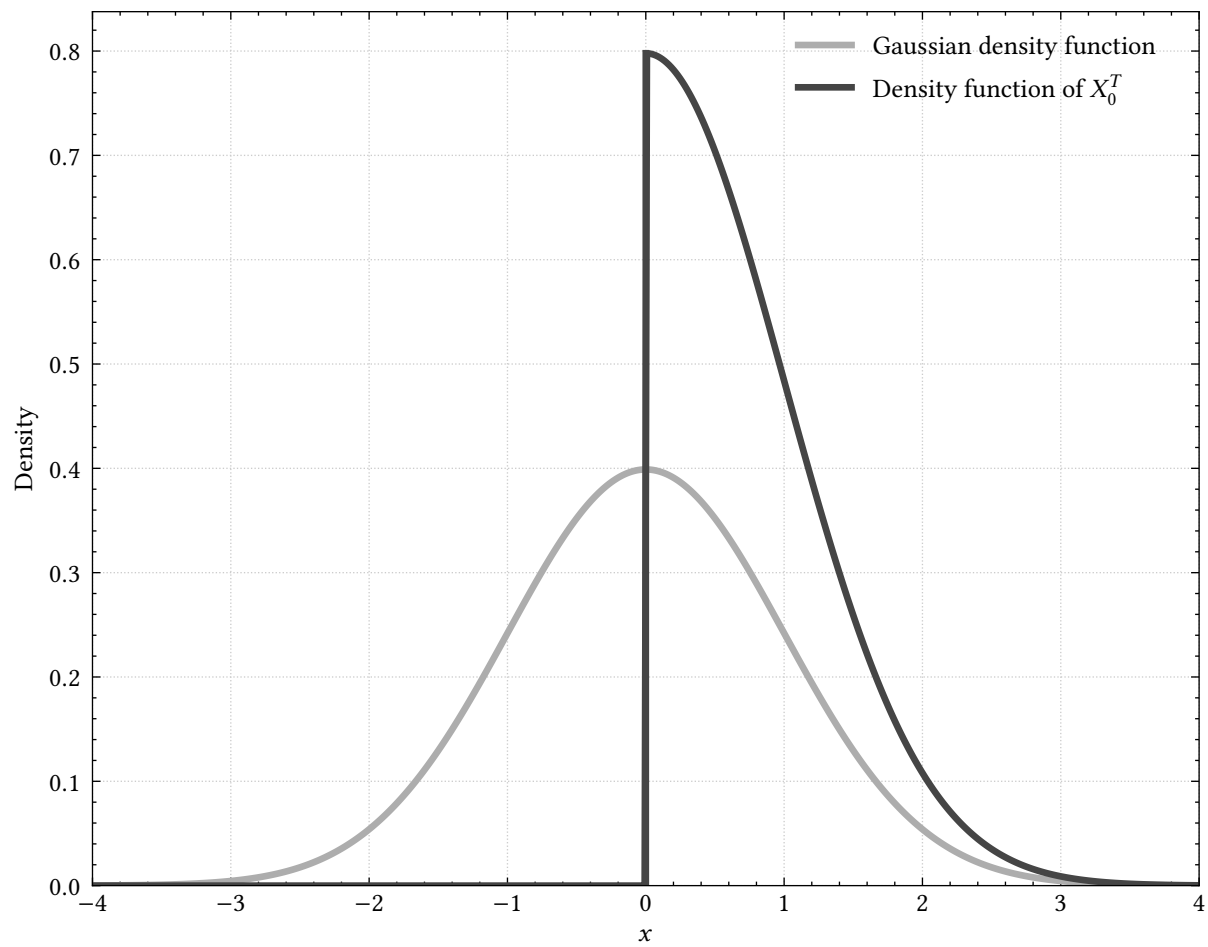


Figure 24: Probability density of the maximum X_0^1 of Brownian motion over $[0, 1]$

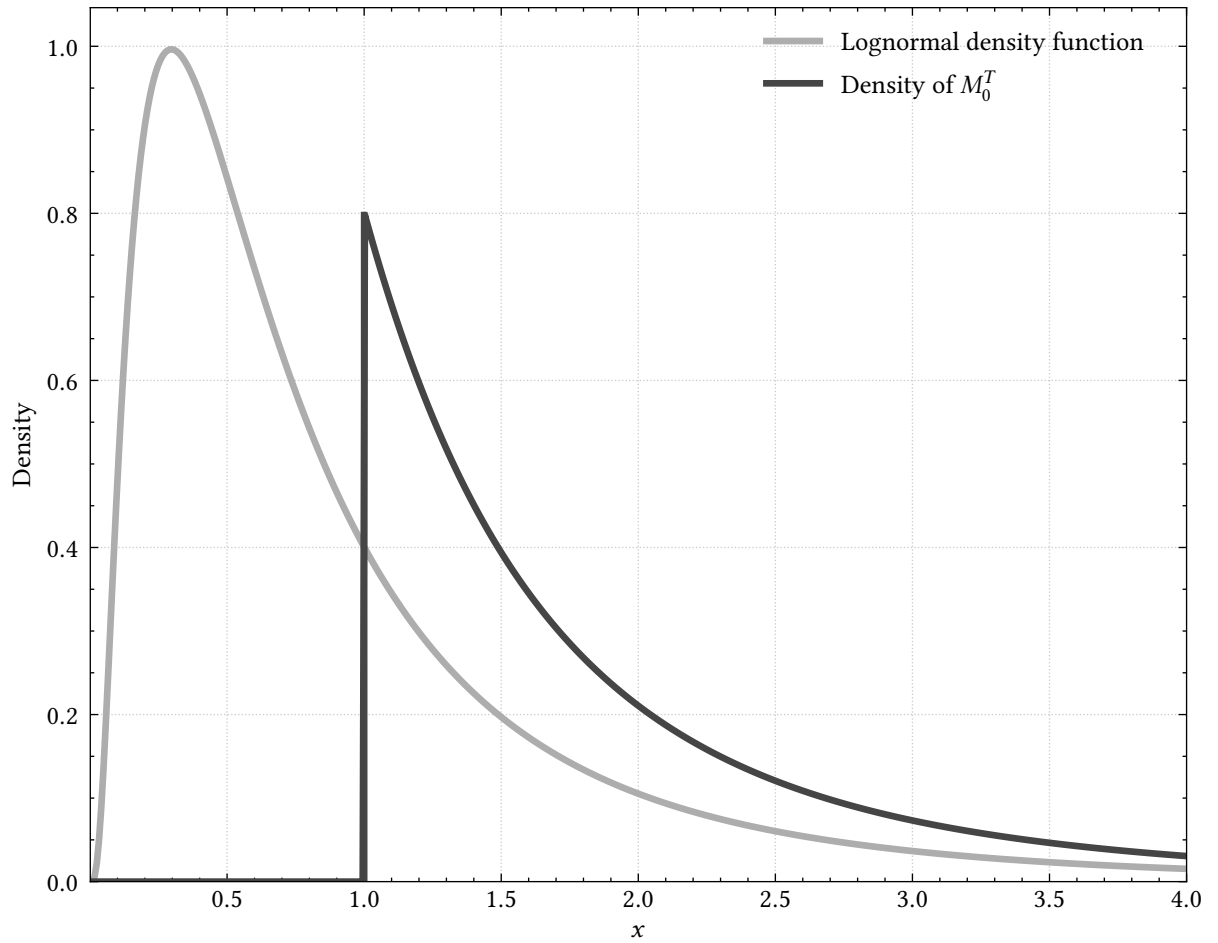


Figure 25: Density of the maximum $M_0^T = \text{Max}_{t \in [0, T]} S_t$ of geometric Brownian motion with $S_0 = 1$.

29.3 Barrier Options

The generic term "barrier options" refers to a class of options whose payoff depends on whether the underlying price hits a prespecified barrier during the option's lifetime.

29.4 Knock-out Barrier

29.4.1 Up-and-out barrier call option

Let us consider an up-and-out barrier call option with maturity T , strike price K , barrier (or call level) B , and payoff

$$C = (S_T - K)^+ \mathbb{1}_{\left\{ \max_{0 \leq t \leq T} S_t < B \right\}} = \begin{cases} S_T - K & \text{if } \max_{0 \leq t \leq T} S_t \leq B, \\ 0 & \text{if } \max_{0 \leq t \leq T} S_t > B, \end{cases}$$

with $B \geq K$.

Proposition 29.1: Up-and-Out Barrier Call Option Pricing

When $K < B$, the price

$$e^{-(T-t)r} \mathbb{1}_{\{M_0^t < B\}} \mathbb{E}^* \left[\left(\frac{S_{T-t}}{S_0} - K \right)^+ \mathbb{1}_{\left\{ x \max_{0 \leq u \leq T-t} \frac{S_u}{S_0} < B \right\}} \middle| x = S_t \right] \quad (29.1)$$

of the up-and-out barrier call option with maturity T , strike price K , and barrier level B is given by the following formula, where $\tau = T - t$:

$$\begin{aligned} C_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ [\Phi(d_1(S_t, K)) - \Phi(d_1(S_t, B))] \right. \\ & \left. - \left(\frac{B}{S_t} \right)^{1 + \frac{2r}{\sigma^2}} \left[\Phi\left(d_1\left(\frac{B^2}{S_t}, K\right)\right) - \Phi\left(d_1\left(\frac{B^2}{S_t}, B\right)\right) \right] \right\} \\ & - e^{-r\tau} K \mathbb{1}_{\{M_0^t < B\}} \left\{ [\Phi(d_2(S_t, K)) - \Phi(d_2(S_t, B))] \right. \\ & \left. - \left(\frac{S_t}{B} \right)^{1 - \frac{2r}{\sigma^2}} \left[\Phi\left(d_2\left(\frac{B^2}{S_t}, K\right)\right) - \Phi\left(d_2\left(\frac{B^2}{S_t}, B\right)\right) \right] \right\}, \end{aligned} \quad (29.2)$$

If $K \geq B$:

$$C_{UO}(t, S_t) = 0. \quad (29.3)$$

where $d_{1,2}(S, X)$ corresponds to the definitions in (29.1) with spot price S , strike price K , and time to maturity τ . The price of the up-and-out barrier call option vanishes when $B \leq K$.

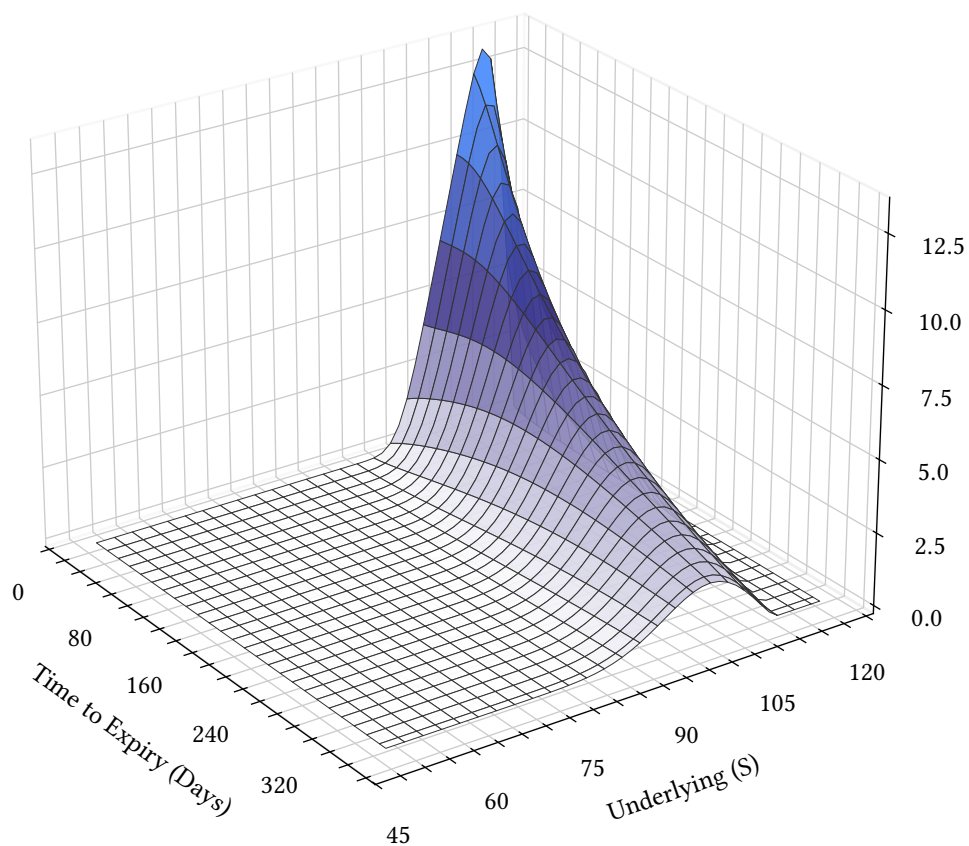


Figure 26: Graph of the up-and-out call option price

29.4.2 Up-and-out barrier put option

This option is also known as a *Callable Bear Contract*, or a Bear CBBC with no residual value, or a turbo warrant with no rebate. In this context, B denotes the call level for pricing Bear CBBCs (up-and-out barrier put options). The payoff is given by

$$P = (K - S_T)^+ \mathbb{1}_{\left\{\max_{0 \leq t \leq T} S_t < B\right\}} = \begin{cases} K - S_T & \text{if } \max_{0 \leq t \leq T} S_t \leq B, \\ 0 & \text{if } \max_{0 \leq t \leq T} S_t > B. \end{cases}$$

Proposition 29.2: Up-and-Out Barrier Put Option Pricing

The price of the up-and-out barrier put option with maturity T , strike price K , and barrier level B is given by the following formulas, where $\tau = T - t$.

If $B < K$, the price is given by

$$\begin{aligned} P_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ -\Phi(-d_1(S_t, B)) + \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \Phi\left(-d_1\left(\frac{B^2}{S_t}, B\right)\right) \right\} \\ & + Ke^{-r\tau} \mathbb{1}_{\{M_0^t < B\}} \left\{ \Phi(-d_2(S_t, B)) - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \Phi\left(-d_2\left(\frac{B^2}{S_t}, B\right)\right) \right\}. \end{aligned} \quad (29.4)$$

If $B \geq K$, the price is given by

$$\begin{aligned} P_{UO}(t) = S_t \mathbb{1}_{\{M_0^t < B\}} & \left\{ -\Phi(-d_1(S_t, K)) + \left(\frac{B}{S_t}\right)^{1+\frac{2r}{\sigma^2}} \Phi\left(-d_1\left(\frac{B^2}{S_t}, K\right)\right) \right\} \\ & + Ke^{-r\tau} \mathbb{1}_{\{M_0^t < B\}} \left\{ \Phi(-d_2(S_t, K)) - \left(\frac{S_t}{B}\right)^{1-\frac{2r}{\sigma^2}} \Phi\left(-d_2\left(\frac{B^2}{S_t}, K\right)\right) \right\}. \end{aligned} \quad (29.5)$$

This corresponds to the value of a vanilla put option minus the value of the "knock-out" component.

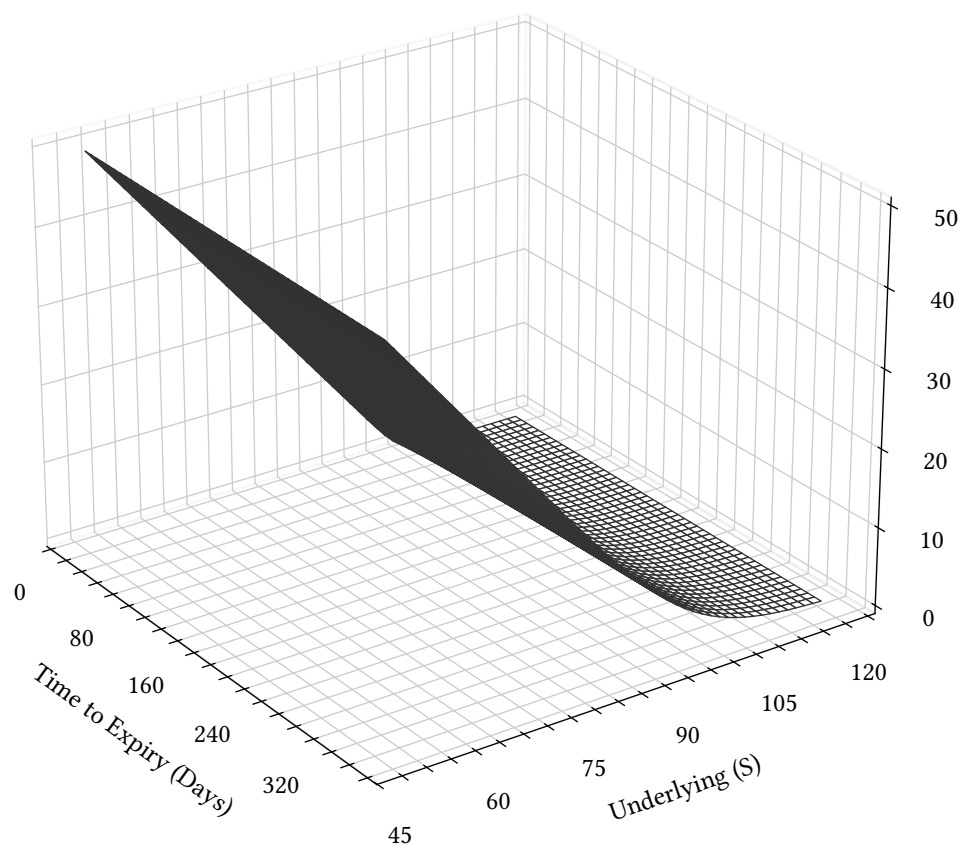


Figure 27: Graph of the up-and-out put option price

29.4.3 Down-and-out barrier call option

Let us now consider a down-and-out barrier call option on the underlying asset price S_t with exercise date T , strike price K , barrier level B , and payoff

$$C = (S_T - K)^+ \mathbb{1}_{\left\{\min_{0 \leq t \leq T} S_t > B\right\}} = \begin{cases} S_T - K & \text{if } \min_{0 \leq t \leq T} S_t > B, \\ 0 & \text{if } \min_{0 \leq t \leq T} S_t \leq B, \end{cases} \quad (29.6)$$

with $0 \leq B \leq K$. The down-and-out barrier call option is also called a *Callable Bull Contract*, or a Bull CBBC with no residual value, or a turbo warrant with no rebate, in which B denotes the call level.

Proposition 29.3: Down-and-Out Barrier Call Option Pricing

The price of the down-and-out barrier call option with maturity T , strike price K , and barrier level B is given by the following formulas, where $\tau = T - t$.

When $B \leq K$, we have:

$$\begin{aligned} C_{DO}(t, S_t) &= \mathbb{1}_{\{m_0^t > B\}} \text{Bl}(S_t, K, r, \tau, \sigma) \\ &\quad - B \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t}\right)^{1+2r/\sigma^2} \Phi\left(d_1\left(\frac{B^2}{KS_t}, \tau\right)\right) \\ &\quad + Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t}\right)^{-1+2r/\sigma^2} \Phi\left(d_2\left(\frac{B^2}{KS_t}, \tau\right)\right). \end{aligned} \quad (29.7)$$

When $B \geq K$, we find:

$$\begin{aligned} C_{DO}(t, S_t) &= S_t \mathbb{1}_{\{m_0^t > B\}} \Phi\left(d_1\left(\frac{S_t}{B}, \tau\right)\right) - Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \Phi\left(d_2\left(\frac{S_t}{B}, \tau\right)\right) \\ &\quad - B \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t}\right)^{2r/\sigma^2} \Phi\left(d_1\left(\frac{B}{S_t}, \tau\right)\right) \\ &\quad + Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{S_t}{B}\right)^{1-2r/\sigma^2} \Phi\left(d_2\left(\frac{B}{S_t}, \tau\right)\right). \end{aligned} \quad (29.8)$$

Here Bl denotes the standard Black-Scholes call price, and $d_{1,2}(x, \tau)$ are defined as in ??? with the spot price term replaced by the ratio argument x . If $m_0^t \leq B$, the price is 0.

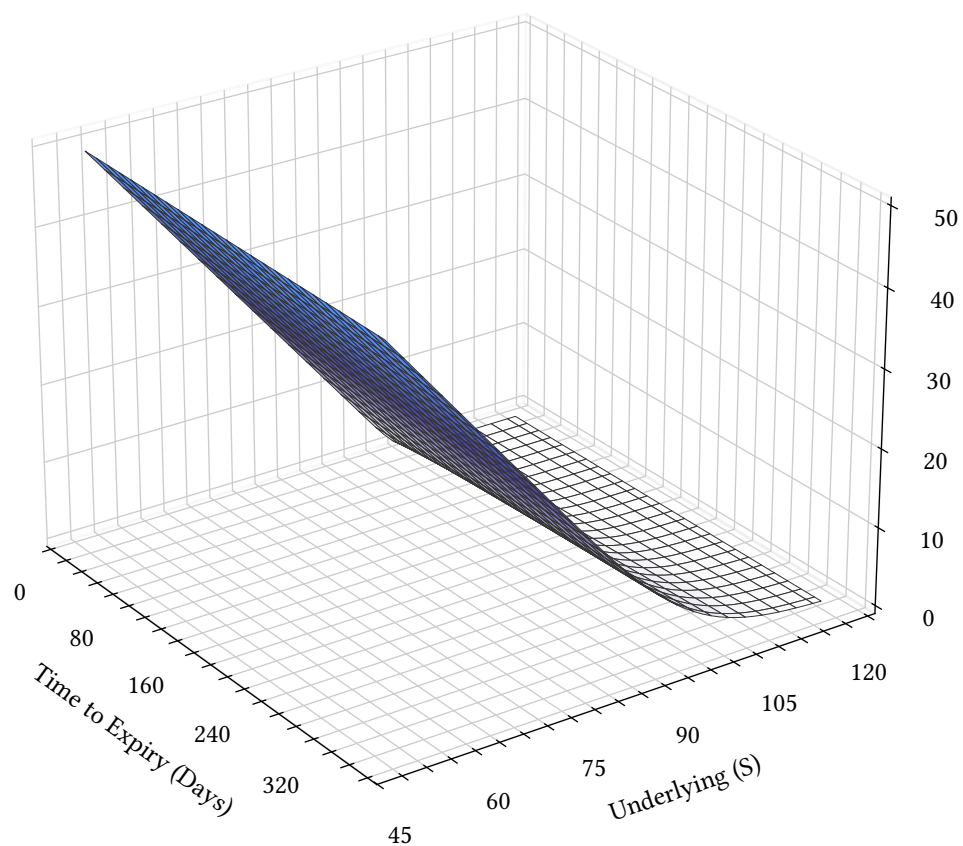


Figure 28: Graph of the down-and-out call option price

29.4.4 Down-and-out barrier put option

The down-and-out barrier put option with exercise date T , strike price K , and barrier level B has the payoff:

$$P = (K - S_T)^+ \mathbb{1}_{\{\min_{0 \leq t \leq T} S_t > B\}} = \begin{cases} K - S_T & \text{if } \min_{0 \leq t \leq T} S_t > B, \\ 0 & \text{if } \min_{0 \leq t \leq T} S_t \leq B. \end{cases} \quad (29.9)$$

For this option to have non-zero value, we typically require $S_0 > B$. Since it is a put option, the intrinsic value is positive when $S_T < K$. If $B \geq K$, the condition $S_T < K$ combined with the knock-out condition $\min S_t > B \geq K$ makes the payoff always zero (as $S_T > B \geq K \implies K - S_T < 0$). Thus, we only consider the case $B < K$.

Proposition 29.4: Down-and-Out Barrier Put Option Pricing

The price of the down-and-out barrier put option with maturity T , strike price K , and barrier level B (where $B \leq K$) is given by

$$\begin{aligned} P_{DO}(t, S_t) = & Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left[\Phi \left(-d_2 \left(\frac{S_t}{K}, \tau \right) \right) - \Phi \left(-d_2 \left(\frac{S_t}{B}, \tau \right) \right) \right] \\ & - S_t \mathbb{1}_{\{m_0^t > B\}} \left[\Phi \left(-d_1 \left(\frac{S_t}{K}, \tau \right) \right) - \Phi \left(-d_1 \left(\frac{S_t}{B}, \tau \right) \right) \right] \\ & - Ke^{-r\tau} \mathbb{1}_{\{m_0^t > B\}} \left(\frac{S_t}{B} \right)^{1-2r/\sigma^2} \left[\Phi \left(d_2 \left(\frac{B^2}{KS_t}, \tau \right) \right) - \Phi \left(d_2 \left(\frac{B}{S_t}, \tau \right) \right) \right] \\ & + S_t \mathbb{1}_{\{m_0^t > B\}} \left(\frac{B}{S_t} \right)^{1+2r/\sigma^2} \left[\Phi \left(d_1 \left(\frac{B^2}{KS_t}, \tau \right) \right) - \Phi \left(d_1 \left(\frac{B}{S_t}, \tau \right) \right) \right]. \end{aligned} \quad (29.10)$$

Here $\tau = T - t$, and $d_{1,2}(x, \tau)$ correspond to the definitions with the spot price term replaced by the ratio argument x .

If $B > K$, the price vanishes: $P_{DO}(t, S_t) = 0$.

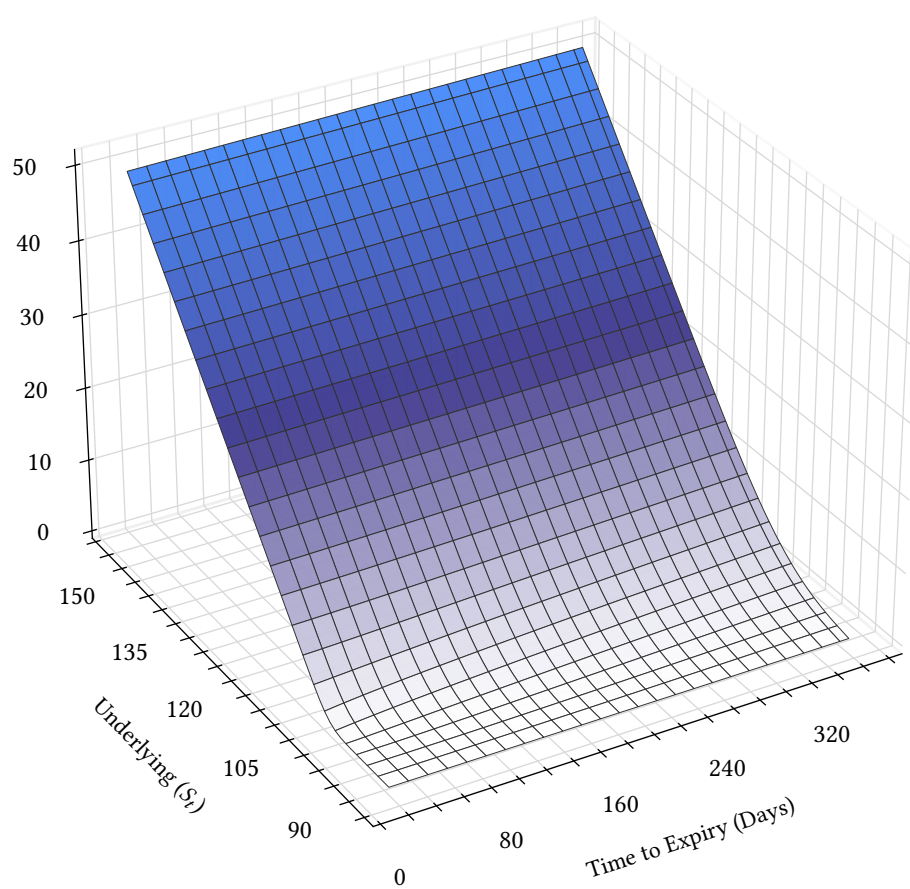


Figure 29: Graph of the down-and-out put option price

29.5 Knock-in Barrier

29.5.1 Up-and-Out Call

29.6 Lookback Options

Lookback call (resp. put) options are financial derivatives that allow their holders to exercise the option by setting the strike price at the minimum (resp. maximum) of the underlying asset price process $(S_t)_{t \in [0, T]}$ over the time interval $[0, T]$. Lookback options can be priced by PDE arguments or by computing the discounted expected values of their claim payoff C , namely $C = S_T - \min_{0 \leq t \leq T} S_t$ in the case of call options, and $C = \max_{0 \leq t \leq T} S_t - S_T$ in the case of put options.

29.7 The Lookback Put Option

The standard lookback put option gives its holder the right to sell the underlying asset at its historically highest price. In this case, the floating strike price is M_0^T and the payoff is given by the terminal value

$$C = M_0^T - S_T \quad (29.11)$$

of the drawdown process $(M_0^T - S_t)_{t \in [0, T]}$. The following pricing formula for lookback put options is a direct consequence of Proposition 10.9.

Proposition 29.5: The Price Of The Lookback Put Option

The price at time $t \in [0, T]$ of the lookback put option with payoff $M_0^T - S_T$ is given by

$$\begin{aligned}
 P_{LB}(t) &= e^{-r\tau} \mathbb{E}^* [M_0^T - S_T \mid \mathcal{F}_t] \\
 &= M_0^t e^{-r\tau} \Phi(-d_2(S_t, M_0^t)) \\
 &\quad + S_t \left(1 + \frac{\sigma^2}{2r}\right) \Phi(d_1(S_t, M_0^t)) \\
 &\quad - S_t e^{-r\tau} \frac{\sigma^2}{2r} \left(\frac{M_0^t}{S_t}\right)^{2r/\sigma^2} \Phi(-d_2(M_0^t, S_t)) - S_t,
 \end{aligned} \tag{29.12}$$

where $\tau = T - t$, and $d_{1,2}(\cdot, \cdot)$ correspond to the definitions in (29.1) with the fixed strike K replaced by the running maximum M_0^t (or S_t where arguments are inverted).

29.8 Floating Strike Lookback Option

We begin with a geometric brownian motion asset price, which may be written as in

29.8.1 Floating Strike Lookback Option

29.9 Asian Options

An *Asian option* (or *average option*) defines a class of options, whether European or American style, whose terminal payoff depends on the average asset value over a specific period within the option's life. Due to this averaging feature, Asian options are particularly suitable for thinly traded assets or commodities. Unlike standard options, they are typically less expensive and offer greater robustness against price manipulation near the expiry date.

Asian options were first traded in Tokyo in 1987, and have become particularly popular in commodities trading.

Let T denote the exercise date, and let $0 \leq T_0 < T$ represent the commencement of the averaging period. The payoff at expiry for an Asian call option is defined as

$$C_T^A := (A_S(T_0, T) - K)^+, \tag{29.13}$$

where

$$A_S(T_0, T) = \frac{1}{T - T_0} \int_{T_0}^T S_u \, du \quad (29.14)$$

represents the arithmetic average of the asset price over the time interval $[T_0, T]$. Here, K denotes the fixed strike price, and the asset price S is assumed to follow a geometric Brownian motion.

The main difficulty in pricing and hedging Asian options is due to the fact that the random variable $A(T_0, T)$ does not have a lognormal distribution. This feature makes the task of finding an explicit formula for the price of an Asian option surprisingly involved.

For this reason, early studies of Asian options were based either on approximations or on the direct application of the Monte Carlo method. The numerical approach to the valuation of Asian options, is based on the approximation of the arithmetic average using the geometric average. Note first that it is natural to substitute for the continuous-time average $A(T_0, T)$ its discrete-time counterpart

$$A_S^n(T_0, T) = \frac{1}{n} \sum_{i=0}^{n-1} S_{T_i}, \quad (29.15)$$

where $T_i = T_0 + i(T - T_0)/n$. Furthermore, the arithmetic average $A_S^n(T_0, T)$ can be replaced with the geometric average, denoted by $G_S^n(T_0, T)$, in what follows. Recall that the random variables $S_{T_i}, i = 1, \dots, n$ are given explicitly by the expression

$$S_{T_i} = S_{T_0} \exp \left(\sigma(W_{T_i}^* - W_{T_0}^*) + \left(r - \frac{1}{2}\sigma^2 \right) (T_i - T_0) \right), \quad (29.16)$$

where W^* is a standard Brownian motion under the martingale measure $\mathbb{P}^*$. Hence the geometric average admits the following representation

$$G_S^n(T_0, T) = \left(\prod_{i=0}^{n-1} S_{T_i} \right)^{1/n} = c S_{T_0} \exp \left(\frac{\sigma}{n} \sum_{i=0}^{n-1} (n - i - 1) (W_{T_{i+1}}^* - W_{T_i}^*) \right) \quad (29.17)$$

for a strictly positive constant c . In view of the independence of increments of the Brownian motion, the last formula makes clear that the geometric average $G_S^n(T_0, T)$ has a lognormal distribution under $\mathbb{P}^*$. The approximate Black-Scholes-like formula for the price of an Asian

call option can thus be easily found by the direct evaluation of the conditional expectation

$$\tilde{C}_{T_0}^n = e^{-r(T-T_0)} \mathbb{E}_{\mathbb{P}^*} \left((G_S^n(T_0, T) - K)^+ \mid \mathcal{F}_{T_0} \right). \quad (29.18)$$

It appears, however, that such an approach significantly underprices Asian call options. To overcome this deficiency, one may directly approximate the true distribution of the arithmetic average using an approximate distribution, typically a lognormal distribution with the appropriate parameters

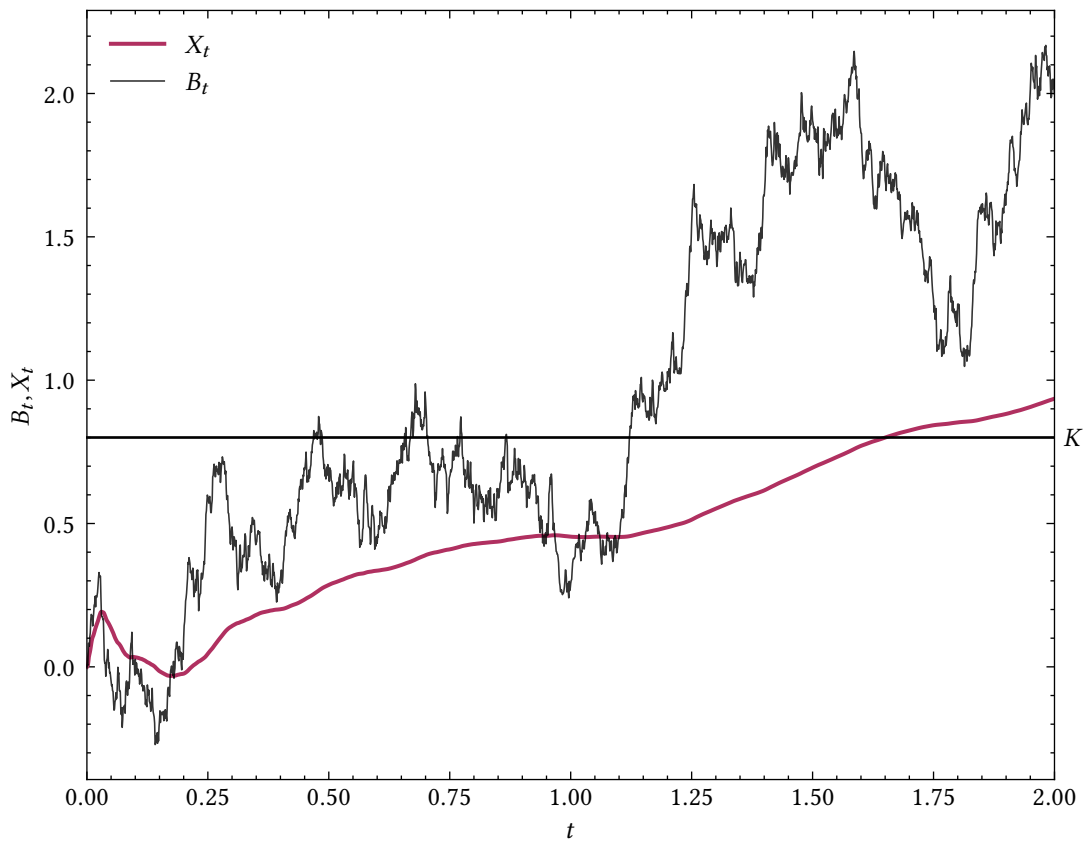


Figure 30: Brownian motion B_t and its moving average X_t

30 American Options

European option contracts specify an expiration date, and if the option is to be exercised at all, the exercise must occur on the expiration date. An option whose owner can choose to exercise at any time up to and including the expiration date is called *American*.

Whereas a European option can be exercised only at a fixed date, an American option can be exercised any time up to its expiration. The value of an American option is the value achieved by exercising optimally. Finding this value entails finding the optimal exercise rule by solving an optimal stopping problem and computing the expected discounted payoff of the option under this rule. The embedded optimisation problem makes this a difficult problem for simulation.

30.1 Approximate Continuation Values

Regression-based methods posit an expression for the continuation value of the form

$$\mathbb{E}[V_{i+1}(X_{i+1}) \mid X_i = x] = \sum_{r=1}^M \beta_{ir} \psi_r(x), \quad (30.1)$$

for some basis functions $\psi_r : \mathbb{R}^d \rightarrow \mathbb{R}$ and constants $\beta_{ir}, r = 1, \dots, M$. Using the notation C_i for the continuation value at time i , we may equivalently write [Equation \(30.1\)](#) as

$$C_i(x) = \beta_i^\top \psi(x), \quad (30.2)$$

with

$$\beta_i^\top = (\beta_{i1}, \dots, \beta_{iM}), \quad \psi(x) = (\psi_1(x), \dots, \psi_M(x))^\top.$$

One could use different basis functions at different exercise dates, but to simplify notation, we suppress any dependence of ψ on i .

Assuming a relation of the form [Equation \(30.1\)](#) holds, the vector β_i is given by

$$\beta_i = (\mathbb{E}[\psi(X_i)\psi(X_i)^\top])^{-1} \mathbb{E}[\psi(X_i)V_{i+1}(X_{i+1})] \equiv B_\psi^{-1} B_{\psi V}. \quad (30.3)$$

Here, B_ψ is the indicated $M \times M$ matrix (assumed nonsingular) and $B_{\psi V}$ the indicated vector of length M . The variables (X_i, X_{i+1}) inside the expectations have a joint distribution corresponding to the state of the underlying Markov chain at dates i and $i + 1$.

The coefficients β_{ir} could be estimated from observations of pairs $(X_{ij}, V_{i+1}(X_{i+1,j}))$, $j = 1, \dots, b$, each consisting of the state at time i and the corresponding option value at time $i + 1$. Consider, in particular, independent paths $(X_{1j}, \dots, X_{mj})$, $j = 1, \dots, b$, and suppose for a moment that the values $V_{i+1}(X_{i+1,j})$ are known. The least-squares estimate of β_i is then given by

$$\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_{\psi V},$$

where $\hat{B}_\psi$ and $\hat{B}_{\psi V}$ are the sample counterparts of B_ψ and $B_{\psi V}$. More explicitly, $\hat{B}_\psi$ is the $M \times M$ matrix with qr entry

$$\frac{1}{b} \sum_{j=1}^b \psi_q(X_{ij}) \psi_r(X_{ij})$$

and $\hat{B}_{\psi V}$ is the M -vector with r th entry

$$\frac{1}{b} \sum_{k=1}^b \psi_r(X_{ik}) V_{i+1}(X_{i+1,k}). \quad (30.4)$$

All of these quantities can be calculated from function values at pairs of consecutive nodes $(X_{ij}, X_{i+1,j})$, $j = 1, \dots, b$. In practice, V_{i+1} is unknown and must be replaced by estimated values $\hat{V}_{i+1}$ at downstream nodes. The estimate $\hat{\beta}_i$ then defines an estimate

$$\hat{C}_i(x) = \hat{\beta}_i^\top \psi(x), \quad (30.5)$$

of the continuation value at an arbitrary point x in the state space $\mathbb{R}^d$. This, in turn, defines a procedure for estimating the option value.

Algorithm 1 Regression-Based Pricing Algorithm

- 1: Simulate b independent paths $\{X_{1j}, \dots, X_{mj}\}$, $j = 1, \dots, b$, of the Markov chain.
- 2: At terminal nodes, set $\hat{V}_{mj} = h_m(X_{mj})$ for $j = 1, \dots, b$.
- 3: **for** $i = m - 1$ **down to** 1 **do**
- 4: Given estimated values $\hat{V}_{i+1,j}$, $j = 1, \dots, b$, use regression to calculate $\hat{\beta}_i = \hat{B}_\psi^{-1} \hat{B}_\psi V$.
- 5: Compute estimated continuation values $\hat{C}_i(X_{ij})$ using Equation (30.5).
- 6: **for** $j = 1, \dots, b$ **do**
- 7: Set value based on immediate exercise versus estimated continuation:

$$\hat{V}_{ij} = \max \{h_i(X_{ij}), \hat{C}_i(X_{ij})\}. \quad (30.6)$$

- 8: **end for**
- 9: **end for**
- 10: Set $\hat{V}_0 = (\hat{V}_{11} + \dots + \hat{V}_{1b})/b$.

Definition 30.1: American Contingent Claim

An *American contingent claim* $X^a = (X, \mathcal{T}_{[0,T]})$ expiring at time T consists of a sequence of payoffs $(X_t)_{t=0}^T$, where X_t is an $\mathcal{F}_t$ -measurable random variable for $t = 0, 1, \dots, T$, and of the set $\mathcal{T}_{[0,T]}$ of admissible exercise policies.

Proposition 30.1: Arbitrage Price of American Put

The arbitrage price P_t^a of an American put option equals

$$P_t^a = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} (K - S_\tau)^+ \mid \mathcal{F}_t \right), \quad \forall t \leq T. \quad (30.7)$$

Moreover, for any $t \leq T$ the stopping time τ_t^* that realizes the maximum in Equation (30.7) is given by the expression (by convention $\min \emptyset = T$)

$$\tau_t^* = \min \{u \in \{t, t+1, \dots, T\} \mid (K - S_u)^+ \geq P_u^a\}. \quad (30.8)$$

Proposition 30.2: Arbitrage Price of American Claim in CRR

For every $t \leq T$, the arbitrage price $\pi(X^a)$ of an arbitrary American claim X^a in the CRR model equals

$$\pi_t(X^a) = \max_{\tau \in \mathcal{T}_{[t,T]}} \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-(\tau-t)} X_\tau \mid \mathcal{F}_t \right). \quad (30.9)$$

The price process $\pi(X^a)$ can be determined using the following recurrence relation, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ X_t, \mathbb{E}_{\mathbb{P}^*} \left(\hat{r}^{-1} \pi_{t+1}(X^a) \mid \mathcal{F}_t \right) \right\} \quad (30.10)$$

subject to the terminal condition $\pi_T(X^a) = X_T$. In the case of a path-independent American claim X^a with the reward function h we have that, for every $t \leq T - 1$,

$$\pi_t(X^a) = \max \left\{ h(S_t, t), \hat{r}^{-1} \left(p_* \pi_{t+1}^u(X^a) + (1 - p_*) \pi_{t+1}^d(X^a) \right) \right\}, \quad (30.11)$$

where, for a generic stock price value S_t at time t , we write $\pi_{t+1}^u(X^a)$ and $\pi_{t+1}^d(X^a)$ to denote the values of the price process $\pi(X^a)$ at time $t + 1$ in the nodes that correspond to the upward and downward movements of the stock price during the time-period $[t, t + 1]$, that is, for the values uS_t and dS_t of the stock price at time $t + 1$, respectively.

31 Basket Options

A *basket option*, as its name suggests, is a type of option contract that serves to hedge against the risk exposure of a basket of assets, i.e., a predetermined portfolio of assets. Generally speaking, a basket option is more cost-effective than a portfolio of single options, as the latter over-hedges the exposure and incurs higher costs than a basket option. An intuitive explanation for this feature is that the basket option considers the correlation between different risk factors. For instance, in the case of a strong negative correlation between two or more underlying assets, the total risk exposure may almost vanish; however, this beneficial feature is not reflected in the payoffs and prices of single options.

From the analytical viewpoint, there is a close analogy between basket options and Asian options. Let us denote by S^i , $i = 1, \dots, k$, the price processes of k underlying assets; they will be referred to as stocks in what follows. In this case, it seems natural to refer to such a basket option as the *stock index option* (in market practice, options on a basket of currencies are also quite common). The payoff at expiry of a basket call option is defined in the following way

$$C_T^B := \left(\sum_{i=1}^k w_i S_T^i - K \right)^+ = (A_T - K)^+, \quad (31.1)$$

where $w_i \geq 0$ is the weight of the i^{th} asset, so that $\sum_{i=1}^k w_i = 1$. Note that A_T stands for the weighted arithmetic average $A_T = \sum_{i=1}^k w_i S_T^i$. We assume that each stock price S^i follows a geometric Brownian motion. More explicitly, under the martingale measure $\mathbb{P}^*$, we have

$$dS_t^i = S_t^i (r dt + \hat{\sigma}_i \cdot dW_t^*) \quad (31.2)$$

for some non-zero vectors $\hat{\sigma}_i \in \mathbb{R}^k$, where $W^* = (W^{1*}, \dots, W^{k*})$ stands for a k -dimensional Brownian motion under $\mathbb{P}^*$, and r is the risk-free interest rate. Observe that for any fixed i , we can find a standard one-dimensional Brownian motion $\tilde{W}^i$ such that

$$dS_t^i = S_t^i (r dt + \sigma_i d\tilde{W}_t^i) \quad (31.3)$$

and $\sigma_i = |\hat{\sigma}_i|$, where $|\hat{\sigma}_i|$ is the Euclidean norm of $\hat{\sigma}_i$. Let us denote by $\rho_{i,j}$ the instantaneous correlation coefficient

$$\rho_{i,j} = \frac{\hat{\sigma}_i \cdot \hat{\sigma}_j}{\sigma_i \sigma_j} = \frac{\hat{\sigma}_i \cdot \hat{\sigma}_j}{|\hat{\sigma}_i| |\hat{\sigma}_j|}.$$

We may thus alternatively assume that the dynamics of price processes S^i are given by (31.3), where $\tilde{W}^i$, $i = 1, \dots, k$ are one-dimensional Brownian motions, whose cross-variations satisfy $\langle \tilde{W}^i, \tilde{W}^j \rangle_t = \rho_{i,j} t$ for every $i, j = 1, \dots, k$. Let us return to the problem of valuation of basket options. For similar reasons as those applying to Asian options, basket options are rather intractable analytically. developed a simple technique of pricing basket options on a bivariate binomial lattice, thus generalising the standard Cox-Ross-Rubinstein methodology. Unfortunately, this numerical method is very time-consuming, especially where there are several underlying assets. To overcome this, a proposed valuation of a basket using an approximation of the weighted arithmetic mean in the form of its geometric counterpart (this follows the approach of Asian options). For a fixed $t \leq T$, let us denote by $\hat{w}_i$ the modified weights

$$\hat{w}_i = \frac{w_i S_t^i}{\sum_{j=1}^k w_j S_t^j} = \frac{w_i F_{S^i}(t, T)}{\sum_{j=1}^k w_j F_{S^j}(t, T)}, \quad (31.4)$$

where $F_{S^i}(t, T)$ is the forward price at time t of the i^{th} asset for the settlement date T .

We may rewrite (31.1) as follows³

$$\begin{aligned} C_T^B &= \left(\sum_{j=1}^k w_j F_{S^j}(t, T) \right) \left(\sum_{i=1}^k \hat{w}_i \tilde{S}_T^i - \tilde{K} \right)^+ \\ &= \left(\sum_{j=1}^k w_j F_{S^j}(t, T) \right) (\tilde{A}_T - \tilde{K})^+, \end{aligned} \quad (31.5)$$

where $\tilde{S}_T^i = S_T^i / F_{S^i}(t, T)$, $\tilde{A}_T = \sum_{i=1}^k \hat{w}_i \tilde{S}_T^i$, and

$$\tilde{K} = \frac{K}{\sum_{j=1}^k w_j F_{S^j}(t, T)} = \frac{e^{-r(T-t)} K}{\sum_{j=1}^k w_j S_t^j}. \quad (31.6)$$

³This representation is introduced because it appears to give a better approximation of the price of a basket option than formula (31.1).

The arbitrage price at time t of a basket call option thus equals

$$C_t^B = e^{-r(T-t)} \left(\sum_{j=1}^k w_j F_{S^j}(t, T) \right) \mathbb{E}_{\mathbb{P}^*}((\tilde{A}_T - \tilde{K})^+ \mid \mathcal{F}_t),$$

or equivalently,

$$C_t^B = \left(\sum_{j=1}^k w_j S_t^j \right) \mathbb{E}_{\mathbb{P}^*}((\tilde{A}_T - \tilde{K})^+ \mid \mathcal{F}_t). \quad (31.7)$$

The next step relies on an approximation of the weighted arithmetic mean $\sum_{i=1}^k \hat{w}_i \tilde{S}_T^i$ using a similarly weighted geometric mean. More specifically, we approximate the price C_0^B of the basket option using the number $\hat{C}_0^B$ given by the formula (for ease of notation, we set $t = 0$ in what follows)

$$\hat{C}_0^B = \left(\sum_{j=1}^k w_j S_0^j \right) \mathbb{E}_{\mathbb{P}^*}(\tilde{G}_T - \hat{K})^+, \quad (31.8)$$

where $\tilde{G}_T = \prod_{i=1}^k (\tilde{S}_T^i)^{\hat{w}_i}$ and

$$\hat{K} = \tilde{K} + \mathbb{E}_{\mathbb{P}^*}(\tilde{G}_T - \tilde{A}_T). \quad (31.9)$$

In view of (31.2), we have (recall that $F_{S^i}(0, T) = e^{rT} S_0^i$)

$$\tilde{S}_T^i = S_T^i / F_{S^i}(0, T) = e^{\hat{\sigma}_i \cdot W_T^* - \sigma_i^2 T / 2},$$

and thus the weighted geometric average $\tilde{G}_T$ equals

$$\tilde{G}_T = e^{c_1 \cdot W_T^* - c_2 T / 2} = e^{\eta_T - c_2 T / 2}, \quad (31.10)$$

with $\eta_T = c_1 \cdot W_T^*$, where we write $c_1 = \sum_{i=1}^k \hat{w}_i \hat{\sigma}_i$ and $c_2 = \sum_{i=1}^k \hat{w}_i \sigma_i^2$. We conclude that the random variable $\tilde{G}_T$ is log-normally distributed under $\mathbb{P}^*$.

More precisely, the random variable η_T in (31.10) has Gaussian distribution with expected value 0 and variance

$$\begin{aligned}\text{Var}_{\mathbb{P}^*}(\eta_T) &= \mathbb{E}_{\mathbb{P}^*} \left\{ \left(\sum_{i,l=1}^k \hat{w}_i \hat{\sigma}_{il} W_T^{l*} \right) \left(\sum_{j,m=1}^k \hat{w}_j \hat{\sigma}_{jm} W_T^{m*} \right) \right\} \\ &= \sum_{i,j,l,m=1}^k \hat{w}_i \hat{w}_j \hat{\sigma}_{il} \hat{\sigma}_{jm} \mathbb{E}_{\mathbb{P}^*}(W_T^{l*} W_T^{m*}) = \sum_{i,j=1}^k \hat{w}_i \hat{w}_j \hat{\sigma}_i \cdot \hat{\sigma}_j T = v^2 T,\end{aligned}$$

where we write $v^2 = \sum_{i,j=1}^k \rho_{i,j} \hat{w}_i \hat{w}_j \sigma_i \sigma_j$. We have used here the equality $\mathbb{E}_{\mathbb{P}^*}(W_T^{l*} W_T^{m*}) = \delta_{lm} T$, where δ_{lm} stands for Kronecker's delta; that is, δ_{lm} equals 1 if $l = m$ and zero otherwise.

Note that

$$\mathbb{E}_{\mathbb{P}^*}(\tilde{A}_T) = \sum_{j=1}^k \hat{w}_j S_0^j \mathbb{E}_{\mathbb{P}^*}(e^{-rT} S_T^j) = \sum_{j=1}^k \hat{w}_j = 1,$$

and

$$\mathbb{E}_{\mathbb{P}^*}(\tilde{G}_T) = e^{(v^2 - c_2)T/2} \mathbb{E}_{\mathbb{P}^*}(e^{\eta_T - \frac{1}{2} \text{Var}(\eta_T)}) = e^{(v^2 - c_2)T/2} := c.$$

We conclude that $\hat{K} = \tilde{K} + c - 1$. The expectation in (??) can now be evaluated explicitly.

Lemma 31.1: Gaussian Expectation

Let ξ be a Gaussian random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ with expected value 0 and variance $\sigma^2 > 0$. For any real numbers $a, b > 0$, we have

$$\mathbb{E}_{\mathbb{P}}(ae^{\xi - \frac{1}{2}\sigma^2} - b)^+ = aN(h) - bN(h - \sigma), \quad (31.11)$$

where $h = \sigma^{-1} \ln(a/b) + \frac{1}{2}\sigma$.

We have

$$\mathbb{E}_{\mathbb{P}^*}(\tilde{G}_T - \hat{K})^+ = \mathbb{E}_{\mathbb{P}^*}(ce^{\eta_T - \frac{1}{2} \text{Var}(\eta_T)} - (\tilde{K} + c - 1))^+,$$

so that we may set $\xi = \eta_T$, $a = c$ and $b = \tilde{K} + c - 1$. In view of Lemma 31.1, the following result is straightforward.

Proposition 31.1: Approximate Basket Option Valuation

The approximate value $\hat{C}_t^B$ of the price C_t^B of a basket call option with strike price K and expiry date T equals

$$\hat{C}_t^B = \left(\sum_{j=1}^k w_j S_t^j \right) (cN(l_1(T-t)) - (\tilde{K} + c - 1)N(l_2(T-t))), \quad (31.12)$$

where

$$c = \exp \left\{ \left(\frac{1}{2} \sum_{i,j=1}^k \rho_{i,j} \hat{w}_i \hat{w}_j \sigma_i \sigma_j - \sum_{j=1}^k \hat{w}_j \sigma_j^2 \right) (T-t) \right\}, \quad (31.13)$$

and where the modified weights $\hat{w}_i$ are given by (31.4), $\tilde{K}$ is given by (31.6), and

$$l_{1,2}(t) = \frac{\ln c - \ln(\tilde{K} + c - 1) \pm \frac{1}{2}v^2t}{v\sqrt{t}}. \quad (31.14)$$

Suppose that $k = 1$. In this case, $w_1 = 1$, and the arithmetic average agrees with the geometric one. Consequently, $c = 1$ and (31.12) reduces to the standard Black-Scholes formula. In this approach, the weighted average $\tilde{A}_T$ in (31.7) is directly replaced by a log-normally distributed random variable. Then the series expansion of the true distribution of $\tilde{A}_T$ is derived in terms of an approximating log-normal distribution. An estimate of the basket option price can thus be obtained by direct integration.

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Appendices

A Existence of Lebesgue Measure

We first show the countable additivity of the Lebesgue pre-measure on $E := (0, 1]$. Denote by I the set of subintervals of E of the form $(c, d]$, and let $\mathcal{E}_0$ be the algebra of sets that are finite unions of intervals in I .

Recall that an *algebra* $\mathcal{E}_0$ on E is a collection of subsets of E that contains E itself, and that is closed under complements and *finite unions*. Recall also that a *pre-measure* is a set function $\lambda : \mathcal{E}_0 \rightarrow [0, \infty]$, with $\lambda(\emptyset) = 0$ that satisfies the *countable additivity property*:

$$\lambda\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \lambda(A_k) \quad (\text{A.1})$$

for every sequence $A_1, A_2, \dots$ of *disjoint sets* in $\mathcal{E}_0$ with $\bigcup_n A_n \in \mathcal{E}_0$.

Define on $\mathcal{E}_0$ the natural “length” pre-measure λ . When λ is applied to intervals, we use the notation $|\cdot|$ instead. Thus, for disjoint intervals $\{(a_k, b_k]\}$:

$$\lambda\left(\bigcup_{k=1}^n (a_k, b_k]\right) = \left|\bigcup_{k=1}^n (a_k, b_k]\right| = \sum_{k=1}^n (b_k - a_k).$$

We want to show that this pre-measure is countably additive. We do this in two steps. First, we prove it for intervals, and then for general sets in $\mathcal{E}_0$. Below, $I = (a, b]$ and $I_k = (a_k, b_k]$ are bounded intervals of lengths $|I| = b - a$ and $|I_k| = b_k - a_k$.

Proposition A.1: Countable Additivity for Intervals

If $I = \bigcup_{k=1}^{\infty} I_k$ and the $\{I_k\}$ are disjoint, then $|I| = \sum_{k=1}^{\infty} |I_k|$.

Proof. We are going to prove the theorem by showing:

(1) If $\bigcup_{k=1}^{\infty} I_k \subseteq I$, and the $\{I_k\}$ are *disjoint*, then

$$\sum_{k=1}^{\infty} |I_k| \leq |I|. \quad (\text{A.2})$$

(2) If $\cup_{k=1}^{\infty} I_k \supseteq I$ (the $\{I_k\}$ need not be disjoint), then

$$\sum_{k=1}^{\infty} |I_k| \geq |I|. \quad (\text{A.3})$$

We first show Equation (A.2) for finite sums (that is, $\sum_{k=1}^n |I_k| \leq |I|$), using induction. The statement is obviously true for $n = 1$. Suppose it is true for $n - 1$. Without loss of generality, we may assume a_n is the largest among $a_1, \dots, a_n$. Then, $\cup_{k=1}^{n-1} (a_k, b_k] \subseteq (a, a_n]$, so that, by the induction hypothesis, $\sum_{k=1}^{n-1} (b_k - a_k) \leq a_n - a$, and hence:

$$\sum_{k=1}^n (b_k - a_k) \leq a_n - a + b_n - a_n \leq b - a. \quad (\text{A.4})$$

The infinite sum case follows directly from this. Namely, from the finite sum case, we have that Equation (A.4) holds for every $n \geq 1$. But this can only be the case if

$$\sum_{k=1}^{\infty} (b_k - a_k) \leq b - a.$$

To prove Equation (A.3), we again consider first the finite sum and then the infinite sum case. We use again induction to show $|I| \leq \sum_{k=1}^n |I_k|$, which is true for $n = 1$. Suppose it is true for $n - 1$. Without loss of generality, we may assume $a_n < b \leq b_n$. If $a_n \leq a$, the result is obvious. If $a_n > a$, then (draw a picture) $(a, a_n] \subseteq \cup_{i=1}^{n-1} (a_i, b_i]$, so that $\sum_{k=1}^{n-1} (b_k - a_k) \geq a_n - a$ by the induction hypothesis, and hence:

$$\sum_{k=1}^n (b_k - a_k) \geq (a_n - a) + (b_n - a_n) \geq b - a.$$

For the infinite sum case, suppose that $(a, b] \subseteq \cup_{k=1}^{\infty} (a_k, b_k]$. If $0 < \varepsilon < b - a$, the open intervals $\{(a_k, b_k + \varepsilon 2^{-k})\}$ cover the closed interval $[a + \varepsilon, b]$. The *Heine-Borel property* of the real-number system states that bounded closed intervals are *compact*; that is, every (infinite) cover of such a set has a finite subcover; see also Example B.6. Thus,

$$[a + \varepsilon, b] \subseteq \bigcup_{k=1}^n (a_k, b_k + \varepsilon 2^{-k})$$

for some n . But then

$$(a + \varepsilon, b] \subseteq \bigcup_{k=1}^n (a_k, b_k + \varepsilon 2^{-k}).$$

Consequently, using the finite case,

$$b - (a + \varepsilon) \leq \sum_{k=1}^n (b_k + \varepsilon 2^{-k} - a_k) \leq \sum_{k=1}^{\infty} (b_k - a_k) + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we have

$$b - a \leq \sum_{k=1}^{\infty} (b_k - a_k).$$

This completes the proof for intervals. □

Next, we prove the countable additivity for arbitrary sets in $\mathcal{E}_0$. Recall that we used the notation $\lambda(A) = |A|$ if A is an interval.

Theorem A.1: Countable Additivity for Sets

Let $A \in \mathcal{E}_0$ be of the form $A = \bigcup_{k=1}^{\infty} A_k$, where the $\{A_k\}$ are disjoint and in $\mathcal{E}_0$. Then,

$$\lambda(A) = \sum_{k=1}^{\infty} \lambda(A_k).$$

Proof. We can write (for certain intervals $\{I_i\}$ and $\{J_{kj}\}$):

$$A = \bigcup_{i=1}^n I_i \quad \text{and} \quad A_k = \bigcup_{j=1}^{m_k} J_{kj}.$$

It follows that

$$A = \bigcup_{i=1}^n \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{m_k} (I_i \cap J_{kj}).$$

Note that the $\{I_i \cap J_{kj}\}$ are disjoint and lie in I . Moreover,

$$\bigcup_{i=1}^n (I_i \cap J_{kj}) = \bigcup_{i=1}^n I_i \cap J_{kj} = \underbrace{A \cap J_{kj}}_{J_{kj}} = J_{kj},$$

as $J_{kj} \subseteq A$. Thus, we have, by the countable additivity for intervals:

$$\lambda(A) = \sum_{i=1}^n |I_i| = \sum_{i=1}^n \sum_{k=1}^{\infty} \sum_{j=1}^{m_k} |I_i \cap J_{kj}| = \sum_{k=1}^{\infty} \sum_{j=1}^{m_k} |J_{kj}| = \sum_{k=1}^{\infty} \lambda(A_k).$$

□

If in [Proposition A.1](#) the $\{A_k\}$ are not disjoint, we still have countable subadditivity:

$$\lambda(A) \leq \sum_{k=1}^{\infty} \lambda(A_k). \quad (\text{A.5})$$

The proof is exactly the same as in Proposition 1.42.

We next show the existence of the Lebesgue measure on $(E, \mathcal{E})$ for $E := (0, 1]$ and $\mathcal{E} := \mathcal{B}_{(0,1]}$. Recall that $\mathcal{E}_0$ is the algebra of sets that are finite unions of disjoint intervals of the form $(c, d]$, and that λ is the pre-measure on $\mathcal{E}_0$, which we have just shown to be countably additive. The proof of existence relies on the definition of two objects: $\bar{\lambda}$ and $\mathcal{M}$.

Definition A.1: Outer Measure $\bar{\lambda}$

The *outer measure* of λ is the set function $\bar{\lambda}$ that assigns to each $A \in 2^E$ (that is, for each $A \subseteq E$) the value

$$\bar{\lambda}(A) := \inf \sum_n \lambda(A_n), \quad (\text{A.6})$$

where the infimum is over all sequences $A_1, A_2, \dots$ of $\mathcal{E}_0$ -sets satisfying $A \subseteq \cup_n A_n$.

Note that $\bar{\lambda}$ is not actually a measure, as the set 2^E is simply too big for it to be countably additive. We can also define an *inner measure*, $\underline{\lambda}(A) := 1 - \bar{\lambda}(A^c)$, $A \in 2^E$.

The outer measure $\bar{\lambda}$ has properties reminiscent of those of a measure, such as positivity $\bar{\lambda}(A) \geq 0$ and monotonicity $\bar{\lambda}(A) \leq \bar{\lambda}(B)$, when $A \subseteq B$. Also, $\bar{\lambda}(\emptyset) = 0$. The following *countable subadditivity* property is important as well:

Lemma A.1: The Outer Measure $\bar{\lambda}$ is Countably Subadditive

For any collection of sets in E (not necessarily disjoint), it holds that

$$\bar{\lambda}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \bar{\lambda}(A_n). \quad (\text{A.7})$$

Proof. Take $\varepsilon > 0$ and choose $B_{nk} \in \mathcal{E}_0$ such that $A_n \subseteq \bigcup_k B_{nk}$ and $\sum_k \lambda(B_{nk}) < \bar{\lambda}(A_n) + \varepsilon 2^{-n}$.

Because $\bigcup_n A_n \subseteq \bigcup_{n,k} B_{nk}$, it follows that

$$\bar{\lambda}\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n,k} \lambda(B_{nk}) < \sum_{n=1}^{\infty} \bar{\lambda}(A_n) + \varepsilon,$$

showing Equation (A.7), as ε is arbitrary. $\square$

Definition A.2: The Collection $\mathcal{M}$

The collection $\mathcal{M}$ consists of all sets $A \in 2^E$ for which

$$\bar{\lambda}(A \cap C) + \bar{\lambda}(A^c \cap C) = \bar{\lambda}(C) \quad \text{for every } C \in 2^E. \quad (\text{A.8})$$

Taking $C = E$, this implies that $\bar{\lambda}(A) = \underline{\lambda}(A)$ for every $A \in \mathcal{M}$. The following lemma shows that $\mathcal{M}$ is an algebra. Later on we will show that it is in fact a σ -algebra.

Lemma A.2: The Collection $\mathcal{M}$ is an Algebra

The collection $\mathcal{M}$ contains E and is closed under complements and finite unions. Hence, it is an algebra on E .

Proof. The collection $\mathcal{M}$ is obviously closed under complements and contains E . To show that it is closed under finite unions, we need to show that $A \cup B \in \mathcal{M}$ for any $A, B \in \mathcal{M}$. Take any $C \in 2^E$. By the countable (and hence finite) subadditivity of $\bar{\lambda}$ in Equation (A.7), we have

$$\bar{\lambda}(C) \leq \bar{\lambda}((A \cap B) \cap C) + \bar{\lambda}((A \cap B)^c \cap C). \quad (\text{A.9})$$

Since the reverse inequality also holds:

$$\begin{aligned}
\bar{\lambda}(C) &= \bar{\lambda}(B \cap C) + \bar{\lambda}(B^c \cap C) \\
&= \bar{\lambda}(A \cap B \cap C) + \bar{\lambda}(A^c \cap B \cap C) + \bar{\lambda}(A \cap B^c \cap C) + \bar{\lambda}(A^c \cap B^c \cap C) \\
&\geq \bar{\lambda}(A \cap B \cap C) + \bar{\lambda}((A^c \cap B \cap C) \cup (A \cap B^c \cap C) \cup (A^c \cap B^c \cap C)) \\
&= \bar{\lambda}((A \cap B) \cap C) + \bar{\lambda}((A \cap B)^c \cap C),
\end{aligned}$$

we have in fact equality in Equation (A.9), which shows that $A \cup B \in \mathcal{M}$, by Definition A.2. $\square$

When the outer measure $\bar{\lambda}$ is restricted to $\mathcal{M}$, we have the following form of countable additivity. Take $C = E$ to get the usual countable additivity.

Lemma A.3: The Outer Measure $\bar{\lambda}$ Restricted to $\mathcal{M}$ is Countably Additive

If $\{A_k\} \in \mathcal{M}$ are disjoint, then for any $C \in 2^E$:

$$\bar{\lambda}((\cup_k A_k) \cap C) = \sum_k \bar{\lambda}(A_k \cap C). \quad (\text{A.10})$$

Proof. We first prove the case for a *finite* collection $A_1, \dots, A_n$ by induction, starting with $n = 2$. In that case, if $A_1 \cup A_2 = E$, then the statement follows from the definition of $\mathcal{M}$ in Equation (A.8). If $A_1 \cup A_2$ is smaller than E , take $C := (A_1 \cup A_2) \cap D$ and $A := A_1$ in Equation (A.8) and use the disjointness of A_1 and A_2 to conclude that

$$\bar{\lambda}(A_1 \cap D) + \bar{\lambda}(A_2 \cap D) = \bar{\lambda}((A_1 \cup A_2) \cap D),$$

showing that the induction statement holds for $n = 2$. Suppose that Equation (A.10) holds for $n - 1$ disjoint sets $A_1, \dots, A_{n-1}$ in $\mathcal{M}$. Then, by the induction hypothesis and the case $n = 2$, we have for any C ,

$$\bar{\lambda}((\cup_{k=1}^n A_k) \cap C) = \bar{\lambda}((\cup_{k=1}^{n-1} A_k) \cap C) + \bar{\lambda}(A_n \cap C) = \sum_{k=1}^n \bar{\lambda}(A_k \cap C),$$

completing the induction step. For an *infinite* sequence $A_1, A_2, \dots$ of disjoint sets, we have, by the monotonicity of $\bar{\lambda}$:

$$\bar{\lambda}((\cup_{k=1}^{\infty} A_k) \cap C) \geq \bar{\lambda}((\cup_{k=1}^n A_k) \cap C) = \sum_{k=1}^n \bar{\lambda}(A_k \cap C)$$

for all n , so that

$$\bar{\lambda}((\cup_{k=1}^{\infty} A_k) \cap C) \geq \sum_{k=1}^{\infty} \bar{\lambda}(A_k \cap C).$$

The proof is completed by observing that the reverse inequality also holds, by the countable subadditivity of $\bar{\lambda}$ in [Equation \(A.7\)](#). $\square$

The following proposition contains all the ingredients needed for the existence proof:

Proposition A.2: Four Main Properties of $\bar{\lambda}$ and $\mathcal{M}$

- (1) $\mathcal{M}$ is a σ -algebra.
- (2) $\bar{\lambda}$ restricted to $\mathcal{M}$ is countably additive.
- (3) $\mathcal{E}_0 \subset \mathcal{M}$.
- (4) For $A \in \mathcal{E}_0$, we have

$$\bar{\lambda}(A) = \lambda(A) = |A|. \quad (\text{A.11})$$

Proof. (1) We have already proved that $\mathcal{M}$ is an algebra in [Lemma A.2](#). To show that it is a σ -algebra, let $A_1, A_2, \dots$ be in $\mathcal{M}$ and $B_1 := A_1$ and $B_n := A_n \setminus A_{n-1}$, $n = 2, 3, \dots$. The $\{B_k\}$ are disjoint and in $\mathcal{M}$. By the countable additivity of $\bar{\lambda}$ on $\mathcal{M}$ (see [Lemma A.3](#)), we have for any $C \in 2^E$:

$$\bar{\lambda}(\cup_k B_k \cap C) + \bar{\lambda}((\cup_k B_k)^c \cap C) = \bar{\lambda}(C).$$

Thus, [Equation \(A.8\)](#) holds for the set $\cup_k B_k$ and the latter therefore lies in $\mathcal{M}$. But since $\cup_{k=1}^n B_k = A_n$ and $\cup_{k=1}^{\infty} B_k = \cup_{k=1}^{\infty} A_k$, the latter also lies in $\mathcal{M}$. In other words, $\mathcal{M}$ is closed under countable unions.

(2) We have already proved this in [Lemma A.3](#).

(3) We need to show that for any $A \in \mathcal{E}_0$ it holds that

$$\bar{\lambda}(A \cap C) + \bar{\lambda}(A^c \cap C) = \bar{\lambda}(C) \quad \text{for all } C \in 2^E.$$

Similar to the proof of [Lemma A.1](#), take $\varepsilon > 0$ and choose sets $\{A_n\}$ in $\mathcal{E}_0$ such that $C \subseteq \cup_n A_n$ and $\sum_n \lambda(A_n) \leq \bar{\lambda}(C) + \varepsilon$. This can always be done, by the way $\bar{\lambda}$ is defined. Define sets $F_n := A_n \cap A$ and $G_n = A_n \cap A^c$, $n = 1, 2, \dots$. These sets lie in $\mathcal{M}$ as it is an algebra. Note that $\cup_n F_n$ contains $C \cap A$ and $\cup_n G_n$ contains $C \cap A^c$. Thus, by the definition of $\bar{\lambda}$ and the finite additivity of λ , we have

$$\bar{\lambda}(A \cap C) + \bar{\lambda}(A^c \cap C) \leq \sum_n \lambda(F_n) + \sum_n \lambda(G_n) = \sum_n \lambda(A_n) \leq \bar{\lambda}(C) + \varepsilon.$$

Since ε is arbitrary, this shows that $\bar{\lambda}(A \cap C) + \bar{\lambda}(A^c \cap C) \leq \bar{\lambda}(C)$. The reverse inequality holds also, by the countable subadditivity of $\bar{\lambda}$ in [Lemma A.1](#).

(4) From the definition of the outer measure ([Equation \(A.6\)](#)) it is clear that $\bar{\lambda}(A) \leq \lambda(A)$ for all $A \in \mathcal{E}_0$. To show that also $\lambda(A) \leq \bar{\lambda}(A)$, we use the countable additivity of λ proved in [Theorem A.1](#). In particular, let $A \subseteq \cup_n A_n$, where A and A_n , $n = 1, 2, \dots$ are in $\mathcal{E}_0$. Then,

$$\lambda(A) \leq \lambda\left(\bigcup_{n=1}^{\infty} (A \cap A_n)\right) \leq \sum_{n=1}^{\infty} \lambda(A \cap A_n) \leq \sum_{n=1}^{\infty} \lambda(A_n). \quad (\text{A.12})$$

The second inequality follows from the countable subadditivity of the pre-measure λ , which can be proved in exactly the same way as for proper measures. As [Equation \(A.12\)](#) holds for any choice of $\{A_n\}$, it must hold that $\lambda(A) \leq \inf \sum_n \lambda(A_n) = \bar{\lambda}(A)$.

□

Using [Proposition A.2](#), the existence of the Lebesgue measure can now be proved with a few brush strokes.

Theorem A.2: Existence of the Lebesgue Measure

The outer measure $\bar{\lambda}$ restricted to $\mathcal{B}_{(0,1]}$ is a probability measure that extends the pre-measure λ .

Proof. From [Proposition A.2](#), the outer measure $\bar{\lambda}$ restricted to $\mathcal{M}$ is a proper measure on $(E, \mathcal{M})$, as it is countably additive and $\mathcal{M}$ is a σ -algebra. Also, we have

$$\mathcal{E}_0 \subset \sigma(\mathcal{E}_0) \subset \mathcal{M} \subset 2^E.$$

So, $\bar{\lambda}$ is also a measure on $(E, \sigma(\mathcal{E}_0))$, where $\sigma(\mathcal{E}_0) = \mathcal{B}_{(0,1]}$. That $\bar{\lambda}$ is an extension of λ follows from [Equation \(A.11\)](#). Finally, the fact that $\bar{\lambda}(E) = \lambda(E) = 1$ shows that it is a probability measure. $\square$

Note that the proofs above hold verbatim for any probability pre-measure λ on $(E, \mathcal{E}_0)$; that is, λ is σ -additive and $\lambda(E) = 1$. [Theorem 1.9](#) then guarantees that λ can be uniquely extended to a measure on $(E, \sigma(\mathcal{E}_0))$.

B Linear Algebra and Functional Analysis

B.1 Vector Spaces, Bases, and Matrices

Linear algebra is the study of vector spaces and linear mappings. Vectors are, by definition, elements of some *vector space* $\mathcal{V}$ and satisfy the usual rules of addition and scalar multiplication, e.g.,

$$\text{if } \mathbf{x} \in \mathcal{V} \text{ and } \mathbf{y} \in \mathcal{V}, \text{ then } \alpha\mathbf{x} + \beta\mathbf{y} \in \mathcal{V} \text{ for all } \alpha, \beta \in \mathbb{R} \text{ (or } \mathbb{C}\text{)}.$$

We will be dealing mostly with vectors in the Euclidean vector space $\mathbb{R}^n$ for some n . That is, we view the points of $\mathbb{R}^n$ as objects that can be added up and multiplied by a scalar, e.g., $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$ for points in $\mathbb{R}^2$. Sometimes it is convenient to work with the complex vector space $\mathbb{C}^n$ instead of $\mathbb{R}^n$.

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are called *linearly independent* if none of them can be expressed as a linear combination of the others; that is, if $\alpha_1\mathbf{v}_1 + \dots + \alpha_n\mathbf{v}_n = \mathbf{0}$, then it must hold that $\alpha_i = 0$ for all $i = 1, \dots, n$.

Definition B.1: Basis of a Vector Space

A set of vectors $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is called a *basis* of the vector space $\mathcal{V}$ if every vector $\mathbf{x} \in \mathcal{V}$ can be written as a unique linear combination of the vectors in $\mathcal{B}$.

$$\mathbf{x} = \alpha_1\mathbf{v}_1 + \dots + \alpha_n\mathbf{v}_n.$$

The (possibly infinite) number n is called the *dimension* of $\mathcal{V}$.

Using a basis $\mathcal{B}$ of $\mathcal{V}$, we can thus represent each vector $\mathbf{x} \in \mathcal{V}$ as a row or column of numbers

$$[\alpha_1, \dots, \alpha_n] \quad \text{or} \quad \begin{bmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{bmatrix}. \quad (\text{B.1})$$

Typically, vectors in $\mathbb{R}^n$ are represented via the *standard basis*, consisting of unit vectors (points) $\mathbf{e}_1 = (1, 0, \dots, 0), \dots, \mathbf{e}_n = (0, 0, \dots, 0, 1)$. As a consequence, any point $(x_1, \dots, x_n) \in \mathbb{R}^n$ can be represented, using the standard basis, as a row or column vector of the form Equation (B.1) above, with $\alpha_i = x_i, i = 1, \dots, n$. We will also write $[x_1, x_2, \dots, x_n]^\top$, for the corresponding column vector, where $^\top$ denotes the *transpose*.

Attention B.1: T

avoid confusion, we will use the convention from now on that a generic vector $\mathbf{x}$ is always represented via the standard basis as a column vector. The corresponding row vector is denoted by $\mathbf{x}^\top$.

A *matrix* can be viewed as an array of m rows and n columns that defines a *linear transformation* from $\mathbb{R}^n$ to $\mathbb{R}^m$ (or for complex matrices, from $\mathbb{C}^n$ to $\mathbb{C}^m$). The matrix is said to be *square* if $m = n$. If $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are the columns of A , that is, $A = [\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n]$, and if $\mathbf{x} = [x_1, \dots, x_n]^\top$, then $A\mathbf{x} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n$. In particular, the standard basis vector $\mathbf{e}_k$ is mapped to the vector $\mathbf{a}_k, k = 1, \dots, n$. We sometimes use the notation $A = [a_{ij}]$, to denote a matrix whose (i, j) -th element is a_{ij} . When we wish to emphasize that a matrix A is real-valued with m rows and n columns, we write $A \in \mathbb{R}^{m \times n}$. The *rank* of a matrix is the number of linearly independent rows or, equivalently, the number of linearly independent columns.

The *transpose* of a matrix $A = [a_{ij}]$ is the matrix $A^\top = [a_{ji}]$; that is, the (i, j) -th element of $A^\top$ is the (j, i) -th element of A . The *trace* of a square matrix is the sum of its diagonal elements. A useful result is the following cyclic property.

Theorem B.1: Cyclic Property

The trace is invariant under cyclic permutations: $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$.

Proof. It suffices to show that $\text{tr}(DE)$ is equal to $\text{tr}(ED)$ for any $m \times n$ matrix $D = [d_{ij}]$ and $n \times m$ matrix $E = [e_{ij}]$. The diagonal elements of DE are $\sum_{j=1}^n d_{ij}e_{ji}, i = 1, \dots, m$ and the diagonal elements of ED are $\sum_{i=1}^m e_{ji}d_{ij}, j = 1, \dots, n$. They sum up to the same number $\sum_{i=1}^m \sum_{j=1}^n d_{ij}e_{ji}$. $\square$

A square matrix has an inverse if and only if its columns (or rows) are *linearly independent*. This is the same as the matrix being of *full rank*; that is, its rank is equal to the number of columns. An equivalent statement is that its determinant is not zero. The *determinant* of an $n \times n$ matrix $A = [a_{i,j}]$ is defined as

$$\det(A) := \sum_{\pi} (-1)^{\zeta(\pi)} \prod_{i=1}^n a_{\pi_i, i}, \quad (\text{A.3})$$

where the sum is over all permutations $\pi = (\pi_1, \dots, \pi_n)$ of $(1, \dots, n)$, and $\zeta(\pi)$ is the number of pairs (i, j) for which $i < j$ and $\pi_i > \pi_j$. For example, $\zeta(2, 3, 4, 1) = 3$ for the pairs $(1, 4), (2, 4), (3, 4)$. The determinant of a *diagonal matrix*, a matrix with only zero elements off the diagonal, is simply the product of its diagonal elements.

Geometrically, the determinant of a square matrix $A = [\mathbf{a}_1, \dots, \mathbf{a}_n]$ is the (signed) *volume* of the parallelepiped (n -dimensional parallelogram) defined by the columns $\mathbf{a}_1, \dots, \mathbf{a}_n$; that is, the set of points $\mathbf{x} = \sum_{i=1}^n \alpha_i \mathbf{a}_i$, where $0 \leq \alpha_i \leq 1, i = 1, \dots, n$.

The easiest way to compute the determinant of a general matrix is to apply simple operations to the matrix that potentially reduce its complexity (as in the number of non-zero elements, for example), while retaining its determinant:

- Adding a multiple of one column (or row) to another does not change the determinant.
- Multiplying a column (or row) by a number multiplies the determinant by the same number.
- Swapping two rows changes the sign of the determinant.

By applying these rules repeatedly, one can reduce any matrix to a diagonal matrix. It follows then that the determinant of the original matrix is equal to the product of the diagonal elements of the resulting diagonal matrix multiplied by a known constant.

Theorem B.2: Useful Matrix Rules

- (1) $(AB)^\top = B^\top A^\top$
- (2) $(AB)^{-1} = B^{-1}A^{-1}$
- (3) $(A^{-1})^\top = (A^\top)^{-1} := A^{-\top}$
- (4) $\det(AB) = \det(A)\det(B)$
- (5) $\mathbf{x}^\top A \mathbf{x} = \text{tr}(A \mathbf{x} \mathbf{x}^\top)$
- (6) $\det(A) = \prod_i a_{ii}$ if $A = [a_{ij}]$ is triangular

Theorem B.2 summarises a number of useful matrix rules for the concepts that we have discussed so far.

Next, consider an $n \times p$ matrix A for which the matrix inverse fails to exist. That is, A is either non-square ($n \neq p$) or its determinant is 0. Instead of the inverse, we can use its so-called pseudo-inverse, which always exists.

Definition B.2: Moore–Penrose Pseudo-Inverse

The *Moore–Penrose pseudo-inverse* of a real matrix $A \in \mathbb{R}^{n \times p}$ is defined as the unique matrix $A^+ \in \mathbb{R}^{p \times n}$ that satisfies the conditions:

- (1) $AA^+A = A$
- (2) $A^+AA^+ = A^+$
- (3) $(AA^+)^\top = AA^+$
- (4) $(A^+A)^\top = A^+A$

We can write A^+ explicitly in terms of A when A has a full column or row rank. For example, we always have

$$A^\top AA^+ = A^\top (AA^+)^\top = ((AA^+)A)^\top = (A)^\top = A^\top. \quad (\text{B.2})$$

If A has a full column rank p , then $(A^\top A)^{-1}$ exists, so that from (B.2) it follows that $A^+ = (A^\top A)^{-1} A^\top$. This is referred to as the *left pseudo-inverse*, as $A^+A = I_p$. Similarly, if A has a full row rank n , that is, $(AA^\top)^{-1}$ exists, then it follows from

$$A^+AA^\top = (A^+A)^\top A^\top = (A(A^+A))^\top = A^\top$$

that $A^+ = A^\top (AA^\top)^{-1}$. This is the *right pseudo-inverse*, as $AA^+ = I_n$. Finally, if A is of full rank and square, then $A^+ = A^{-1}$.

B.2 Inner Product

The (Euclidean) *inner product* of two real vectors $\mathbf{x} = [x_1, \dots, x_n]^\top$ and $\mathbf{y} = [y_1, \dots, y_n]^\top$ is defined as the number

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^n x_i y_i = \mathbf{x}^\top \mathbf{y}.$$

Here $\mathbf{x}^\top \mathbf{y}$ is the matrix multiplication of the $(1 \times n)$ matrix $\mathbf{x}^\top$ and the $(n \times 1)$ matrix $\mathbf{y}$.

The inner product induces a geometry on the linear space $\mathbb{R}^n$, allowing for the definition of length, angle, and so on. The inner product satisfies the following properties:

- (1) $\langle \alpha \mathbf{x} + \beta \mathbf{y}, \mathbf{z} \rangle = \alpha \langle \mathbf{x}, \mathbf{z} \rangle + \beta \langle \mathbf{y}, \mathbf{z} \rangle$;
- (2) $\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$;
- (3) $\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$;
- (4) $\langle \mathbf{x}, \mathbf{x} \rangle = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Vectors $\mathbf{x}$ and $\mathbf{y}$ are called *perpendicular* (or *orthogonal*) if $\langle \mathbf{x}, \mathbf{y} \rangle = 0$. The *Euclidean norm* (or length) of a vector $\mathbf{x}$ is defined as

$$\|\mathbf{x}\| = \sqrt{x_1^2 + \dots + x_n^2} = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

If $\mathbf{x}$ and $\mathbf{y}$ are perpendicular, then *Pythagoras' theorem* holds:

$$\|\mathbf{x} + \mathbf{y}\|^2 = \langle \mathbf{x} + \mathbf{y}, \mathbf{x} + \mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{x} \rangle + 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2. \quad (\text{A.5})$$

A basis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ of $\mathbb{R}^n$ in which all the vectors are pairwise perpendicular and have norm 1 is called an *orthonormal* (short for orthogonal and normalized) basis. For example, the standard basis is orthonormal.

Theorem B.3: Orthonormal Basis Representation

If $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is an orthonormal basis of $\mathbb{R}^n$, then any vector $\mathbf{x} \in \mathbb{R}^n$ can be expressed as

$$\mathbf{x} = \langle \mathbf{x}, \mathbf{v}_1 \rangle \mathbf{v}_1 + \dots + \langle \mathbf{x}, \mathbf{v}_n \rangle \mathbf{v}_n. \quad (\text{A.6})$$

Proof. Observe that, because the $\{\mathbf{v}_i\}$ form a basis, there exist unique $\alpha_1, \dots, \alpha_n$ such that $\mathbf{x} = \alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n$. By the linearity of the inner product and the orthonormality of the $\{\mathbf{v}_i\}$ it follows that $\langle \mathbf{x}, \mathbf{v}_j \rangle = \langle \sum_i \alpha_i \mathbf{v}_i, \mathbf{v}_j \rangle = \alpha_j$. $\square$

An $n \times n$ matrix V whose columns form an orthonormal basis is called an *orthogonal matrix*.⁴ Note that for an orthogonal matrix $V = [\mathbf{v}_1, \dots, \mathbf{v}_n]$, we have

$$V^T V = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_n^T \end{bmatrix} [\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n] = \begin{bmatrix} \mathbf{v}_1^T \mathbf{v}_1 & \mathbf{v}_1^T \mathbf{v}_2 & \dots & \mathbf{v}_1^T \mathbf{v}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{v}_n^T \mathbf{v}_1 & \mathbf{v}_n^T \mathbf{v}_2 & \dots & \mathbf{v}_n^T \mathbf{v}_n \end{bmatrix} = I_n.$$

Hence, $V^{-1} = V^T$. Note also that an orthogonal transformation is *length preserving*; that is, $V\mathbf{x}$ has the same length as $\mathbf{x}$. This follows from

$$\|V\mathbf{x}\|^2 = \langle V\mathbf{x}, V\mathbf{x} \rangle = \mathbf{x}^T V^T V \mathbf{x} = \mathbf{x}^T \mathbf{x} = \|\mathbf{x}\|^2.$$

B.3 Complex Vectors and Matrices

Instead of the vector space $\mathbb{R}^n$ of n -dimensional real vectors, it is sometimes useful to consider the vector space $\mathbb{C}^n$ of n -dimensional complex vectors. In this case the *adjoint* or *conjugate transpose* operation ($*$) replaces the transpose operation ($\top$). This involves the usual transposition of the matrix or vector with the additional step that any complex

⁴The qualifier “orthogonal” for such matrices has been fixed by history. A better term would have been “orthonormal”.

number $z = x + iy$ is replaced by its complex conjugate $\bar{z} = x - iy$. For example, if

$$\mathbf{x} = \begin{bmatrix} a_1 + ib_1 \\ a_2 + ib_2 \end{bmatrix} \quad \text{and} \quad \mathbf{A} = \begin{bmatrix} a_{11} + ib_{11} & a_{12} + ib_{12} \\ a_{21} + ib_{21} & a_{22} + ib_{22} \end{bmatrix},$$

then

$$\mathbf{x}^* = [a_1 - ib_1, a_2 - ib_2] \quad \text{and} \quad \mathbf{A}^* = \begin{bmatrix} a_{11} - ib_{11} & a_{21} - ib_{21} \\ a_{12} - ib_{12} & a_{22} - ib_{22} \end{bmatrix}.$$

The (Euclidean) inner product of $\mathbf{x}$ and $\mathbf{y}$ (viewed as column vectors) is now defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{y}^* \mathbf{x} = \sum_{i=1}^n x_i \bar{y}_i,$$

which is no longer symmetric: $\langle \mathbf{x}, \mathbf{y} \rangle = \overline{\langle \mathbf{y}, \mathbf{x} \rangle}$. Note that this generalizes the real-valued inner product. The determinant of a complex matrix $\mathbf{A}$ is defined exactly as in (??). As a consequence, $\det(\mathbf{A}^*) = \overline{\det(\mathbf{A})}$.

A complex matrix is said to be *Hermitian* or *self-adjoint* if $\mathbf{A}^* = \mathbf{A}$, and *unitary* if $\mathbf{A}^* \mathbf{A} = \mathbf{I}$ (that is, if $\mathbf{A}^* = \mathbf{A}^{-1}$). For real matrices “Hermitian” is the same as “symmetric”, and “unitary” is the same as “orthogonal”.

B.4 Orthogonal Projections

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be a set of linearly independent vectors in $\mathbb{R}^n$. The set

$$\mathcal{V} = \text{span } \mathbf{u}_1, \dots, \mathbf{u}_k = \{\alpha_1 \mathbf{u}_1 + \dots + \alpha_k \mathbf{u}_k, \alpha_1, \dots, \alpha_k \in \mathbb{R}\},$$

is called the *linear subspace* spanned by $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$. The *orthogonal complement* of $\mathcal{V}$, denoted by $\mathcal{V}^\perp$, is the set of all vectors $\mathbf{w}$ that are orthogonal to $\mathcal{V}$, in the sense that $\langle \mathbf{w}, \mathbf{v} \rangle = 0$ for all $\mathbf{v} \in \mathcal{V}$. The matrix $\mathbf{P}$ such that $\mathbf{P}\mathbf{x} = \mathbf{x}$, for all $\mathbf{x} \in \mathcal{V}$, and $\mathbf{P}\mathbf{x} = \mathbf{0}$, for all $\mathbf{x} \in \mathcal{V}^\perp$ is called the *orthogonal projection matrix* onto $\mathcal{V}$. Suppose that $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_k]$ has full rank, in which case $\mathbf{U}^\top \mathbf{U}$ is an invertible matrix. The orthogonal projection matrix

P onto $\mathcal{V} = \text{span } \mathbf{u}_1, \dots, \mathbf{u}_k$ is then given by

$$P = U(U^\top U)^{-1}U^\top.$$

Namely, since $PU = U$, the matrix P projects any vector in $\mathcal{V}$ onto itself. Moreover, P projects any vector in $\mathcal{V}^\perp$ onto the zero vector. Using the pseudo-inverse, it is possible to specify the projection matrix also for the case where U not of full rank, leading to the following theorem.

Theorem B.4: Orthogonal Projection

Let $U = [\mathbf{u}_1, \dots, \mathbf{u}_k]$. Then, the orthogonal projection matrix P onto $\mathcal{V} = \text{span } \mathbf{u}_1, \dots, \mathbf{u}_k$ is given by

$$P = UU^+, \quad (\text{A.7})$$

where U^+ is the (right) pseudo-inverse of U .

Proof. By Property 1 of Definition B.2 we have $PU = UU^+U = U$, so that P projects any vector in $\mathcal{V}$ onto itself. Moreover, P projects any vector in $\mathcal{V}^\perp$ onto the zero vector. $\square$

Note that in the special case where $\mathbf{u}_1, \dots, \mathbf{u}_k$ above form an orthonormal basis of $\mathcal{V}$, then the projection onto $\mathcal{V}$ is very simple to describe, namely we have

$$P\mathbf{x} = UU^\top \mathbf{x} = \sum_{i=1}^k \langle \mathbf{x}, \mathbf{u}_i \rangle \mathbf{u}_i. \quad (\text{A.8})$$

For any point $\mathbf{x} \in \mathbb{R}^n$, the point in $\mathcal{V}$ that is closest to $\mathbf{x}$ is its orthogonal projection $P\mathbf{x}$, as the following theorem shows.

Theorem B.5: Orthogonal Projection and Minimal Distance

Let $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ be an orthonormal basis of subspace $\mathcal{V}$ and let P be the orthogonal projection matrix onto $\mathcal{V}$. The solution to the minimization program

$$\min_{\mathbf{y} \in \mathcal{V}} \|\mathbf{x} - \mathbf{y}\|^2$$

is $\mathbf{y} = P\mathbf{x}$. That is, $P\mathbf{x} \in \mathcal{V}$ is closest to $\mathbf{x}$.

Proof. We can write each point $\mathbf{y} \in \mathcal{V}$ as $\mathbf{y} = \sum_{i=1}^k \alpha_i \mathbf{u}_i$. Consequently,

$$\|\mathbf{y} - \mathbf{x}\|^2 = \left\langle \mathbf{x} - \sum_{i=1}^k \alpha_i \mathbf{u}_i, \mathbf{x} - \sum_{i=1}^k \alpha_i \mathbf{u}_i \right\rangle = \|\mathbf{x}\|^2 - 2 \sum_{i=1}^k \alpha_i \langle \mathbf{x}, \mathbf{u}_i \rangle + \sum_{i=1}^k \alpha_i^2.$$

Minimizing this with respect to the $\{\alpha_i\}$ gives $\alpha_i = \langle \mathbf{x}, \mathbf{u}_i \rangle, i = 1, \dots, k$. In view of Equation (A.8), the optimal $\mathbf{y}$ is thus $P\mathbf{x}$. $\square$

B.5 Eigenvalues and Eigenvectors

Let $\mathbf{A}$ be an $n \times n$ matrix. If $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$ for some number λ and non-zero vector $\mathbf{v}$, then λ is called an *eigenvalue* of $\mathbf{A}$ with *eigenvector* $\mathbf{v}$.

If $(\lambda, \mathbf{v})$ is an (eigenvalue, eigenvector) pair, the matrix $\lambda\mathbf{I} - \mathbf{A}$ maps any multiple of $\mathbf{v}$ to the zero vector. Consequently, the columns of $\lambda\mathbf{I} - \mathbf{A}$ are linearly dependent, and hence their determinant is 0. This provides a way to identify the eigenvalues, namely as the $r \leq n$ different roots $\lambda_1, \dots, \lambda_r$ of the *characteristic polynomial*

$$\det(\lambda\mathbf{I} - \mathbf{A}) = (\lambda - \lambda_1)^{\alpha_1} \cdots (\lambda - \lambda_r)^{\alpha_r},$$

where $\alpha_1 + \cdots + \alpha_r = n$. The integer α_i is called the *algebraic multiplicity* of λ_i . The eigenvectors that correspond to an eigenvalue λ_i lie in the *kernel* or *null space* of the matrix $\lambda_i\mathbf{I} - \mathbf{A}$; that is, the linear space of vectors $\mathbf{v}$ such that $(\lambda_i\mathbf{I} - \mathbf{A})\mathbf{v} = \mathbf{0}$. This space is called the *eigenspace* of λ_i . Its dimension, $d_i \in \{1, \dots, n\}$, is called the *geometric multiplicity* of λ_i . It always holds that $d_i \leq \alpha_i$. If $\sum_i d_i = n$, then we can construct a basis for $\mathbb{R}^n$ consisting of eigenvectors.

A matrix for which the algebraic and geometric multiplicities of all its eigenvalues are the same is called *semi-simple*. This is equivalent to the matrix being *diagonalizable*, meaning that there is a matrix V and a diagonal matrix D such that

$$A = VDV^{-1}.$$

To see that this so-called *eigen-decomposition* holds, suppose A is a semi-simple matrix with eigenvalues

$$\underbrace{\lambda_1, \dots, \lambda_1}_{d_1}, \dots, \underbrace{\lambda_r, \dots, \lambda_r}_{d_r}.$$

Let D be the diagonal matrix whose diagonal elements are the eigenvalues of A , and let V be a matrix whose columns are linearly independent eigenvectors corresponding to these eigenvalues. Then, for each (eigenvalue, eigenvector) pair (λ, v) , we have $Av = \lambda v$. Hence, in matrix notation, we have $AV = VD$, and so $A = VDV^{-1}$.

B.6 Matrix Decompositions

Matrix decompositions are frequently employed in linear algebra to simplify proofs, mitigate numerical instability, and expedite computations. We mention three important matrix decompositions: (P)LU, QR, and SVD.

B.6.1 (P)LU Decomposition

Every invertible matrix A can be written as the product of three matrices:

$$A = PLU, \tag{B.3}$$

where L is a lower triangular matrix, U an upper triangular matrix, and P a *permutation matrix*. A permutation matrix is a square matrix with a single 1 in each row and column, and zeros otherwise. The matrix product PB simply permutes the rows of a matrix B and, likewise, BP permutes its columns. A decomposition of the form Equation (B.3) is called a *PLU decomposition*. As a permutation matrix is orthogonal, its transpose is equal to its

inverse, and so we can write Equation (B.3) as

$$P^T A = LU.$$

The decomposition is not unique, and in many cases P can be taken to be the identity matrix, in which case we speak of the *LU decomposition* of A , also called the LR for left-right (triangular) decomposition.

A PLU decomposition of an invertible $n \times n$ matrix A_0 can be obtained recursively as follows. The first step is to swap the rows of A_0 such that the element in the first column and first row of the pivoted matrix is as large as possible in absolute value. Write the resulting matrix as

$$\tilde{P}_0 A_0 = \begin{bmatrix} a_1 & \mathbf{b}_1^T \\ \mathbf{c}_1 & D_1 \end{bmatrix},$$

where $\tilde{P}_0$ is the permutation matrix that swaps the first and k -th row, where k is the row that contains the largest element in the first column. Next, add the matrix $-\mathbf{c}_1[1, \mathbf{b}_1^T/a_1]$ to the last $n - 1$ rows of $\tilde{P}_0 A_0$, to obtain the matrix

$$\begin{bmatrix} a_1 & \mathbf{b}_1^T \\ \mathbf{0} & D_1 - \mathbf{c}_1 \mathbf{b}_1^T / a_1 \end{bmatrix} := \begin{bmatrix} a_1 & \mathbf{b}_1^T \\ \mathbf{0} & A_1 \end{bmatrix}.$$

In effect, we add some multiple of the first row to each of the remaining rows in order to obtain zeros in the first column, except for the first element.

We now apply the same procedure to A_1 as we did to A_0 and then to subsequent smaller matrices $A_2, \dots, A_{n-1}$:

- (1) Swap the first row with the row having the maximal absolute value element in the first column.
- (2) Make every other element in the first column equal to 0 by adding appropriate multiples of the first row to the other rows.

Suppose that A_t has a PLU decomposition $P_t L_t U_t$. Then it is easy to check that

$$\underbrace{\widetilde{P}_{t-1}^\top \begin{bmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & P_t \end{bmatrix}}_{P_{t-1}} \underbrace{\begin{bmatrix} 1 & \mathbf{0}^\top \\ P_t^\top c_t/a_t & L_t \end{bmatrix}}_{L_{t-1}} \underbrace{\begin{bmatrix} a_t & b_t^\top \\ \mathbf{0} & U_t \end{bmatrix}}_{U_{t-1}} \quad (\text{A.10})$$

is a PLU decomposition of A_{t-1} . Since the PLU decomposition for the scalar A_{n-1} is trivial, by working backwards, we obtain a PLU decomposition $P_0 L_0 U_0$ of A .

B.6.2 Woodbury Identity

LU (or more generally PLU) decompositions can also be applied to block matrices. A starting point is the following LU decomposition for a general 2×2 matrix:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a & 0 \\ c & d - bc/a \end{bmatrix} \begin{bmatrix} 1 & b/a \\ 0 & 1 \end{bmatrix},$$

which holds as long as $a \neq 0$; this can be seen by simply writing out the matrix product.

The block matrix generalization for matrices $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times k}$, $C \in \mathbb{R}^{k \times n}$, $D \in \mathbb{R}^{k \times k}$ is

$$\Sigma := \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} A & O_{n \times k} \\ C & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I_n & A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix}, \quad (\text{B.4})$$

provided that A is invertible (again, write out the block matrix product). Here, we use the notation $O_{p \times q}$ to denote the $p \times q$ matrix of zeros. We can further rewrite this as:

$$\Sigma = \begin{bmatrix} I_n & O_{n \times k} \\ CA^{-1} & I_k \end{bmatrix} \begin{bmatrix} A & O_{n \times k} \\ O_{k \times n} & D - CA^{-1}B \end{bmatrix} \begin{bmatrix} I_n & A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix}.$$

Thus, inverting both sides, we obtain

$$\Sigma^{-1} = \begin{bmatrix} I_n & A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix}^{-1} \begin{bmatrix} A & O_{n \times k} \\ O_{k \times n} & D - CA^{-1}B \end{bmatrix}^{-1} \begin{bmatrix} I_n & O_{n \times k} \\ CA^{-1} & I_k \end{bmatrix}^{-1}.$$

Inversion of the above block matrices gives (again write out)

$$\Sigma^{-1} = \begin{bmatrix} I_n & -A^{-1}B \\ O_{k \times n} & I_k \end{bmatrix} \begin{bmatrix} A^{-1} & O_{n \times k} \\ O_{k \times n} & (D - CA^{-1}B)^{-1} \end{bmatrix} \begin{bmatrix} I_n & O_{n \times k} \\ -CA^{-1} & I_k \end{bmatrix}. \quad (\text{B.5})$$

Assuming that D is invertible, we could also perform a block UL (as opposed to LU) decomposition:

$$\Sigma = \begin{bmatrix} A - BD^{-1}C & B \\ O_{k \times n} & D \end{bmatrix} \begin{bmatrix} I_n & O_{n \times k} \\ D^{-1}C & I_k \end{bmatrix}, \quad (\text{B.6})$$

which, after a similar calculation as the one above, yields

$$\Sigma^{-1} = \begin{bmatrix} I_n & O_{n \times k} \\ -D^{-1}C & I_k \end{bmatrix} \begin{bmatrix} (A - BD^{-1}C)^{-1} & O_{n \times k} \\ O_{k \times n} & D^{-1} \end{bmatrix} \begin{bmatrix} I_n & -BD^{-1} \\ O_{k \times n} & I_k \end{bmatrix}. \quad (\text{B.7})$$

The upper-left block of Σ^{-1} from Equation (B.7) must be the same as the upper-left block of Σ^{-1} from Equation (B.5), leading to the *Woodbury identity*:

$$(A - BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1}. \quad (\text{B.8})$$

From Equation (B.4) and the fact that the determinant of a product is the product of the determinants, we see that $\det(\Sigma) = \det(A) \det(D - CA^{-1}B)$. Similarly, from Equation (B.6) we have $\det(\Sigma) = \det(A - BD^{-1}C) \det(D)$, leading to the identity

$$\det(A - BD^{-1}C) \det(D) = \det(A) \det(D - CA^{-1}B). \quad (\text{B.9})$$

B.6.3 Cholesky Decomposition

If A is a real-valued positive definite matrix (and therefore symmetric), e.g., a covariance matrix, then an LU decomposition can be achieved with matrices L and $U = L^T$.

Theorem B.6: Cholesky Decomposition

A real-valued positive definite matrix $A = [a_{ij}] \in \mathbb{R}^{n \times n}$ can be decomposed as

$$A = LL^\top, \quad (\text{B.10})$$

where the real $n \times n$ lower triangular matrix $L = [l_{kj}]$ satisfies the recursive formula

$$l_{kj} = \frac{a_{kj} - \sum_{i=1}^{j-1} l_{ji} l_{ki}}{\sqrt{a_{jj} - \sum_{i=1}^{j-1} l_{ji}^2}}, \quad \text{where} \quad \sum_{i=1}^0 l_{ji} l_{ki} := 0 \quad (\text{B.11})$$

for $k = 1, \dots, n$ and $j = 1, \dots, k$.

Proof. The proof is by inductive construction. For $k = 1, \dots, n$, let A_k be the left-upper $k \times k$ submatrix of $A = A_n$. With $\mathbf{e}_1 := [1, 0, \dots, 0]^\top$, we have $A_1 = a_{11} = \mathbf{e}_1^\top A \mathbf{e}_1 > 0$ by the positive-definiteness of A . It follows that $l_{11} = \sqrt{a_{11}}$. Suppose that A_{k-1} has a Cholesky factorization $L_{k-1} L_{k-1}^\top$ with L_{k-1} having strictly positive diagonal elements, we can construct a Cholesky factorization of A_k as follows. First write

$$A_k = \begin{bmatrix} L_{k-1} L_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} \quad (\text{B.12})$$

and propose L_k to be of the form

$$L_k = \begin{bmatrix} L_{k-1} & \mathbf{0} \\ \mathbf{l}_k^\top & l_{kk} \end{bmatrix} \quad (\text{B.13})$$

for some vector $\mathbf{l}_k \in \mathbb{R}^{k-1}$ and scalar l_{kk} , for which it must hold that

$$\begin{bmatrix} L_{k-1} L_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} = \begin{bmatrix} L_{k-1} & \mathbf{0} \\ \mathbf{l}_k^\top & l_{kk} \end{bmatrix} \begin{bmatrix} L_{k-1}^\top & \mathbf{l}_k \\ \mathbf{0}^\top & l_{kk} \end{bmatrix}. \quad (\text{B.14})$$

To establish that such an $\mathbf{l}_k$ and l_{kk} exist, we must verify that the set of equations

$$\begin{aligned} \mathbf{L}_{k-1}\mathbf{l}_k &= \mathbf{a}_{k-1} \\ \mathbf{l}_k^\top \mathbf{l}_k + l_{kk}^2 &= a_{kk} \end{aligned} \tag{B.15}$$

has a solution. The system $\mathbf{L}_{k-1}\mathbf{l}_k = \mathbf{a}_{k-1}$ has a unique solution, because (by assumption) $\mathbf{L}_{k-1}$ is lower diagonal with strictly positive entries down the main diagonal and we can solve for $\mathbf{l}_k$ using forward substitution: $\mathbf{l}_k = \mathbf{L}_{k-1}^{-1}\mathbf{a}_{k-1}$. We can solve the second equation as $l_{kk} = \sqrt{a_{kk} - \|\mathbf{l}_k\|^2}$, provided that the term within the square root is positive. We demonstrate this using the fact that $\mathbf{A}$ is a positive definite matrix. In particular, for $\mathbf{x} \in \mathbb{R}^n$ of the form $[\mathbf{x}_1^\top, x_2, \mathbf{0}^\top]^\top$, where $\mathbf{x}_1$ is a non-zero $(k-1)$ -dimensional vector and x_2 a non-zero number, we have

$$0 < \mathbf{x}^\top \mathbf{A} \mathbf{x} = [\mathbf{x}_1^\top, x_2] \begin{bmatrix} \mathbf{L}_{k-1}\mathbf{L}_{k-1}^\top & \mathbf{a}_{k-1} \\ \mathbf{a}_{k-1}^\top & a_{kk} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ x_2 \end{bmatrix} = \|\mathbf{L}_{k-1}^\top \mathbf{x}_1\|^2 + 2\mathbf{x}_1^\top \mathbf{a}_{k-1} x_2 + a_{kk} x_2^2. \tag{B.16}$$

Now take $\mathbf{x}_1 = -x_2 \mathbf{L}_{k-1}^{-\top} \mathbf{l}_k$ to obtain $0 < \mathbf{x}^\top \mathbf{A} \mathbf{x} = x_2^2 (a_{kk} - \|\mathbf{l}_k\|^2)$. Therefore, (B.15) can be uniquely solved. As we have already solved it for $k = 1$, we can solve it for any $k = 1, \dots, n$, leading to the recursive formula (B.11) and Algorithm 1 below. $\square$

An implementation of Cholesky's decomposition, using the notation from the proof of B.6, is as follows. The running cost of this algorithm is $O(n^3)$.

Algorithm 1 Cholesky Decomposition**Require:** Positive-definite $n \times n$ matrix A_n with entries $\{a_{ij}\}$.**Ensure:** Lower triangular L_n such that $L_n L_n^\top = A_n$.

```

1:  $L_1 \leftarrow \sqrt{a_{11}}$ 
2: for  $k = 2, \dots, n$  do
3:    $\mathbf{a}_{k-1} \leftarrow [a_{1k}, \dots, a_{k-1,k}]^\top$ 
4:    $\mathbf{l}_{k-1} \leftarrow L_{k-1}^{-1} \mathbf{a}_{k-1}$ 
5:    $l_{kk} \leftarrow \sqrt{a_{kk} - \mathbf{l}_{k-1}^\top \mathbf{l}_{k-1}}$ 
6:    $L_k \leftarrow \begin{bmatrix} L_{k-1} & \mathbf{0} \\ \mathbf{l}_{k-1}^\top & l_{kk} \end{bmatrix}$ 
7: end for
8: return  $L_n$ 

```

B.7 Functional Analysis

Much of the previous theory on Euclidean vector spaces can be generalized to vector spaces of *functions*. Every element of a (real-valued) *function space* $\mathcal{H}$ is a function from some set $\mathcal{X}$ to $\mathbb{R}$, and elements can be added and scalar multiplied as if they were vectors. In other words, if $f \in \mathcal{H}$ and $g \in \mathcal{H}$, then $\alpha f + \beta g \in \mathcal{H}$ for all $\alpha, \beta \in \mathbb{R}$. On $\mathcal{H}$ we can impose an inner product as a mapping $\langle \cdot, \cdot \rangle$ from $\mathcal{H} \times \mathcal{H}$ to $\mathbb{R}$ that satisfies

- (1) $\langle \alpha f_1 + \beta f_2, g \rangle = \alpha \langle f_1, g \rangle + \beta \langle f_2, g \rangle$;
- (2) $\langle f, g \rangle = \langle g, f \rangle$;
- (3) $\langle f, f \rangle \geq 0$;
- (4) $\langle f, f \rangle = 0$ if and only if $f = 0$ (the zero function).

We focus on real-valued function spaces, although the theory for complex-valued function spaces is similar (and sometimes easier), under suitable modifications (e.g., $\langle f, g \rangle = \overline{\langle g, f \rangle}$).

We say that two elements f and g in $\mathcal{H}$ are *orthogonal* to each other with respect to this inner product if $\langle f, g \rangle = 0$. Given an inner product, we can measure distances between elements of the function space $\mathcal{H}$ using the *norm*

$$\|f\| := \sqrt{\langle f, f \rangle}.$$

For example, the distance between two functions f_m and f_n is given by $\|f_m - f_n\|$. The space $\mathcal{H}$ is said to be *complete* if every sequence of functions $f_1, f_2, \dots \in \mathcal{H}$ for which

$$\|f_m - f_n\| \rightarrow 0 \text{ as } m, n \rightarrow \infty, \quad (\text{B.17})$$

converges to some $f \in \mathcal{H}$; that is, $\|f - f_n\| \rightarrow 0$ as $n \rightarrow \infty$. A sequence that satisfies (B.17) is called a *Cauchy sequence*.

A complete inner product space is called a *Hilbert space*. The most fundamental Hilbert space of functions is the space L^2 . An in-depth introduction to L^2 requires some measure theory [6]. For our purposes, it suffices to assume that $\mathcal{X} \subseteq \mathbb{R}^d$ and that on $\mathcal{X}$ a *measure* μ is defined which assigns to each suitable⁵ set A a positive number $\mu(A) \geq 0$ (e.g., its volume). In many cases of interest μ is of the form

$$\mu(A) = \int_A w(x) dx \quad (\text{B.18})$$

where $w \geq 0$ is a positive function on $\mathcal{X}$ which is called the *density* of μ with respect to the Lebesgue measure (the natural volume measure on $\mathbb{R}^d$). We write $\mu(dx) = w(x)dx$ to indicate that μ has density w . Another important case is where

$$\mu(A) = \sum_{x \in A \cap \mathbb{Z}^d} w(x), \quad (\text{B.19})$$

where $w \geq 0$ is again called the density of μ , but now with respect to the counting measure on $\mathbb{R}^d$ (which counts the points of $\mathbb{Z}^d$). Integrals with respect to measures μ in (B.18) and (B.19) can now be defined as

$$\int f(x) \mu(dx) = \int f(x) w(x) dx,$$

and

$$\int f(x) \mu(dx) = \sum_x f(x) w(x),$$

⁵Not all sets have a measure. Suitable sets are Borel sets, which can be thought of as countable unions of rectangles.

respectively. We assume for simplicity that μ has the form (B.18). For measures of the form (B.19) (so-called discrete measures), replace integrals by sums in what follows.

Definition B.3: L^2 Space

Let $\mathcal{X}$ be a subset of $\mathbb{R}^d$ with measure $\mu(dx) = w(x)dx$. The Hilbert space $L^2(\mathcal{X}, \mu)$ is the linear space of functions from $\mathcal{X}$ to $\mathbb{R}$ that satisfy

$$\int_{\mathcal{X}} f(x)^2 w(x) dx < \infty, \quad (\text{B.20})$$

and with inner product

$$\langle f, g \rangle = \int_{\mathcal{X}} f(x)g(x)w(x)dx. \quad (\text{B.21})$$

Let $\mathcal{H}$ be a Hilbert space. A set of functions $\{f_i, i \in \mathcal{I}\}$ is called an *orthonormal system* if

- (1) the norm of every f_i is 1; that is, $\langle f_i, f_i \rangle = 1$ for all $i \in \mathcal{I}$,
- (2) the $\{f_i\}$ are orthogonal; that is, $\langle f_i, f_j \rangle = 0$ for $i \neq j$.

It follows then that the $\{f_i\}$ are linearly independent; that is, the only linear combination $\sum_j \alpha_j f_j(x)$ that is equal to $f_i(x)$ for all x is the one where $\alpha_i = 1$ and $\alpha_j = 0$ for $j \neq i$. An orthonormal system $\{f_i\}$ is called an *orthonormal basis* if there is no other $f \in \mathcal{H}$ that is orthogonal to all the $\{f_i, i \in \mathcal{I}\}$ (other than the zero function). Although the general theory allows for uncountable bases, in practice the set $\mathcal{I}$ is taken to be countable.

We conclude with a number of key results in functional analysis. The first one is the celebrated *Cauchy–Schwarz inequality*.

Theorem B.7: Cauchy–Schwarz

Let $\mathcal{H}$ be a Hilbert space. For every $f, g \in \mathcal{H}$ it holds that

$$|\langle f, g \rangle| \leq \|f\| \|g\|.$$

Proof. The inequality is trivially true for $g = 0$ (zero function). For $g \neq 0$, we can write $f = \alpha g + h$, where $h \perp g$ and $\alpha = \langle f, g \rangle / \|g\|^2$. Consequently, $\|f\|^2 = |\alpha|^2 \|g\|^2 + \|h\|^2 \geq |\alpha|^2 \|g\|^2$. The result follows after rearranging this last inequality. $\square$

Let $\mathcal{V}$ and $\mathcal{W}$ be two linear vector spaces (for example, Hilbert spaces) on which norms $\|\cdot\|_{\mathcal{V}}$ and $\|\cdot\|_{\mathcal{W}}$ are defined. Suppose $A : \mathcal{V} \rightarrow \mathcal{W}$ is a mapping from $\mathcal{V}$ to $\mathcal{W}$. When $\mathcal{W} = \mathcal{V}$, such a mapping is often called an *operator*; when $\mathcal{W} = \mathbb{R}$ it is called a *functional*. Mapping A is said to be *linear* if $A(\alpha f + \beta g) = \alpha A(f) + \beta A(g)$. In this case we write Af instead of $A(f)$. If there exists $\gamma < \infty$ such that

$$\|Af\|_{\mathcal{W}} \leq \gamma \|f\|_{\mathcal{V}}, \quad f \in \mathcal{V}, \quad (\text{B.22})$$

then A is said to be a *bounded mapping*. The smallest γ for which (B.22) holds is called the *norm* of A ; denoted by $\|A\|$. A (not necessarily linear) mapping $A : \mathcal{V} \rightarrow \mathcal{W}$ is said to be *continuous* at f if for any sequence $f_1, f_2, \dots$ converging to f the sequence $A(f_1), A(f_2), \dots$ converges to $A(f)$. That is, if

$$\forall \varepsilon > 0, \exists \delta > 0 : \forall g \in \mathcal{V}, \|f - g\|_{\mathcal{V}} < \delta \implies \|A(f) - A(g)\|_{\mathcal{W}} < \varepsilon. \quad (\text{B.23})$$

If the above property holds for every $f \in \mathcal{V}$, then the mapping A itself is called *continuous*.

Theorem B.8: Continuity and Boundedness for Linear Mappings

For a linear mapping, continuity and boundedness are equivalent.

Proof. Let A be linear and bounded. We may assume that A is non-zero (otherwise the statement holds trivially), and that therefore $0 < \|A\| < \infty$. Taking $\delta < \varepsilon / \|A\|$ in (B.23) now ensures that $\|Af - Ag\|_{\mathcal{W}} \leq \|A\| \|f - g\|_{\mathcal{V}} < \|A\| \delta < \varepsilon$. This shows that A is continuous. Conversely, suppose A is continuous. In particular, it is continuous at $f = 0$ (the zero element of $\mathcal{V}$). Thus, take $f = 0$ and let ε and δ be as in (B.23). For any $g \neq 0$, let $h = \delta / (2\|g\|_{\mathcal{V}})g$. As $\|h\|_{\mathcal{V}} = \delta/2 < \delta$, it follows from (B.23) that

$$\|Ah\|_{\mathcal{W}} = \frac{\delta}{2\|g\|_{\mathcal{V}}} \|Ag\|_{\mathcal{W}} < \varepsilon.$$

Rearranging the last inequality gives $\|Ag\|_{\mathcal{W}} < 2\varepsilon/\delta\|g\|_{\mathcal{V}}$, showing that A is bounded. $\square$

Theorem B.9: Riesz Representation Theorem

Any bounded linear functional ϕ on a Hilbert space $\mathcal{H}$ can be represented as $\phi(h) = \langle h, g \rangle$, for some $g \in \mathcal{H}$ (depending on ϕ).

Proof. Let P be the projection of $\mathcal{H}$ onto the nullspace $\mathcal{N}$ of ϕ ; that is, $\mathcal{N} = \{g \in \mathcal{H} : \phi(g) = 0\}$. If ϕ is not the 0-functional, then there exists a $g_0 \neq 0$ with $\phi(g_0) \neq 0$. Let $g_1 = g_0 - Pg_0$. Then $g_1 \perp \mathcal{N}$ and $\phi(g_1) = \phi(g_0)$. Take $g_2 = g_1/\phi(g_1)$. For any $h \in \mathcal{H}$, $f := h - \phi(h)g_2$ lies in $\mathcal{N}$. As $g_2 \perp \mathcal{N}$ it holds that $\langle f, g_2 \rangle = 0$, which is equivalent to $\langle h, g_2 \rangle = \phi(h)\|g_2\|^2$. By defining $g = g_2/\|g_2\|^2$ we have found our representation. $\square$

B.8 Fourier Transforms

A central concern in mathematical physics and engineering mathematics is the transformation of equations into a coordinate system where expressions simplify, decouple, and become amenable to computation and analysis.

We will now briefly introduce the Fourier transform. Before doing so, we will extend the concept of L^2 space of real-valued functions as follows.

Definition B.4: L^p Space

Let $\mathcal{X}$ be a subset of $\mathbb{R}^d$ with measure $\mu(dx) = w(x)dx$ and $p \in [1, \infty)$. Then $L^p(\mathcal{X}, \mu)$ is the linear space of functions from $\mathcal{X}$ to $\mathbb{C}$ that satisfy

$$\int_{\mathcal{X}} |f(x)|^p w(x) dx < \infty. \quad (\text{B.24})$$

When $p = 2$, $L^2(\mathcal{X}, \mu)$ is in fact a Hilbert space equipped with inner product

$$\langle f, g \rangle = \int_{\mathcal{X}} f(x) \overline{g(x)} w(x) dx. \quad (\text{B.25})$$

Definition B.5: (Multivariate) Fourier Transform

The *Fourier transform* $\mathcal{F}[f]$ of a (real- or complex-valued) function $f \in L^1(\mathbb{R}^d)$ is the function $\tilde{f}$ defined as

$$\tilde{f}(t) := \int_{\mathbb{R}^d} e^{-i2\pi t^\top x} f(x) dx, \quad t \in \mathbb{R}^d.$$

The Fourier transform $\tilde{f}$ is continuous, uniformly bounded (since $f \in L^1(\mathbb{R}^d)$ implies that $|\tilde{f}(t)| \leq \int_{\mathbb{R}^d} |f(x)| dx < \infty$), and satisfies $\lim_{\|t\| \rightarrow \infty} \tilde{f}(t) = 0$ (a result known as the *Riemann–Lebesgue lemma*). However, $|\tilde{f}|$ does not necessarily have a finite integral. A simple example in $\mathbb{R}^1$ is the Fourier transform of $f(x) = \mathbb{1}_{\{-1/2 < x < 1/2\}}$. Then $\tilde{f}(t) = \sin(\pi t)/(\pi t) = \text{sinc}(\pi t)$, which is not absolutely integrable.

Definition B.6: (Multivariate) Inverse Fourier Transform

The *inverse Fourier transform* $\mathcal{F}^{-1}[\tilde{f}]$ of a (real- or complex-valued) function $\tilde{f} \in L^1(\mathbb{R}^d)$ is the function $\check{f}$ defined as

$$\check{f}(x) := \int_{\mathbb{R}^d} e^{i2\pi t^\top x} \tilde{f}(t) dt, \quad x \in \mathbb{R}^d.$$

As one would hope, it holds that if f and $\mathcal{F}[f]$ are both in $L^1(\mathbb{R}^d)$, then $f = \mathcal{F}^{-1}[\mathcal{F}[f]]$ almost everywhere. The Fourier transform enjoys many interesting and useful properties, some of which we list below.

- (1) *Linearity*: For $f, g \in L^1(\mathbb{R}^d)$ and constants $a, b \in \mathbb{R}$,

$$\mathcal{F}[af + bg] = a\mathcal{F}[f] + b\mathcal{F}[g].$$

- (2) *Space Shifting and Scaling*: Let $A \in \mathbb{R}^{d \times d}$ be an invertible matrix and $b \in \mathbb{R}^d$ a constant vector. Let $f \in L^1(\mathbb{R}^d)$ and define $h(x) := f(Ax + b)$. Then

$$\mathcal{F}[h](t) = e^{i2\pi(A^{-1}t)^\top b} \tilde{f}(A^{-\top}t)/|\det(A)|,$$

where $A^{-\top} := (A^\top)^{-1} = (A^{-1})^\top$.

- (3) *Frequency Shifting and Scaling:* Let $A \in \mathbb{R}^{d \times d}$ be an invertible matrix and $\mathbf{b} \in \mathbb{R}^d$ a constant vector. Let $f \in L^1(\mathbb{R}^d)$ and define

$$h(\mathbf{x}) := e^{-i2\pi \mathbf{b}^\top A^{-\top} \mathbf{x}} f(A^{-\top} \mathbf{x}) / |\det(A)|.$$

Then $\mathcal{F}[h](t) = \tilde{f}(At + \mathbf{b})$.

- (4) *Differentiation:* Let $f \in L^1(\mathbb{R}^d) \cap C^1(\mathbb{R}^d)$ and let $f_k := \partial f / \partial x_k$ be the partial derivative of f with respect to x_k . If $f_k \in L^1(\mathbb{R}^d)$ for $k = 1, \dots, d$, then

$$\mathcal{F}[f_k](t) = (i2\pi t_k) \tilde{f}(t).$$

- (5) *Convolution:* Let $f, g \in L^1(\mathbb{R}^d)$ be real or complex valued functions. Their convolution, $f * g$, is defined as

$$(f * g)(x) = \int_{\mathbb{R}^d} f(y)g(x - y)dy,$$

and is also in $L^1(\mathbb{R}^d)$. Moreover, the Fourier transform satisfies

$$\mathcal{F}[f * g] = \mathcal{F}[f]\mathcal{F}[g].$$

- (6) *Duality:* Let f and $\mathcal{F}[f]$ both be in $L^1(\mathbb{R}^d)$. Then $\mathcal{F}[\mathcal{F}[f]](t) = f(-t)$.

- (7) *Product Formula:* Let $f, g \in L^1(\mathbb{R}^d)$ and denote by $\tilde{f}, \tilde{g}$ their respective Fourier transforms. Then $\tilde{f}\tilde{g}, f\tilde{g} \in L^1(\mathbb{R}^d)$, and

$$\int_{\mathbb{R}^d} \tilde{f}(z)g(z)dz = \int_{\mathbb{R}^d} f(z)\tilde{g}(z)dz.$$

There are many additional properties which hold if $f \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$. In particular, if $f, g \in L^1(\mathbb{R}^d) \cap L^2(\mathbb{R}^d)$, then $\tilde{f}, \tilde{g} \in L^2(\mathbb{R}^d)$ and $\langle \tilde{f}, \tilde{g} \rangle = \langle f, g \rangle$, a result often known as *Parseval's formula*. Putting $g = f$ gives the result often referred to as *Plancherel's theorem*.

The Fourier transform can be extended in several ways, in the first instance to functions in $L^2(\mathbb{R}^d)$ by continuity. A substantial extension of the theory is realized by replacing integration with respect to the Lebesgue measure (i.e., $\int_{\mathbb{R}^d} \cdots dx$) with integration with

respect to a (finite Borel) measure μ (i.e., $\int_{\mathbb{R}^d} \cdots \mu(dx)$). Moreover, there is a close connection between the Fourier transform and characteristic functions arising in probability theory. Indeed, if X is a random vector with pdf f , then its characteristic function ψ satisfies

$$\psi(t) := \mathbb{E}e^{it^\top X} = \mathcal{F}[f](-t/(2\pi)).$$

B.8.1 Discrete Fourier Transform

Here, we introduce the (univariate) discrete Fourier transform, which can be viewed as a special case of the Fourier transform introduced in Definition A.6, where $d = 1$, integration is with respect to the counting measure, and $f(x) = 0$ for $x < 0$ and $x > (n-1)$.

Definition B.7: Discrete Fourier Transform

The *discrete Fourier transform* (DFT) of a vector $\mathbf{x} = [x_0, \dots, x_{n-1}]^\top \in \mathbb{C}^n$ is the vector $\tilde{\mathbf{x}} = [\tilde{x}_0, \dots, \tilde{x}_{n-1}]^\top$ whose elements are given by

$$\tilde{x}_t = \sum_{s=0}^{n-1} \omega^{st} x_s, \quad t = 0, \dots, n-1, \quad (\text{B.26})$$

where $\omega = \exp(-i2\pi/n)$.

In other words, $\tilde{\mathbf{x}}$ is obtained from $\mathbf{x}$ via the linear transformation

$$\tilde{\mathbf{x}} = F\mathbf{x},$$

where

$$F = \begin{bmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \omega & \omega^2 & \dots & \omega^{n-1} \\ 1 & \omega^2 & \omega^4 & \dots & \omega^{2(n-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{n-1} & \omega^{2(n-1)} & \dots & \omega^{(n-1)^2} \end{bmatrix}.$$

The matrix F is a so-called *Vandermonde matrix*, and is clearly symmetric (i.e., $F = F^\top$). Moreover, $F/\sqrt{n}$ is in fact a unitary matrix and hence its inverse is simply its complex

conjugate $\bar{F}/\sqrt{n}$. Thus, $F^{-1} = \bar{F}/n$ and we have that the *inverse discrete Fourier transform* (IDFT) is given by

$$x_t = \frac{1}{n} \sum_{s=0}^{n-1} \omega^{-st} \tilde{x}_s, \quad t = 0, \dots, n-1, \quad (\text{B.27})$$

or in terms of matrices and vectors,

$$\mathbf{x} = \bar{F} \tilde{\mathbf{x}}/n.$$

Observe that the IDFT of a vector $\mathbf{y}$ is related to the DFT of its complex conjugate $\bar{\mathbf{y}}$, since

$$\bar{F} \mathbf{y}/n = \overline{F \bar{\mathbf{y}}}/n.$$

Consequently, an IDFT can be computed via a DFT. There is a close connection between circulant matrices C and the DFT. To make this connection concrete, let C be the circulant matrix corresponding to the vector $\mathbf{c} \in \mathbb{C}^n$ and denote by $\mathbf{f}_t$ the t -th column of the discrete Fourier matrix F , $t = 0, 1, \dots, n-1$. Then, the s -th element of $C \mathbf{f}_t$ is

$$\sum_{k=0}^{n-1} c_{(s-k) \bmod n} \omega^{tk} = \sum_{y=0}^{n-1} c_y \omega^{t(s-y)} = \underbrace{\omega^{ts}}_{s\text{-th element of } \mathbf{f}_t} \underbrace{\sum_{y=0}^{n-1} c_y \omega^{-ty}}_{\lambda_t}.$$

Hence, the eigenvalues of C are

$$\lambda_t = \mathbf{c}^\top \bar{\mathbf{f}}_t, \quad t = 0, 1, \dots, n-1,$$

with corresponding eigenvectors $\mathbf{f}_t$. Collecting the eigenvalues into the vector $\boldsymbol{\lambda} = [\lambda_0, \dots, \lambda_{n-1}]^\top = \bar{F} \mathbf{c}$, we therefore have the eigen-decomposition

$$C = F \text{diag}(\boldsymbol{\lambda}) \bar{F}/n.$$

Consequently, one can compute the *circular convolution* of a vector $\mathbf{a} = [a_1, \dots, a_n]^\top$ and $\mathbf{c} = [c_0, \dots, c_{n-1}]^\top$ by a series of DFTs as follows. Construct the circulant matrix C corre-

sponding to $\mathbf{c}$. Then, the circular convolution of $\mathbf{a}$ and $\mathbf{c}$ is given by $\mathbf{y} = \mathbf{C}\mathbf{a}$. Proceed in four steps:

- (1) Compute $\mathbf{z} = \overline{\mathbf{F}}\mathbf{a}/n$.
- (2) Compute $\boldsymbol{\lambda} = \overline{\mathbf{F}}\mathbf{c}$.
- (3) Compute $\mathbf{p} = \mathbf{z} \odot \boldsymbol{\lambda} = [z_1\lambda_0, \dots, z_n\lambda_{n-1}]^\top$.
- (4) Compute $\mathbf{y} = \mathbf{F}\mathbf{p}$.

Steps 1 and 2 are (up to constants) in the form of an IDFT, and step 4 is in the form of a DFT. These are computable via the FFT in $O(n \ln n)$ time. Step 3 is a dot product computable in $O(n)$ time. Thus, the circular convolution can be computed with aid of the FFT in $O(n \ln n)$ time.

One can also efficiently compute the product of an $n \times n$ Toeplitz matrix T and an $n \times 1$ vector $\mathbf{a}$ in $O(n \ln n)$ time by embedding T into a circulant matrix C of size $2n \times 2n$. Namely, define

$$C = \begin{bmatrix} T & B \\ B & T \end{bmatrix},$$

where

$$B = \begin{bmatrix} 0 & t_{n-1} & \dots & t_2 & t_1 \\ t_{-(n-1)} & 0 & t_{n-1} & \dots & t_2 \\ \vdots & t_{-(n-1)} & 0 & \ddots & \vdots \\ t_{-2} & \ddots & \ddots & \ddots & t_{n-1} \\ t_{-1} & t_{-2} & \dots & t_{-(n-1)} & 0 \end{bmatrix}.$$

Then a product of the form $\mathbf{y} = T\mathbf{a}$ can be computed in $O(n \ln n)$ time, since we may write

$$C \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} T & B \\ B & T \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} T\mathbf{a} \\ B\mathbf{a} \end{bmatrix}.$$

The left-hand side is a product of a $2n \times 2n$ circulant matrix with vector of length $2n$, and so can be computed in $O(n \ln n)$ time via the FFT, as previously discussed.

Conceptually, one can also solve equations of the form $C\mathbf{x} = \mathbf{b}$ for a given vector $\mathbf{b} \in \mathbb{C}^n$ and circulant matrix C (corresponding to $\mathbf{c} \in \mathbb{C}^n$, assuming all its eigenvalues are non-zero) via the following four steps:

- (1) Compute $\mathbf{z} = \overline{F}\mathbf{b}/n$.
- (2) Compute $\boldsymbol{\lambda} = \overline{F}\mathbf{c}$.
- (3) Compute $\mathbf{p} = \mathbf{z}/\boldsymbol{\lambda} = [z_1/\lambda_0, \dots, z_n/\lambda_{n-1}]^\top$.
- (4) Compute $\mathbf{x} = F\mathbf{p}$.

Once again, Steps 1 and 2 are (up to constants) in the form of an IDFT, and Step 4 is in the form of a DFT, all of which are computable via the FFT in $O(n \ln n)$ time, and Step 3 is computable in $O(n)$ time, meaning the solution $\mathbf{x}$ can be computed using the FFT in $O(n \ln n)$ time.

B.8.2 Fast Fourier Transform

The *fast Fourier transform* (FFT) is a numerical algorithm for the fast evaluation of (B.26) and (B.27). By using a divide-and-conquer strategy, the algorithm reduces the computational complexity from $O(n^2)$ (for the naïve evaluation of the linear transformation) to $O(n \ln n)$.

The essence of the algorithm lies in the following observation. Suppose $n = r_1 r_2$. Then one can express any index t appearing in (B.26) via a pair (t_0, t_1) , with $t = t_1 r_1 + t_0$, where $t_0 \in \{0, 1, \dots, r_1 - 1\}$ and $t_1 \in \{0, 1, \dots, r_2 - 1\}$. Similarly, one can express any index s appearing in (B.26) via a pair (s_0, s_1) , with $s = s_1 r_2 + s_0$, where $s_0 \in \{0, 1, \dots, r_2 - 1\}$ and $s_1 \in \{0, 1, \dots, r_1 - 1\}$.

Identifying $\tilde{x}_t \equiv \tilde{x}_{t_1, t_0}$ and $x_s \equiv x_{s_1, s_0}$, we may re-express (B.26) as

$$\tilde{x}_{t_1, t_0} = \sum_{s_0=0}^{r_2-1} \omega^{s_0 t} \sum_{s_1=0}^{r_1-1} \omega^{s_1 r_2 t} x_{s_1, s_0}, \quad t_0 = 0, 1, \dots, r_1 - 1, t_1 = 0, 1, \dots, r_2 - 1. \quad (\text{B.28})$$

Observe that $\omega^{s_1 r_2 t} = \omega^{s_1 r_2 t_0}$ (because $\omega^{r_1 r_2} = 1$), so that the inner sum over s_1 depends only on s_0 and t_0 . Define

$$y_{t_0, s_0} := \sum_{s_1=0}^{r_1-1} \omega^{s_1 r_2 t_0} x_{s_1, s_0}, \quad t_0 = 0, 1, \dots, r_1 - 1, s_0 = 0, 1, \dots, r_2 - 1.$$

Computing each y_{t_0, s_0} requires $O(nr_1)$ operations. In terms of the $\{y_{t_0, s_0}\}$, (B.28) can be written as

$$\tilde{x}_{t_1, t_0} = \sum_{s_0=0}^{r_2-1} \omega^{s_0 t} y_{t_0, s_0}, \quad t_1 = 0, 1, \dots, r_2 - 1, t_0 = 0, 1, \dots, r_1 - 1,$$

requiring $O(nr_2)$ operations to compute. Thus, calculating the DFT using this two-step procedure requires $O(n(r_1 + r_2))$ operations, rather than $O(n^2)$.

Now supposing $n = r_1 r_2 \cdots r_m$, repeated application the above divide-and-conquer idea yields an m -step procedure requiring $O(n(r_1 + r_2 + \cdots + r_m))$ operations. In particular, if $r_k = r$ for all $k = 1, 2, \dots, m$, we have that $n = r^m$ and $m = \log_r n$, so that the total number of operations is $O(rnm) \equiv O(rn \log_r(n))$. Typically, the radix r is a small (not necessarily prime) number, for instance $r = 2$.

C Multivariate Differentiation and Optimisation

The purpose of this appendix is to review various aspects of multivariate differentiation and optimisation. We assume the reader is familiar with differentiating a real-valued function.

D Multivariate Differentiation

For a multivariate function f that maps a vector $\mathbf{x} = [x_1, \dots, x_n]^\top$ to a real number $f(\mathbf{x})$, the *partial derivative* with respect to x_i , denoted $\frac{\partial f}{\partial x_i}$, is the derivative taken with respect to x_i while all other variables are held constant. We can write all the n partial derivatives neatly using the “scalar/vector” derivative notation:

$$\text{scalar/vector:} \quad \frac{\partial f}{\partial \mathbf{x}} := \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}. \quad (\text{D.1})$$

This vector of partial derivatives is known as the *gradient* of f at $\mathbf{x}$ and is sometimes written as $\nabla f(\mathbf{x})$.

Next, suppose that $\mathbf{f}$ is a multivalued (vector-valued) function taking values in $\mathbb{R}^m$, defined by

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \mapsto \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} = \mathbf{f}(\mathbf{x}).$$

We can compute each of the partial derivatives $\partial f_i / \partial x_j$ and organize them neatly in a “vector/vector” derivative notation:

$$\text{vector/vector:} \quad \frac{\partial \mathbf{f}}{\partial \mathbf{x}} := \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_2}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_1} \\ \frac{\partial f_1}{\partial x_2} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_2} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial f_1}{\partial x_n} & \frac{\partial f_2}{\partial x_n} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}. \quad (\text{D.2})$$

The transpose of this matrix is known as the *matrix of Jacobi* of f at $\mathbf{x}$ (sometimes called the *Fréchet derivative* of f at $\mathbf{x}$); that is,

$$J_f(\mathbf{x}) := \left[\frac{\partial f}{\partial \mathbf{x}} \right]^\top = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}. \quad (\text{D.3})$$

If we define $\mathbf{g}(\mathbf{x}) := \nabla f(\mathbf{x})$ and take the “vector/vector” derivative of $\mathbf{g}$ with respect to $\mathbf{x}$, we obtain the matrix of second-order partial derivatives of f :

$$\mathbf{H}_f(\mathbf{x}) := \frac{\partial \mathbf{g}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_m} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_m} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial^2 f}{\partial x_m \partial x_1} & \frac{\partial^2 f}{\partial x_m \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_m^2} \end{bmatrix}, \quad (\text{D.4})$$

which is known as the *Hessian matrix* of f at $\mathbf{x}$, also denoted as $\nabla^2 f(\mathbf{x})$. If these second-order partial derivatives are *continuous* in a region around $\mathbf{x}$, then $\frac{\partial f}{\partial x_i \partial x_j} = \frac{\partial f}{\partial x_j \partial x_i}$ and, hence, the Hessian matrix $\mathbf{H}_f(\mathbf{x})$ is symmetric.

Finally, note that we can also define a “scalar/matrix” derivative of y with respect to $\mathbf{X} \in \mathbb{R}^{m \times n}$ with (i, j) -th entry x_{ij} :

$$\frac{\partial y}{\partial \mathbf{X}} := \begin{bmatrix} \frac{\partial y}{\partial x_{11}} & \frac{\partial y}{\partial x_{12}} & \cdots & \frac{\partial y}{\partial x_{1n}} \\ \frac{\partial y}{\partial x_{21}} & \frac{\partial y}{\partial x_{22}} & \cdots & \frac{\partial y}{\partial x_{2n}} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial y}{\partial x_{m1}} & \frac{\partial y}{\partial x_{m2}} & \cdots & \frac{\partial y}{\partial x_{mn}} \end{bmatrix}$$

and a “matrix/scalar” derivative:

$$\frac{\partial X}{\partial y} := \begin{bmatrix} \frac{\partial x_{11}}{\partial y} & \frac{\partial x_{12}}{\partial y} & \cdots & \frac{\partial x_{1n}}{\partial y} \\ \frac{\partial x_{21}}{\partial y} & \frac{\partial x_{22}}{\partial y} & \cdots & \frac{\partial x_{2n}}{\partial y} \\ \vdots & \vdots & \cdots & \vdots \\ \frac{\partial x_{m1}}{\partial y} & \frac{\partial x_{m2}}{\partial y} & \cdots & \frac{\partial x_{mn}}{\partial y} \end{bmatrix}.$$

D.1 Optimisation Theory

Optimisation is concerned with finding minimal or maximal solutions of a real-valued *objective function* f in some set $\mathcal{X}$:

$$\min_{x \in \mathcal{X}} f(x) \quad \text{or} \quad \max_{x \in \mathcal{X}} f(x). \quad (\text{D.5})$$

Since any maximisation problem can easily be converted into a minimisation problem via the equivalence $\max_x f(x) = -\min_x -f(x)$, we focus only on minimisation problems. We use the following terminology. A *local minimiser* of $f(x)$ is an element $x^* \in \mathcal{X}$ such that $f(x^*) \leq f(x)$ for all x in some neighborhood of x^* . If $f(x^*) \leq f(x)$ for all $x \in \mathcal{X}$, then x^* is called a *global minimiser* or *global solution*. The set of global minimiser is denoted by

$$\arg \min_{x \in \mathcal{X}} f(x).$$

The function value $f(x^*)$ corresponding to a local/global minimiser x^* is referred to as the *local/global minimum* of $f(x)$.

Optimisation problems may be classified by the set $\mathcal{X}$ and the objective function f . If $\mathcal{X}$ is countable, the optimisation problem is called *discrete* or *combinatorial*. If instead $\mathcal{X}$ is a nondenumerable set such as $\mathbb{R}^n$ and f takes values in a nondenumerable set, then the problem is said to be *continuous*. Optimisation problems that are neither discrete nor continuous are said to be *mixed*.

The search set $\mathcal{X}$ is often defined by means of *constraints*. A standard setting for *constrained optimisation* (minimisation) is the following:

$$\begin{aligned} & \min_{\mathbf{x} \in \mathcal{Y}} f(\mathbf{x}) \\ & \text{subject to: } h_i(\mathbf{x}) = 0, \quad i = 1, \dots, m, \\ & \quad \quad \quad g_i(\mathbf{x}) \leq 0, \quad i = 1, \dots, k. \end{aligned} \tag{D.6}$$

Here, f is the objective function, and $\{g_i\}$ and $\{h_i\}$ are given functions so that $h_i(\mathbf{x}) = 0$ and $g_i(\mathbf{x}) \leq 0$ represent the *equality* and *inequality constraints*, respectively. The region $\mathcal{X} \subseteq \mathcal{Y}$ where the objective function is defined and where all the constraints are satisfied is called the *feasible region*. An optimisation problem without constraints is said to be an *unconstrained problem*.

For an unconstrained continuous optimisation problem, the search space $\mathcal{X}$ is often taken to be (a subset of) $\mathbb{R}^n$, and f is assumed to be a C^k function for sufficiently high k (typically $k = 2$ or 3 suffices); that is, its k -th order derivative is continuous. For a C^1 function the standard approach to minimising $f(\mathbf{x})$ is to solve the equation

$$\nabla f(\mathbf{x}) = \mathbf{0}, \tag{D.7}$$

where $\nabla f(\mathbf{x})$ is the *gradient* of f at $\mathbf{x}$. The solutions $\mathbf{x}^*$ to [eq. \(D.7\)](#) are called *stationary points*. Stationary points can be local/global minimisers, local/global maximisers, or *saddle points* (which are neither). If, in addition, the function is C^2 , the condition

$$\mathbf{y}^\top (\nabla^2 f(\mathbf{x}^*)) \mathbf{y} > 0 \quad \text{for all } \mathbf{y} \neq \mathbf{0} \tag{D.8}$$

ensures that the stationary point $\mathbf{x}^*$ is a local minimiser of f . The condition [eq. \(D.8\)](#) states that the *Hessian matrix* of f at $\mathbf{x}^*$ is *positive definite*. Recall that we write $\mathbf{H} \succ 0$ to indicate that a matrix $\mathbf{H}$ is positive definite.

D.2 Numerical Root-Finding and Minimisation

In order to minimize a C^1 function $f : \mathbb{R}^n \rightarrow \mathbb{R}$ one may solve

$$\nabla f(\mathbf{x}) = \mathbf{0},$$

which gives a stationary point of f . As a consequence, any technique for root-finding can be transformed into an unconstrained optimization method by attempting to locate roots of the gradient. However, as noted in [subsection D.1](#), not all stationary points are minima, and so additional information (such as is contained in the Hessian, if f is C^2) needs to be considered in order to establish the type of stationary point.

Alternatively, a root of a continuous function $\mathbf{g} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ may be found by minimizing the norm of $\mathbf{g}(\mathbf{x})$ over all $\mathbf{x}$; that is, by solving $\min_{\mathbf{x}} f(\mathbf{x})$, with $f(\mathbf{x}) := \|\mathbf{g}(\mathbf{x})\|_p$, where for $p \geq 1$ the p -norm of $\mathbf{y} = [y_1, \dots, y_n]^\top$ is defined as

$$\|\mathbf{y}\|_p := \left(\sum_{i=1}^n |y_i|^p \right)^{1/p}.$$

Hence, any (un)constrained optimization method can be transformed into a technique for locating the roots of a function.

Starting with an initial guess $\mathbf{x}_0$, most minimization and root-finding algorithms create a sequence $\mathbf{x}_0, \mathbf{x}_1, \dots$ using the iterative updating rule:

$$\mathbf{x}_{t+1} = \mathbf{x}_t + \alpha_t \mathbf{d}_t, \quad t = 0, 1, 2, \dots, \tag{D.9}$$

where $\alpha_t > 0$ is a (typically small) step size, called the *learning rate*, and the vector $\mathbf{d}_t$ is the *search direction* at step t . The iteration [eq. \(D.9\)](#) continues until the sequence $\{\mathbf{x}_t\}$ is deemed to have converged to a solution, or a computational budget has been exhausted. The performance of all such iterative methods depends crucially on the quality of the initial guess $\mathbf{x}_0$.

There are two broad categories of iterative optimization algorithms of the form [eq. \(D.9\)](#):

- Those of *line search* type, where at iteration t we first compute a direction $\mathbf{d}_t$ and then determine a reasonable step size α_t along this direction. For example, in the case of minimization, $\alpha_t > 0$ may be chosen to approximately minimize $f(\mathbf{x}_t + \alpha \mathbf{d}_t)$ for fixed $\mathbf{x}_t$ and $\mathbf{d}_t$.
- Those of *trust region* type, where at each iteration t we first determine a suitable step size α_t and then compute an approximately optimal direction $\mathbf{d}_t$.

In the following sections, we review several widely-used root-finding and optimization algorithms of the line search type.

D.2.1 Newton-Like Methods

Suppose we wish to find roots of a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$. If f is in C^1 , we can approximate f around a point $\mathbf{x}_t$ as

$$f(\mathbf{x}) \approx f(\mathbf{x}_t) + J_f(\mathbf{x}_t)(\mathbf{x} - \mathbf{x}_t),$$

where J_f is the *matrix of Jacobi*, the matrix of partial derivatives of f ; see eq. (D.3). When $J_f(\mathbf{x}_t)$ is invertible, this linear approximation has root $\mathbf{x}_t - J_f^{-1}(\mathbf{x}_t)f(\mathbf{x}_t)$. This gives the iterative updating formula eq. (D.9) for finding roots of f with direction $\mathbf{d}_t = -J_f^{-1}(\mathbf{x}_t)f(\mathbf{x}_t)$ and learning rate $\alpha_t = 1$. This is known as *Newton's method* (or the *Newton–Raphson method*) for root-finding.

Instead of a unit learning rate, sometimes it is more effective to use an α_t that satisfies the *Armijo inexact line search condition*:

$$\|f(\mathbf{x}_t + \alpha_t \mathbf{d}_t)\| < (1 - \varepsilon_1 \alpha_t) \|f(\mathbf{x}_t)\|,$$

where ε_1 is a small heuristically chosen constant, say $\varepsilon_1 = 10^{-4}$. For C^1 functions, such an α_t always exists by continuity and can be computed as in the following algorithm.

Algorithm 1 Newton–Raphson for Finding Roots of $f(\mathbf{x}) = 0$

Require: An initial guess $\mathbf{x}$ and stopping error $\varepsilon > 0$.

Ensure: The approximate root of $f(\mathbf{x}) = 0$.

```

1: while  $\|f(\mathbf{x})\| > \varepsilon$  and budget is not exhausted do
2:   Solve the linear system  $J_f(\mathbf{x})\mathbf{d} = -f(\mathbf{x})$ .
3:    $\alpha \leftarrow 1$ 
4:   while  $\|f(\mathbf{x} + \alpha\mathbf{d})\| > (1 - 10^{-4}\alpha)\|f(\mathbf{x})\|$  do
5:      $\alpha \leftarrow \alpha/2$ 
6:   end while
7:    $\mathbf{x} \leftarrow \mathbf{x} + \alpha\mathbf{d}$ 
8: end while
9: return  $\mathbf{x}$ 

```

We can adapt a root-finding Newton-like method in order to minimise a differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. We simply try to locate a zero of the gradient of f . When f is a C^2 function, the function $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is continuous, and so the root of ∇f leads to the search direction

$$\mathbf{d}_t = -\mathbf{H}_t^{-1}\nabla f(\mathbf{x}_t), \quad (\text{D.10})$$

where $\mathbf{H}_t$ is the Hessian matrix at $\mathbf{x}_t$ (the matrix of Jacobi of the gradient is the Hessian). When the learning rate α_t is equal to 1, the update $\mathbf{x}_t - \mathbf{H}_t^{-1}\nabla f(\mathbf{x}_t)$ can alternatively be derived by assuming that $f(\mathbf{x})$ is approximately quadratic and convex in the neighborhood of $\mathbf{x}_t$, that is,

$$f(\mathbf{x}) \approx f(\mathbf{x}_t) + (\mathbf{x} - \mathbf{x}_t)^\top \nabla f(\mathbf{x}_t) + \frac{1}{2}(\mathbf{x} - \mathbf{x}_t)^\top \mathbf{H}_t(\mathbf{x} - \mathbf{x}_t), \quad (\text{D.11})$$

and then minimizing the right-hand side of eq. (D.11) with respect to $\mathbf{x}$.

D.3 Constrained Minimisation via Penalty Functions

E Probability and Statistics

W^E review basic concepts, including the sum and product rules of probability, random variables and their probability distributions, expectations, independence, conditional probability, transformation rules, limit theorems, and Markov chains. The properties of the multivariate normal distribution are discussed in more detail. The main ideas from statistics are also reviewed, including estimation techniques (such as maximum likelihood estimation), confidence intervals, and hypothesis testing.

E.1 Random Experiments and Probability Spaces

The basic notion in probability theory is that of a *random experiment*: an experiment whose outcome cannot be determined in advance. Mathematically, a random experiment is modeled via a triplet $(\Omega, \mathcal{H}, \mathbb{P})$, where:

- Ω is the set of all possible outcomes of the experiment, called the *sample space*.
- $\mathcal{H}$ is the collection of all subsets of Ω to which a probability can be assigned; such subsets are called *events*.
- $\mathbb{P}$ is a *probability measure*, which assigns to each event A a number $\mathbb{P}[A]$ between 0 and 1, indicating the likelihood that the outcome of the random experiment lies in A .

Any probability measure $\mathbb{P}$ must satisfy the following *Kolmogorov axioms*:

- (1) $\mathbb{P}[A] \geq 0$ for every event A .
- (2) $\mathbb{P}[\Omega] = 1$.
- (3) For any sequence $A_1, A_2, \dots$ of events,

$$\mathbb{P}\left[\bigcup_i A_i\right] \leq \sum_i \mathbb{P}[A_i], \quad (\text{E.1})$$

with strict equality whenever the events are *disjoint* (that is, non-overlapping).

In many applications the sample space is *countable*; that is, $\Omega = \{a_1, a_2, \dots\}$. In this case the easiest way to specify a probability measure $\mathbb{P}$ is to first assign a number p_i to each

elementary event $\{a_i\}$, with $\sum_i p_i = 1$, and then to define

$$\mathbb{P}[A] = \sum_{i: a_i \in A} p_i \quad \text{for all } A \subseteq \Omega.$$

Here the collection of events $\mathcal{H}$ can be taken to be equal to the collection of *all subsets* of Ω . The triple $(\Omega, \mathcal{H}, \mathbb{P})$ is called a *discrete probability space*. This idea is graphically represented in [fig. 31](#). Each element a_i , represented by a dot, is assigned a weight (that is, probability) p_i , indicated by the size of the dot. The probability of the event A is simply the sum of the weights of all the outcomes in A .

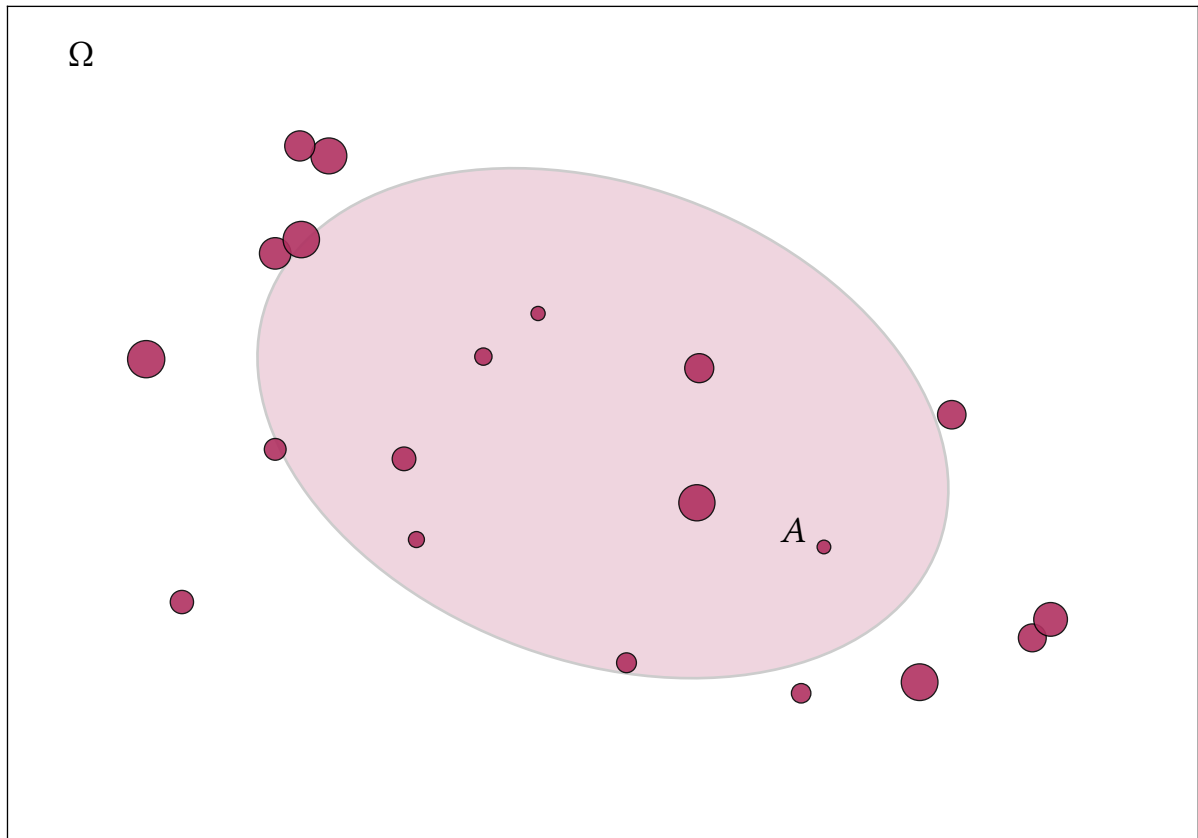


Figure 31: A discrete probability space.

E.2 Random Variables and Probability Distributions

It is often convenient to describe a random experiment via “random variables”, representing numerical measurements of the experiment. Random variables are usually denoted by capital letters from the last part of the alphabet. From a mathematical point of view, a

random variable X is a function from Ω to $\mathbb{R}$ such that sets of the form $\{a < X \leq b\} := \{\omega \in \Omega : a < X(\omega) \leq b\}$ are events (and so can be assigned a probability).

A cdf F_c is called *continuous*⁶, if there exists a positive function f such that for all x

$$F_c(x) = \int_{-\infty}^x f(u)du. \quad (\text{E.2})$$

Note that such an F_c is differentiable (and hence continuous) with derivative f . The function f is called the *probability density function* (continuous pdf). By the fundamental theorem of integration, we have

$$\mathbb{P}[a < X \leq b] = F(b) - F(a) = \int_a^b f(x)dx.$$

Thus, calculating probabilities reduces to integration, as illustrated in ??.

Table 3: Commonly used discrete distributions.

Name	Notation	$f(x)$	$x \in$	Parameters
Bernoulli	$\text{Ber}(p)$	$p^x(1-p)^{1-x}$	$\{0, 1\}$	$0 \leq p \leq 1$
Binomial	$\text{Bin}(n, p)$	$\binom{n}{x}p^x(1-p)^{n-x}$	$\{0, 1, \dots, n\}$	$0 \leq p \leq 1, n \in \mathbb{N}$
Discrete uniform	$\mathcal{U}\{1, \dots, n\}$	$\frac{1}{n}$	$\{1, \dots, n\}$	$n \in \{1, 2, \dots\}$
Geometric	$\text{Geom}(p)$	$p(1-p)^{x-1}$	$\{1, 2, \dots\}$	$0 \leq p \leq 1$
Poisson	$\text{Poi}(\lambda)$	$e^{-\lambda} \frac{\lambda^x}{x!}$	$\mathbb{N}$	$\lambda > 0$

⁶In advanced probability, we would say “*absolutely continuous with respect to the Lebesgue measure*”.

Table 4: Commonly used continuous distributions.

Name	Notation	$f(x)$	$x \in$	Parameters
Uniform	$\mathcal{U}[\alpha, \beta]$	$\frac{1}{\beta - \alpha}$	$[\alpha, \beta]$	$\alpha < \beta$
Normal	$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$\mathbb{R}$	$\sigma > 0, \mu \in \mathbb{R}$
Gamma	$\text{Gamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{\alpha-1} e^{-\lambda x}}{\Gamma(\alpha)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Inverse Gamma	$\text{InvGamma}(\alpha, \lambda)$	$\frac{\lambda^\alpha x^{-\alpha-1} e^{-\lambda x^{-1}}}{\Gamma(\alpha)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Exponential	$\text{Exp}(\lambda)$	$\lambda e^{-\lambda x}$	$\mathbb{R}_+$	$\lambda > 0$
Beta	$\text{Beta}(\alpha, \beta)$	$\frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1} (1-x)^{\beta-1}$	$[0, 1]$	$\alpha, \beta > 0$
Weibull	$\text{Weib}(\alpha, \lambda)$	$\alpha \lambda (\lambda x)^{\alpha-1} e^{-(\lambda x)^\alpha}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Pareto	$\text{Pareto}(\alpha, \lambda)$	$\alpha \lambda (1 + \lambda x)^{-(\alpha+1)}$	$\mathbb{R}_+$	$\alpha, \lambda > 0$
Student	t_ν	$\frac{\Gamma(\frac{\nu+1}{2})}{\sqrt{\nu\pi}\Gamma(\frac{\nu}{2})} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}$	$\mathbb{R}$	$\nu > 0$
F	$F(m, n)$	$\frac{\Gamma(\frac{m+n}{2})(m/n)^{m/2} x^{(m-2)/2}}{\Gamma(\frac{m}{2})\Gamma(\frac{n}{2})[1+(m/n)x]^{(m+n)/2}}$	$\mathbb{R}_+$	$m, n \in \mathbb{N}_+$

E.3 Expectation

It is often useful to consider different kinds of numerical characteristics of a random variable. One such quantity is the expectation, which measures the “average” value of the distribution.

The *expectation* (or expected value or mean) of a random variable X with pdf f , denoted by $\mathbb{E}X$ or⁷ $\mathbb{E}[X]$ (and sometimes μ), is defined by

$$\mathbb{E}X = \begin{cases} \sum_x x f(x) & \text{discrete case,} \\ \int_{-\infty}^{\infty} x f(x) dx & \text{continuous case.} \end{cases}$$

If X is a random variable, then a function of X , such as X^2 or $\sin(X)$, is again a random variable. Moreover, the expected value of a function of X is simply a weighted average of the possible values that this function can take. That is, for any real function h

$$\mathbb{E}h(X) = \begin{cases} \sum_x h(x) f(x) & \text{discrete case,} \\ \int_{-\infty}^{\infty} h(x) f(x) dx & \text{continuous case,} \end{cases}$$

provided that the sum or integral are well-defined.

The *variance* of a random variable X , denoted by $\text{Var } X$ (and sometimes σ^2), is defined by

$$\text{Var } X = \mathbb{E}(X - \mathbb{E}[X])^2 = \mathbb{E}X^2 - (\mathbb{E}X)^2.$$

The square root of the variance is called the *standard deviation*. [table 5](#) lists the expectations and variances for some well-known distributions. Both variance and standard deviation measure the spread or dispersion of the distribution. Note, however, that the standard deviation measures the dispersion in the same units as the random variable, unlike the variance, which uses squared units.

⁷We only use brackets in an expectation if it is unclear with respect to which random variable the expectation is taken.

Table 5: Expectations and variances for some well-known distributions.

Dist.	$\mathbb{E}X$	$\text{Var } X$	Dist.	$\mathbb{E}X$	$\text{Var } X$
$\text{Bin}(n, p)$	np	$np(1-p)$	$\text{Gamma}(\alpha, \lambda)$	$\frac{\alpha}{\lambda}$	$\frac{\alpha}{\lambda^2}$
$\text{Geom}(p)$	$\frac{1}{p}$	$\frac{1-p}{p^2}$	$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2
$\text{Poi}(\lambda)$	λ	λ	$\text{Beta}(\alpha, \beta)$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(1+\alpha+\beta)}$
$\mathcal{U}[\alpha, \beta]$	$\frac{\alpha+\beta}{2}$	$\frac{(\beta-\alpha)^2}{12}$	$\text{Weib}(\alpha, \lambda)$	$\frac{\Gamma(1/\alpha)}{\alpha\lambda}$	$\frac{2\Gamma(2/\alpha)}{\alpha\lambda^2} - \left(\frac{\Gamma(1/\alpha)}{\alpha\lambda}\right)^2$
$\text{Exp}(\lambda)$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$	$F(m, n)$	$\frac{n}{n-2} \ (n > 2)$	$\frac{2n^2(m+n-2)}{m(n-2)^2(n-4)} \ (n > 4)$
t_ν	$0 \ (\nu > 1)$	$\frac{\nu}{\nu-2} \ (\nu > 2)$			

It is sometimes useful to consider the *moment generating function* of a random variable X . This is the function M defined by

$$M(s) = \mathbb{E}e^{sX}, \quad s \in \mathbb{R}. \quad (\text{E.3})$$

The moment generating functions of two random variables coincide if and only if the random variables have the same distribution

E.4 Joint Distributions

Distributions for random vectors and stochastic processes can be specified in much the same way as for random variables. In particular, the distribution of a random vector $\mathbf{X} =$

$[X_1, \dots, X_n]^\top$ is completely determined by specifying the *joint cdf* F , defined by

$$F(x_1, \dots, x_n) = \mathbb{P}[X_1 \leq x_1, \dots, X_n \leq x_n], \quad x_i \in \mathbb{R}, i = 1, \dots, n.$$

Similarly, the distribution of a *stochastic process*, that is, a collection of random variables $\{X_t, t \in \mathcal{T}\}$, for some index set $\mathcal{T}$, is completely determined by its finite-dimensional distributions; specifically, the distributions of the random vectors $[X_{t_1}, \dots, X_{t_n}]^\top$ for every choice of n and $t_1, \dots, t_n$.

By analogy to the one-dimensional case, a random vector $X = [X_1, \dots, X_n]^\top$ taking values in $\mathbb{R}^n$ is said to have a pdf f if, in the continuous case,

$$\mathbb{P}[X \in B] = \int_B f(\mathbf{x}) d\mathbf{x}, \quad (\text{E.4})$$

for all n -dimensional rectangles B . Replace the integral with a sum for the discrete case. The pdf is also called the *joint pdf* of $X_1, \dots, X_n$. The pdfs of the individual components, called *marginal pdfs*, can be recovered from the joint pdf by “integrating out the other variables”. For example, for a continuous random vector $[X, Y]^\top$ with pdf f , the pdf f_X of X is given by

$$f_X(x) = \int f(x, y) dy.$$

E.5 Conditioning and Independence

Conditional probabilities and conditional distributions are used to model additional information on a random experiment. Independence is used to model *lack* of such information.

E.5.1 Conditional Probability

Suppose some event $B \subseteq \Omega$ occurs. Given this fact, event A will occur if and only if $A \cap B$ occurs, and the relative chance of A occurring is therefore $\mathbb{P}[A \cap B]/\mathbb{P}[B]$, provided $\mathbb{P}[B] > 0$. This leads to the definition of the *conditional probability* of A given B :

$$\mathbb{P}[A | B] = \frac{\mathbb{P}[A \cap B]}{\mathbb{P}[B]}, \quad \text{if } \mathbb{P}[B] > 0. \quad (\text{E.5})$$

The above definition breaks down if $\mathbb{P}[B] = 0$. Such conditional probabilities must be treated with more care [11]. Three important consequences of the definition of conditional probability are:

- (1) *Product rule*: For any sequence of events $A_1, A_2, \dots, A_n$,

$$\mathbb{P}[A_1 \cdots A_n] = \mathbb{P}[A_1] \mathbb{P}[A_2 | A_1] \mathbb{P}[A_3 | A_1 A_2] \cdots \mathbb{P}[A_n | A_1 \cdots A_{n-1}], \quad (\text{E.6})$$

using the abbreviation $A_1 A_2 \cdots A_k := A_1 \cap A_2 \cap \cdots \cap A_k$.

- (2) *Law of total probability*: If $\{B_i\}$ forms a partition of Ω (that is, $B_i \cap B_j = \emptyset, i \neq j$ and $\cup_i B_i = \Omega$), then for any event A

$$\mathbb{P}[A] = \sum_i \mathbb{P}[A | B_i] \mathbb{P}[B_i]. \quad (\text{E.7})$$

- (3) *Bayes' rule*: Let $\{B_i\}$ form a partition of Ω . Then, for any event A with $\mathbb{P}[A] > 0$,

$$\mathbb{P}[B_j | A] = \frac{\mathbb{P}[A | B_j] \mathbb{P}[B_j]}{\sum_i \mathbb{P}[A | B_i] \mathbb{P}[B_i]}. \quad (\text{E.8})$$

E.5.2 Independence

Two events A and B are said to be *independent* if the knowledge that B has occurred does not change the probability that A occurs. That is, A, B independent $\iff \mathbb{P}[A | B] = \mathbb{P}[A]$. Since $\mathbb{P}[A | B] \mathbb{P}[B] = \mathbb{P}[A \cap B]$, an alternative definition of independence is

$$A, B \text{ independent} \iff \mathbb{P}[A \cap B] = \mathbb{P}[A] \mathbb{P}[B].$$

This definition covers the case where $\mathbb{P}[B] = 0$ and can be extended to arbitrarily many events: events $A_1, A_2, \dots$ are said to be (mutually) independent if for any k and any choice of distinct indices $i_1, \dots, i_k$,

$$\mathbb{P}[A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_k}] = \mathbb{P}[A_{i_1}] \mathbb{P}[A_{i_2}] \cdots \mathbb{P}[A_{i_k}].$$

The concept of independence can also be formulated for random variables. Random variables $X_1, X_2, \dots$ are said to be *independent* if the events $\{X_{i_1} \leq x_{i_1}\}, \dots, \{X_{i_n} \leq x_{i_n}\}$ are independent for all finite choices of n distinct indices $i_1, \dots, i_n$ and values $x_{i_1}, \dots, x_{i_n}$.

An important characterization of independent random variables is the following.

Theorem E.1: Independence Characterization

Random variables $X_1, \dots, X_n$ with marginal pdfs $f_{X_1}, \dots, f_{X_n}$ and joint pdf f are independent if and only if

$$f(x_1, \dots, x_n) = f_{X_1}(x_1) \cdots f_{X_n}(x_n) \quad \text{for all } x_1, \dots, x_n. \quad (\text{E.9})$$

Many probabilistic models involve random variables $X_1, X_2, \dots$ that are *independent and identically distributed*, abbreviated as i.i.d..

E.5.3 Expectation and Covariance

Similar to the univariate case, the expected value of a real-valued function h of a random vector $X \sim f$ is a weighted average of all values that $h(X)$ can take. Specifically, in the continuous case, $\mathbb{E}h(X) = \int h(\mathbf{x})f(\mathbf{x})d\mathbf{x}$. In the discrete case replace this multidimensional integral with a sum. Using this result, it is not difficult to show that for any collection of dependent or independent random variables $X_1, \dots, X_n$,

$$\mathbb{E}[a + b_1X_1 + b_2X_2 + \cdots + b_nX_n] = a + b_1\mathbb{E}X_1 + \cdots + b_n\mathbb{E}X_n \quad (\text{E.10})$$

for all constants $a, b_1, \dots, b_n$. Moreover, for *independent* random variables,

$$\mathbb{E}[X_1X_2 \cdots X_n] = \mathbb{E}X_1\mathbb{E}X_2 \cdots \mathbb{E}X_n. \quad (\text{E.11})$$

We leave the proofs as an exercise.

The *covariance* of two random variables X and Y with expectations μ_X and μ_Y , respectively, is defined as

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (\text{E.12})$$

This measures the degree of linear dependency between the variables. Let $\sigma_X^2 = \text{Var } X$ and $\sigma_Y^2 = \text{Var } Y$. A scaled version of the covariance is given by the *correlation coefficient*,

$$\varrho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}. \quad (\text{E.13})$$

The following properties follow directly from the definitions of variance and covariance.

1. $\text{Var } X = \mathbb{E}X^2 - \mu_X^2$.
2. $\text{Var}[aX + b] = a^2 \sigma_X^2$.
3. $\text{Cov}(X, Y) = \mathbb{E}[XY] - \mu_X \mu_Y$.
4. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
5. $-\sigma_X \sigma_Y \leq \text{Cov}(X, Y) \leq \sigma_X \sigma_Y$.
6. $\text{Cov}(aX + bY, Z) = a \text{Cov}(X, Z) + b \text{Cov}(Y, Z)$.
7. $\text{Cov}(X, X) = \sigma_X^2$.
8. $\text{Var}[X + Y] = \sigma_X^2 + \sigma_Y^2 + 2 \text{Cov}(X, Y)$.
9. If X and Y are independent, then $\text{Cov}(X, Y) = 0$.

As a consequence of Properties 2 and 8 we have that for any sequence of *independent* random variables $X_1, \dots, X_n$ with variances $\sigma_1^2, \dots, \sigma_n^2$,

$$\text{Var}[a_1 X_1 + a_2 X_2 + \dots + a_n X_n] = a_1^2 \sigma_1^2 + a_2^2 \sigma_2^2 + \dots + a_n^2 \sigma_n^2, \quad (\text{E.14})$$

for any choice of constants $a_1, \dots, a_n$.

For random column vectors, such as $\mathbf{X} = [X_1, \dots, X_n]^\top$, it is convenient to write the expectations and covariances in vector and matrix notation. For a random vector $\mathbf{X}$ we define its *expectation vector* as the vector of expectations

$$\boldsymbol{\mu} = [\mu_1, \dots, \mu_n]^\top = [\mathbb{E}X_1, \dots, \mathbb{E}X_n]^\top. \quad (\text{E.15})$$

The *covariance matrix* Σ is defined as the matrix whose (i, j) -th element is

$$\text{Cov}(X_i, X_j) = \mathbb{E}[(X_i - \mu_i)(X_j - \mu_j)]. \quad (\text{E.16})$$

If we define the expectation of a vector (matrix) to be the vector (matrix) of expectations, then we can compactly write

$$\boldsymbol{\mu} = \mathbb{E}X \quad (\text{E.17})$$

and

$$\Sigma = \mathbb{E}[(X - \boldsymbol{\mu})(X - \boldsymbol{\mu})^\top]. \quad (\text{E.18})$$

A useful application of the cyclic property of the trace of a matrix (see B.1) is the following.

Theorem E.2: Expectation of a Quadratic Form

Let A be an $n \times n$ matrix and X an n -dimensional random vector with expectation vector $\boldsymbol{\mu}$ and covariance matrix Σ . The random variable $Y := X^\top A X$ has expectation $\text{tr}(A\Sigma) + \boldsymbol{\mu}^\top A \boldsymbol{\mu}$.

Proof. Since Y is a scalar, it is equal to its trace. Now, using the cyclic property: $\mathbb{E}Y = \mathbb{E}\text{tr}(Y) = \mathbb{E}\text{tr}(X^\top A X) = \mathbb{E}\text{tr}(A X X^\top) = \text{tr}(A \mathbb{E}[X X^\top]) = \text{tr}(A(\Sigma + \boldsymbol{\mu} \boldsymbol{\mu}^\top)) = \text{tr}(A\Sigma) + \text{tr}(A \boldsymbol{\mu} \boldsymbol{\mu}^\top) = \text{tr}(A\Sigma) + \boldsymbol{\mu}^\top A \boldsymbol{\mu}$. $\square$

E.6 Markov Chains

The index t in X_t is usually seen as a “time” or “step” parameter. The index set $\{0, 1, 2, \dots\}$ in the definition above was chosen out of convenience. It can be replaced by any countable index set. We restrict ourselves to *time-homogeneous Markov chains*, Markov chains for which the conditional pdfs $f_{X_{t+1}|X_t}(y|x)$ do not depend on t ; we abbreviate these as $q(y|x)$. The $\{q(y|x)\}$ are called the (one-step) *transition densities* of the Markov chain. Note that the random variables or vectors $\{X_t\}$ may be *discrete* (e.g., taking values in some set $\{1, \dots, r\}$) or *continuous* (e.g., taking values in an interval $[0, 1]$ or $\mathbb{R}^d$). In particular, in the *discrete* case, each $q(y|x)$ is a probability: $q(y|x) = \mathbb{P}[X_{t+1} = y | X_t = x]$.

Definition E.1: Markov Chain

A *Markov chain* is a collection $\{X_t, t = 0, 1, 2, \dots\}$ of random variables (or random vectors) whose futures are conditionally independent of their pasts given their present values. That is,

$$\mathbb{P}[X_{t+1} \in A | X_s, s \leq t] = \mathbb{P}[X_{t+1} \in A | X_t] \quad \text{for all } t. \quad (\text{E.19})$$

In other words, the conditional distribution of the future variable X_{t+1} , given the entire past $\{X_s, s \leq t\}$, is the same as the conditional distribution of X_{t+1} given only the present X_t . eq. (E.19) is called the *Markov property*.

The distribution of X_0 is called the *initial distribution* of the Markov chain. The one-step transition densities and the initial distribution completely specify the distribution of the random vector $[X_0, X_1, \dots, X_t]^\top$. Namely, we have by the product rule eq. (22.4) and the Markov property that the joint pdf is given by

$$\begin{aligned} f_{X_0, \dots, X_t}(x_0, \dots, x_t) &= f_{X_0}(x_0) f_{X_1|X_0}(x_1|x_0) \cdots f_{X_t|X_{t-1}, \dots, X_0}(x_t|x_{t-1}, \dots, x_0) \\ &= f_{X_0}(x_0) f_{X_1|X_0}(x_1|x_0) \cdots f_{X_t|X_{t-1}}(x_t|x_{t-1}) \\ &= f_{X_0}(x_0) q(x_1|x_0) q(x_2|x_1) \cdots q(x_t|x_{t-1}). \end{aligned}$$

A Markov chain is said to be *ergodic* if the probability distribution of X_t converges to a fixed distribution as $t \rightarrow \infty$. Ergodicity is a property of many Markov chains. Intuitively, the probability of encountering the Markov chain in a state x at a time t far into the future should not depend on the t , provided that the Markov chain can reach every state from any other state, such Markov chains are said to be *irreducible*, and does not “escape” to infinity. Thus, for an ergodic Markov chain the pdf $f_{X_t}(x)$ converges to a fixed *limiting pdf* $f(x)$ as $t \rightarrow \infty$, irrespective of the starting state. For the discrete case, $f(x)$ corresponds to the long-run fraction of times that the Markov process visits x .

Under mild conditions (such as irreducibility) the limiting pdf $f(x)$ can be found by solving the *global balance equations*:

$$f(x) = \begin{cases} \sum_y f(y)q(x|y) & \text{(discrete case),} \\ \int f(y)q(x|y)dy & \text{(continuous case).} \end{cases} \quad (\text{E.20})$$

For the discrete case the rationale behind this is as follows. Since $f(x)$ is the long-run proportion of time that the Markov chain spends in x , the proportion of transitions *out* of x is $f(x)$. This should be balanced with the proportion of transitions *into* state x , which is $\sum_y f(y)q(x|y)$.

One is often interested in a stronger type of balance equations. Imagine that we have taken a video of the evolution of the Markov chain, which we may run in forward and reverse time. If we cannot determine whether the video is running forward or backward (we cannot determine any systematic “looping”, which would indicate in which direction time is flowing), the chain is said to be time-reversible or simply *reversible*.

Although not every Markov chain is reversible, each ergodic Markov chain, when run backwards, gives another Markov chain, the *reverse Markov chain*, with transition densities $\tilde{q}(y|x) = f(y)q(x|y)/f(x)$. To see this, first observe that $f(x)$ is the long-run proportion of time spent in x for both the original and reverse Markov chain. Secondly, the “probability flux” from x to y in the reversed chain must be equal to the probability flux from y to x in the original chain, meaning $f(x)\tilde{q}(y|x) = f(y)q(x|y)$, which yields the stated transition probabilities for the reversed chain. In particular, for a *reversible* Markov chain we have

$$f(x)q(y|x) = f(y)q(x|y) \quad \text{for all } x, y. \quad (\text{E.21})$$

These are the *detailed* (or *local*) *balance equations*. Note that the detailed balance equations imply the global balance equations. Hence, if a Markov chain is irreducible and there exists a pdf such that [eq. \(E.21\)](#) holds, then $f(x)$ must be the limiting pdf. In the discrete state space case an additional condition is that the chain must be *aperiodic*, meaning that the return times to the same state cannot always be a multiple of some integer ≥ 2 .

F Python Primer

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